

Ophthalmology AIIMS 2017

Question 1:

A 62-year-old presented with diminished vision 1 year after cataract surgery. The findings are shown below. What is the most likely diagnosis?



- a) After cataract
- b) Irvine-Gass syndrome
- c) End ophthalmitis
- d) Uveitis-glaucoma-hyphema syndrome

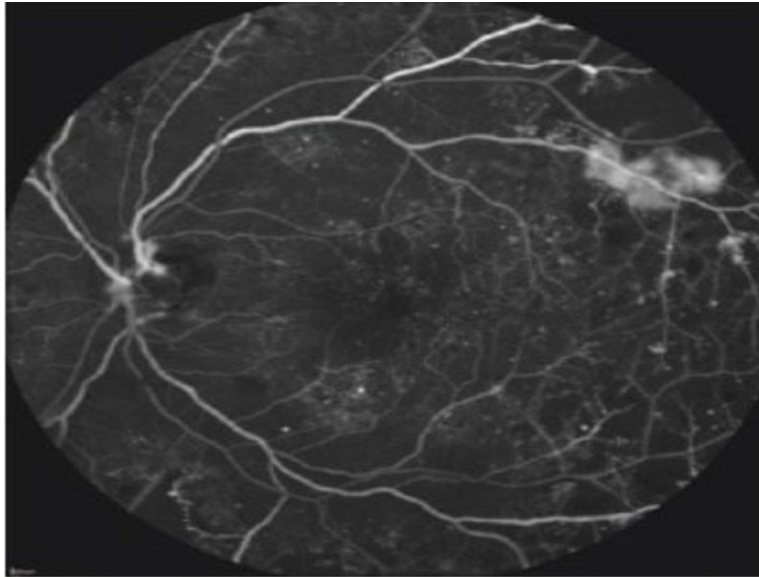
Question 2:

Defect at the level of the optic chiasma will result in which of the following visual field defects?

- a) Central scotoma
- b) Bilateral hemianopia
- c) Binasal hemianopia
- d) Bitemporal hemianopia

Question 3:

A 60-year-old diabetic male presents with a history of decreased vision. Fluorescein angiography showed the following image. What is the most likely diagnosis?



- a) Mild non-proliferative diabetic retinopathy
- b) Severe non-proliferative diabetic retinopathy
- c) Bird shot choroidopathy
- d) Proliferative diabetic retinopathy

Question 4:

What is the most probable diagnosis based on the given image?



- a) 3rd cranial palsy
- b) 6th nerve palsy
- c) Bell's palsy
- d) Horner's syndrome

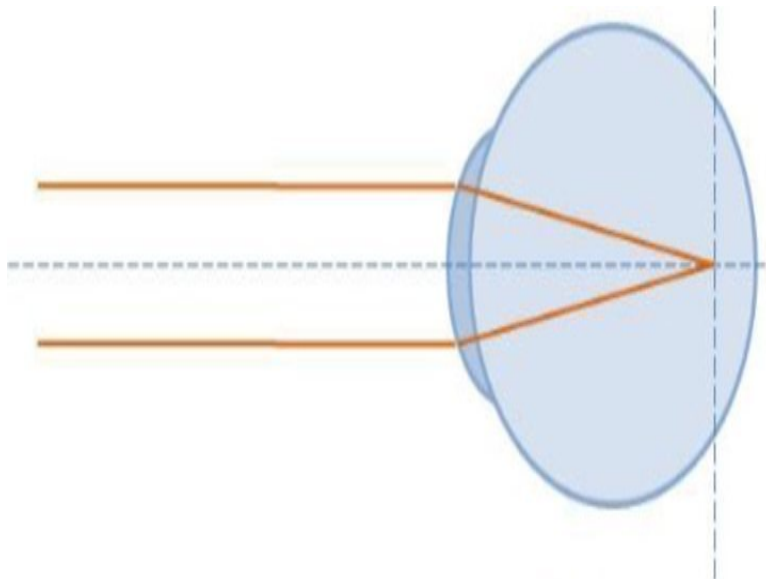
Question 5:

A patient presents with endophthalmitis 5 days after cataract surgery. All of the following treatments can be used, except _____.

- a) Intravitreal antibiotic injection
- b) Intravenous steroids
- c) Pars plana vitrectomy
- d) Topical antibiotics

Question 6:

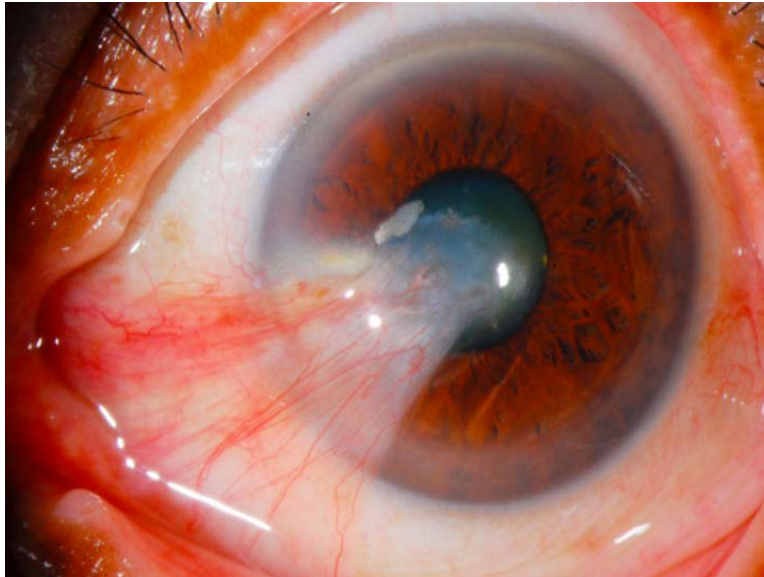
Identify the refractive error.



- a) Myopia
- b) Presbyopia
- c) Hypermetropia
- d) Astigmatism

Question 7:

What is the diagnosis of the condition seen in the image?



- a) Pinguecula
- b) Pterygium
- c) Chemical burn
- d) Thermal burn

Question 8:

A patient with glaucoma presents with a bulging cornea. What is the most likely diagnosis?

- a) Keratoconus
- b) Keratomalacia
- c) Staphyloma
- d) Granular dystrophy

Question 9:

A patient presents with unilateral ptosis with hypotropia in the same eye. He was given an IV injection, following which the symptoms go away. Which of the following is the most likely diagnosis?

- a) Myasthenia gravis

- b) 6th nerve palsy
- c) 3rd nerve palsy
- d) Horner syndrome

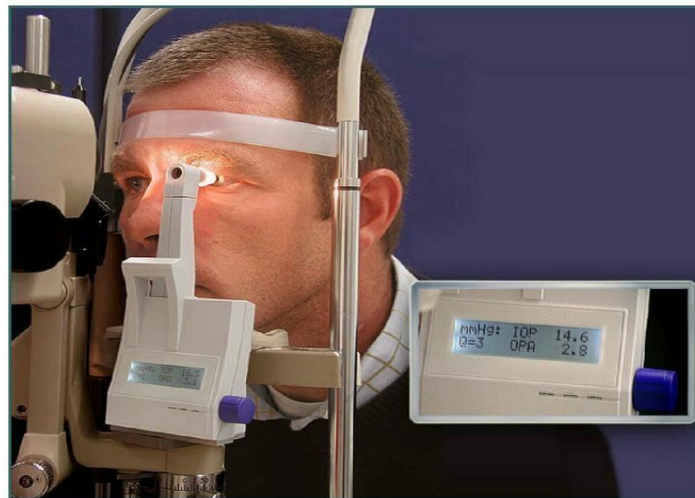
Question 10:

Which is the most common lacrimal gland tumor?

- a) Adenoid cystic carcinoma
- b) Pleomorphic adenoma
- c) Mucoepidermoid carcinoma
- d) Malignant mixed tumor

Question 11:

A patient recently underwent LASIK surgery for his myopia. He is now complaining of headaches and eye pain. Identify the test being performed on him.



- a) Pachymetry
- b) Tonometry
- c) Auto Refractometry
- d) A Scan

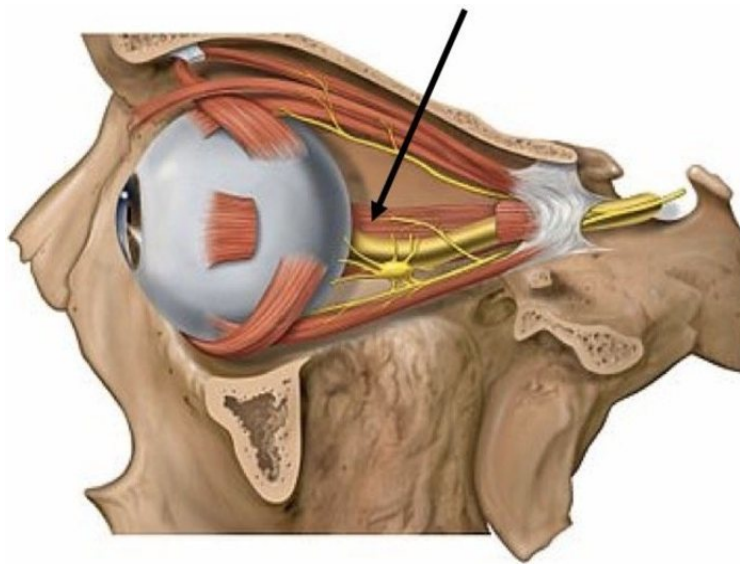
Question 12:

A patient presents to the emergency department and is found to have a corneal abrasion, corneal ulcer, corneal perforation, and foreign body. Which of the following will not cause hypotonic maculopathy in this patient?

- a) Suprachoroidal haemorrhage
- b) Cyclodialysis of uvea
- c) Corneal perforation
- d) Filtration site leak

Question 13:

Name the location of the nucleus supplying the muscle marked by the arrow in the image given below.



- a) Superior olivary
- b) Inferior colliculus
- c) Superior colliculus
- d) Facial colliculus

Question 14:

Silent choroid on FFA is a feature of which of the following disorders?

- a) Stargardt's disease
- b) Best's disease
- c) Age related macular degeneration
- d) Cystoid macular edema

Question 15:

The physician noticed the following finding in a patient that came to the OPD for an eye exam. What is the most likely diagnosis?



- a) Entropion of right eye
- b) Lagophthalmos of right eye
- c) Ptosis of right eye
- d) Exophthalmos of left eye

Answer Key

Question No.	Correct Option
1	a
2	d
3	d
4	b

5	b
6	a
7	b
8	c
9	a
10	b
11	b
12	a
13	c
14	a
15	c

Detailed Explanations

Solution to Question 1:

Diminished vision 1 year after cataract surgery, supported with the image of vacuolated clusters seen behind IOL is suggestive of Elschnig's pearls. It is a type of after cataract.

After cataract, or secondary cataract or posterior capsular opacification, is a late postoperative complication that follows extracapsular extraction of the lens. It is the most common late postoperative complication. It may be caused by residual opaque lens matter or proliferation of left out anterior epithelial cells. Different types of after cataract include:

- Elschnig's pearls: The subcapsular epithelial cells proliferate and develop into vacuolated soap bubble like clusters along the posterior capsule.
- Soemmering's ring: A peripheral dense ring is formed behind the iris and is enclosed between layers of the capsule.
- Dense membranous after-cataract: It is formed in the pupillary area impairing the vision.
- Thin membranous after-cataract: It is thickened posterior capsule.

Types of after-cataract



Dense, membranous after-cataract



Soemmering's ring

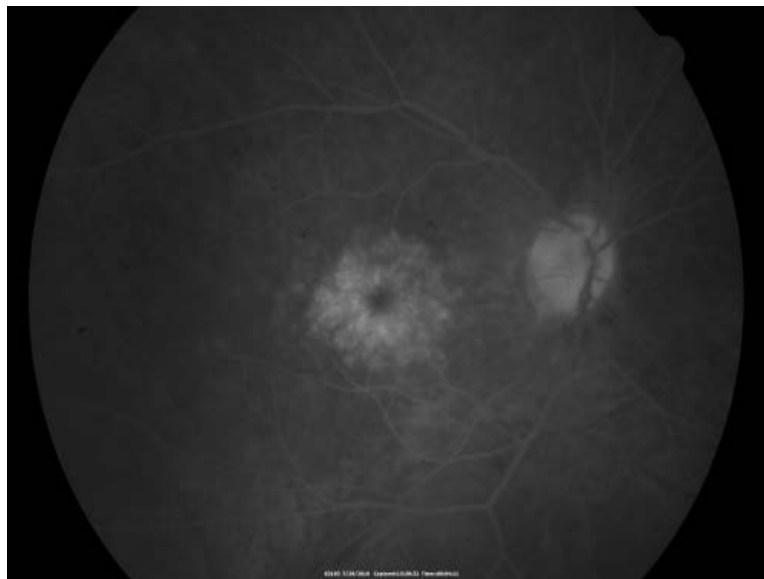


Elschnig's pearls

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Treatment of after cataract is by Nd-YAG laser capsulotomy of the posterior capsule, which works on the principle of photodisruption, or surgical posterior capsulotomy.

Option B: Irvine–Gass syndrome is cystoid macular edema (CME) that develops in post-cataract surgery patients. Though it is mostly clinically insignificant, some cases can present with blurring of vision, one to three months after surgery. In CME, on fundoscopy a honeycomb appearance and on fundus fluorescein angiography(FFA) flower petal pattern can be seen. The below image depicts the flower petal pattern seen on FFA.



Option C: Post operative endophthalmitis presents as ocular pain, diminished vision and hypopyon within months to years after the surgery. Fungal endophthalmitis can rarely occur with puff ball exudates in the vitreous cavity.

Option D: UGH syndrome is where uveitis, glaucoma and hyphema occurs simultaneously. It is seen commonly with rigid anterior chamber intraocular lens (IOL) implantation.

Solution to Question 2:

Lesions at the level of the optic chiasma cause bitemporal hemianopia.

The Nasal fibers of each retina decussate at the optic chiasma. Hence, a lesion at this level affects the temporal quadrants of both eyes.

The optic chiasma lies directly above the tuberculum and diaphragma sellae & is affected in

- Pituitary adenoma
- Craniopharyngioma
- Meningioma
- Gliomas
- Multiple sclerosis, sarcoidosis

Option A: Central scotoma is seen in lesions affecting the optic disc (intraocular part of optic nerve) such as ethambutol toxic neuropathy, macular degeneration and lesions of visual cortex with bilateral homonymous macular defects.

Option C: Binasal hemianopia is seen in lesions affecting bilateral lateral chiasma where the temporal fibers of both the retina continue forward without crossing. This affects vision in the nasal quadrants of both eyes.

Solution to Question 3:

The given clinical vignette and Fluorescein angiography findings of neovascularization are suggestive of proliferative diabetic retinopathy (PDR).

Neovascularization is a hallmark of PDR. So this rules out Non-proliferative diabetic retinopathy-NPDR (Option A,B).

Option C: Birdshot retinochoroidopathy is an idiopathic bilateral chorioretinitis characterized by creamy yellow spots of chorioretinal hypopigmentation resembling a birdshot scatter. It most commonly affects middle aged females who are HLA-A29 positive.

PDR is most common in patients with Type ■ diabetes after a period of 15-20 years and is characterized by proliferation of new vessels at the optic disc (NVD) or elsewhere in the fundus (NVE).

(NVD- neovascularization at the optic disc ; NVE-neovascularization elsewhere in the fundus)

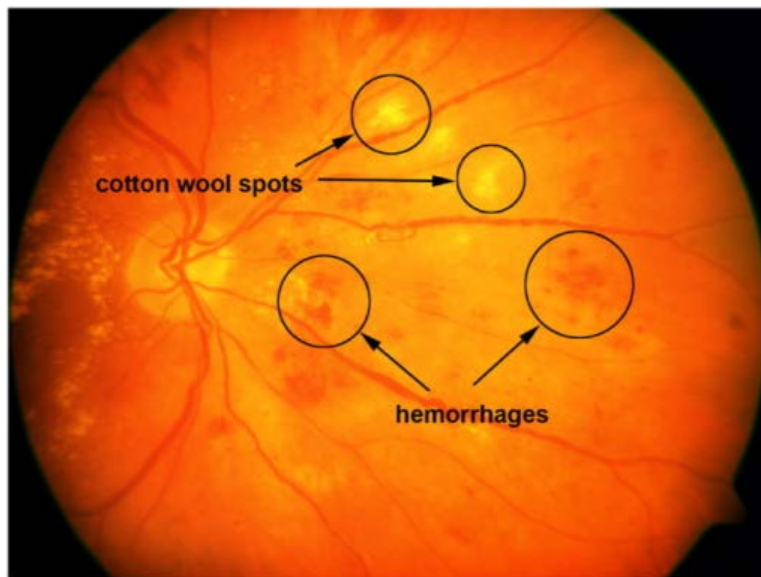
Complications of PDR include recurrent vitreous hemorrhage, tractional retinal detachment and pre retinal hemorrhages.

It is treated by administration of Anti-VEGF drugs followed by pan retinal photocoagulation which destroys the hypoxic retina that has been producing the vasoformative factors responsible for neovascularization.

Duration of diabetes mellitus is the most important factor for the development of diabetic retinopathy.

Ophthalmoscopic features of NPDR include:

- Microaneurysms: due to loss of pericytes around the capillaries
- Retinal hemorrhages: both superficial (flame shaped) and deep (dot & blot) hemorrhages.
- Hard exudates: composed of lipoproteins and lipid laden macrophages. They are yellow waxy looking and arranged in clumps or circinate patterns.
- Cotton wool spots: which are small whitish fluffy lesions which represent areas of nerve infarct.
- Intra retinal microvascular abnormalities(IRMA): are arteriovenous shunts in retina
- Venous abnormalities such as beading,looping and dilatation.



The below given image is that of Hard Exudates.



Solution to Question 4:

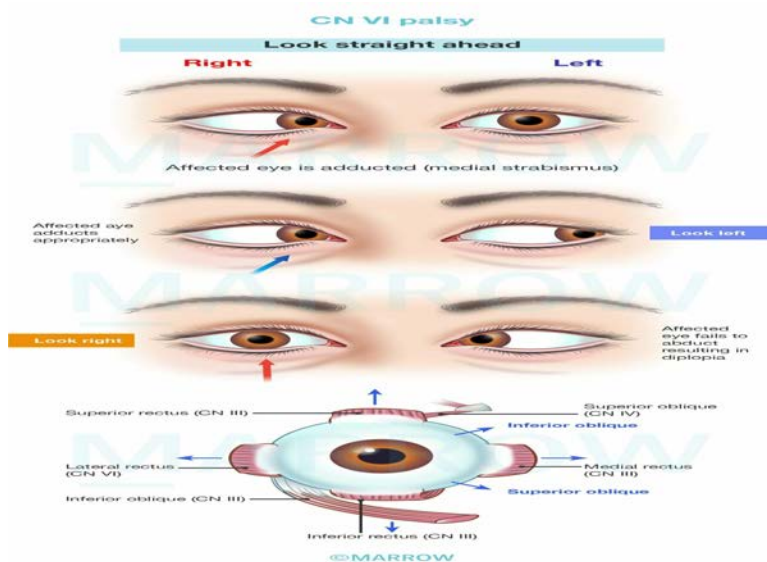
The given image depicts restricted lateral movement of the left eye when the person is in left lateral gaze. This is suggestive of left lateral rectus muscle paralysis due to isolated 6th cranial nerve palsy.

Isolated abducens(6th cranial) nerve palsy presents with esotropia / convergent squint in primary position with limitation of abduction on the side of the lesion.

This is due to unopposed action of the medial rectus muscle in the affected eye. There is also a compensatory head turn towards the side of the lesion to reduce the symptom of diplopia. The adduction is normal in 6th nerve palsy.

The various causes of 6th nerve palsy includes:

- Raised intracranial pressure
- Nasopharyngeal tumors
- Gradenigo syndrome as seen in mastoiditis or acute petrositis
- Basal skull fracture



Option A: 3rd Cranial nerve palsy presents with

- ptosis due to weakness of levator palpebrae superioris muscle(LPS)
- Abduction and depression in the primary position(down and out) due to unopposed action of the lateral rectus(6th nerve supply) and superior oblique muscles(4th nerve supply)
- Dilated pupil and defective accommodation due to parasympathetic palsy(involvement of Edinger Westphal parasympathetic supply)
- Limited adduction, elevation due to ipsilateral paralysis of medial rectus, inferior oblique and inferior rectus .

Signs of Oculomotor Nerve Palsy	
Description	Image
Profound ptosis due to weakness of levator	
Abduction & depression in 1 st gaze due to unopposed action of lateral rectus & superior oblique	
Intorsion due to intact superior oblique	
Limited depression due to weak inferior rectus	
Limited elevation due to weak superior rectus & inferior oblique	
Dilated pupil due to parasympathetic palsy	
Normal abduction	

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Option D: Bell's palsy is an acute onset, idiopathic lower motor neuron lesion of the facial nerve which involves the ipsilateral upper and lower halves of the face.

It presents with an inability to close the eye and any attempts to close the eye results in the eye turning upwards and outwards which is known as the Bell's phenomenon. Other features of Bell's palsy include drooling of saliva, deviation of angle of the mouth and facial asymmetry. Additionally there may be noise intolerance and a loss of taste sensation.

The following image shows a patient with Bell's palsy:



Option D: Horner syndrome, also called oculosympathetic paresis, is produced by a lesion anywhere along the sympathetic pathway supplying the head, neck and eyes. It is characterized by the classical triad of ptosis, miosis and anhidrosis(reduced ipsilateral sweating). Other features include mild enophthalmos, hypochromic heterochromia and anisocoria that is accentuated in dim light.

Tests for confirmation of Horner's syndrome -

- Cocaine drops (2-4%)

Horner pupil does not dilate

Normal pupil dilates

- Apraclonidine drops (0.5-1%)

Horner pupil dilates - due to upregulation of α_1 receptors in the dilator pupillae.

Normal pupil does not dilate

Solution to Question 5:

Intravenous steroids are not used for the treatment of endophthalmitis.

Endophthalmitis is defined as an intraocular inflammation that predominantly affects the inner spaces of the eye and their contents, i.e. the vitreous and/or the anterior chamber. The cardinal features of endophthalmitis are pain, swelling of the lid and diminished vision. Once the endophthalmitis sets in it is important to treat all the cases empirically with a combination of antibiotics and steroids from the beginning.

Steroids are given in endophthalmitis to reduce the inflammation of the tissues. The preferred route of administration is either by intravitreal injections, topical concentrations or oral route. Intravenous route of administration of steroids is not used in endophthalmitis.

Acute postoperative bacterial endophthalmitis occurs within 1-7 days after cataract surgery and the most important factor to prevent it is preoperative administration of 5% povidone iodine.

The various drugs, routes and surgical management used to treat endophthalmitis are as follows:

- The mainstay of treatment is intravitreal injections of antibiotics.

First choice: Vancomycin 1 mg in 0.1 mL plus ceftazidime 2.25mg in 0.1ml.

Second choice: Vancomycin 1 mg in 0.1 mL plus amikacin 0.4mg in 0.1ml.

If there is no improvement, a repeat intravitreal injection should be given after 48 hours

- Topical Solutions of antibiotics are started immediately and used frequently

Vancomycin (50mg/ml) or cefazolin (50mg/ml)

Amikacin (20mg/ml) or tobramycin

- Systemic antibiotics have a limited role in the treatment but are commonly used in practice.

- Steroid therapy includes

Intravitreal injection of Dexamethasone(0.4mg in 0.1ml) which is started along with antibiotics.

Topical Dexamethasone (0.1%)

Oral steroids such as prednisolone are started after 24 hours of intensive antibiotic therapy.

- Supportive therapy are

Cycloplegics such as Atropine 1%

Anti-Glaucoma drugs such as oral acetazolamide or timolol in patients with raised intraocular pressure

- Vitrectomy- Pars Plana Vitrectomy is done if the patient does not improve with the above therapy in 48-72 hours or when the visual acuity is reduced to hand movements near the eye.



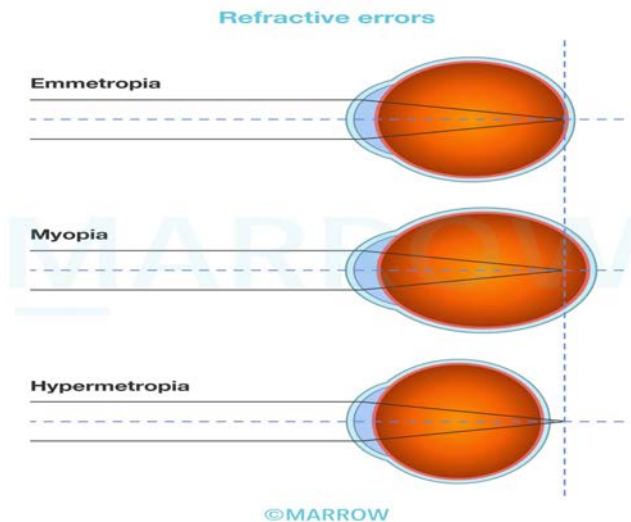
Solution to Question 6:

In the above image, parallel rays are focused in front of the retina and hence the refractive error shown is Myopia.

Myopia, also known as short sightedness, is that dioptric condition of the eye in which, with the accommodation at rest, incident parallel rays come to a focus anterior to the light-sensitive layer of the retina. In myopia the defect in vision is more for objects present at a distance.

American Optometric Association (AOA) Grading of Myopia:

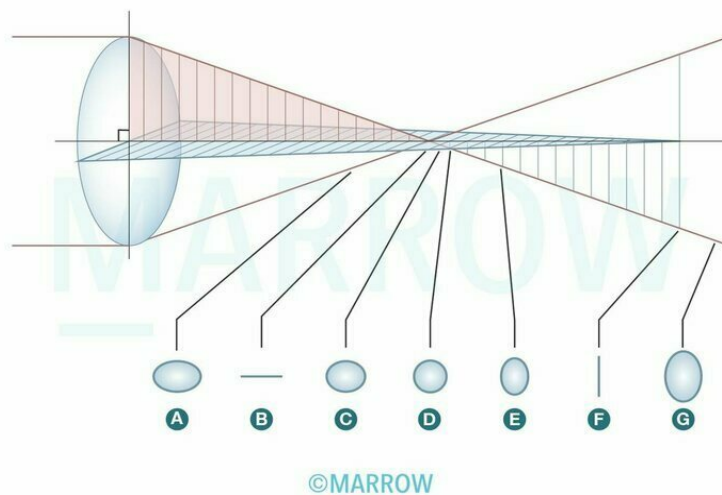
- Low myopia, when the error is $\leq -3D$.
- Moderate myopia, when the error is between $-3D$ to $-6D$.
- High myopia, when the error is $\geq -6D$.



Option B: Presbyopia is used to define physiological insufficiency of accommodation that causes progressive loss of near vision. It is commonly seen in elderly due to age-related decline in ciliary muscle power. Bifocals spectacles are prescribed in presbyopia.

Option C: In hypermetropia, parallel rays of light are focussed behind the retina causing defective vision which is more for near objects.

Option D: In astigmatism, the rays of light entering the eye fail to converge at a point focus but they form focal lines. Strum's conoid of astigmatism represents the configuration of a bundle of light rays formed by a regularly astigmatic(toric) surface and helps in understanding the optics of regular astigmatism.



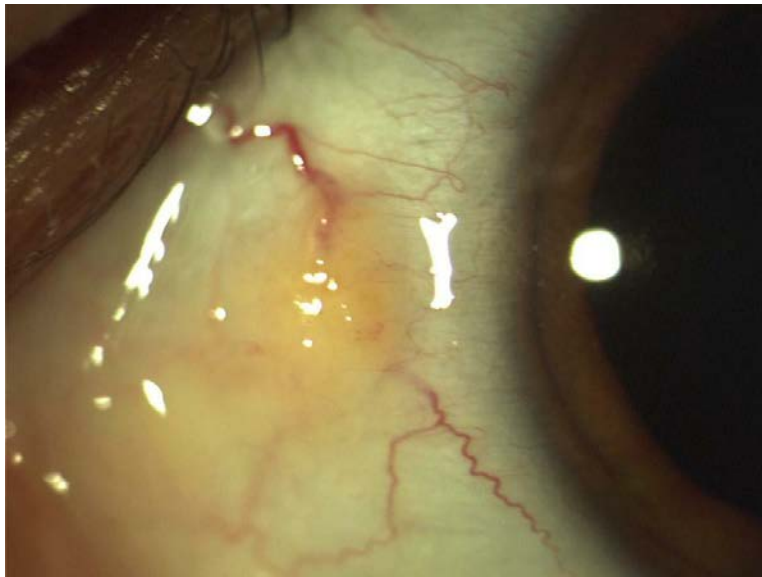
Solution to Question 7:

The above image depicting a wing shaped encroachment of conjunctiva onto the cornea is suggestive of Pterygium.

Pterygium is a triangular fibrovascular ingrowth that is seen following an elastotic degeneration of the subconjunctival tissue. The apex of the pterygium lies towards the cornea. The pterygium is usually seen on the nasal side in the palpable aperture and can give rise to symptoms such as foreign body sensation in the eye, diplopia and defective vision due to corneal astigmatism. In some cases, iron deposition can be seen in the corneal epithelium anterior to the advancing head of the pterygium known as the stocker's line.

Pterygium can be left untreated if there is no visual impairment. In cases with impaired vision, procedures such as surgical excision with free conjunctival limbal graft & surgical excision with amniotic membrane graft and mitomycin-C can be done.

Option A: Pinguecula is an asymptomatic yellow-white mound of degenerated collagen fibers of the conjunctival stroma. It more commonly starts on the nasal limbus and can later be seen on the temporal side. The apex of the Pinguecula is directed away from the cornea. It is mostly a stationary condition, so no treatment is usually required. If inflamed, topical steroids can be applied.



Option C: Chemical burns are caused by alkalis and acids. In mild-moderate burns, the end result is corneal ulceration followed by opacification. Severe burns result in sloughing of the whole cornea followed by staphyloma formation. Complications of chemical burns include symblepharon where the palpebral & bulbar conjunctiva are fused.

Option D: Thermal burns from fire & hot fluids mainly affect the lids. Cornea & conjunctiva will be affected in severe cases.

Solution to Question 8:

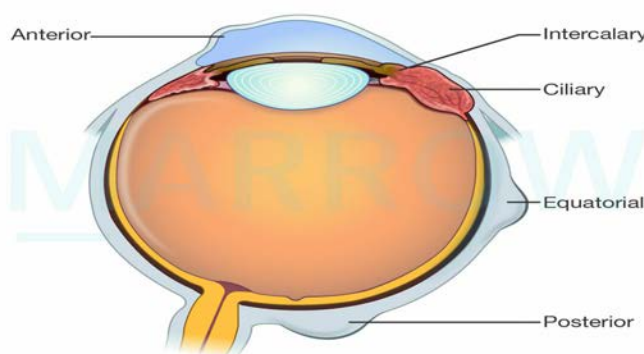
High intraocular pressure leads to the thinning of the outer tunics of the eyeball such as cornea and sclera, thereby resulting in their ectasia (bulging) which is known as

staphyloma. Ciliary and equatorial staphylomas are most commonly associated with glaucoma. Staphyloma is an abnormal protrusion of the uveal tissue through a weak and thin portion of the cornea or sclera. So, staphyloma is lined internally by uveal tissue and externally by weak cornea or sclera.

Types of Staphyloma:

Types	Pathology	Causes
Anterior staphyloma	Due to bulging of the pseudo cornea, internally lined by the iris & externally by the newly formed epithelium.	Sloughing corneal ulcer
Intercalary staphyloma	Due to bulging of weakened limbal tissue at the limbus, internally lined by the root of iris & the anterior most part of the ciliary body	Developmental glaucoma Secondary angle closure glaucoma Healing of perforating injury Peripheral corneal ulcer
Ciliary staphyloma	Due to bulging of the weakened sclera, internally lined by the ciliary body, can extend up to 8mm behind the limbus	Scleritis End stage glaucoma Perforating injuries
Equatorial staphyloma	Due to bulging of sclera with choroidal lining, seen around 14 mm behind the limbus, in the equatorial region	Scleritis Chronic uncontrolled glaucoma Scleral degeneration in pathological myopia
Posterior staphyloma	Due to bulging of sclera lined by the choroid, seen behind the equator	Posterior scleritis Pathological myopia Perforating injuries

Anatomical types of staphyloma



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Keratoconus (Option A) is a progressive bilateral corneal ectatic disorder due to defective synthesis of mucopolysaccharide and collagen tissue. It usually starts at puberty and progresses slowly, not associated with risk factors like glaucoma. Therefore, we can rule out keratoconus in

this patient.

Solution to Question 9:

Only Myasthenia gravis among the conditions mentioned above, would show dramatic improvement after administration of an IV injection like Anti-Cholinesterase, edrophonium.

Myasthenia gravis is an autoimmune disease that is characterized by generalized muscle weakness and rapidly progressing fatigue that improves after short periods of rest. Ocular myasthenia gravis causes progressive external ophthalmoplegia which affects only the extraocular muscles. Early ocular manifestations include failure of convergence & ptosis which is mostly bilateral and increases on prolonged fixation or attempts to look upwards. In the initial phase, there can be unilateral myogenic ptosis, which improves after short rest rapidly.

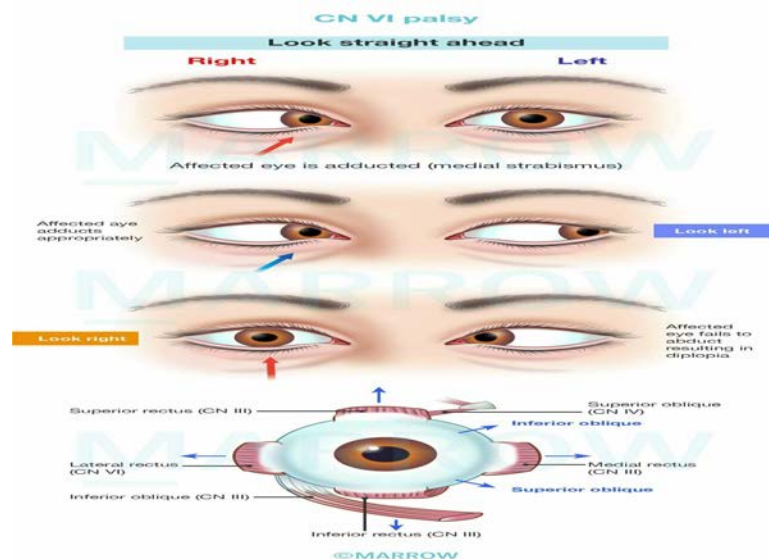
Diagnosis of myasthenia gravis can be made by performing the tensilon test where the symptoms improve for a short period on administering a short acting Anticholinesterase drug such as edrophonium.

Note:

Single most sensitive test for Myasthenia Gravis - Single fiber Electromyography

The single most specific test for Myasthenia Gravis - AChR antibodies

Option B: 6th nerve palsy- isolated 6th nerve palsy presents with convergent squint with limitation of abduction in the affected eye, due to the lateral rectus muscle palsy.



Option A: 3rd Cranial nerve palsy presents with

- ptosis due to weakness of levator palpebrae superioris muscle(LPS)
- Abduction and depression in the primary position(down and out) due to unopposed action of the lateral rectus(6th nerve supply) and superior oblique muscles(4th nerve supply)

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Horner pupil dilates - due to upregulation of α1 receptors in the dilator pupillae.

Normal pupil does not dilate

Solution to Question 10:

Pleomorphic adenoma is the most common lacrimal gland tumor. It is also the most common epithelial tumor of the lacrimal gland.

Pleomorphic adenoma, or benign mixed tumor, presents commonly in middle age as a slowly progressive painless swelling in the upper outer quadrant of the eyelid. The bone is usually not invaded in these tumors. They are treated by surgical removal along with their intact pseudo-capsule.

Malignant tumors of the lacrimal gland such as adenoid cystic carcinoma (option A), mucoepidermoid carcinoma (option C) and malignant mixed tumor (option D) generally occur in the elderly with a short history of appearance, rapid progression and pain. These tumors also present in the upper eyelid and are known to invade the adjacent bones. Radical surgical procedures and high dose radiotherapy are necessary for the removal of the tumor. Adenoid cystic carcinoma is the most common malignant epithelial tumor of the lacrimal gland.

Both malignant and benign tumors are known to push the eyeball downwards and inwards while restricting the upwards and outwards movement of the eyeball.

The most common epithelial tumors, in order of frequency, are as follows:

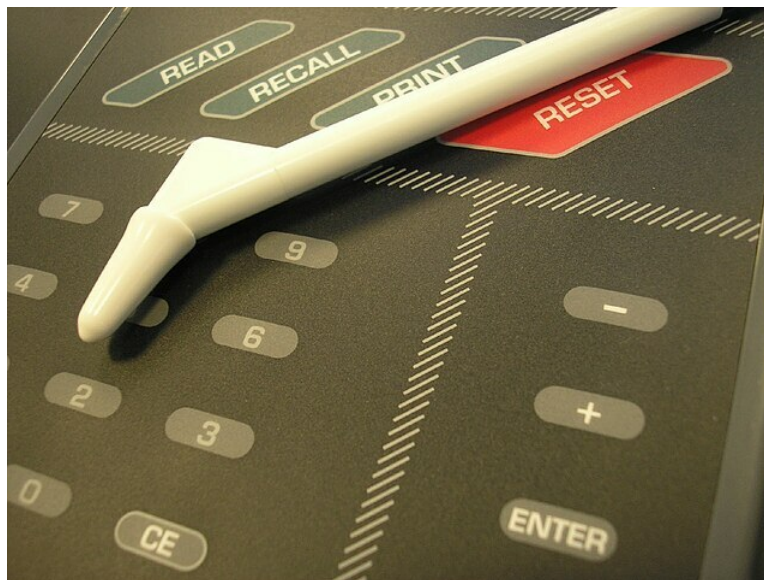
- Benign mixed tumor (BMT), also known as a pleomorphic adenoma (PA)
- Adenoid cystic carcinoma (ACC)
- Pleomorphic adenocarcinoma (ACA)
- Mucoepidermoid carcinoma

Solution to Question 11:

The above image is of the dynamic contour tonometer such as the Pascal tonometer. The dynamic contour tonometer(Pascal) offers a non-invasive and direct method of intraocular pressure (IOP) measurement with an additional measurement of the ocular pulse amplitude. This type of tonometer is unaffected by the corneal thickness and it is the tonometer of choice in post-Lasik patients.

Option A: Pachymetry is used to measure the corneal thickness.

The below image is of pachymetry.



Option C: Auto Refractotometry is a computerized refractometer that is used to find out the error of refraction. This gives rapid information about the kind of refractive error in terms of sphere, cylinder with axis and inter pupillary distance.

Option D: A-Scan ultrasonography is a useful diagnostic tool for intraocular evaluation especially in the setting of opaque media. The report of the A scan is displayed as a unidimensional display of spikes along a baseline where the distance between the spikes gives the indirect measurement of the tissues. A-scan is used to measure

- The axial length of the eyeball for Intraocular lens (IOL) power calculation
- Anterior chamber depth
- Thickness of intraocular mass.

A-scan ultrasonography is shown in the image below.



Solution to Question 12:

Suprachoroidal hemorrhage will not cause hypotonic maculopathy in this patient.

Maculopathy is common in eyes developing hypotony, which is defined as IOP ≤ 6 mmHg. The most common cause is excessive drainage following glaucoma filtration surgery (Option D). Other causes include trauma such as cyclodialysis of the uvea (Option B), penetrating injury (Option C), chronic uveitis, retinal detachment, and other systemic causes such as dehydration, uncontrolled diabetes, uremia, and treatment with hyperosmotic agents. They usually cause bilateral hypotonic maculopathy.

Suprachoroidal hemorrhage is a bleed into the suprachoroidal space that results from the rupture of the posterior ciliary artery. When the bleed is severe it causes extrusion of intraocular contents that is termed as expulsive hemorrhage. It is a rare but dreaded complication of the phacoemulsification procedure. Signs seen in suprachoroidal hemorrhage are:

- Progressive increase in intraocular pressure
- Prolapse of iris
- Loss or partial obscuration of the red reflex

- In severe cases there can be extrusion of the posterior segment contents into the anterior chamber and through the incision such as vitreous extrusion

Immediate treatment involves rapid closure of the incision followed by administration of IOP-lowering medications such as oral acetazolamide and topical or systemic steroids to reduce inflammation. Standard post-operative antibiotics can be added in these patients.

Solution to Question 13:

The muscle marked is the medial rectus. The medial rectus is supplied by the oculomotor nerve, the nucleus of which is situated at the level of the superior colliculus in the posterior midbrain.

The oculomotor (3rd cranial) nerve nucleus consists of

- Single central nuclei for the levator palpebrae muscle of both the eyes
- Bilateral subnuclei with crossed projections supplying the contralateral superior rectus
- Bilateral subnuclei with uncrossed projections supplying the ipsilateral inferior rectus, inferior oblique, medial rectus in the dorso-ventral order.
- Bilateral subnuclei known as the Edinger-Westphal nucleus which provides parasympathetic input to the ciliary body and the iris sphincter.

Option A: Superior olivary nucleus is an important relay station in the auditory pathway, it is responsible for detection of direction and localization of sound.

Option B: Inferior colliculus- The trochlear(4th cranial) nerve nucleus lies in the midbrain at the level of the inferior colliculus. The trochlear nerves leave the midbrain from its posterior surface and decussate before supplying the contralateral superior oblique muscle of the eyeball.

Option D: Facial colliculus- the facial colliculus is located in the cranial part of the pons. It is produced by the facial nerve(7th cranial nerve) winding around the nucleus of the abducens nerve (6th cranial nerve).

Solution to Question 14:

Silent/Dark choroid on FFA is a feature of Stargardt's disease. The classic feature of 'silent choroid' is due to the masking of background choroidal fluorescence by lipofuscin deposits in the subretinal space.

Stargardt disease constitutes the most common type of macular dystrophy. It has mutations in a photoreceptor cell-specific ATP-binding cassette gene (ABC4) that is usually inherited in an autosomal recessive manner. The disease usually starts in the juvenile group with progressive central vision loss and color vision loss that occurs due to the atrophy of the macular retinal pigment epithelium with an accumulation of discrete pisciform flecks of lipofuscin.

Signs of Stargardt's disease:

- Beaten bronze appearance of the macula that is caused by the atrophic changes in the RPE.
- Silent/Dark choroid on FFA.

- Visual field testing shows central vision loss.
- The electroretinogram (ERG) is normal early in the disease but may be reduced in more advanced cases.

Fundus flavimaculatus is genetically considered to be a part of the same spectrum as Stargardt's disease with late onset i.e. usually after 40 years of age.

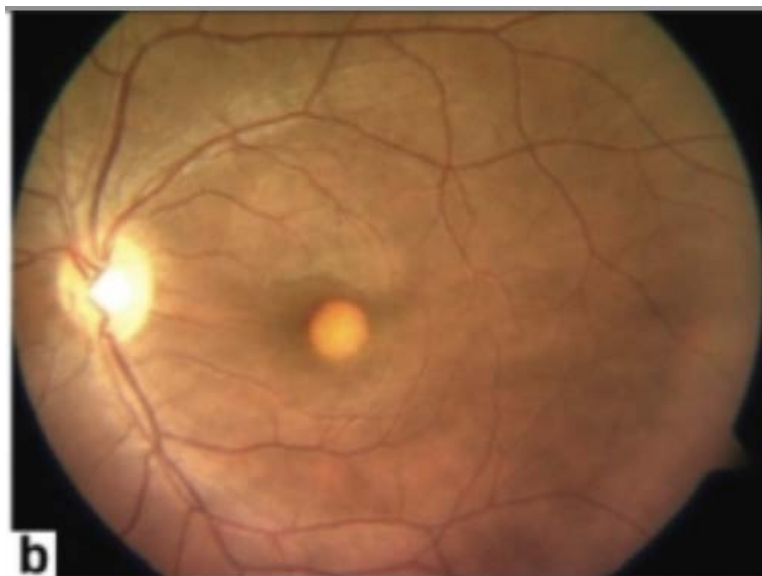


Other options:

Option B: Best's disease also known as vitelliform macular dystrophy is an autosomal dominant macular dystrophy in which lipofuscin accumulates in the central macula, causing progressive central vision loss. This disease is linked to mutations in the bestrophin (VMD2) gene and they do not exhibit silent choroid on FFA. Findings seen in Best's disease are:

- Vitelliform(yolk like) appearance of the fundus
- Relatively normal ERG results associated with abnormal EOG

The given below is the fundus picture of Best's disease showing an egg yolk lesion in the macula.

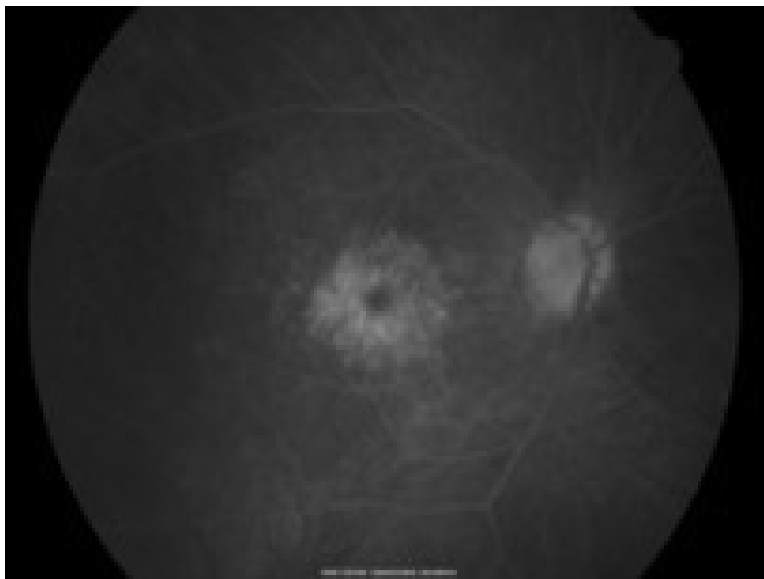


Option C: Age related Macular degeneration(ARMD) also known as senile macular degeneration is a chronic and progressive bilateral degenerative disorder of the macula that affects older individuals >50 yrs. ARMD is classified as dry and wet ARMD where the hallmarks are the presence of drusen and choroidal neovascularization respectively. Drusen are extracellular eosinophilic deposits between RPE and Bruch's membrane.

Clinical features include loss of central vision, distorted vision (lines seen as wavy), and sudden painless loss of vision as seen in wet ARMD. The image below shows Drusens in the macular area in dry macular degeneration.



Option D: Cystoid Macular edema (CME) is a collection of fluid between the outer plexiform layer and the inner nuclear layer of the retina following a break in the blood-retinal barrier. In CME on fundoscopy, a honeycomb appearance and on FFA flower petal pattern can be seen. The below image shows the flower petal pattern on FFA as seen in CME.



Solution to Question 15:

The given image of the patient shows ptosis of the right eye. Ptosis is abnormal drooping of the upper eyelid to a level that covers more than 2mm of the superior cornea, it commonly occurs due to paralysis or defective development of the levator palpebrae superioris (LPS).

Types:

- Congenital ptosis
- Acquired ptosis
- Neurogenic ptosis is caused by innervational defects such as third nerve palsy, Horner's syndrome.
- Myogenic ptosis occurs due to acquired disorders of the LPS muscle or the myoneural junction-Myasthenia gravis.
- Aponeurotic ptosis develops due to defects of the levator aponeurosis.
- Mechanical ptosis may result from the excessive weight on the upper lid as seen in patients with lid tumors, multiple chalazia, and lid edema.

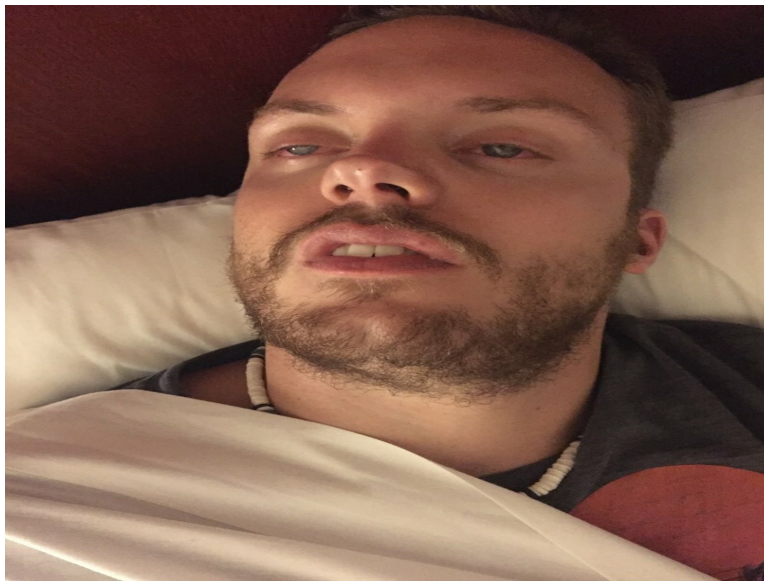
Treatment of ptosis:

- The Fasanella–Servat operation is indicated for cases of minimal ptosis of 1.5 –2 mm with good function of the levator and a good lid fold.
- In moderate-to-severe ptosis with a moderate levator action, levator resection can be carried out.

Note: Pseudoptosis is apparent but not true drooping of the eyelid. Causes include lack of support of the lids by the globe (orbital volume deficit like an artificial eye, microphthalmos, or enophthalmos), contralateral lid retraction, ipsilateral hypotropia or brow ptosis.

Option A: Entropion is defined as inward rolling of the lid margin. The constant rubbing of the eyelashes onto the cornea can cause irritation, punctate corneal erosions and in severe cases ulceration and pannus formation. Weiss operation is done in senile entropion.

Option B: Lagophthalmos- in lagophthalmos there is incomplete closure of the palpebral aperture when the patient attempts to close the eyes due to congenital deformity, paralysis of the orbicularis muscle, ectropion or contraction of the lid. Treatment includes the management of exposure keratopathy with frequent instillation of lubricants.



Option D: Exophthalmos- Although exophthalmos and proptosis are used interchangeably, the term exophthalmos is used to describe the protrusion/prominence of the eyes due to thyroid diseases while proptosis can be due to any other causes. Most common causes of proptosis:

Unilateral proptosis	Bilateral proptosis
In children: orbital cellulitis	Neuroblastoma & leukemia
In adults: Thyroid ophthalmopathy	Thyroid ophthalmopathy

Ophthalmology AIIMS 2018

Question 1:

All of the following are true about oculomotor nerve palsy except _____.

- a) Ptosis
- b) Mid-dilated pupil
- c) Loss of light reflex
- d) Lateral and upward gaze

Question 2:

The center for vertical gaze is _____

- a) Pontine Paramedian Reticular Formation
- b) Raphe Nuclues
- c) Rostral Interstitial Nucleus of Cajal
- d) Nucleus of Perlia

Question 3:

You can use a stenopaeic slit for all of the following except _____.

- a) Fincham's test
- b) Corneal tattooing
- c) Optical Iridectomy
- d) To find out the axis in astigmatism

Question 4:

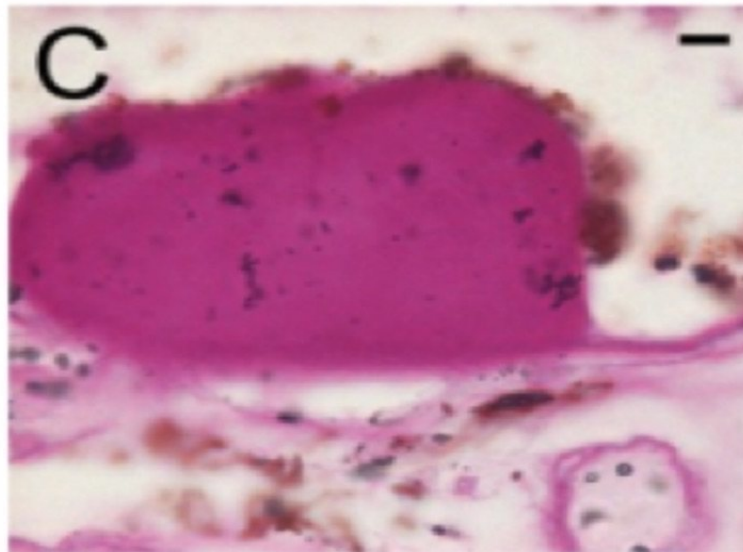
A 55-year-old male patient came to OPD with the complaints of glare during night drives. He had undergone cataract surgery 1 year back. The best-corrected visual acuity is RE = 6/12 and LE = 6/9. No improvement with pinhole. What is the probable diagnosis?

- a) Cystoid macular edema
- b) Pseudophakic bullous keratopathy

- c) ARMD
- d) Posterior capsular opacification

Question 5:

Identify the lesion in the histopathological image of retina.



- a) Drusen
- b) Hard exudates
- c) Cotton wool spots
- d) Macular Edema

Question 6:

Identify the picture



- a) Macular Hole
- b) Cystoid Macular Edema
- c) Smoke stack pattern
- d) Age related Macular degeneration

Question 7:

What is the adjuvant therapy in fungal corneal ulcer?

- a) Pilocarpine e/d
- b) Lignocaine e/d
- c) Dexamethasone e/d
- d) Atropine Sulfate eye ointment

Question 8:

All are ocular manifestations of dengue except _____.

- a) Cataract
- b) Subconjunctival hemorrhage
- c) Uveitis
- d) Retinal hemorrhage

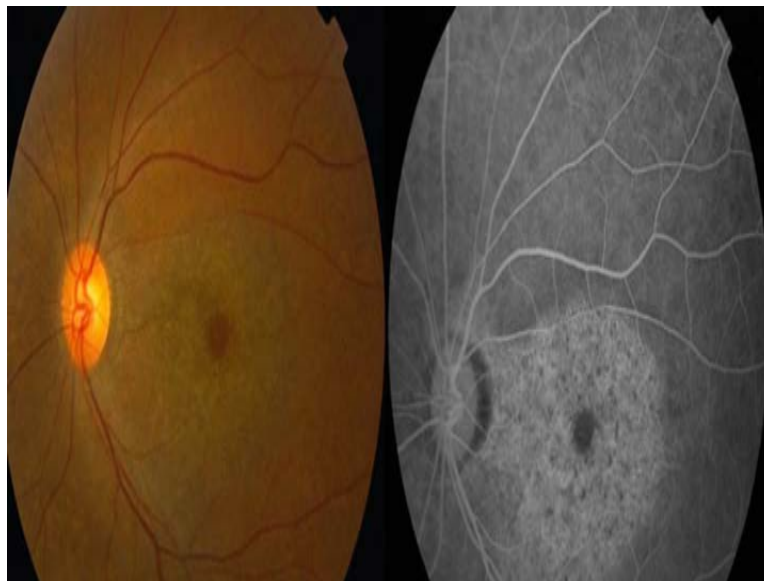
Question 9:

A 5-year-old boy was brought with the complaints of protrusion of the right eye with no h/o fever. CT Scan showed a well-defined mass in the orbit with an irregular border and adjacent bony destruction. Biopsy showed small, round cells which were positive for Desmin in Immunohistochemistry. What is the probable diagnosis?

- a) Retinoblastoma
- b) Cavernous Hemangioma
- c) Orbital Cellulitis
- d) Rhabdomyosarcoma

Question 10:

A 65-year-old female patient with a history of chronic arthritis on treatment came with complaints of defective vision in both eyes. On examination, vision in RE = 6/12, LE = 6/18, IOP = 18 mmHg in both eyes, the anterior segment was normal, fundus examination and fundus fluorescein angiography (FFA) findings given in the figure. What is the probable diagnosis?



- a) Choroiditis
- b) Chloroquine toxicity
- c) Steroid-induced glaucoma
- d) Macular edema

Question 11:

A 30-year-old man presents with a history of injury to the eye caused by a leaf 5 days ago, resulting in pain, photophobia, and redness of the eye for 2 days. What would be the most likely pathology?

- a) Anterior uveitis
- b) Conjunctivitis
- c) Fungal corneal ulcer
- d) Corneal laceration

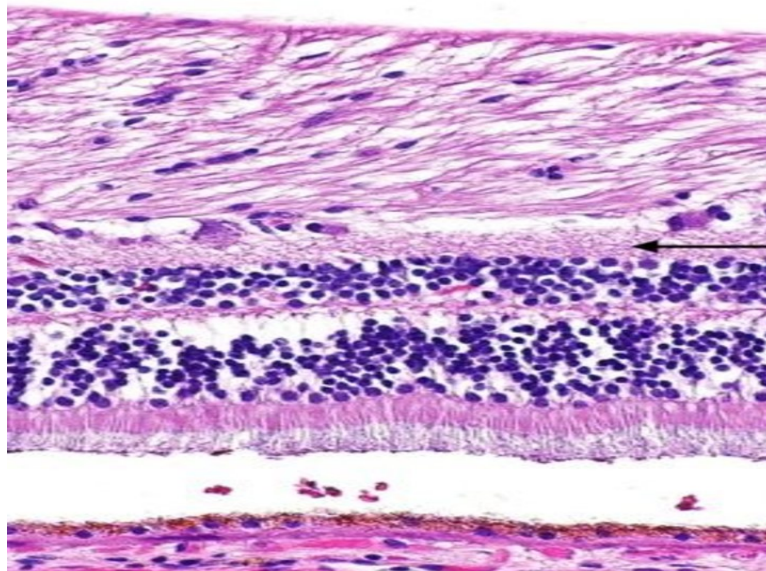
Question 12:

What is the angle subtended by the largest alphabet on the Snellen's chart at 60m distance?

- a) 5
- b) 30
- c) 50
- d) 10

Question 13:

Identify the marked layer of the retina.



- a) Inner plexiform layer

- b) External limiting layer
- c) Internal limiting layer
- d) Outer plexiform layer

Question 14:

The spectacles shown in the image are not used for the management of which of the following?



- a) Presbyopia
- b) Aphakia
- c) Pediatric pseudophakia
- d) Hypermetropia

Question 15:

A 6-year-old girl is admitted for evaluation for recurrent seizures. She has a large port-wine stain on the right side of her face as shown below. The MRI of the head shows a leptomeningeal angioma. What is the most common ocular manifestation of the underlying condition?



- a) Uveitis
- b) Retinitis pigmentosa
- c) Keratitis
- d) Glaucoma

Question 16:

The child in the picture has large corneas, photophobia, and lacrimation. What is the likely diagnosis?



- a) Congenital glaucoma

- b) Retinoblastoma
- c) Congenital hereditary endothelial dystrophy
- d) Megalocornea

Question 17:

The following image indicates:



- a) Trochlear nerve palsy
- b) Lateral rectus palsy
- c) Internuclear ophthalmoplegia
- d) Occulomotor palsy

Question 18:

A 26-year-old female patient presents with 6/60 visual acuity in one eye and 6/18 in the other eye. She has recurrent episodes of diminution of vision which recovers on treatment with steroids. She also has a history of transverse myelitis. What is the most probable diagnosis?

- a) Hereditary spastic quadriplegia
- b) Subacute combined degeneration
- c) Neuromyelitis optica
- d) Multiple sclerosis

Question 19:

Which of the following is the site of lesion in internuclear ophthalmoplegia?

- a) Medial Longitudinal Fasciculus
- b) 6th Nerve nucleus
- c) Pontine paramedian reticular formation
- d) A & C

Answer Key

Question No.	Correct Option
1	d
2	c
3	b
4	d
5	a
6	b
7	d
8	a
9	d
10	b
11	c
12	a
13	a
14	d
15	d
16	a
17	d
18	c
19	a

Detailed Explanations

Solution to Question 1:

Lateral and upward gaze is a sign of trochlear (4th) nerve palsy and not of oculomotor palsy. Oculomotor (3rd) nerve palsy leads to lateral and downward gaze.

3rd Cranial nerve palsy presents with

- Ptosis(option A) due to weakness of levator palpebrae superioris muscle (LPS)
- Abduction and depression in the primary position ("down and out") due to unopposed action of the lateral rectus (6th nerve supply) and superior oblique muscles (4th nerve supply)
- Dilated pupil(option B), loss of light reflex(option C) and defective accommodation due to parasympathetic palsy (involvement of Edinger Westphal parasympathetic supply)
- Limited adduction, elevation and depression due to ipsilateral paralysis of medial rectus, inferior oblique and inferior rectus .

Description	Image
Profound ptosis due to weakness of levator	
Abduction & depression in 1° gaze due to unopposed action of lateral rectus & superior oblique	
Intorsion due to intact superior oblique	
Limited depression due to weak inferior rectus	
Limited elevation due to weak superior rectus & inferior oblique	
Dilated pupil due to parasympathetic palsy	
Normal abduction	

4th cranial nerve palsy presents with superior oblique muscle paralysis.

Clinical findings of trochlear nerve palsy include:

- Hypertropia (elevation) in the primary position
- Extorsion that is greatest in abduction
- Compensatory head posture with contralateral face turn and head tilt with chin slightly depressed. This position is assumed to avoid Diplopia.
- Limitation of depression
- Normal abduction and elevation

The below given image depicts 4th cranial nerve palsy.



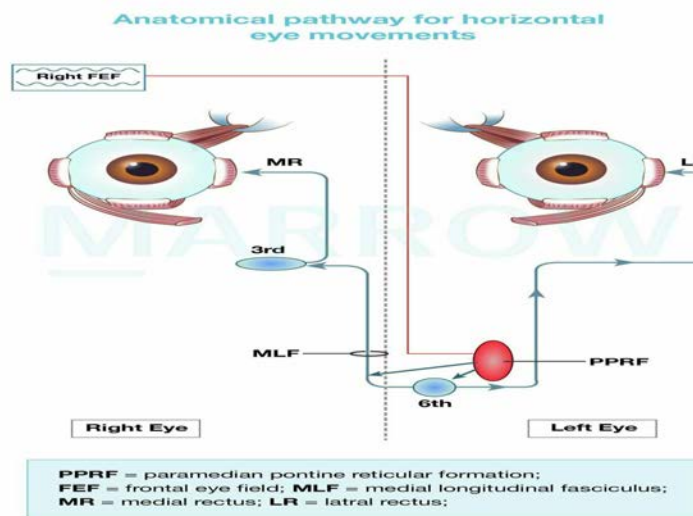
Solution to Question 2:

The Rostral Interstitial Nucleus of Cajal is the center for vertical gaze.

Vertical eye movement initiation in the frontal lobes is mediated by the impulses generated via the vertical gaze center called Rostral interstitial Nucleus of Cajal. This nucleus is a prominent group of cells present within the Medial Longitudinal Fasciculus (MLF) in the midbrain.

Pontine Paramedian Reticular Formation (Option A) is the center for horizontal gaze. Classically, initiation occurs in the premotor cortex frontal eye fields from where impulses for horizontal movement pass to the contralateral paramedian pontine reticular formation (PPRF).

Both horizontal and vertical gaze are saccadic types of conjugate eye movements. Conjugate eye movements, also known as versions, are binocular movements in which the eyes move synchronously in the same direction.



Solution to Question 3:

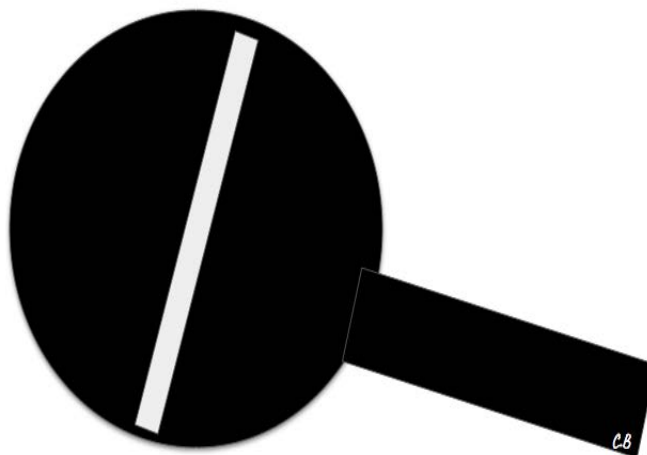
Stenopaic slit is not used for corneal tattooing.

Stenopaic slit is an ophthalmological instrument that is used in:

- Fincham's test (Option A) : This test is used to differentiate between the colored halos due to immature cataract and corneal abnormalities such as corneal edema due to glaucoma.
- To find out the axis in astigmatism (option D) for prescribing the correction glasses.
- To find out the optimal site for optical iridectomy (option C) for treating cases of central macular or leucomatous corneal opacities.

Corneal tattooing was done mainly in the past for cosmetic purposes. It is commonly done for firm scars of corneal opacities in a eye without any useful vision.

The below given image is of stenopaic slit.



Solution to Question 4:

The given clinical picture of a post cataract surgery patient coming with symptoms of glare and reduced visual acuity which doesn't improve on pinhole testing is suggestive of posterior capsular opacification (PCO).

Pinhole testing is done to differentiate between the various causes of reduced visual acuity. On pinhole testing if the visual acuity remains the same or worsens it is suggestive of either a macular disease or posterior capsular opacification.

Posterior capsular opacification also known as secondary cataract or After cataract is a late postoperative complication that follows extracapsular extraction of the lens.

Different types of after cataract include:

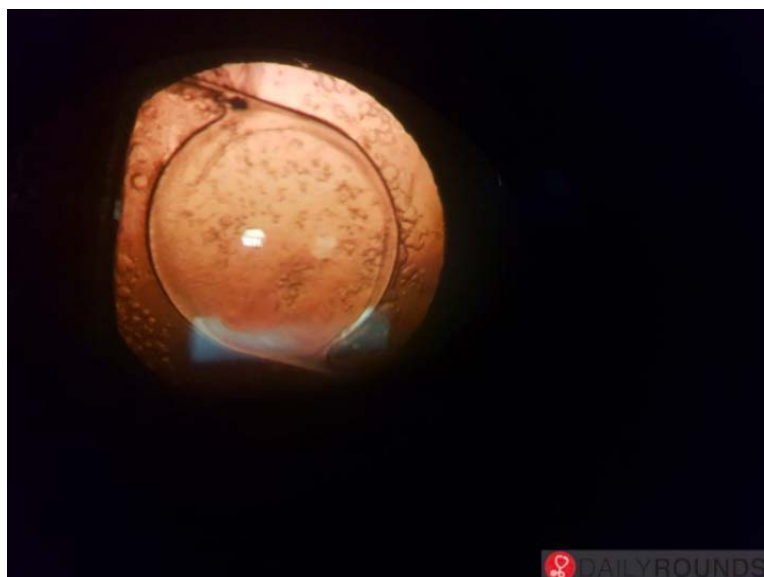
- Elschinig's pearls: where the subcapsular epithelial cells proliferate and develop into vacuolated soap bubble like clusters along the posterior capsule
- Soemmering's ring: A peripheral dense ring is formed behind the iris and is enclosed between layers of the capsule.
- Dense membranous after cataract: formed in the pupillary area impairing the vision.
- Thin membranous after cataract (thickened posterior capsule)

Treatment of after cataract is by:

- Nd-YAG laser capsulotomy of the posterior capsule whose principle is photodisruption.
- Surgical posterior capsulotomy.



The below given image is of Elschinig's pearls type of after cataract.



Option A: Cystoid Macular edema (CME) is a collection of fluid between the outer plexiform layer and the inner nuclear layer of the retina following a break in the blood-retina barrier. In CME, on fundoscopy a honeycomb appearance and on fundus fluorescein angiography (FFA) flower petal pattern can be seen.

The below given image shows the flower petal pattern on FFA as seen in CME



Option B: Pseudophakic bullous keratopathy is a postoperative complication of cataract surgery where the patient presents with corneal edema. This occurs due to a surgical or chemical insult to the corneal endothelium during the cataract surgery. It is treated by penetrating keratoplasty.

Option C: Age related Macular degeneration (ARMD) also known as senile macular degeneration is a chronic and progressive bilateral degenerative disorder of the macula that affects older individuals >50 yrs. ARMD is classified as dry and wet ARMD where the hallmarks are presence of drusen and choroidal neovascularization respectively.

Drusen are extracellular eosinophilic deposits between RPE and Bruch's membrane. Clinical features include loss of central vision, distorted vision (lines seen as wavy) and sudden painless loss of vision as seen in wet ARMD. The below given image shows Drusens in the macular area in Dry macular degeneration.



Solution to Question 5:

The below given image shows extracellular eosinophilic deposits between retinal pigment epithelium(RPE) and bruch's membrane which is suggestive of drusens.

Drusens are seen in dry age related macular degeneration(ARMD) and are composed of lipids, complement factors and other cellular components.

Age related Macular degeneration(ARMD) also known as senile macular degeneration is a chronic and progressive bilateral degenerative disorder of the macula that affects older individuals >50 yrs.

It is the leading cause of blindness in elderly in developed countries and its risk factors include smoking, exposure to sunlight, hypertension, female gender and nutritional deficiencies.

ARMD generally starts as dry ARMD which is characterized by the presence of drusen and gradually progresses to the advanced form known as wet ARMD whose hallmark is choroidal neovascularization with additional findings such as geographic atrophy of retinal pigmentary epithelium (RPE) and photoreceptors at the macula.

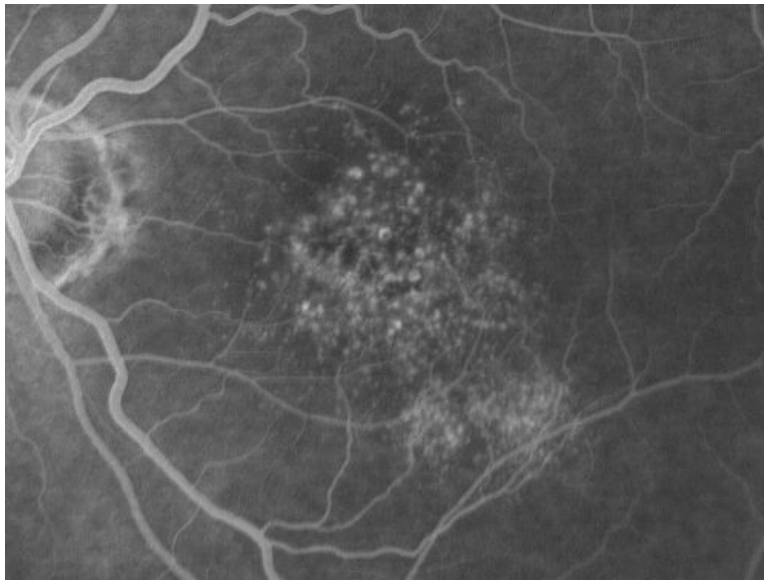
Clinical features include loss of central vision, distorted vision(lines seen as wavy) and sudden painless loss of vision as seen in wet ARMD.

On fundus fluorescein angiography (FFA) the Drusens shows hyperfluorescent spots arranged in 'stars in the sky' appearance.

Disease management is by:

- Preventative measures to delay progression such as cessation of smoking, intake of good nutrition and supplementation of vitamins C,E and zinc.
- Treatment modalities with intravitreal Anti-VEGF injection- Bevacizumab (avastin) and Ranibizumab (Lucentis), photodynamic therapy (PDT) and submacular surgery.

The below given images show drusens in macular area and FFA depicting the 'stars in sky appearance' of drusens respectively.

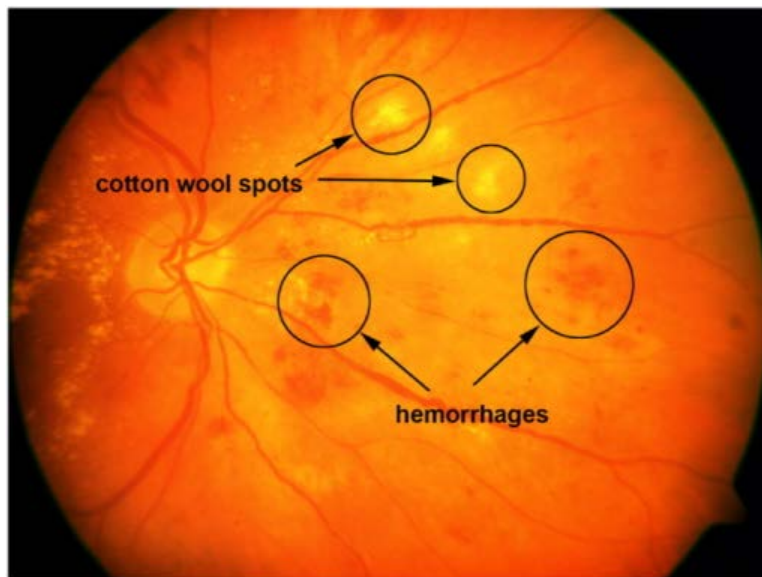


Option B: Hard exudates are composed of lipoproteins and lipid laden macrophages. They are yellow waxy looking and arranged in clumps or circinate patterns. They are seen in diabetic retinopathy, hypertensive retinopathy and coat's disease.

The below given image is that of Hard Exudates.



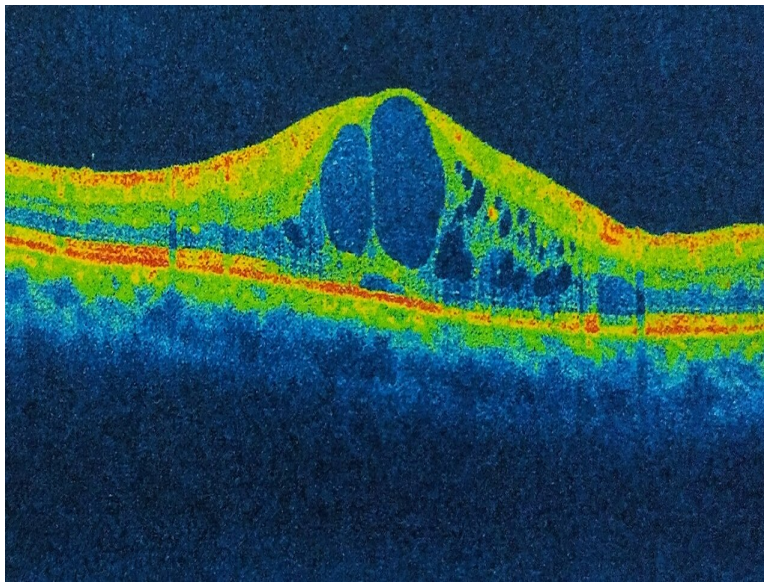
Option C: Cotton wool spots also known as soft exudates are small whitish fluffy lesions which represent areas of nerve infarcts in the nerve fiber layer of the retina. They are seen in central retinal vein occlusion (CRVO), hypertensive retinopathy, diabetic retinopathy, toxæmic retinopathy of pregnancy and papilledema.



Option D: Cystoid Macular edema (CME) is a collection of fluid between the outer plexiform layer (Henle's layer) and the inner nuclear layer of the retina.

In CME, on fundoscopy a honeycomb appearance and on FFA, flower petal pattern can be seen. Optical coherence tomography (OCT) of CME shows loss of foveal depression and intraretinal cystoid spaces.

The below given OCT image shows intraretinal cystoid spaces seen in CME.



Solution to Question 6:

The given Fundus fluorescein angiography(FFA) image depicts the flower petal pattern that is seen in Cystoid macular edema.

Cystoid Macular edema(CME) is a collection of fluid between the outer plexiform(Henle's) layer and the inner nuclear layer of the retina following a break in the blood-retina barrier.

CME can manifest in the following conditions

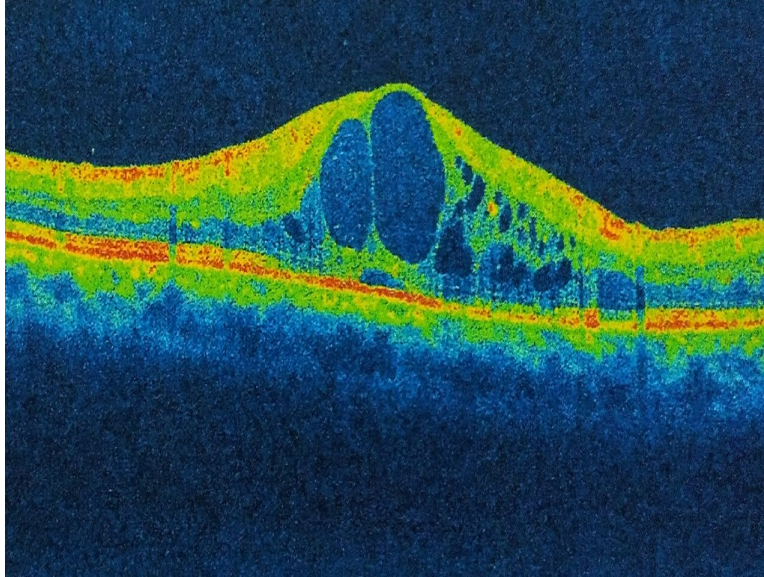
- Post cataract surgery- Pseudophakics (called Irvine-Gass syndrome)
- Topical prostaglandins side effect(especially in diabetics)
- Niacin(Vitamin B3) toxicity
- Epinephrine(especially in Aphakia)
- Uveitis
- Retinitis pigmentosa
- Central and branched retinal venous occlusion
- Diabetic retinopathy
- Retinopathy of prematurity

Clinical features include sudden painless diminution of vision, blurring of vision and metamorphosis such as micropsia.

In CME, on fundoscopy, a honeycomb appearance and on FFA, flower petal pattern can be seen. On Optical coherence tomography(OCT), loss of foveal depression and intraretinal cystoid spaces can be seen.

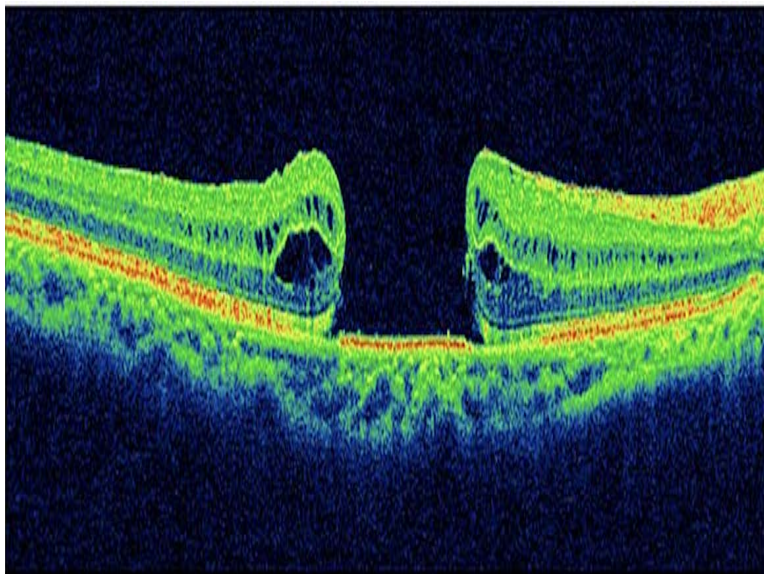
Treatment of CME is by treatment of causative factor, administration of anti-prostaglandins such as NSAIDS(ketorolac) preoperatively and topical or systemic steroids.

The below given OCT image shows intraretinal cystoid spaces seen in CME.



Option A: Macular hole is a partial or full-thickness hole seen in the neurosensory retina of the foveal region. It could be idiopathic, senile or caused due to trauma or CME. There is a decrease in vision and metamorphopsia seen in this condition. Amsler's grid testing shows a central scotoma. FFA shows absent foveal reflex with a macular hole.

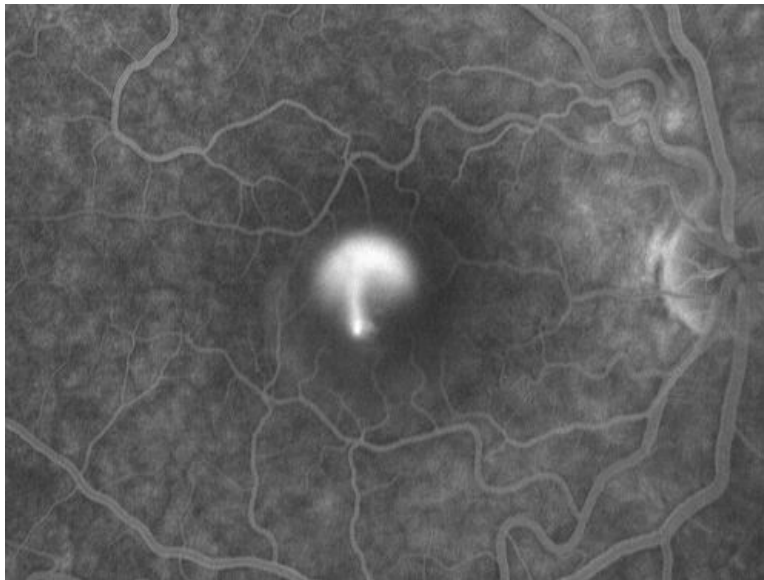
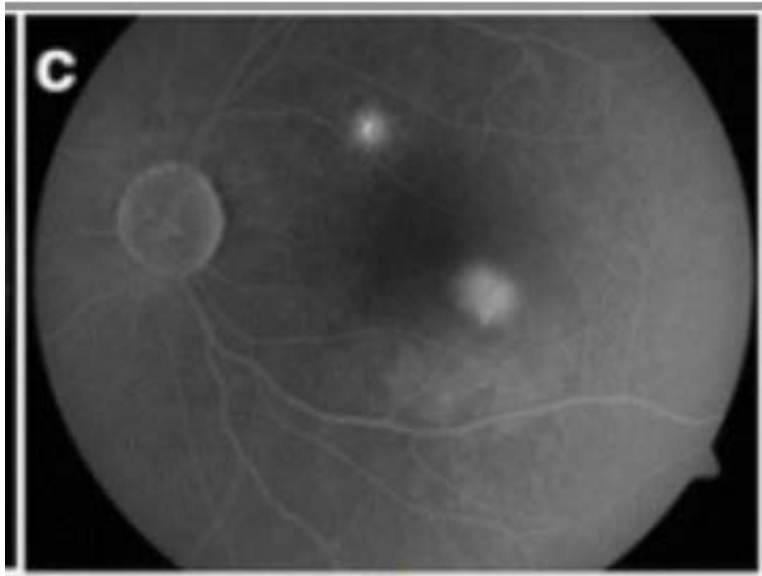
An OCT image of a full-thickness macular hole is shown below:



Option B: smoke stack pattern is seen on FFA in central serous retinopathy(CSR). CSR is the spontaneous serous detachment of the sensory neuroretina in the foveal region with or without retinal pigment epithelium (RPE) detachment occurring due to choroidal vascular hyperpermeability.

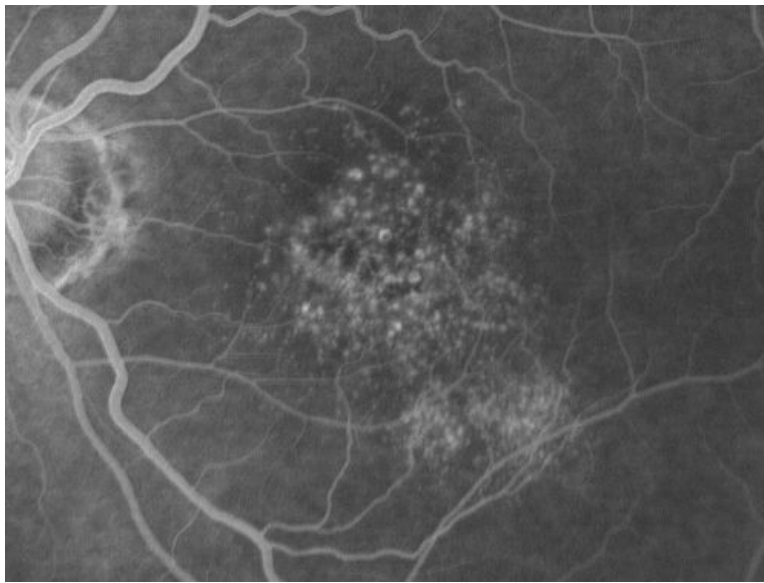
On FFA it is initially seen as an ink-blot pattern or smoke stack pattern of hyperfluorescence which then spreads to take the configuration of mushroom or umbrella pattern.

The below given images depict the smoke stack pattern and mushroom pattern respectively.



Option D: Age related Macular degeneration (ARMD) also known as senile macular degeneration is a chronic and progressive bilateral degenerative disorder of the macula that affects older individuals >50 yrs. ARMD generally starts as dry ARMD which is characterized by the presence of drusen and gradually progresses to the advanced form known as wet ARMD whose hallmark is choroidal neovascularization with additional findings such as geographic atrophy.

Drusen are extracellular eosinophilic deposits between retinal Pigment Epithelium and Bruch's membrane. The below given FFA image shows stars in sky appearance of Drusen.



Solution to Question 7:

Atropine sulfate (1%) eye ointment or eye drops is used as the nonspecific supportive treatment in both bacterial as well as a fungal corneal ulcer.

Atropine sulphate (1%) is a cycloplegic drug that is used as an adjunct in the treatment of corneal ulcers to

- reduce pain from ciliary spasm
- Prevent posterior synechiae formation that can develop from secondary iridocyclitis
- Increase blood supply to the anterior chamber that gets more antibodies into aqueous humor
- Reduce exudation by decreasing hyperpermeability of the vessels

Another cycloplegic that can be used as an adjunct is homatropine 2%.

Topical steroids such as Dexamethasone(option C) will increase fungal replication and corneal invasion and hence are contraindicated in the treatment of fungal corneal ulcers.

Management of fungal corneal ulcer includes

Definitive treatment:

- Topical antifungal eye drops: used for 6-8 weeks
- Filamentous fungi such as Aspergillus and Fusarium- Drug of choice (DOC) is natamycin (5%). Others include amphotericin B (0.1-0.3%), fluconazole (0.2%) or voriconazole 10%.
- Yeast such as Candida- Drug of choice (DOC) is Amphotericin B. Others include Nystatin 3.5%.
- Intracameral or intracorneal administration of Voriconazole in cases with intraocular extension or anterior chamber involvement.
- Systemic antifungals such as tablet fluconazole are given in severe cases of deeper fungal keratitis.

- Therapeutic penetrating keratoplasty for unresponsive cases.

Adjunct therapy:

- Cycloplegics : Atropine sulphate 1% or homatropine 2%
- Systemic analgesics and anti-inflammatory drugs such as paracetamol, ibuprofen to decrease pain and edema.
- Vitamins A,B-complex,C- to aid healing of the ulcer.
- Hot fomentation, dark goggles, rest, good diet.

Solution to Question 8:

Cataract is not an ocular manifestation of dengue.

Dengue has varied manifestations, in different organ systems, mainly of inflammation and bleeding. The mechanisms for ocular manifestations in dengue include a direct viral effect on the eye, an immune-mediated vasculitis, and a variety of secondary effects related to dengue-induced vascular leakage and hemostatic abnormalities.

Ocular manifestations of dengue are:

- visual impairment which can be either painless or painful
- Retro orbital pain
- Conjunctival injection and subconjunctival hemorrhage(option B)
- Uveitis(option C)
- Complications such as retinal hemorrhage(option D), retinal edema, macular ischemia, macular edema, vitreous hemorrhage, vitritis, and optic neuritis or papillitis.

Depending on the severity of the disease, gradual improvement in vision occurs, although in rare cases permanent visual impairment may be seen.

Solution to Question 9:

The clinical scenario of a patient with complaints of protrusion of eye with well-defined mass in the orbit and adjacent bony destruction on CT scan and biopsy showing small, round cells positive for Desmin favors rhabdomyosarcoma.

Cavernous hemangioma(option B) and orbital cellulitis(option C) usually are associated with fever, hence ruled out. Retinoblastoma(option A) is associated with Flexner–Wintersteiner rosettes on microscopy, therefore its not correct.

Rhabdomyosarcoma is the most common soft tissue sarcoma of childhood usually in the head and neck region, most common primary orbital malignancy in children. They are malignant tumors arising from voluntary muscle and produce a rapidly increasing proptosis. It extends in all directions. Diagnosis is done using biopsy in which cross-striations in the tumor cells are pathognomonic. Myoglobulin is a specific marker but it can also be positive for desmin.

Subtypes include embryonal(majority), alveolar, botryoid and pleomorphic.

Treatment includes combination of chemotherapy and radiotherapy.

- Two injections of vincristine, cyclophosphamide, and actinomycin D are given at weekly intervals before radiotherapy.
- Radiotherapy is given during which vincristine and cyclophosphamide are administered weekly.
- After radiotherapy, a combination of vincristine, cyclophosphamide, and doxorubicin is given for a year or longer in those patients in whom metastases were detected.

Other options:

Option A: Retinoblastoma is the most common intraocular malignancy in children. It is associated with mutations in the RB1 gene at chromosome 13q14. White pupillary reflex or leukocoria is the most common presenting feature, followed by a squint which is usually convergent.

Histopathology of the tumor shows small round cells with large nuclei. Microscopic features includes Flexner-Wintersteiner rosettes(highly specific of retinoblastoma), Homer-Wright rosettes, pseudorosettes, and fleurettes formation. Areas of necrosis and calcification can also be seen.

Option B: Cavernous hemangioma is the most common primary benign orbital tumor in adults. They are composed of large, dilated vascular channels; are more infiltrative; frequently involve deep structures; and do not spontaneously regress. On histologic examination, the mass is sharply delineated but unencapsulated with large, cavernous blood-filled vascular spaces, separated by connective tissue stroma.

Option C: Orbital cellulitis is the most common cause of unilateral proptosis in children. It refers to the acute infection of the contents of the orbit behind the orbital septum. Features include pain, lid edema, erythema, chemosis, and axial proptosis. Increased pain with movement of eyes and ophthalmoplegia(decreased ocular motility) is a classic symptom. With posterior exenteration cavernous sinus thrombosis, subdural empyema, and intracranial abscess develop.

Solution to Question 10:

The given clinical vignette of female patient on treatment for chronic arthritis which is most likely the drug chloroquine given for rheumatoid arthritis, now presenting with decreased visual acuity, fundoscopy with a typical bull's eye sign and Fundus fluorescein angiography showing increased fluorescence in the macular area is suggestive of chloroquine toxicity.

Chloroquine and its derivative hydroxychloroquine (HCQ) are antimalarial drugs which is also used to treat connective tissue disorders such as Rheumatoid arthritis, Systemic Lupus Erythematosus (SLE) and Discoid Lupus.

The maximum dose allowed for chloroquine is 6 mg/kg in 24 hours while for hydroxychloroquine it is 4.5 mg/kg. Hydroxychloroquine has a lower risk of ocular toxicity and hence is perceived to be a safer drug but with weaker anti-inflammatory effect.

Manifestations of Chloroquine toxicity:

- Reversible on stopping the drug: difficulty in night vision, blurring of vision, diplopia, keratopathy

- Central and paracentral scotomas
- Earliest fundus findings: blunting of foveal reflex, irregularity in the macular pigmentation.
- Advanced stages: 'Bull's eye maculopathy' where the central irregular pigmentation may become surrounded by a concentric zone of hypopigmentation
- End-stage appearance includes bullseye maculopathy, pigmentary degeneration of retina and bone spicule formation, vascular attenuation, and optic atrophy.

Screening should include a baseline examination for all patients before starting these drugs followed by annual screening starting after 5 years of use.

Pre-existing maculopathy is considered a contraindication to treatment with chloroquine and HCQ because underlying abnormalities may mask early toxicity.

Screening examinations include visual acuity testing, dilated fundus examination, spectral-domain optical coherence tomography (SD-OCT) testing, and functional assessment with Humphrey test.

Various causes of Bull's eye maculopathy are:

- Chloroquine and HCQ toxicity
- Cone-Rod dystrophy
- Advanced stargardt's disease
- Clofazimine retinopathy

Solution to Question 11:

The given clinical vignette of the patient presenting with pain, photophobia, and redness of the eye following trauma by a vegetative matter is suggestive of a fungal corneal ulcer.

Fungal corneal ulcer also known as mycotic keratitis is usually caused by trauma from vegetative matter such as branches of trees, leaves and animal tails. Injudicious use of topical steroids and antibiotics increases the predisposition to develop secondary fungal keratitis.

Mycotic keratitis can be caused by

- Filamentous fungi such as *Aspergillus* and *Fusarium*
- Yeasts such as *Candida* and *Cryptococcus*
- Dimorphic fungi such as *Histoplasma* and *Blastomyces*

Typically in fungal corneal ulcers, the signs are more prominent than the symptoms. The symptoms of fungal keratitis are similar to that of bacterial ulcers which include pain, lacrimation, photophobia, blurred vision, and redness of the eyes.

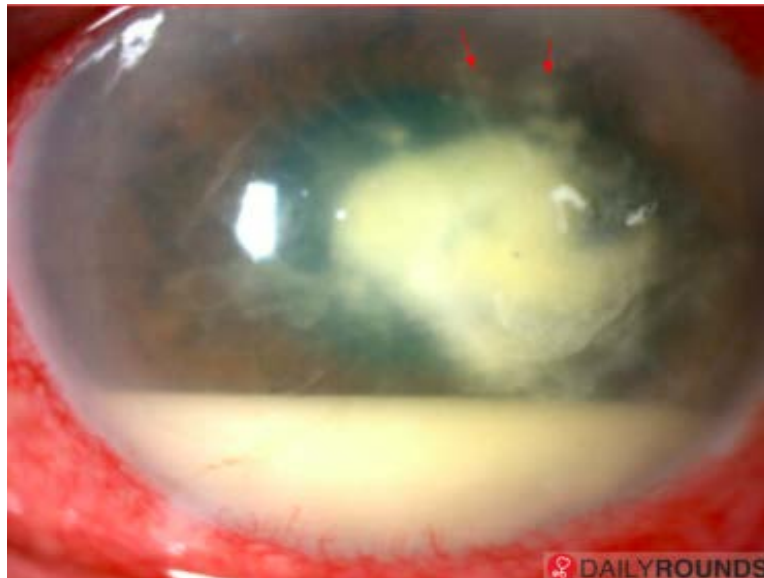
A typical fungal ulcer has the following signs:

- Greyish white looking dry corneal ulcer with rolled out margin.
- Feathery finger-like extensions into the corneal stroma under intact epithelium

- Wessley ring which is a sterile immune ring that occurs due to the deposition of immune complexes
- Multiple small satellite lesions around the ulcer
- Hypopyon: The hypopyon in fungal ulcer is immobile and unsterile unlike bacterial ulcer (sterile hypopyon)
- Perforation is rare
- Corneal vascularization is usually absent.

A diagnosis of fungal ulcer is made by a typical history of trauma with vegetative matter followed by clinical examination and KOH mount or SDA agar cultures from corneal scrappings.

The image below depicts a fungal corneal ulcer with the arrowhead indicating a satellite lesion.



Other options:

Option A: Anterior uveitis is the inflammation of the uveal tissue from the iris up to the pars plicata of the ciliary body. Infectious uveitis can be caused by the exogenous entry of the organisms following penetrating injuries, perforation of corneal ulcer, and post-intraocular operations.

Acute anterior uveitis presents with symptoms of pain, photophobia, lacrimation, blurring of vision and redness of the eye. Signs of anterior uveitis include:

- Aqueous flare - It is the earliest sign of acute anterior uveitis, it occurs following the leakage of protein particles into the aqueous humor.
- Keratic precipitates(KP) - These are pathognomonic for anterior uveitis. They are proteinaceous cellular deposits at the back of the cornea arranged in a triangular fashion. These include mutton fat KP's as seen in granulomatous uveitis and granular KP's of non granulomatous uveitis.
- Iris nodules: typically occur in granulomatous uveitis. Koppe's nodules are at the pupillary border and Busacca's nodules are situated near the collarette.

Option C: Conjunctivitis is the inflammation of the conjunctiva of which the bacterial conjunctivitis is the most common. Acute bacterial conjunctivitis is characterized by marked

conjunctival hyperemia, swelling of the conjunctiva(chemosis), edematous eyelids, sticking together of lid margins and mucopurulent discharge from the eye.

The image shows a patient with bacterial conjunctivitis.



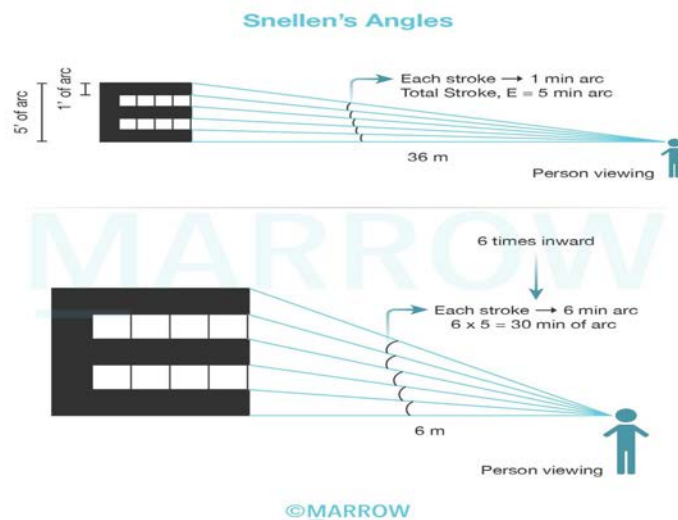
Option D: Corneal laceration: Partial corneal laceration or lamellar corneal laceration occurs from a blunt trauma and is treated by patching and antibiotics.

Solution to Question 12:

The angle subtended by the largest alphabet on the Snellens chart at 60m distance is 5 minutes.

Visual acuity is a measure of the spatial resolution of the eye or an estimation of its ability to discriminate between two points. Snellens chart is used for Visual acuity assesment.

In snellens chart the alphabets decrease in size as we move from above-downwards and these lines, from above-downwards should be read at 60, 36, 24, 18, 12, 9, 6, 5 and 4 m respectively. At these distances, the respective letters subtend a visual angle of 5 minutes. Hence, the largest alphabet that should be read at 60 m subtends a visual angle of 5 minutes in the eye.



Solution to Question 13:

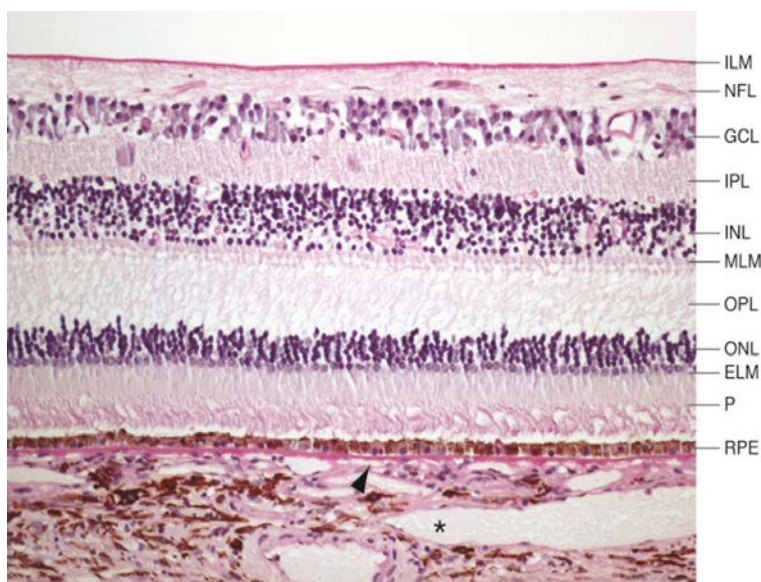
The inner plexiform layer is the marked layer in the given histological image of retina, identified as the fourth layer from the uppermost (innermost) layer.

The retina consists of 10 layers. The retinal pigment epithelium (RPE) is the outermost layer - below the Bruch's membrane of the choroid - and the internal limiting membrane is the innermost layer - separating the retina from the vitreous.

The retinal pigment epithelium consists of a single layer of cells containing pigment. It is separated from the neurosensory retinal layers by the potential space between them which is known as interphotoreceptor matrix.

The inner plexiform layer consists of synapsing nerve fibers. The axons of bipolar cells synapse with dendrites of ganglion cells; and both these processes synapse with processes

of amacrine cells. It also contains some horizontally placed internal plexiform cells; and also a few ganglion cells.



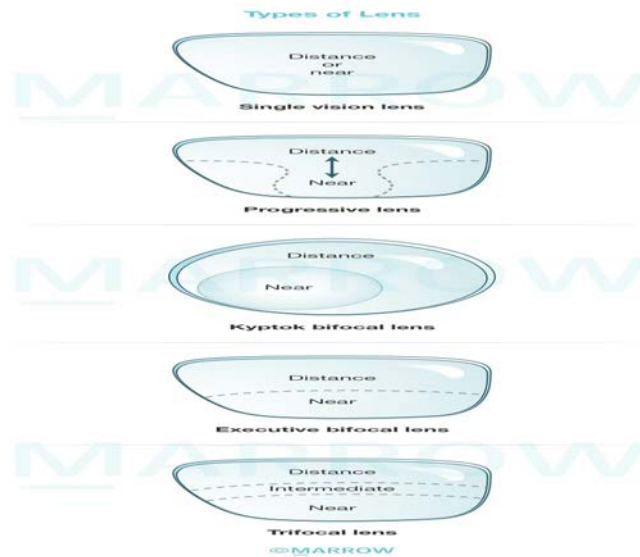
Solution to Question 14:

The given image shows spectacles made of lenses with two different optical power. Thus, it is a pair of bifocal spectacles which is not used in hypermetropia or myopia.

Bifocal glasses have different corrective powers in the upper segment for distant vision and lower segment for near vision. Thus, they are useful in conditions where both distance and near vision are affected, like the following:

1. Presbyopia (option A): It is a definitive indication. It is used to correct age-related near and far vision defects.
2. Aphakia (option B): It provides both near and far vision correction when intra-ocular lens insertion is contraindicated.
3. Paediatric pseudophakia (option C): It is used to correct refractive errors or accommodative fluctuations post intra-ocular lens (IOL) insertion.
4. Non refractive accommodative esotropia

In hypermetropia there is defect in near vision only and hence a single vision glasses having uniform corrective power through the entire surface is used instead of bifocal glasses.



Solution to Question 15:

The given clinical picture of a young child presenting with unilateral port wine stain in the region of trigeminal nerve supply and leptomeningeal angioma as seen on MRI is indicative of Sturge-Webber syndrome (SWS). Glaucoma is the most common ocular manifestation of SWS.

Sturge weber syndrome is a neurocutaneous disorder(phakomatoses) that is characterized by unilateral capillary malformation at the region of trigeminal supply known as port wine stain or nevus flammeus, ipsilateral diffuse cavernous hemangioma of the choroid and ipsilateral leptomeningeal hemangiomatosis, which causes seizures and mental retardation.

Ocular manifestation of SWS include:

- Glaucoma - most common ocular manifestation. It develops anywhere between infancy to adulthood. When developed in infancy it presents as buphthalmos.
- Diffuse choroidal hemangioma
- Serous retinal detachment
- Visual field defects- homonymous hemianopia due to the leptomeningeal hemangiomas affecting the optic tract or occipital lobe.
- Heterochromia of the iris
- Episcleral and conjunctival lesions

Option A: Uveitis is the inflammation of the uveal tissue of the eyes. It can be associated with a number of systemic conditions such as sarcoidosis, behcet's disease, ankylosing spondylitis, reiter's syndrome, Toxoplasmosis and Posner–Schlossman syndrome (PSS).

Option B: Retinitis pigmentosa is an inherited or acquired pigmentary retinal dystrophy affecting the rods more. The earliest manifestation is worsening of night blindness which gradually progresses to tunnel vision and ultimately loss of central vision. The classic triad of findings in RP comprises bone-spicule retinal pigmentation, arteriolar attenuation and waxy disc

pallor.

Systemic Associations of RP include:

- Laurence-Moon-Biedl syndrome
- Cockayne's syndrome
- Refsum's syndrome
- Usher's syndrome
- Bassen-Kornzweig syndrome

The below given image shows the classical triad findings of Retinitis pigmentosa.



Option C: Keratitis is the inflammation of cornea which is characterised by corneal edema, cellular infiltration and ciliary congestion. It can be either infectious or non-infectious in nature which includes allergic keratitis, neurotrophic keratitis, exposure keratitis and peripheral ulcerative keratopathies such as Rosacea keratitis and Mooren's ulcer.

Solution to Question 16:

The given clinical scenario of an infant with large corneas, lacrimation and photophobia supported with the image is suggestive of congenital glaucoma.

Primary congenital glaucoma (PCG) also known as developmental glaucoma refers to abnormally high intraocular pressure (IOP) which results from the developmental anomaly of the angle of the anterior chamber.

It can be classified as:

- True congenital or newborn glaucoma in which IOP is elevated during intrauterine life and the baby is born with ocular enlargement.
- Infantile glaucoma: which manifests prior to age 3 years.

- Juvenile glaucoma: IOP rises between 3 and 16 years of age.

Clinical findings include:

- Blepharospasm
- Buphthalmos: Large eyeball due to stretching of the globe from elevated IOP prior to the age of 3 years.
- Corneal enlargement:
- Corneal haze is due to corneal edema.
- Blue sclera - due to the shining of the underlying uveal tissue through the thinned-out sclera
- Haab striae are healed breaks in Descemet's membrane

Treatment:

- Goniotomy - when the angle of the anterior chamber is visible through the clear cornea.
- Trabeculectomy when the corneal clouding prevents visualization of the angle.
- Filtration procedures such as Trabeculectomy or combined trabeculectomy with trabeculectomy if the above procedures fail.
- Glaucoma drainage devices in unresponsive cases.

Other options:

Option A: Retinoblastoma is the most common primary intraocular malignancy of childhood that is seen between 1-2 years.

Findings of Retinoblastoma include:

- Leukocoria (white pupillary reflex)- the most common manifestation
- Strabismus
- Painful red eye with secondary glaucoma
- An endophytic tumor grows into the vitreous as a white mass that may 'seed' into the gel. If calcification is present it gives a "cottage cheese appearance". An exophytic tumor grows outwards and causes overlying exudative retinal detachment.

The image below shows the leucocoria of the left eye.



Option C: Congenital hereditary endothelial dystrophy (CHED) is an inherited corneal dystrophy that manifests since birth with endothelial degeneration. It appears as diffuse bilateral corneal edema in the absence of elevated IOP. The clinical features of the disease include corneal clouding, visual impairment, and nystagmus.

Option D: Megalocornea is a bilateral non-progressive X-linked condition that most commonly affects males. In this condition, the horizontal corneal diameter is $\geq 12\text{mm}$ (< 2 years) and is $\geq 13\text{mm}$ (≥ 2 years). It is characterized by a clear cornea, normal corneal thickness, and no breaks in the descemet's membrane. Systemic associations of Megalocornea include Marfan's syndrome, Apert syndrome, Ehlers-Danlos, and Down syndrome.

Solution to Question 17:

The given image is suggestive of oculomotor nerve palsy.

The central image shows the primary position of gaze where the patient's left eye is deviated down and out with ptosis of the left eyelid. From the eye movements in other positions, it can be seen that adduction and elevation are absent and depression is reduced in the left eye. All of these findings are indicative of oculomotor nerve palsy.

3rd Cranial nerve(oculomotor nerve) palsy presents with:

- Ptosis due to weakness of levator palpebrae superioris muscle (LPS)
- Abduction and depression in the primary position (down and out) due to unopposed action of the lateral rectus(6th nerve supply) and superior oblique muscles(4th nerve supply)
- Dilated pupil and defective accommodation due to parasympathetic palsy (involvement of Edinger Westphal parasympathetic supply)
- Limited adduction, elevation and depression due to ipsilateral paralysis of medial rectus, inferior oblique and inferior rectus .

Signs of Oculomotor Nerve Palsy	
Description	Image
Profound ptosis due to weakness of levator	
Abduction & depression in 1 st gaze due to unopposed action of lateral rectus & superior oblique	
Intorsion due to intact superior oblique	
Limited depression due to weak inferior rectus	
Limited elevation due to weak superior rectus & inferior oblique	
Dilated pupil due to parasympathetic palsy	
Normal abduction	

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Option A: Trochlear(4th cranial) nerve palsy presents with superior oblique muscle paralysis.

Clinical findings of trochlear nerve palsy include:

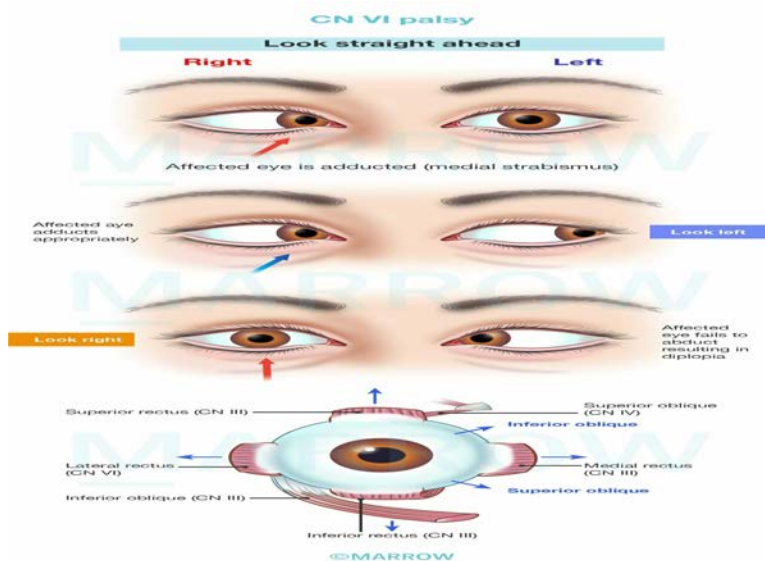
- Hypertropia (elevation) in the primary position
- Extorsion that is greatest in abduction
- Compensatory head posture with contralateral face turn and head tilt with chin slightly depressed. This position is assumed to avoid Diplopia.
- Limitation of depression
- Normal abduction and elevation.

The below given image depicts 4th cranial nerve palsy.



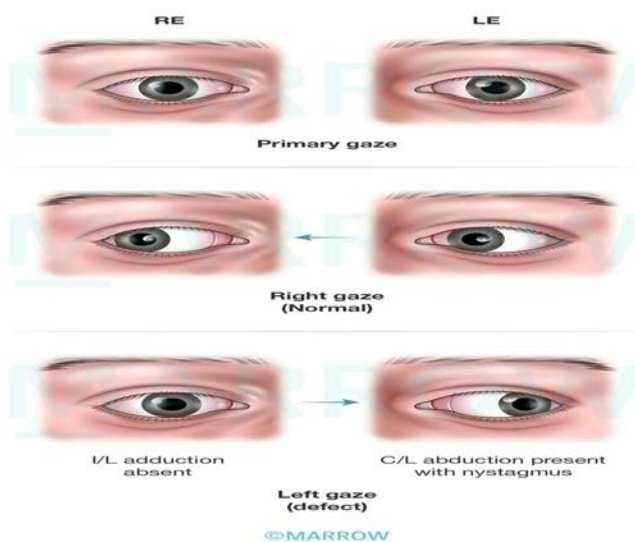
Option B: Lateral rectus palsy is seen in cases of isolated Abducens nerve palsy. These patients present with esotropia/convergent squint in primary position with limitation of abduction on the

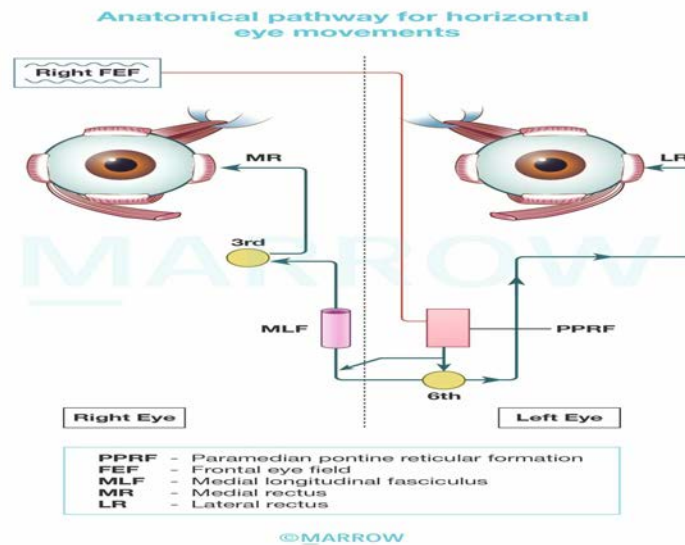
side of the lesion. This is due to unopposed action of the medial rectus muscle in the affected eye. There is also a compensatory head turn towards the side of the lesion to reduce the symptom of Diplopia. The adduction is normal in lateral rectus palsy.



Option C: Internuclear ophthalmoplegia occurs following an injury to the medial longitudinal fasciculus (MLF). A unilateral lesion to the MLF leads to the failure of adduction in the ipsilateral eye with a normal abduction of the contralateral eye.

The below given image depicts the pathogenesis of Inter nuclear ophthalmoplegia in detail.





Solution to Question 18:

The given clinical vignette of a patient presenting with decreased visual acuity and a history of transverse myelitis which is treated with steroids is indicative of Neuromyelitis optica (NMO).

Neuromyelitis optica also known as Devic's disease is an autoimmune demyelinating disease in which antibodies are formed against Aquaporin 4 present in the cell membranes of Astrocytes. It is characterized by the findings of unilateral or bilateral optic neuritis developing within a few weeks of transverse myelitis (demyelination of the spinal cord). The disease may be monophasic or recurrent.

Diagnostic Criteria for Neuromyelitis Optica Spectrum Disorder (NMOSD):

Diagnostic Criteria for NMOSD associated with AQP4-IgG

- At least 1 core clinical characteristic
- A positive test for AQP4-IgG using the best available detection method (cell-based assay)
- Exclusion of alternative diagnoses

Core Clinical Characteristics:

- Optic neuritis is usually bilateral and causes severe visual loss.
- Acute myelitis- may be severe, transverse and typically longitudinally extensive involving three or more contiguous vertebral segments.
- Area postrema syndrome: episodes of otherwise unexplained hiccups or nausea or vomiting
- Acute brainstem syndrome
- Symptomatic narcolepsy associated with NMOSD-typical diencephalic MRI lesions

For acute cases of NMO, high-dose glucocorticoids or plasma exchange is used. Mycophenolate mofetil, rituximab, and glucocorticoids with azathioprine are used for prophylaxis against relapses.

The FDA has approved satralizumab, a humanized IgG2 monoclonal antibody, for the treatment of NMO spectrum disorder with AQP4 antibody positivity.

Eculizumab, a complement inhibitor, and inebilizumab, a B-cell depletion agent, are two more new drugs that have recently been approved for use in NMO attack prevention.

A history of transverse myelitis with optic neuritis(decreased visual acuity) rules out Multiple sclerosis(option D). It is an autoimmune inflammatory and demyelinating disease of the CNS that affects both upper motor neurons (UMN) and lower motor neurons (LMN) with involvement of sensory system, ocular system and bowel and bladder function. Ocular manifestations of MS include Optic neuritis(usually retrobulbar)- decreased visual acuity, internuclear ophthalmoplegia and nystagmus.

Solution to Question 19:

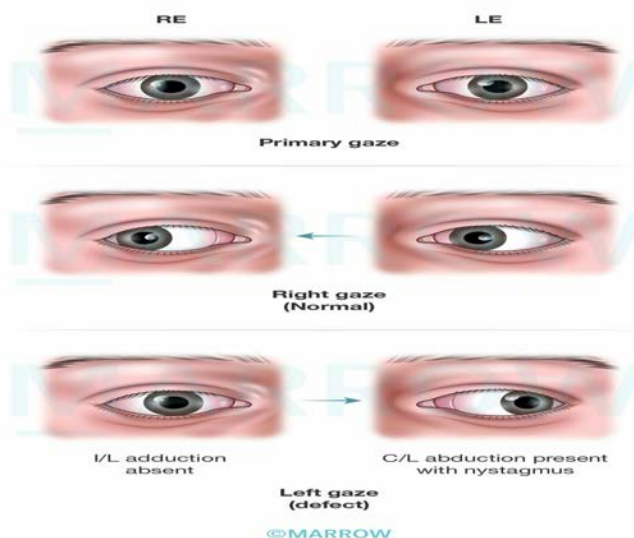
In internuclear ophthalmoplegia (INO) the lesion is at the medial longitudinal fasciculus (MLF).

In INO, there is impaired adduction on the ipsilateral side of the lesion, and nystagmus on abduction on the contralateral side. When the gaze is initiated in the motor cortex, the horizontal gaze center or the paramedian pontine reticular formation (PPRF) generates the eye movements.

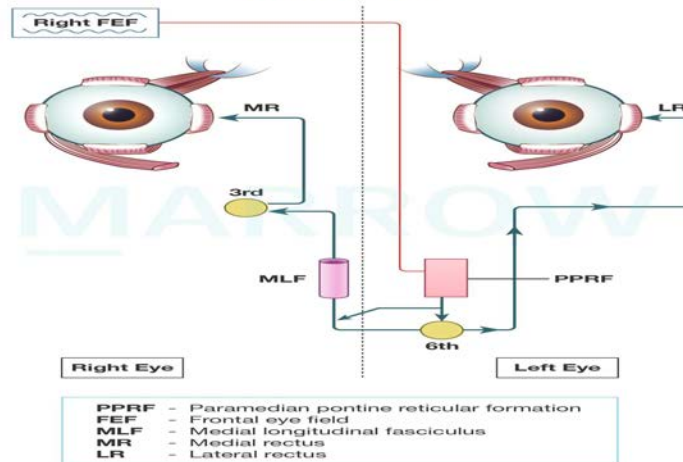
The PPRF has two outputs. The first is to the ipsilateral 6th cranial nerve nucleus which causes abduction of the ipsilateral eye. The second output is to the contralateral MLF which sends an impulse to the contralateral 3rd cranial nerve nucleus causing contralateral adduction. These two outputs from PPRF lead to conjugate movement of the eyes to the same side.

Therefore, a lesion in the right MLF leads to failure of the right eye adduction. The left eye, however, abducts normally. This is called internuclear ophthalmoplegia.

The following image shows the pathway for horizontal eye movements as described above.



Anatomical pathway for horizontal eye movements

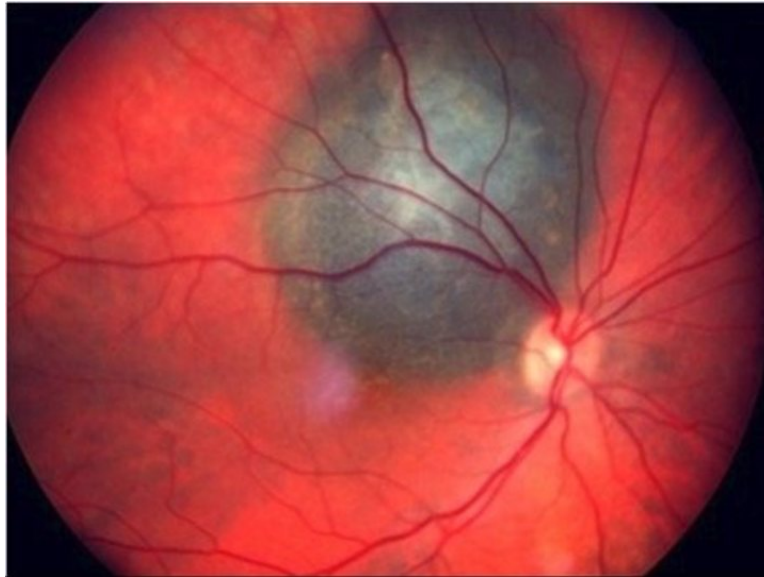


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Ophthalmology AIIMS 2019 (May & Nov)

Question 1:

A patient presented with painless decrease in vision and metamorphopsia. Fundus examination revealed the lesion shown in the image. What is the expected finding on angiography?



- a) Smoke-stack appearance
- b) Inkblot appearance
- c) Collarstud appearance
- d) Double circulation

Question 2:

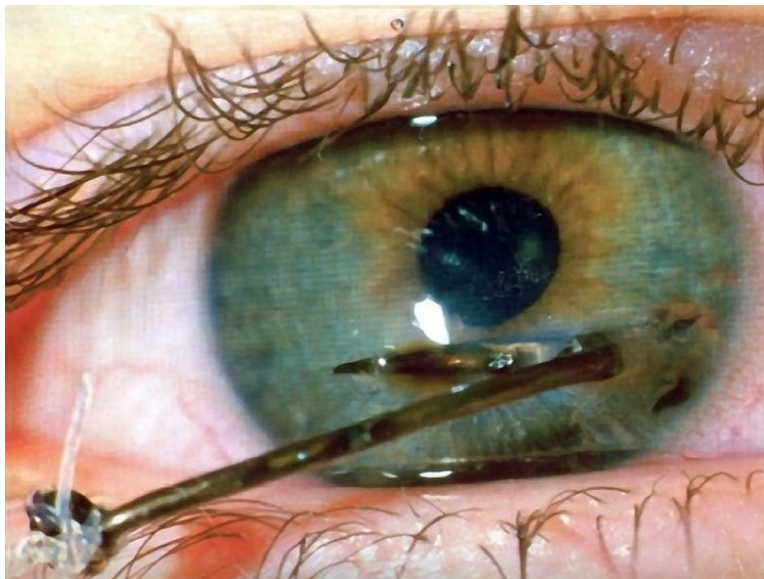
The test given below in the image is used for:



- a) Ocular deviation
- b) Stereopsis test
- c) Refractive errors
- d) Colour Blindness

Question 3:

A patient presents to the emergency and his ocular examination reveals the following. Which of the following is not done initially?



- a) Removal of the foreign body

- b) Check for visual acuity
- c) Give antibiotics
- d) Do primary survey

Question 4:

An Intraocular lens was placed uneventfully in a young patient during cataract surgery. What should be done then?

- a) IOL replaced every 10 years
- b) Never replaced
- c) Replaced when secondary cataract develops
- d) Replaced when presbyopia develops

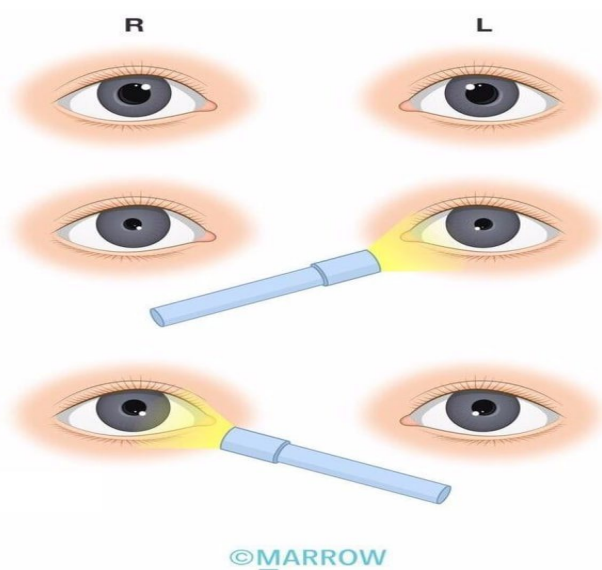
Question 5:

What is the first step in the management of an 18-month-old child with both eyes deviated medially:

- a) Forced ductions test
- b) Examination under anaesthesia
- c) Refractive error testing
- d) Examination of fundus

Question 6:

Which of the following is correct regarding the image given below:



- a) It is called Argyll Robertson pupil
- b) Both eyes are abnormal
- c) The vision is normal in right eye
- d) Seen in optic neuritis, the pupil can paradoxically dilate in the both eyes

Question 7:

A person is diagnosed to have diabetes on his 45th birthday. When will you recommend a dilated fundus examination for him?

- a) Immediately
- b) Before his 50th birthday
- c) After his 50th birthday
- d) When he complains of decrease in vision

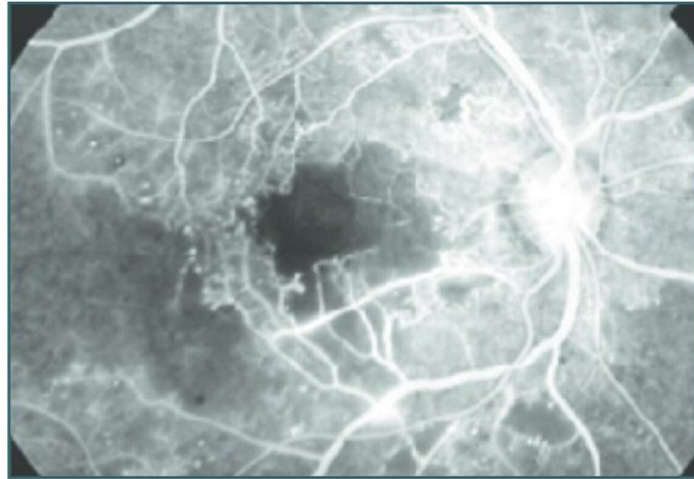
Question 8:

ETDRS study was related to?

- a) Central retinal artery occlusion
- b) Diabetic macular edema
- c) Primary open angle glaucoma
- d) Non-infectious uveitis

Question 9:

A patient with a history of diabetes and hypertension underwent fundus fluorescent angiography (FFA) to evaluate his decreasing vision. The FFA yields an image as shown below, which is characteristic of:



- a) Macular ischemia
- b) Macular edema
- c) Submacular haemorrhage
- d) Premacular haemorrhage

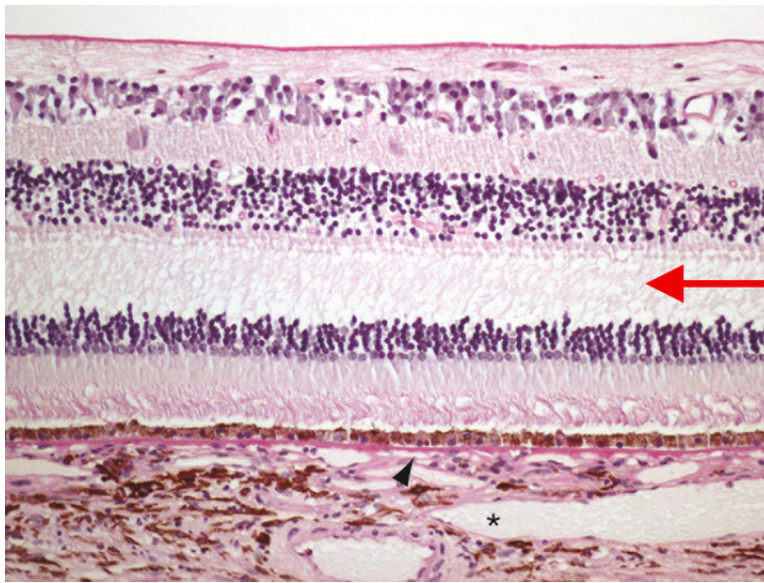
Question 10:

Causes for expanding blind spot are all except _____

- a) POAG
- b) Hypoplasia of optic disc
- c) Papilledema
- d) Medullated nerve fibres

Question 11:

Identify the structure marked with red arrow



- a) Outer limiting membrane
- b) Inner plexiform layer
- c) Outer plexiform layer
- d) Outer nuclear layer

Answer Key

Question No.	Correct Option
1	d
2	b
3	a
4	b
5	c
6	d
7	a
8	b
9	a
10	b
11	c

Detailed Explanations

Solution to Question 1:

The clinical vignette is of a patient presenting with painless diminution of vision and metamorphopsia with the right fundus showing choroidal melanoma superotemporal to the optic disc and involving the macula.

In this patient a “Double circulation” pattern consisting of an internal circulation within the lesion and the normal vascularity of the overlying retina will be seen on angiography.

Although collar stud (Option C) appearance is pathognomonic for choroidal melanoma, it can be seen on B-scan and not on angiography.

Choroidal melanoma is a primary intraocular malignancy in adults that peaks around 60 years of age.

Symptoms include a range of visual disturbances such as painless decrease in vision and metamorphopsia when macula is involved.

Other features of Choroidal melanoma includes:

- A solitary elevated subretinal grey-brown dome shaped mass
- Clumps of orange pigment are common over the mass
- Collar stud appearance on B-scan (option C): It is pathognomonic of choroidal melanoma. It is seen if the tumour breaks through the Bruch's membrane into the subretinal space .
- Associated haemorrhage

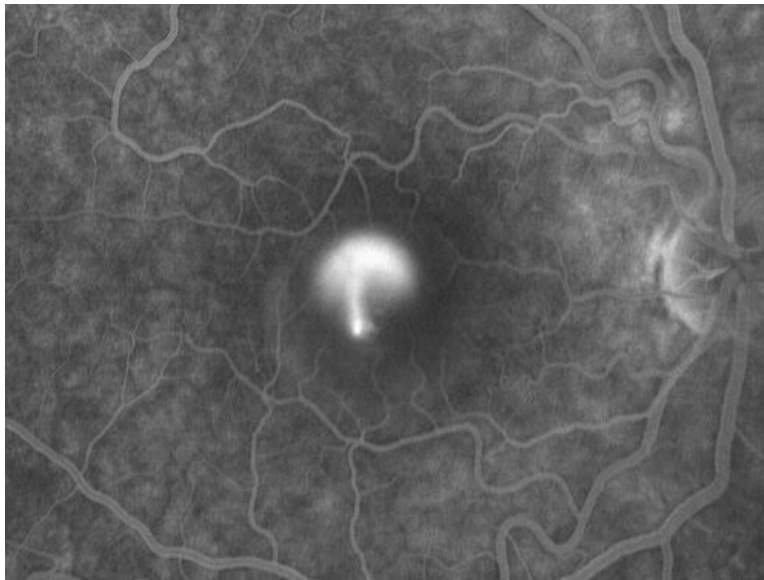
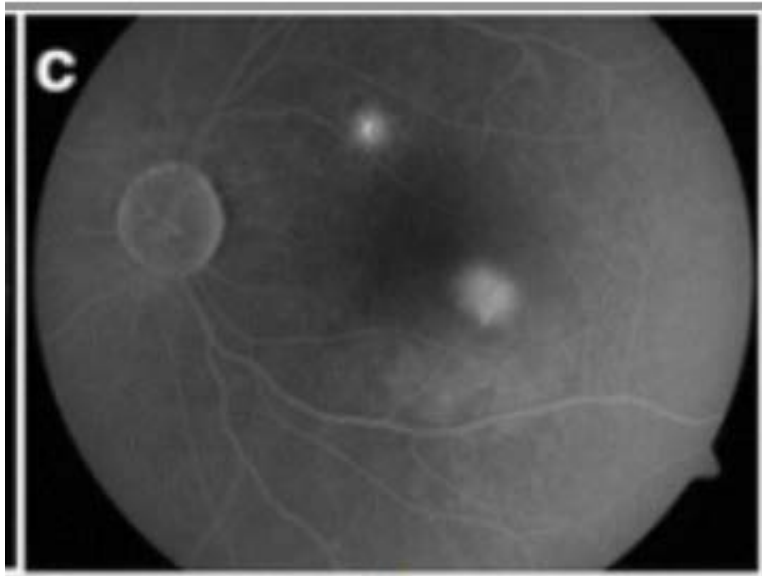
Smoke stack pattern (option A) and ink blot (option B) pattern is seen on FFA in central serous retinopathy (CSR).

CSR is the spontaneous serous detachment of the sensory neuroretina in the foveal region with or without retinal pigment epithelium (RPE) detachment.

This occurs due to Choroidal vascular hyperpermeability.

On FFA it is initially seen as an ink-blot pattern or smoke stack pattern of hyperfluorescence which then spreads to take the configuration of mushroom or umbrella pattern.

The below given images depict the smoke stack pattern and mushroom pattern respectively



Solution to Question 2:

The given image is of Titmus stereotest, which is done for the assesment of stereopsis (perception of depth).

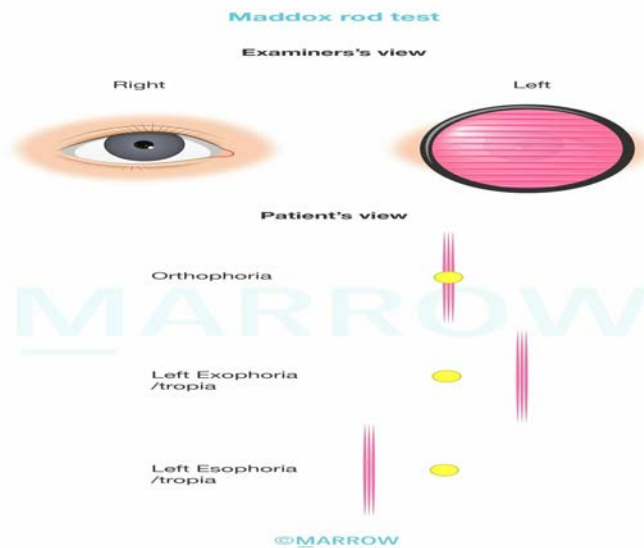
- The pictures consist of a housefly for children and circular patterns for adults.
- The picture is held in front of the eyes of the patient at a distance of about 16 inches and the patient is asked to wear polarized glasses.
- If the patient's perception of the depth is normal, the housefly appears to rise above the plane, like a 3D image. If the patient is asked to pinch the wings of the housefly, he/she will try to pinch in the air, above the plane of the image.

- If the patient's perception of depth in either eye is defective, then the image appears just like normal 2D image.

Option A: Tests for ocular deviation include

- Cover tests
- Prism cover test- This will tell the amount of deviation in prism dioptres.
- Maddox rod test
- Hirschberg corneal reflex test

The below given image shows Maddox rod test.



Option C: Test for refractive error includes

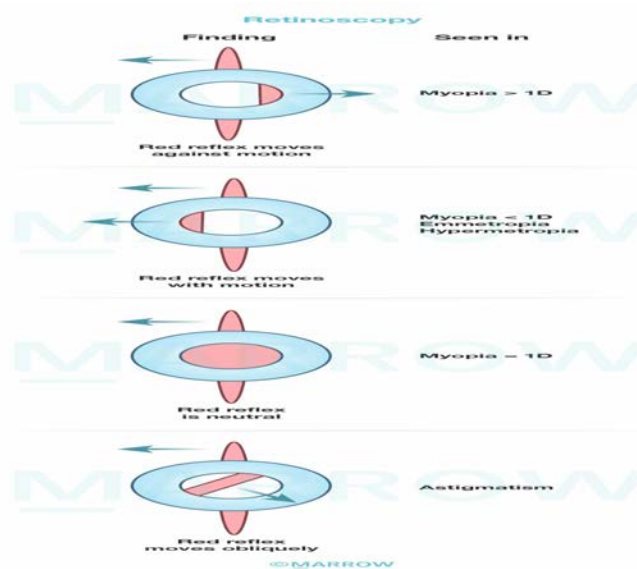
- Retinoscopy
- Auto refractometry

The below given image shows the findings of Retinoscopy in different refractive errors.

Priestley–Smith retinoscope



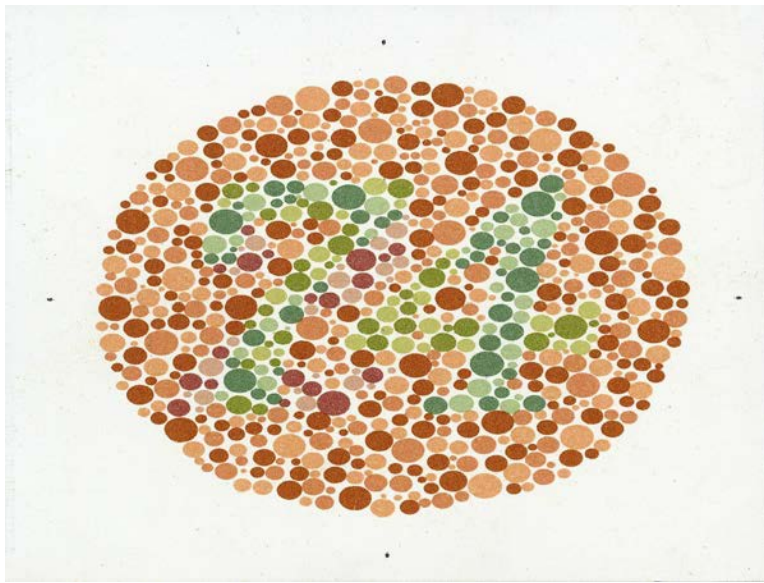
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Option D: Tests for colour blindness include

- Ishihara test: It is frequently used to screen for a red–green colour vision deficit. It consists of a test plate followed by 16 plates, each with a series of dots arranged to show a shape or number that the subject is asked to identify.
- Hardy–Rand–Rittler test: It is similar to Ishihara test but is used to detect all three color vision defects.
- Farnsworth–Munsell 100-hue test: Is the most sensitive test for both acquired and congenital colour blindness. It consists of 85 caps of different hue that need to be rearranged.

The below given image depicts the Ishihara test.



Solution to Question 3:

The above image shows penetrating injury by a metallic fish-hook.

In cases of penetrating injuries, the foreign body is removed only after ascertaining its location through a CT scan.

Hence, Removal of the foreign body is not done initially in these injuries. It is surgically explored & repaired under OT settings.

Management of penetrating injuries of the eye include:

- Primary survey (Option D) of the patient and the wound
- Visual acuity assessment (Option B) in both normal and injured eye
- A CT scan is performed to localize the foreign body and detect any intraorbital or intracranial extensions.
- MRI is contraindicated for metallic foreign bodies.
- Prophylactic intravitreal antibiotics (option C) are prescribed

Solution to Question 4:

Uneventful intraocular lens placement, is not an indication for replacing the IOL.

An IOL replacement (IOL exchange) may be done under the following conditions:

- Incorrect IOL placed
- Posterior or anterior dislocation of IOL
- Hyphema

- Asymmetrical Haptic Placement
- Sunset Syndrome
- Lens-Bag Decentration

Option C: In the event that secondary cataract develops, it is treated by surgical or laser (Nd:YAG) posterior capsulotomy.

Option D: Presbyopia is a disorder of accommodation seen in old age. It is usually corrected by prescription unifocal or Bifocal or multifocal glasses or contact lenses.

Solution to Question 5:

In children, who present with a convergent squint, the first step is to check the visual acuity and estimate the error of refraction.

Convergent squint also known as Esotropia is commonly seen in children with hyperopia (also known as hypermetropia) of which the accommodative type of esotropia is more common in the age group below 3 years.

The first line of management in the treatment of esotropia includes refractive error correction with glasses.

If a child with accommodative issues, is unable to use spectacles (such as a child ≤ 2 years) or is non compliant with wearing glasses we can eliminate accommodation by temporarily treating him with miotic drugs such as echothiophate.

Further treatment may include orthoptic training with specific exercises or surgical correction when the angle of squint is 10° or more with correcting lenses and when orthoptic training doesn't help in reasonable correction of the squint.

Mnemonic:

The algorithm for the management of accommodative esotropia is as follows:

- R: Refractive correction: Glasses/Miotics
- O: Orthoptics/Occlusion: for amblyopia
- O: Operative correction: for non-accommodative component
- P: Prism correction: for remaining error

Classification of Accommodative esotropia:

The below image shows accommodative esotropia before and after correction.

Type	Criteria
Refractive	Associated with high hypermetropia (+4 to +7 D) Typically esotropia is for both near and distance. Fully correctable by spectacles

Type	Criteria
Non-refractive	Occurs independent of refractive errors Abnormal accommodative convergence/accommodation (AC/A) ratio Typically esotropia is greater for near than for distance Fully corrected by bifocal glasses



Option A: Forced Duction Test (FDT) is performed to differentiate between the paralytic and restrictive type of incomitant squint.

FDT is positive in restrictive squint where a resistance is encountered during passive rotation.

Solution to Question 6:

The given image of "swinging flashlight test" shows both pupils constricting when the left eye is stimulated (normal response) followed by both pupils abnormally dilating when the right eye is stimulated (abnormal response).

This finding is suggestive of Marcus Gunn pupil which is also known as Relative afferent pupillary defect (RAPD).

In this patient the right eye is defective (Option A) and the left eye is normal (rules out option B).

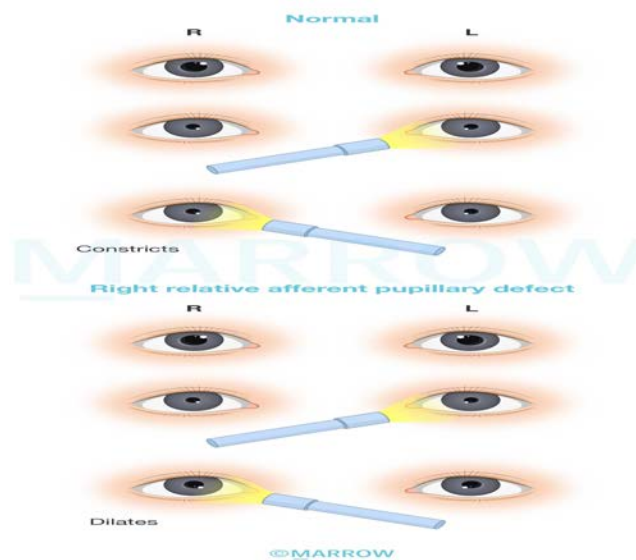
Marcus Gunn pupil refers to the paradoxical response of the pupil to light and it is the earliest sign of optic nerve disease such as optic neuritis or optic nerve damage in glaucoma. It is caused due to a defect in the afferent pathway (optic nerve) of the pupillary light reflex in one eye.

Normal response to stimulation of the eye is constriction of the pupil of the stimulated eye (direct light reflex) and also of the contralateral eye (indirect or consensual light reflex).

Causes of RAPD:

- Optic nerve lesions (e.g. All types of optic neuritis, glaucoma)
- Optic tract lesions (e.g., infarction)
- Severe retinal disease (e.g., central retinal artery occlusion, extensive retinal detachment)
- Maculopathy (macular degeneration)

The below given image depicts the findings seen in Marcus Gunn pupil (RAPD).



Option A: Argyll Robertson pupil is seen in neurosyphilis. In this condition, on light stimulation both the pupils will not constrict, but on accommodation (near target) both the pupils will constrict.

Argyll Robertson pupil is attributed to a dorsal midbrain lesion of neurosyphilis that affects the pupillary light pathway but spares the more ventrally placed pupillary near reflex pathway (Accommodation pathway), thereby causing a light-near dissociation.

Features of Argyll Robertson pupil (ARP) include:

- Bilateral small, irregular pupils in dim light
- Accommodation Reflex Present (ARP)- The eye accommodates by constricting the pupil for a near target.
- Light Reflex absent: pupils do not constrict when exposed to bright light.
- Cocaine eye drops(mydriatics) induced mydriasis.

Solution to Question 7:

A 45 year old diagnosed with diabetes mellitus is likely to have type 2 diabetes and thus, he should undergo screening for diabetic retinopathy immediately.

Duration of diabetes is the most important risk factor for developing diabetic retinopathy, especially proliferative diabetic retinopathy. The recommendations for first fundus examination in diabetic retinopathy is:

- Within 3-5 years of diagnosis in type 1 diabetes
- Immediately in type 2 diabetes.

Later, the screening is continued once every year in both Type 1 and Type 2 diabetes. In the event proliferative diabetic retinopathy develops, screening is done more frequently - once every 2 months.

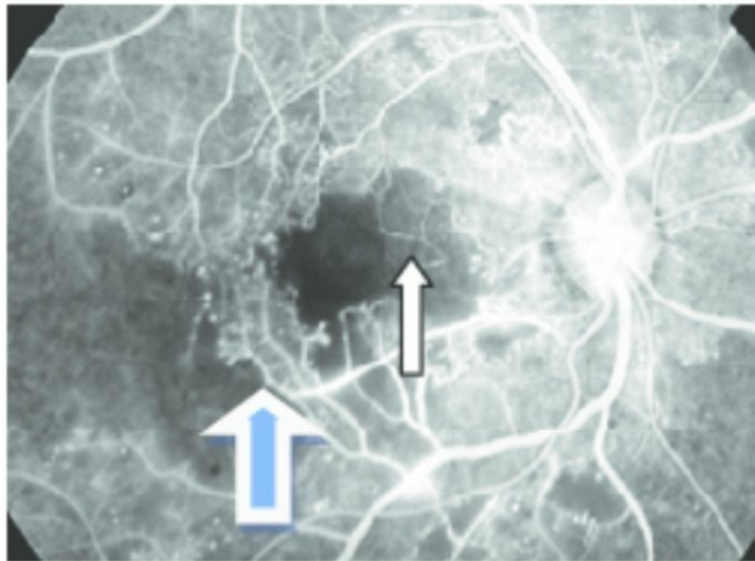
Solution to Question 8:

Early Treatment Diabetic Retinopathy Study (ETDRS) was a randomized clinical trial that studied the effect of photocoagulation on diabetic macular edema.

The results of the study demonstrated the beneficial effects of focal photocoagulation on diabetic macular edema. The study resulted in the recommendation that all eyes with clinically significant diabetic macular edema should be considered for focal photocoagulation.

Solution to Question 9:

The fundus fluorescent angiography (FFA) image shows enlarged foveal avascular zone, seen as a hypofluorescent patch in the macula (white arrow), with surrounding capillary non perfused areas (blue arrow). Both of these are suggestive of macular ischemia.



The above mentioned findings when combined with the history of diabetes in this patient points towards the diagnosis of diabetic macular ischemia.

FFA images in a case of diabetic retinopathy are given below:

- Non proliferative diabetic retinopathy (NPDR)
- Punctate or pinpoint areas of hyperfluorescence - microaneurysms
- Abnormal branching of arteriovenous shunt - intraretinal microvascular abnormalities (IRMA)
- Patchy filling defects of ischemia due to capillary atrophy or non perfusion
- Increase in the size of hypoperfusion in foveal avascular zone in macular ischemia

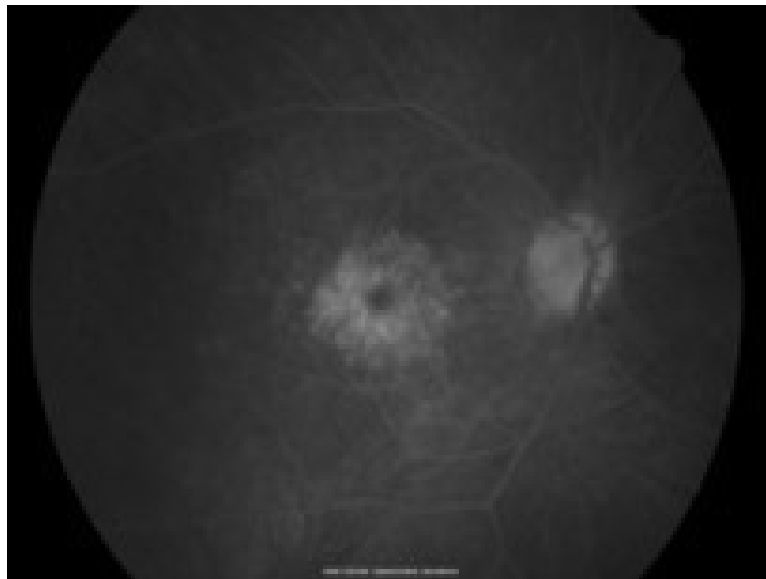
2. Proliferative diabetic retinopathy(PDR)

- Fluorescein leakage - in retinal neovascularization.
- Leakage in the disc region suggests neovascularization of the disc (NVD) and leakage in other regions suggests neovascularization elsewhere (NVE).

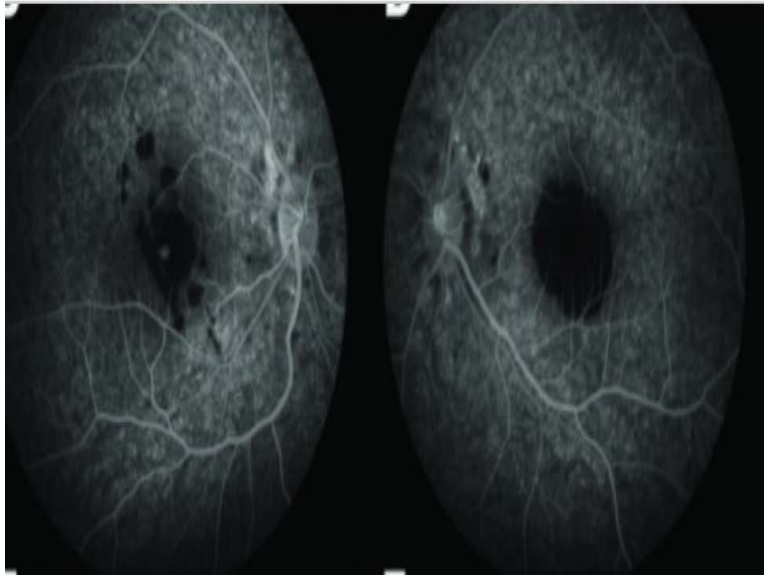
The image below shows pinpoint areas of hyperfluorescence and areas of fluorescein leakage in the fundus.



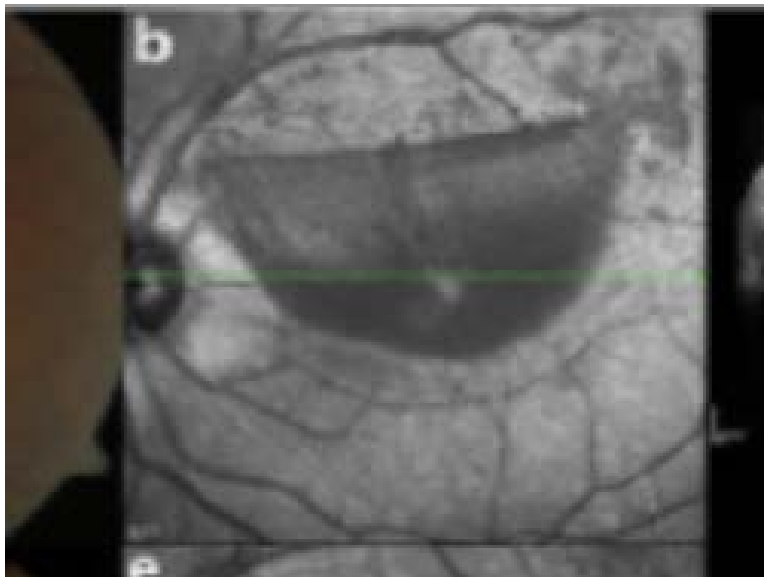
Option B: Macular edema also known as cystoid macular edema (CME) is a collection of fluid between the outer plexiform layer and the inner nuclear layer of the retina following a break in the blood-retina barrier. On fundoscopy, a honeycomb appearance is seen, and on FFA, flower petal pattern can be seen. The below image shows the flower petal pattern on FFA as seen in CME.



Option C: Submacular hemorrhage also known as subretinal hemorrhage will obscure the choroidal dye visualization and appears as ill defined areas of hypoperfusion in the corresponding area. The retinal vessels overlying the area would, however, be visualized. There are no associated features of pinpoint hyperfluorescence and vascular leakage that are usually seen with diabetic macular ischemia. The below given images show fluorescein angiography which reveal blocked fluorescence in the maculae of both eyes due to subretinal hemorrhages.



Option D: Premacular hemorrhage is a term used to define vitreous hemorrhage that occurs in the subhyaloid space. It is seen as a region of hypofluorescence with no overlying retinal vasculature visualization on FFA.



Solution to Question 10:

Optic disc hypoplasia does not cause enlargement of the blind spot.

The physiological blind spot is an area of absolute scotoma in the visual field, seen in the region of optic disc. It is due to the absence of photoreceptors (rods and cones) in that region. Enlarged blind spot is seen in primary open-angle glaucoma (POAG) (Option A), papilledema (Option C), medullated nerve fibers (Option D), optic disc drusen, coloboma of optic disc and myopic disc with a crescent.

Optic nerve hypoplasia is the most common congenital optic disc anomaly. There is underdevelopment of the optic disc in the disease and it typically presents with a small disc that is surrounded by a yellowish pigmented ring, called the 'double ring sign'.

Option A: In primary open-angle glaucoma(POAG), the optic disc enlargement is seen secondary to increased intra-ocular pressure (IOP) in the eye.

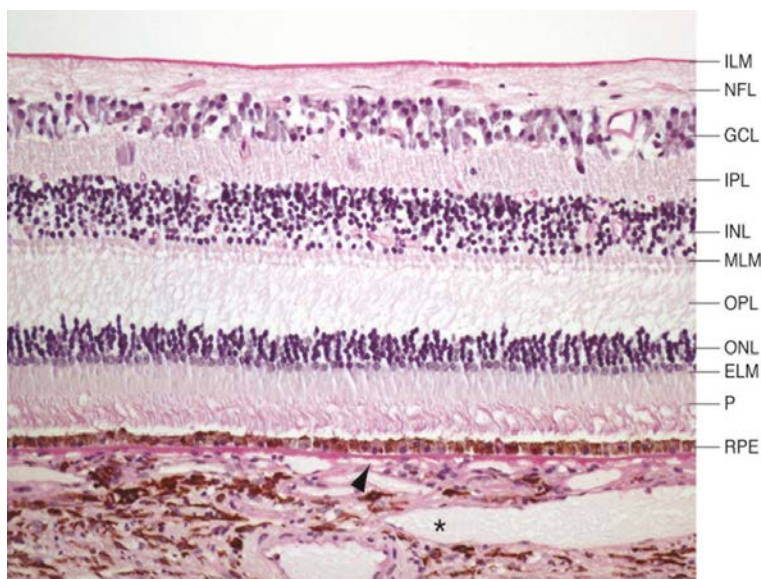
Option C: Papilledema is a passive disc swelling seen with raised intracranial pressure. The blind spot is enlarged due to the edema of the optic disc.

Option D: Medullated or opaque nerve fibers are the abnormal myelination of nerve fibers of the retina. Normally, optic nerve myelination stops at the level of lamina cribrosa. If the myelination continues beyond the optic disc it leads to the expansion of the blind spot. On ophthalmoscopic examination, these are seen as a whitish patch with feathery margins.

Solution to Question 11:

The marked structure in the slide is the outer plexiform layer of the retina.

The outer plexiform layer (outer synaptic layer) is present between the outer nuclear layer and the inner nuclear layer. This layer stains lightly. This layer consists of the axons of rods and cones which synapse with dendrites of bipolar neurons and horizontal cells.



Ophthalmology AIIMS 2020 (May and Nov INI-CET)

Question 1:

A patient presents with left-sided head tilt which on straightening leads to right hypotropia. This increases on dextroversion and right head tilt. Paralysis of which of the following muscles is involved?

- a) Right superior oblique
- b) Right superior rectus
- c) Right inferior oblique
- d) Left inferior oblique

Question 2:

Which of the following procedure is shown in the image given below?



- a) Nasolacrimal duct syringing
- b) Anterior ethmoidal nerve block
- c) Infraorbital nerve block
- d) Trephination of frontal sinus

Question 3:

What is the correct order of using eye drops and eye ointment?

- a) Eye drops first, ointment 15 min later
- b) Ointment first, eye drops 15 min later
- c) Eyedrops first, then ointment immediately
- d) Ointment first, then eyedrops immediately

Question 4:

Which of the following statements is false about corneal epithelium?

- a) Regeneration is limited to the limbus
- b) Bowman's membrane regenerates
- c) Lined by stratified squamous epithelium
- d) Apical cells have microvilli

Question 5:

A patient underwent LASIK for myopia and now is undergoing cataract surgery. What is the formula used for lens power calculation in this patient?

- a) SRK 1
- b) SRK 2
- c) Hoffer Q
- d) Haigis L

Question 6:

Which of the following pupillary abnormalities is most likely to be observed in a patient with primary open-angle glaucoma?

- a) Marcus Gunn pupil
- b) Adie's tonic pupil
- c) Argyll-Robertson pupil
- d) Hutchinson's pupil

Question 7:

Which of the following helps to differentiate a fixed dilated pupil? A. Phenylephrine B. 1% Pilocarpine C. 0.1% Pilocarpine D. Cocaine 0.5% E. Epinephrine

- a) A, B, C, D and E
- b) B and C only
- c) A, D and E only
- d) A, C and D

Question 8:

True regarding POAG:

- a) A, B, D and E
- b) A, B and C
- c) B and E only
- d) B, C and D

Answer Key

Question No.	Correct Option
1	b
2	b
3	a
4	b
5	d
6	a
7	b
8	c

Detailed Explanations

Solution to Question 1:

The given clinical scenario is suggestive of right superior rectus palsy.

The Park's-Bielschowsky three-step test is employed to diagnose the muscle paralysis responsible for vertical eye deviation. In this clinical scenario:

-

Identify the hypotropic eye in primary gaze

-

Right eye is hypotropic.

-

This could be due to either right eye elevator weakness or left eye depressor weakness.

-

Possible muscles: Right superior rectus, right inferior oblique, left inferior rectus, or left superior oblique

-

Observe if hypotropia worsens in specific gazes

-

Hypotropia worsens in right gaze (dextroversion).

-

Paralyzed muscle is one that elevates the right eye in right gaze or depresses the left eye in right gaze.

-

Possible muscles: Left superior oblique and right superior rectus.

-

Determine if hypotropia worsens in head tilts

-

Hypotropia worsens during right head tilt.

-

The right superior rectus is the muscle involved.

Mnemonic for superior oblique paralysis:

- Step 2: Vertical deviation (hypertropia/hypotropia) is Worse On Opposite Gaze – WOOG

- Step 3: Vertical deviation (hypertropia/hypotropia) deviation is Better On Opposite Tilt – BOOT

Muscle	Primary action	Secondary action	Tertiary action
Medial rectus (MR)	Adduction	-	-
Lateral rectus (LR)	Abduction	-	-
Superior rectus (SR)	Elevation(bestwhen the eye is inabduction)	Intorsion (mnemonic: SIN)	Adduction (mnemonic: R Add)

Muscle	Primary action	Secondary action	Tertiary action
Inferior rectus (IR)	Depression(bestwhen the eye is inabduction)	Extorsion	Adduction (mnemonic: R Add)
Superior oblique (S O)	Intorsion (mnemonic: SIN)	Depression(bestwhen eye is inadduction)	Abduction
Inferior oblique (IO)	Extorsion	Elevation(bestwhen eye is in adduction)	Abduction

Solution to Question 2:

The procedure shown in the given image is an anterior ethmoidal nerve block.

The anterior ethmoidal nerve supplies the roof of the nasal cavity. It is a branch of the nasociliary nerve which in turn is a branch of the ophthalmic nerve (V1). It is sometimes used in endonasal surgery.

Other options:

Option A: In nasolacrimal duct syringing, the puncta of the nasolacrimal duct near the medial canthus of the eye is probed using a lacrimal cannula. The below image shows lacrimal syringing.



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Option C: Infraorbital nerve block causes anesthesia in the distribution of infraorbital nerve which includes lower lid, cheek, side of nose, upper lip, and upper teeth. It is most commonly done for dental procedures.

Option D: Trephination of the frontal sinus is a surgical procedure done for the drainage of pus in acute frontal sinusitis. In this procedure, the incision is made on the floor of the frontal sinus.

Solution to Question 3:

Eye drops first followed by ointment 15 minutes later is the correct way of usage.

An ointment is thicker in consistency and needs to stay on the eyes for a longer time for absorption of the medication. Hence, it is generally put in the eyes at bedtime.

Eyedrops are watery thin and can be absorbed faster. Hence eyedrops are put first, and after 10-15 minutes, once they are completely absorbed, eye ointment can be applied.

Eye ointments are usually avoided during the daytime as they cause marked blurring of vision.

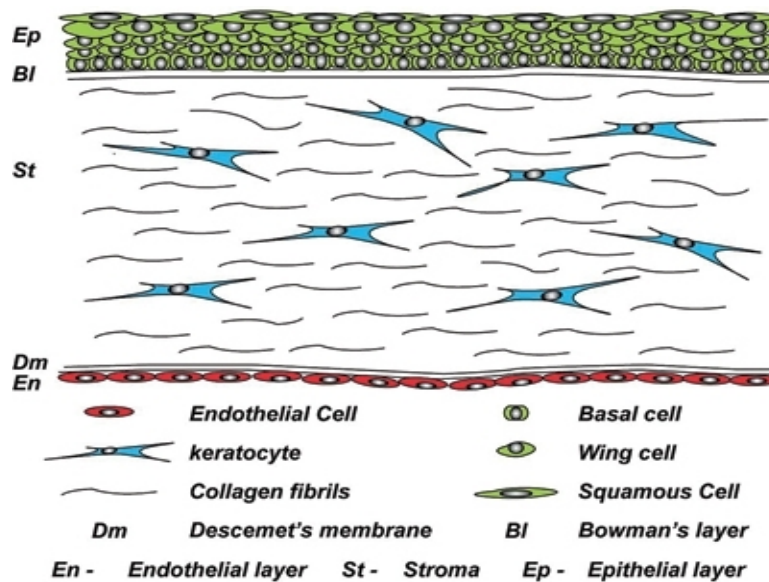
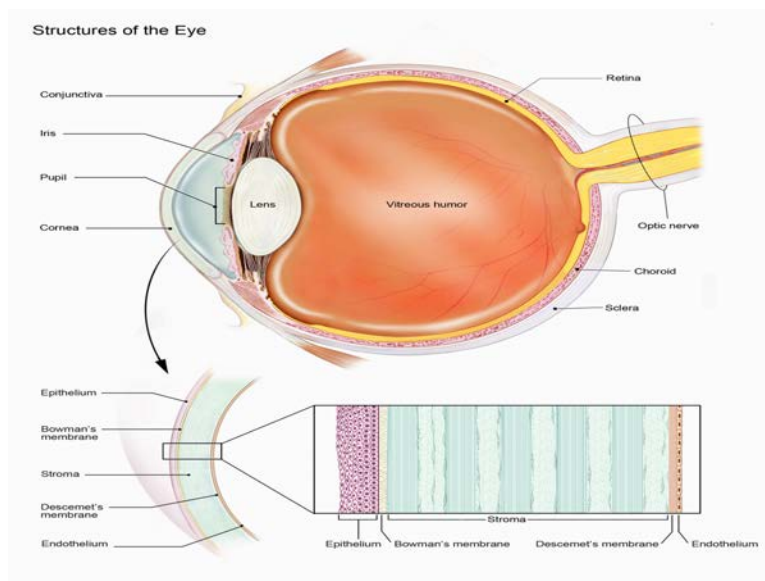
Solution to Question 4:

Bowman's membrane is acellular. Hence, it will not regenerate but heals by scarring when damaged.

Corneal stem cells are located at the corneoscleral limbus (Option A), so the regeneration is limited to the limbus.

The cornea comprises distinct layers:

- Corneal epithelium is lined by non-keratinized stratified squamous epithelium (Option C), with microvilli on the apical layer (Option D) to enhance surface area and tear film attachment. It consists of basal, columnar, wing, and squamous cells.
- Bowman's membrane, a superficial stromal layer, is acellular, collagen-based, and incapable of regeneration.
- The stroma, the thickest layer constituting 90% of corneal thickness, houses collagen fibrils with ground substances such as chondroitin sulfate and keratan sulfate, maintaining transparency. It cannot regenerate and heals with scarring.
- Dua's layer or Pre-Descemet's membrane is the toughest.
- Descemet's membrane extends anteriorly to Schwalbe's line and possesses regenerative potential.
- Endothelium, a metabolically active monolayer, lacks regenerative ability. It maintains deturgescence through active pumps, ensuring transparency with a usual cell density of approximately 3000 cells/mm² measured by specular microscopy.



Solution to Question 5:

Among the given options, the Haigis-L regression formula is used for calculating the lens power in a patient who is undergoing cataract surgery with a previous history of refractive surgery.

Haigis formula uses Axial Length (L), Corneal curvature (K), and Anterior Chamber Depth (ACD) for the calculation of IOL power (biometry).

- Keratometry measures the curvature of the anterior corneal surface.
- A-scan ultrasonography is used to determine the axial length.

Standard IOL calculations are unsuitable in any type of corneal refractive surgery as they are likely to result in residual refractive errors even after IOL implantation in these patients. In these patients, several other methods of calculating corneal refractive power are used which are then

added to the standard and specific formulas such as the Masket, modified masket, Haigis L formula, and double-K method.

Other options:

Options A and B: SRK 1 and SRK 2 are the basic formulas used for IOL power calculations.

SRK formula: $P = A - 2.5L - 0.9K$ where P is the power of IOL, A is a constant for the lens, L is the axial length and K is corneal curvature.

Option C: Hoffer Q formula is used for IOL power calculation in short eyes (hypermetropia). It uses only two variables; Axial length (L) and corneal curvature (K).

Solution to Question 6:

Primary open-angle glaucoma commonly causes bilateral and asymmetric optic nerve damage. Marcus Gunn pupil also known as relative afferent pupillary defect (RAPD) is the earliest sign of optic nerve disease as seen in optic nerve neuritis or primary open-angle glaucoma (POAG).

RAPD is caused due to a defect in the afferent pathway (optic nerve) of the pupillary light reflex in one eye.

It is diagnosed by the swinging flashlight test where the flashlight is swung to and fro, and the pupillary response is observed. Normally, both pupils constrict when any one eye is stimulated (due to direct and consensual light reflex). In the presence of RAPD, both pupils will abnormally dilate when the affected eye is stimulated. This paradoxical response of the pupil to light is called Marcus Gunn pupil.

Causes of RAPD:

- Optic nerve lesions (e.g. All types of optic neuritis, glaucoma)
- Optic tract lesions (e.g., infarction)
- Severe retinal disease (e.g., central retinal artery occlusion, extensive retinal detachment)
- Maculopathy (macular degeneration)

Other options:

Option B: Adie tonic pupil is a unilateral dilated pupil with light-near dissociation due to postganglionic parasympathetic pupillomotor nerve supply damage. In these patients the light reflex is absent and the accommodation (near) reflex is sluggish. It is commonly seen in young females and is associated with absent knee jerk.

Option C: Argyll Robertson pupils are bilateral small pupils with light-near dissociation. It is commonly seen in patients with neurosyphilis. In these patients the pupillary reflex is absent but the accommodation reflex is present (ARP).

Option D: Hutchinson's pupil is a clinical sign where the pupil on the side of an intracranial mass lesion is dilated and unreactive to light, due to compression of the oculomotor nerve on that side.

Solution to Question 7:

Among the given options, 1% pilocarpine and 0.1% pilocarpine can be used to differentiate fixed dilated pupil.

Fixed dilated pupil is seen in conditions such as third nerve palsy, Hutchinson's pupil (increased cranial pressure), Adie's pupil, and due to pharmacological mydriasis.

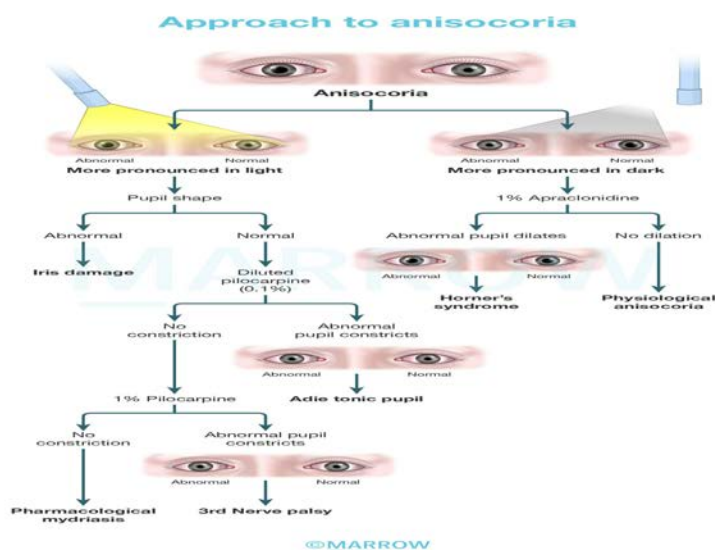
Pilocarpine acts as an agonist at cholinergic muscarinic receptors and ganglion stimulator to cause miosis of the pupil.

Normal strength pilocarpine (1-2%) constricts the dilated pupil of 3rd nerve palsy, whereas there will be no response if the dilation is due to pharmacological mydriasis (anticholinergic causes) such as accidental application of atropine.

0.1% pilocarpine is used to diagnose Adie's pupil. Adie's pupil constricts with weak concentrations of pilocarpine (0.1%), while the normal pupil does not. This is because the denervated sphincter of the iris is very sensitive to topical parasympathomimetics such as pilocarpine. 0.1% pilocarpine also helps differentiate Adie's pupil from Argyll Robertson pupil (ARP).

Note:

Cocaine test (2 to 4%) is done in Horner's syndrome. In these patients, the small pupil fails to dilate on instillation of mydriatics such as Cocaine which is in contrast to the normal pupil which dilates on instilling cocaine. Alternatively, the small pupils of Horner's syndrome dilate with 1% apraclonidine.



Solution to Question 8:

In primary open-angle glaucoma (POAG), family members are likely to develop steroid responsiveness (statement B) and visual field defect can exist with a normal C:D ratio (statement E).

Statement A: Gonioscopy cannot visualize the specific abnormalities that occur in the trabecular meshwork in POAG. Gonioscopy diagnoses POAG because of the visibility of all the structures of the angle of the anterior chamber in contrast to the non-visibility of different structures in angle

closure glaucoma.

Statement C: Primary angle-closure glaucoma is associated with exacerbation of IOP with pupillary dilatation and not POAG.

Statement D: There is a 10% increased risk of POAG in siblings and a 4% increased risk in children of patients with POAG.

POAG also known as chronic simple glaucoma of adult-onset is characterized by the presence of any 2 of the following 3 categories of changes:

- Increased intraocular pressure:
- IOP ≥ 21 mmHg on more than one occasion, and/or
- Circadian variation in IOP of ≥ 8 mmHg, or
- Asymmetry in IOP ≥ 5 mmHg between both eyes
- Optic nerve head changes suggestive of glaucomatous damage. These may include one or more of the following changes:
- Vertically oval cup
- Cup-disc (C:D) ratio ≥ 0.6 (average cup: disc ratio 0.3-0.4)
- Asymmetry in C:D ratio ≥ 0.2 between both eyes.
- Narrowing, notching, or pallor of the neuroretinal rim.
- Disc haemorrhages
- Nasal shifting of retinal vessels-Bayonetting sign
- Lamellar dot sign
- Visual field changes - isopter contraction, barring of blind spot, paracentral scotoma, Seidel's scotoma, Bjerrum's scotoma, double arcuate scotoma, and Roenne's central nasal step.

A patient presenting with just raised IOP and visual field defects with normal optic nerve head findings such as a normal C:D ratio can still be diagnosed as POAG (statement E).

Ophthalmology NEET 2018

Question 1:

Which of the following would be prescribed for simple myopic astigmatism?

- a) +1.00 DS
- b) -1.00 DC X 180 degree
- c) -1.00 DS
- d) -1.00 DS -1.00 DC X 180 degree

Question 2:

What is the causative organism of angular conjunctivitis?

- a) Moraxella lacunata
- b) Haemophilus influenzae
- c) Adeno virus type 32
- d) Coxsackie B virus

Question 3:

What is the most common cause of ophthalmia neonatorum in developed countries?

- a) Chlamydia
- b) Gonorrhoea
- c) Chemical conjunctivitis
- d) Staphylococcus aureus

Question 4:

What is the cause of acute hemorrhagic conjunctivitis?

- a) Enterovirus 68
- b) Enterovirus 69
- c) Enterovirus 70

d) Enterovirus 71

Question 5:

100-day glaucoma is seen in which of the following conditions?

- a) Central retinal vein occlusion
- b) Eale's disease
- c) Central retinal artery occlusion
- d) Steroid-induced glaucoma

Question 6:

Yoke muscle of the right lateral rectus in dextroversion is the _____.

- a) Left superior oblique
- b) Left medial rectus
- c) Left inferior rectus
- d) Left superior rectus

Question 7:

A patient, 36 hours after cataract surgery, presents with complaints of pain and visual disturbance. His eye is as shown below. What is the management of this condition?



- a) Intravenous steroids
- b) Intravenous antibiotics
- c) Bandage and padding of the eye
- d) Intravitreal antibiotics

Question 8:

Which of the following is the most common feature of third nerve palsy, in a case of long-standing, uncontrolled diabetes?

- a) Proptosis
- b) Pseudoptosis
- c) Ptosis
- d) Pupillary dysfunction

Question 9:

What is the most common site of an orbital blowout fracture?

- a) Medial wall
- b) Lateral wall
- c) Roof
- d) Floor

Question 10:

A male patient came with a history of unprotected sexual intercourse around 20 years ago. Now, he complains of neurological symptoms and visual disturbances. Which of the following will be seen on exposure of his pupils to light?

- a) Pupils react but do not accommodate.
- b) Pupils accommodate but do not react.
- c) Pupils neither react nor accommodate.
- d) Pupils react and accommodate.

Question 11:

Roth spots are seen in _____.

- a) Acute leukemia
- b) Uveal melanoma
- c) Uveal lymphoma
- d) All of the above

Question 12:

A 50-year-old man is presbyopic. The power of his prescription glasses should be _____.

- a) +1.75D
- b) +1.5D
- c) +2D
- d) +2.25D

Answer Key

Question No.	Correct Option
1	b
2	a
3	a
4	c
5	a
6	b
7	d
8	c
9	d
10	b
11	a
12	b

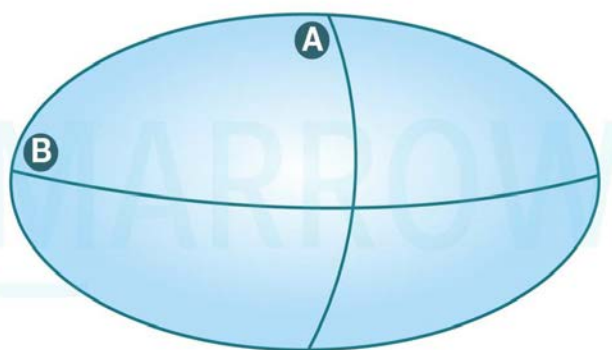
Detailed Explanations

Solution to Question 1:

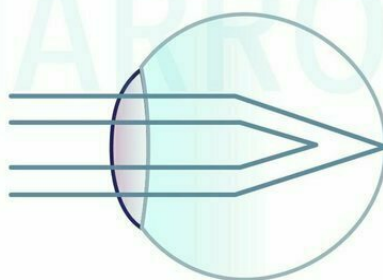
Among the above options, $-1.00 \text{ DC} \times 180$ degrees is the correct prescription for simple myopic astigmatism (SMA).

There are 3 components for prescription of glasses.

1. The first number is the refractive power of the spherical component. It is represented as convex (+) or concave (-) dioptre sphere (DS).
2. The second number denotes the power of the cylinder, represented as convex (+) or concave (-) dioptre cylinder (DC).
3. The above two components are followed by an x representing the axis (orientation) of the cylinder.



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In the most common type of SMA - with-the-rule astigmatism - meridian A (represented by 90 degrees) is more curved than meridian B (represented by 180 degrees). The light rays refracting

from B focus on the retina, whereas those refracting from A focus in front of the retina.

To correct this error, a negative cylindrical lens (DC) is used to focus the myopic rays of meridian A on the retina. This correction needs to occur at 90 degrees (meridian A). The axis of a cylindrical lens is perpendicular to the required correction and thus, the spectacles will have an axis of 180 degrees.

There is no spherical refractive error since the light rays from meridian B lies on the retina. Thus DS is 0.

Solution to Question 2:

Moraxella species, a type of gram-negative diplobacilli, are involved in causing angular conjunctivitis which is a type of chronic conjunctivitis.

Angular conjunctivitis is a chronic type of conjunctivitis where there is inflammation of the conjunctiva at the lid margins and the inner and outer canthi of the eyes with red, scaly, excoriated, macerated and fissured skin at the canthi. This is because of the proteolytic enzymes secreted by the Moraxella species.

Patients with angular conjunctivitis usually experience eye irritation, a burning sensation and collection of dirty-white, foamy discharge at the angles. These patients can also have associated blepharitis and papillary and follicular conjunctivitis.

Causative organisms are Moraxella lacunata, Moraxella axenfeld and Staphylococcus aureus (Rare cause). The source of infection is typically the nasal tract or the nasal discharge of patients with angular conjunctivitis. The infection spreads from the nasal cavity to the eyes through contaminated hands or handkerchiefs. Treatment includes oxytetracycline 1% ointment for 10 to 14 days, and zinc oxide as it inhibits proteolytic enzymes.

The image given below shows the eye of a patient infected with angular conjunctivitis.



Other options:

Option B: Haemophilus influenzae causes acute conjunctivitis.

Option C: Adenovirus is the most common cause of viral conjunctivitis and it most commonly causes follicular conjunctivitis, pharyngoconjunctival fever, and epidemic keratoconjunctivitis.

Option D: Coxsackie B virus is known to cause aseptic meningitis, encephalitis, pericarditis, and myocarditis. There are no known mentions of ocular disease caused by the virus.

Note: Pathogens that cause bacterial conjunctivitis

The below-given table elaborates on the type of conjunctivitis caused by different serotypes of adenovirus.

Hyperacute(symptoms develop within 24 hours; aggressive course)	Acute(resolving in less than 4 weeks)	Chronic(of greater than 4 weeks duration)
Neisseria gonorrhoeae	Staphylococcus aureus	Staphylococcus aureus
Neisseria meningitidis	Streptococcus pneumoniae	Moraxella lacunata
	Haemophilus influenzae	Enteric bacteria

Conjunctivitis type	Adenovirus serotype
Follicular Conjunctivitis	Adenovirus serotypes 1-11 and 19
Pharyngoconjunctival Fever	Adenovirus serotypes 3, 4, and 7
Epidemic Keratoconjunctivitis	Adenovirus serotypes 8, 19, and 37

Solution to Question 3:

Chlamydia is at present the most common cause of ophthalmia neonatarum in developed countries.

Ophthalmia neonatarum also called neonatal conjunctivitis is inflammation of the conjunctiva occurring in an infant <30 days after birth. It should be ruled out in any infant having discharge or watering within the first week of birth.

Causes include:

- Chlamydia(Option A): neonatal inclusion conjunctivitis caused by serotypes D-K is at present the most common cause in developed countries.
- Gonorrhoea(Option B): effective prophylaxis against gonococcal infection and its subsequent decline has almost eliminated its incidence in developed countries.
- Staphylococcus aureus(Option D), Haemophilus species, Streptococcus haemolyticus, and Streptococcus pneumoniae are other less commonly seen causes of ophthalmia neonatarum

- Herpes simplex ophthalmia neonatrum rarely occurs and is caused by Herpes simplex type 2 virus from the mother.
- Chemical conjunctivitis(Option C) is caused by the application of prophylactic antibiotics or silver nitrate at the time of birth.

Time of presentation	Causative agent
Within 48 h	Chemical, Gonorrhoea
48–72 h	Staphylococcus aureus,Streptococcus pneumoniae
5–7 days	Herpes simplex II
>1 week	Chlamydia

Prophylactic treatment: 1% tetracycline or erythromycin eye ointment is applied to the eyes, if the mother is suspected to have gonococcal or chlamydia infection.

Curative treatment:

- Chlamydia trachomatis (D– K)- Erythromycin eye ointment
- Neisseria gonorrhoeae- Ceftriaxone injection, gentamicin drops, bacitracin eye ointment

The below image shows an infant presenting with ophthalmia neonatarum.



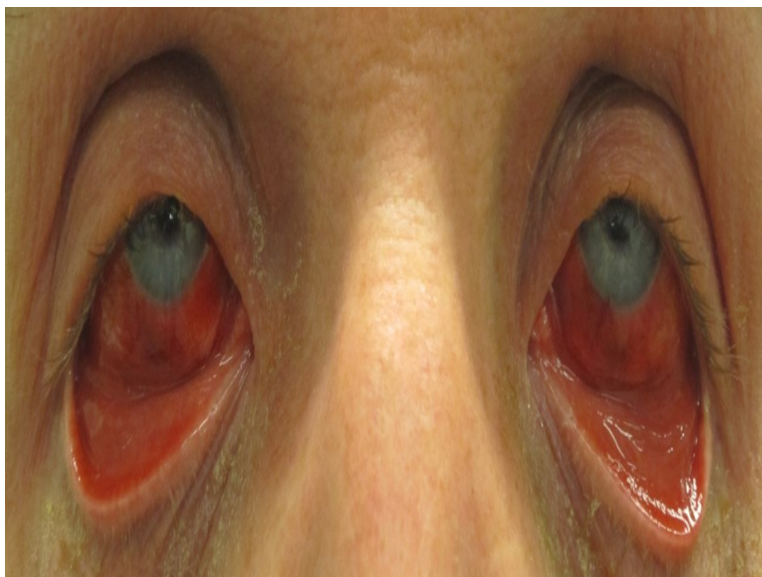
Solution to Question 4:

Acute hemorrhagic conjunctivitis is caused by picornoviruses namely enterovirus 70.

Acute hemorrhagic conjunctivitis also known as "Apollo conjunctivitis" is caused by picornoviruses such as enterovirus 70 and coxsackievirus A24. Rarely it is also caused by adenovirus type 11.

It manifests with severe conjunctival congestion, chemosis, subconjunctival hemorrhages, painful follicular conjunctivitis, tearing, lid edema and pre-auricular lymphadenopathy. The cornea may demonstrate a fine punctate keratopathy. The disease is usually self-limiting, running a course of 3-7 days.

The image given below shows the eyes of a patient with acute hemorrhagic conjunctivitis.



Enterovirus 71 (Option D) is associated with fatal meningitis, encephalitis, and paralysis. It is also implicated in Hand-Foot-Mouth disease (HFMD) and Herpangia which is papulo-vesiculo-ulcerative rash in oral cavity.

Solution to Question 5:

100-day glaucoma is a neovascular glaucoma that occurs in ischemic type of central retinal vein occlusion (CRVO).

It is known as 100-day glaucoma because it occurs 3 months after the episode of CRVO. Although neovascular glaucoma occurs as a complication in Eale's disease (Option B) and central retinal artery occlusion (Option C) and long-term use of topical steroid therapy causes steroid-induced glaucoma (Option D) none of them are termed as 100-day glaucoma.

CRVO is further classified into two types:

- Non-ischemic CRVO: occurs secondary to venous stasis. It is the most common type of CRVO of the two.
- Ischemic CRVO: occurs due to acute occlusion of the central retinal vein. It is also termed as hemorrhagic retinopathy.

Ischemic CRVO presents with sudden loss of vision and Relative afferent pupil defect (RAPD).

Fundus examination shows massive enlargement of retinal veins and massive retinal hemorrhages giving a "splashed tomato appearance". Other findings include cotton wool spots, optic disc edema and neovascularization at the disc and in the periphery.

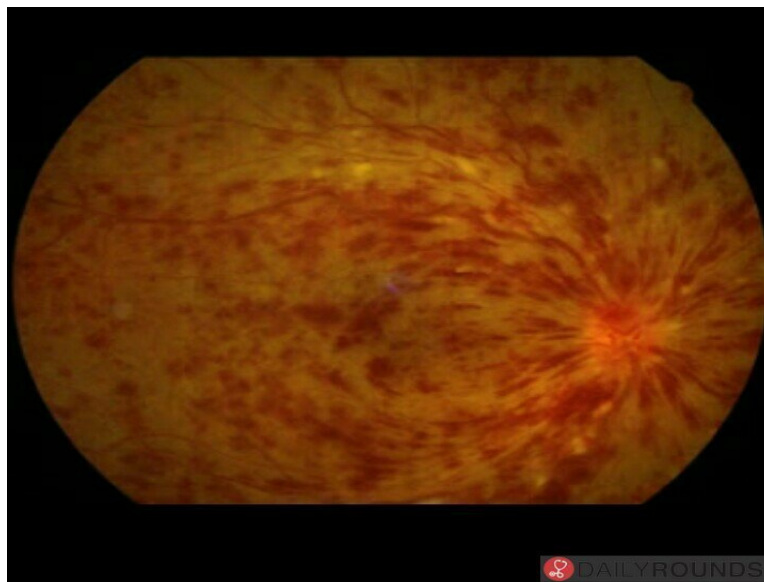
Pathognomic features of ischemic CRVO that differentiates it from non-ischemic CRVO are:

- RAPD
- Reduced amplitude of B-wave on electroretinogram
- Visual field defects

Complications of the disease:

- Neovascular Glaucoma known as 100 day glaucoma
- Neovascularisation of iris known as Rubeosis iridis
- Vitreous hemorrhage
- Proliferative retinopathy

The below-given fundus image is of the "splashed tomato" appearance of CRVO.



Option B: Eales disease is an idiopathic inflammatory disease of the peripheral retinal veins. It is considered to be a hypersensitivity reaction to the tubercular proteins and it is commonly seen in young males. It is characterized by recurrent vitreous hemorrhage.

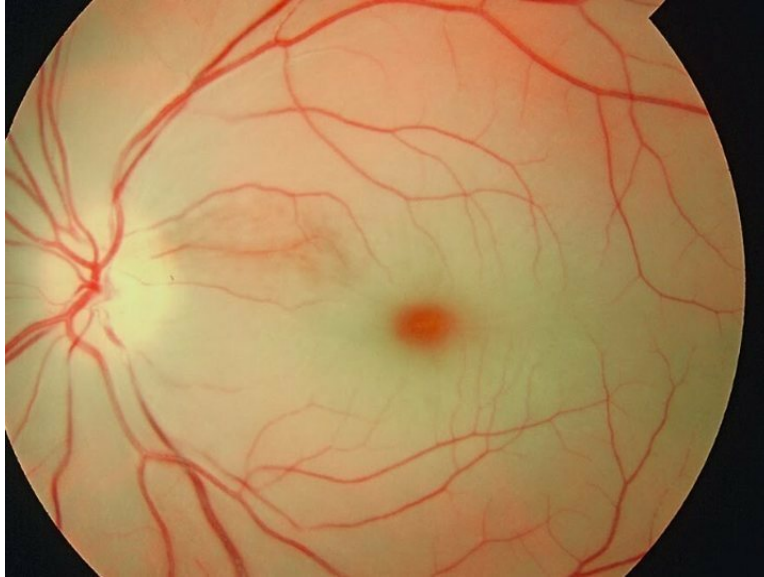
Complications include:

- Neovascular glaucoma
- Proliferative vitreoretinopathy
- Tractional retinal detachment,
- Rubeosis iridis

Option C: Central retinal artery occlusion (CRAO) is most often caused due to an embolus from an atherosclerotic plaque. It presents as sudden painless loss of vision and amaurosis fugax (transient monocular blindness that is due to ischemia). Fundus examination shows an opaque, milky-white retina with a cherry-red spot.

Neovascular glaucoma may occur as a delayed complication of CRAO but it is not termed as 100 day glaucoma.

The below image depicts the fundus image seen in CRAO with the cherry red spot.



Option D: Steroid induced glaucoma is a type of secondary open angle glaucoma that usually follows weeks-months of topical steroid therapy.

Solution to Question 6:

The yoke muscle of the Right lateral rectus in right dextroversion is the left medial rectus.

Yoke muscles also known as contralateral synergists include a pair of muscles, one from each eye, which contract simultaneously during version movements.

Versions are simultaneous movements of both eyes in the same direction, such as dextroversion in which both the eyes move to the right.

Hence, in dextroversion, the right eye is in abduction due to the action of right lateral rectus and the left eye is in adduction by the action of the left medial rectus.

An easy method to determine yoke muscle pairs:

Just take the opposite of each word-

Left → Right

Superior → Inferior

Oblique → Rectus

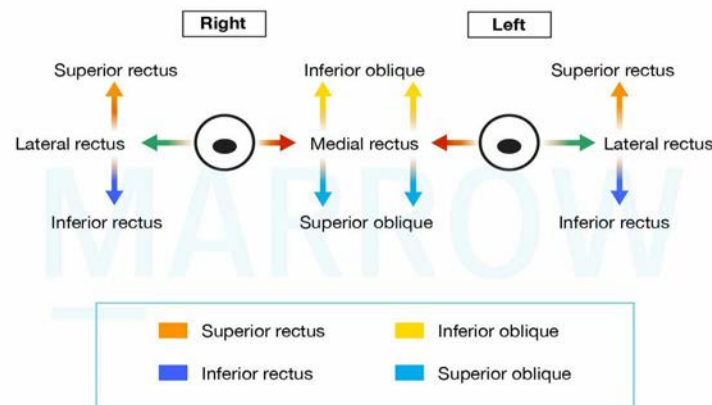
“Left SO and Right IR” are a yoke muscle pair.

Other examples of yoke muscles are:

- Right MR and Left LR for levoversion
- Right IO and Left SR - Levoelevation
- Right SR and Left IO - Dextroelevation

- Right IR and Left SO - Dextrodepression

[IO-inferior oblique;SO-superior oblique;SR-superior rectus;IR-inferior rectus;MR-medial rectus;LR-lateral rectus]



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Solution to Question 7:

A post cataract surgery patient presenting within 36 hours of surgery with inflammation, congestion and hypopyon as seen in the image, is suggestive of acute bacterial endophthalmitis. The treatment of acute bacterial endophthalmitis is intravitreal antibiotics.

Endophthalmitis is an intraocular inflammation affecting the inner structures of the eyeball. Acute postoperative bacterial endophthalmitis occurs within 1-7 days after cataract surgery and the most important factor to prevent it is preoperative administration of 5% povidone iodine. The cardinal features of endophthalmitis are pain, swelling of the lid, and a decrease in vision. The patient classically shows inflammation of the eye, corneal edema and a hypopyon due to the pouring of exudates in the vitreous cavity, anterior chamber, and posterior chamber.

Once the endophthalmitis sets in it is important to treat all the cases empirically with a combination of antibiotics and steroids from the beginning. The various drugs, routes and surgical management used to treat endophthalmitis are as follows:

- The mainstay of treatment is intravitreal injections of antibiotics.
- First choice: Vancomycin 1 mg in 0.1 mL plus ceftazidime 2.25mg in 0.1ml.
- Second choice: Vancomycin 1 mg in 0.1 mL plus amikacin 0.4mg in 0.1ml.
- If there is no improvement, a repeat intravitreal injection should be given after 48 hours
- Topical antibiotics are started immediately and used frequently.
- Vancomycin (50mg/ml) or cefazolin (50mg/ml)
- Amikacin (20mg/ml) or tobramycin

- Systemic antibiotics have a limited role in the treatment but are commonly used in practice.
- Steroid therapy includes:
 - Intravitreal injection of dexamethasone (0.4mg in 0.1ml) which is started along with antibiotics.
 - Topical dexamethasone (0.1%)
 - Oral steroids such as prednisolone are started after 24 hours of intensive antibiotic therapy.
- Supportive therapy includes:
 - Cycloplegics such as atropine 1%
 - Anti-glaucoma drugs such as oral acetazolamide or timolol in patients with raised intraocular pressure
 - Pars plana vitrectomy is done if the patient does not improve with the above therapy in 48-72 hours or when the visual acuity is reduced to hand movements near the eye

Other options:

Option A: Steroids are given in endophthalmitis to reduce the inflammation of the tissues. The preferred route of administration is either by intravitreal injections, topical concentrations or oral route. Intravenous route (option A) of administration of steroids is not used in endophthalmitis.

Option B: Systemic administration (intravenous) of antibiotics have a limited role in the treatment but are commonly used in practice as an add on.

Solution to Question 8:

The most common feature seen in diabetic oculomotor palsy among the given options is ptosis.

In the anatomical arrangement of the nerve fibers of the 3rd cranial nerve, the fibers controlling the pupillary reflex are present superficially and are thus spared from diabetic ischemia-induced injury which starts from the central nerve fibres. Thus, pupil sparing is commonly seen in diabetic patients.

The etiology of diabetic neuropathy is hyperglycemia-induced damage to nerve cells and neuronal ischemic change due to microvascular angiopathy.

Involvement of the third cranial nerve is most common and it often presents with early features of ptosis and diplopia with pupil sparing.

The involvement of levator palpebrae superioris muscle of the eyelid manifests as ptosis and the involvement of extraocular muscles such as medial rectus, inferior rectus and inferior oblique presents as ophthalmoplegia with diplopia. All the muscles mentioned above are supplied by the Oculomotor nerve.

Note: Aneurysm of the posterior communicating artery is an important cause of isolated third nerve palsy with involvement of the pupil. This is because the aneurysm impinges on the superficial pupil nerve fibres of the 3rd cranial nerve.

Solution to Question 9:

The most common site of an orbital blowout fracture is the floor of the orbit.

Blow-out fractures are isolated comminuted fractures caused from direct trauma to the orbit that results in a backward displacement of the eyeball and an increase in the intraorbital pressure. This causes a fracture at the weakest point of the orbital wall which is usually the posteromedial portion of the maxillary bone in the orbital floor.

The second most common site for orbital blowout fracture is the medial wall of the orbit (Option A).

The fractures of the roof (Option C) are relatively rare and most often do not require any treatment. The lateral wall (Option B) of the orbit is the most solid orbital wall and a fracture here is usually seen in injuries with extensive facial damage.

The patient with blow out fracture usually presents with immediate periorbital edema and blood extravasation in and around the orbit. Proptosis of the eyeball secondary to orbital edema or hemorrhage with paresthesia along the infraorbital nerve are also very common.

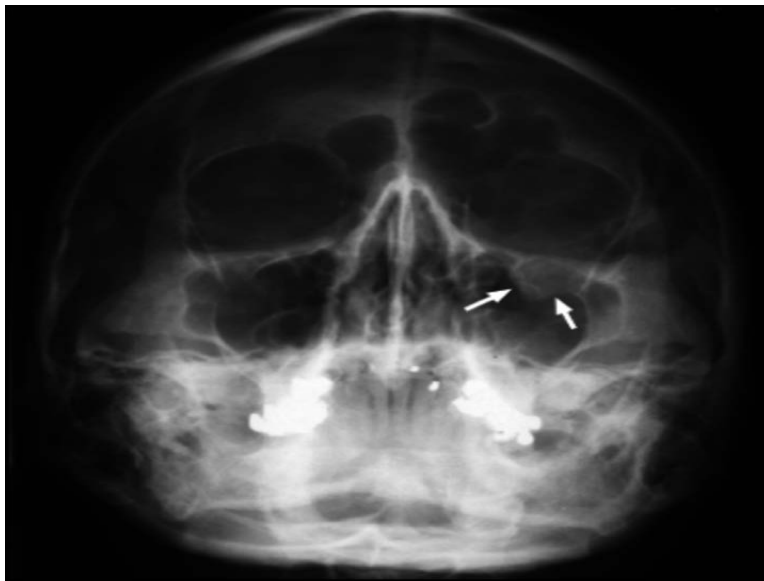
Imaging by a plain x-ray in water's view or by CT demonstrates depression of bony fragments and orbital contents herniating through the orbital floor into the maxillary antrum. This is called as the "hanging drop sign" or "tear drop sign".

Definitive treatment is by surgical repair 10-14 days after injury with initial supportive management with analgesics and antibiotics.

The below given CT image shows hanging drop sign.



The below given X-ray image in water's view shows "hanging drop sign".



Solution to Question 10:

The patient most likely has acquired syphilis due to unprotected sex and has developed neurological and visual disturbances secondarily as a complication of neurosyphilis. Argyll-Robertson pupils (ARP) are seen associated with neurosyphilis where the pupils accommodate but do not react to light.

Argyll Robertson pupil is attributed to a dorsal midbrain lesion of neurosyphilis that affects the pupillary light pathway but spares the more ventrally placed pupillary near reflex pathway (Accommodation pathway), thereby causing a light-near dissociation.

Features of Argyll Robertson pupil (ARP) include:

- Bilateral small, irregular pupils in dim light
- Accommodation Reflex Present (ARP)- The eye accommodates by constricting the pupil for a near target.
- Light Reflex absent: pupils do not constrict when exposed to bright light.
- Cocaine eye drops (mydriatics) induced mydriasis.

Other clinical features in neurosyphilis:

- Fleeting and repetitive lancinating pains, primarily in the legs
- acute abdominal pain with vomiting (visceral crisis)
- Tabes dorsalis with Ataxia and gait disturbances due to loss of position sense
- Loss of reflexes in the legs
- Paresthesias, bladder disturbances

Other causes of light-near dissociation are:

In unilateral eye

- Adie pupil
- Herpes zoster ophthalmicus
- Aberrant regeneration of the third cranial nerve

In bilateral eyes

- Neurosyphilis
- Type 1 diabetes mellitus
- Myotonic dystrophy

Note: Adie's pupil more commonly manifests in young women with unilateral large regular pupil (Anisocoria) with absent light reflex and sluggish and tonic accommodation reflex.

It is caused by post-ganglionic parasympathetic pupillary supply damage and is usually associated with absent knee jerk.

Adie's pupil constricts with weak pilocarpine (0.1%) drops, while normal pupil and Argyll Robertson pupil does not.

Solution to Question 11:

Roth's spots are seen in acute leukemia.

Roth's spots are retinal hemorrhages with white or pale center. The white center due to coagulated fibrin-platelet plug is surrounded by red rim that is caused by hemorrhage. Roth's spots may be seen in:

- Subacute bacterial endocarditis
- Leukemia
- Diabetes
- Sickle cell retinopathy
- Hypertensive retinopathy
- HIV retinopathy
- Systemic lupus erythematosus

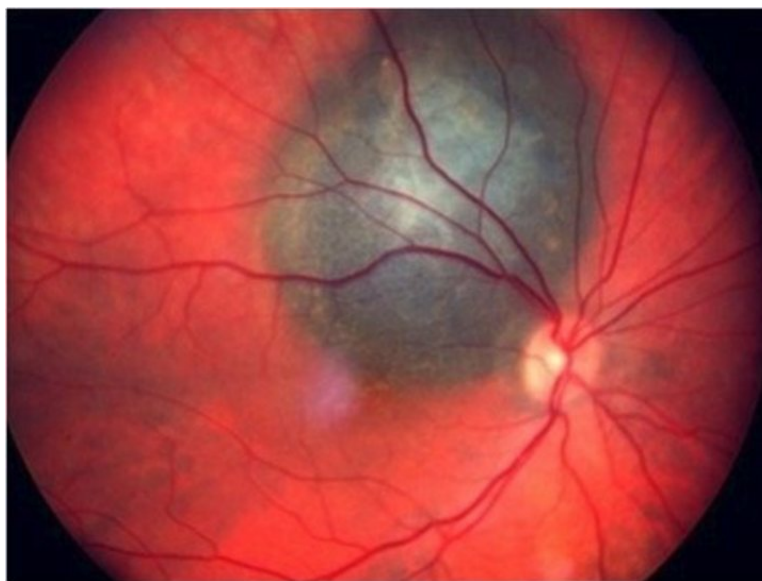
The fundus image below shows Roth's spots.



Option B: Uveal melanoma develops due to malignant transformation of the normal melanocytes present in the uveal tissue. Primary posterior uveal melanoma (choroidal melanoma) which forms the majority of uveal melanoma manifests as:

- A solitary elevated subretinal grey-brown dome shaped mass
- "Collar stud" appearance on B-scan - when the tumor breaks through the Bruch's membrane and protrudes into subretinal space
- "Double circulation" sign

Primary anterior uveal melanoma frequently involves the iris and presents as nodular melanocytic mass on the iris. The below given fundus image shows the "double circulation sign" of choroidal melanoma.



Option C: Uveal lymphoma is characterized by localized or diffuse uveal infiltrates of malignant lymphoma cells. Fundus examination in these patients shows diffuse creamy thickening of posterior uvea with nodular accentuation.

Solution to Question 12:

The power of prescription glasses for a 50-year-old woman should be +1.5D.

Presbyopia is the condition of decreasing near vision due to an age-related decrease in the magnitude of accommodation.

Basic principles for correction in presbyopic patients are given below:

- Correct any refractive error for distance present in the patient first.
- Add the presbyopic correction(correction for near vision) of each eye to any distant correction that is present.
- For presbyopic correction, the weakest convex lens(plus lens) with which the patient can see clearly at the near point is prescribed, since over-correction will also result in eye fatigue symptoms.

Presbyopic spectacles may be unifocal, bifocal or Multifocal.

A rough prescription of presbyopic glasses can be made by the age of the patient in an emmetrope as follows:

Age	Power
45 ye ars	+1 to +1.25 D
50 ye ars	+1.5 to 1.75 D
55 ye ars	+2 to +2.25 D
60 ye ars	+2.5 to +3D

Ophthalmology NEET 2019

Question 1:

Which of the following would be prescribed for simple myopic astigmatism?

- a) +1.00 DS
- b) -1.00 DC X 180 degree
- c) -1.00 DS
- d) -1.00 DS -1.00 DC X 180 degree

Question 2:

Treatment of ptosis caused by Horner's syndrome is:

- a) Levator resection
- b) Frontalis sling
- c) Fasanella servat operation
- d) None of the above

Question 3:

Which of the following is not true about retinitis pigmentosa?

- a) Retinal pigmentation
- b) Pale waxy disc
- c) Narrowing of vessels
- d) ERG – normal

Question 4:

A 45-year-old patient comes with complaints of progressive weakness as the day progresses. He is found to have antibodies against nicotinic acetylcholine receptors. Which is the most common ocular finding in this disease?

- a) Diplopia
- b) Lagophthalmos

- c) Exophthalmos
- d) Ptosis

Question 5:

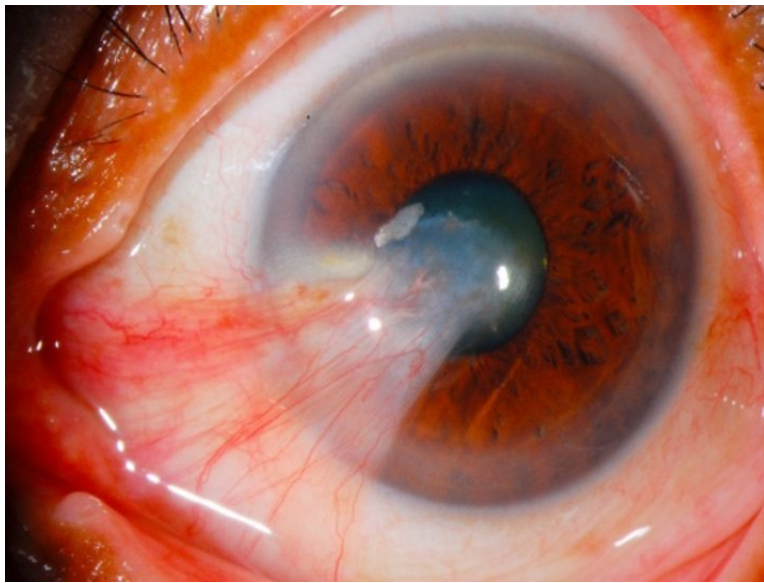
A patient with a history of trauma, presented with diplopia, epistaxis, and restricted eye movements. Looking at the CT image given below, what will be your diagnosis?



- a) Le- fort fracture
- b) Fracture maxilla
- c) Fracture zygomatic
- d) Blow out fracture

Question 6:

What is the likely diagnosis from the image given below?



- a) Pinguecula
- b) Concretions
- c) Pterygium
- d) Bitots spots

Question 7:

Which of the following statements is false regarding accommodative esotropia?

- a) It is the most common type of squint.
- b) Miotics are used as the treatment strategy in the high AC/A ratio
- c) It is associated with high myopia.
- d) Refractive type can be fully corrected by use of spectacles.

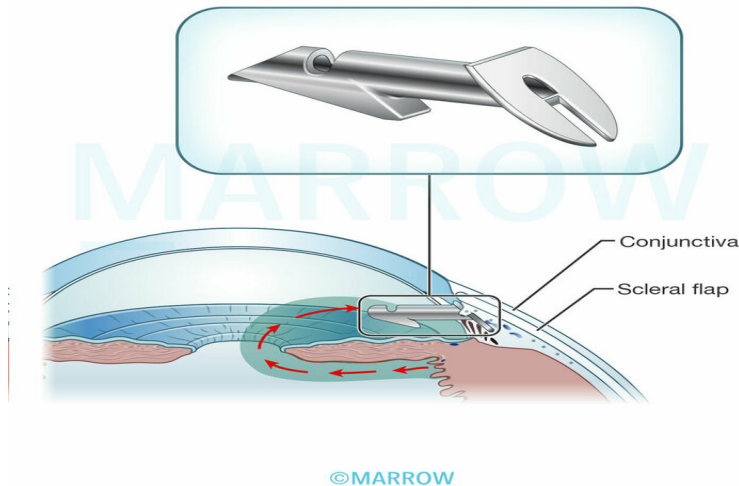
Question 8:

Which of the following regimen is commonly used in the treatment of retinoblastoma?

- a) Vinblastine, etoposide and bleomycin
- b) Vinblastine, vincristine and etoposide
- c) Vinblastine, vincristine and cisplatin
- d) Vincristine, carboplatin and etoposide

Question 9:

A 50-year-old male presented with progressive blurred vision, photophobia, and eye pain. On examination, there was conjunctival redness, a mid-dilated pupil, and a shallow anterior chamber in the right eye. He has been treated both medically and surgically for the condition but showed no improvement. The following procedure incorporates a device to aid drainage. Identify the device.



- a) Ex-Press™ implant
- b) Ahmed glaucoma valve
- c) Molteno implant
- d) Baerveldt implant

Question 10:

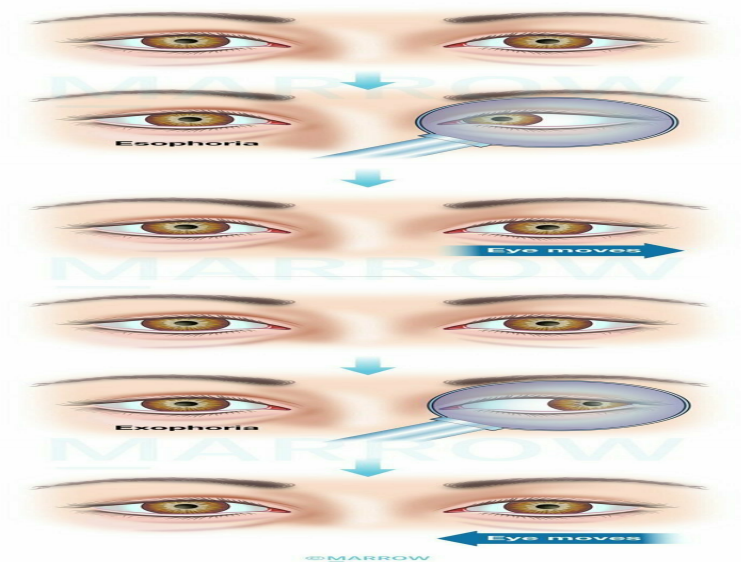
Identify the instrument.



- a) Maddox wing
- b) Maddox glass
- c) Maddox rod
- d) Phoropter

Question 11:

Identify the test being shown below.



- a) Obscure test
- b) Prism test

- c) Cover uncover test
- d) Maddox rod test

Answer Key

Question No.	Correct Option
1	b
2	c
3	d
4	d
5	d
6	c
7	c
8	d
9	a
10	c
11	c

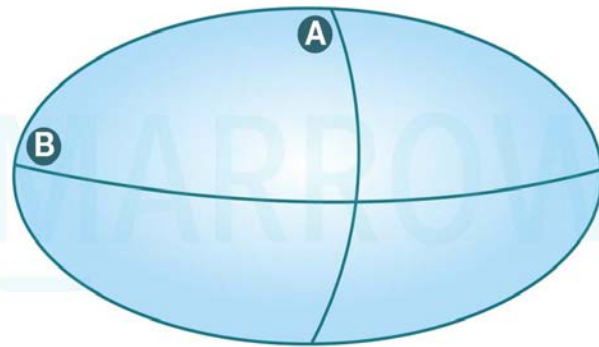
Detailed Explanations

Solution to Question 1:

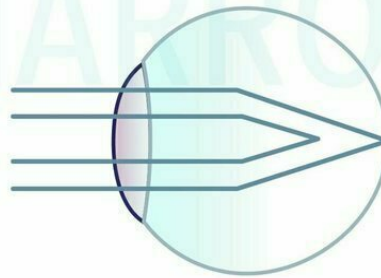
Among the above options, -1.00 DC × 180 degrees is the correct prescription for simple myopic astigmatism (SMA).

There are 3 components for prescription of glasses.

1. The first number is the refractive power of the spherical component. It is represented as convex (+) or concave (-) dioptre sphere (DS).
2. The second number denotes the power of the cylinder, represented as convex (+) or concave (-) dioptre cylinder (DC).
3. The above two components are followed by an x representing the axis (orientation) of the cylinder.



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In the most common type of SMA - with-the-rule astigmatism - meridian A (represented by 90 degrees) is more curved than meridian B (represented by 180 degrees). The light rays refracting from B focus on the retina, whereas those refracting from A focus in front of the retina.

To correct this error, a negative cylindrical lens (DC) is used to focus the myopic rays of meridian A on the retina. This correction needs to occur at 90 degrees (meridian A). The axis of a cylindrical lens is perpendicular to the required correction and thus, the spectacles will have an axis of 180 degrees.

There is no spherical refractive error since the light rays from meridian B lie on the retina. Thus DS is 0.

Solution to Question 2:

Fasanella servat operation is done for cases of mild ptosis (1.5-2 mm) with retained levator palpebrae superioris muscle function as seen in Horner's syndrome.

Ptosis in Horner's syndrome is due to the loss of sympathetic supply to the muller's muscle of the lid and not due to levator muscle palsy. Fasanella servat operation is not done in patients with severe ptosis and poor levator function.

Option A: Levator resection is performed for moderate and severe grades of ptosis but with retained levator function. This operation is contraindicated in patients with poor levator function.

Option B: Frontalis sling also known as Brow suspension is done in patients with severe ptosis with poor or absent levator muscle function. Here, the lid is anchored to the frontalis muscle via a sling.

Surgical treatment of various types of ptosis

Surgery	Amount of ptosis that can be treated	Indications
Fasanella Servat	< 2 mm -mild ptosis	Horner Syndrome
Levator Resection– Anterior approach	Any amount (but with retained levator function)	Larger resection in ptosis
Levator resection – Conjunctival approach	Any amount (but with retained levator function)	Moderate resection in ptosis
Levator resection with aponeurotic reinsertion	Any amount (but with retained levator function)	Acquired ptosis
Frontalis suspension– Fasciata	>4 mm -severe ptosis	Congenital ptosis– Marcus Gunn ptosis

Solution to Question 3:

Electroretinogram (ERG) is abnormal in retinitis pigmentosa (RP).

ERG is a sensitive diagnostic test in retinitis pigmentosa that shows reduced scotopic, subnormal responses in early disease and becomes completely extinguished in advanced disease.

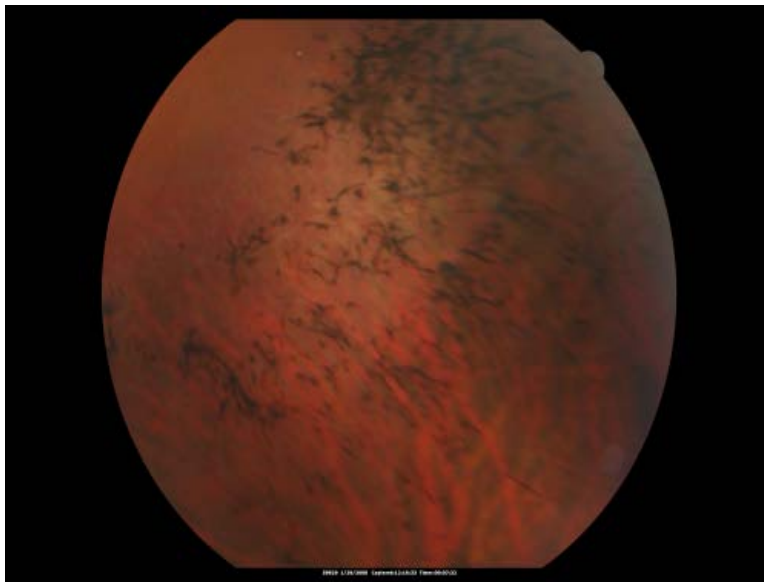
Electrooculogram (EOG) is also abnormal in retinitis pigmentosa.

Retinitis pigmentosa (RP) is an inherited or acquired pigmentary retinal dystrophy affecting the rods more than the cones. The earliest manifestation is worsening of night blindness and delayed adaptation which gradually progresses to tunnel vision and ultimately loss of central vision.

The classic triad of findings in RP comprises:

- Bone-spicule retinal pigmentation
- Arteriolar attenuation
- 'Waxy' disc pallor

The image below shows the classical triad findings of retinitis pigmentosa.



Solution to Question 4:

The given clinical scenario is suggestive of myasthenia gravis and ptosis is the most common ocular feature of the disease.

Myasthenia gravis (MG) is an autoimmune disorder of the neuromuscular junction most commonly caused by anti-acetylcholine receptor antibodies. The disorder is characterized by weakness and fatigability of skeletal muscles where the weakness increases during repeated use and may improve following rest.

Ptosis and diplopia are common initial complaints due to early involvement of the lids and extraocular muscles (EOMs). Of them, Ptosis is a more common manifestation than diplopia. Ptosis in myasthenia gravis is insidious, bilateral, and frequently asymmetrical.

Investigations of myasthenia gravis include:

- Ice pack test -This tests for a transient improvement of the ptotic eyelid on placing an ice pack for 2 minutes, as the cold inhibits the breakdown of acetylcholine by acetylcholinesterase.
- Tensilon (Edrophonium) test - This tests for a transient improvement in the weakness on administration of a short-acting anticholinesterase such as edrophonium.
- Antibody testing - for Acetylcholine receptor (AChR) antibodies, MuSK protein antibodies helps support a diagnosis of myasthenia gravis.

Solution to Question 5:

The given clinical scenario plus the supportive CT image showing the "hanging drop sign" is suggestive of a blowout fracture of the orbital wall.

Blow-out fractures are isolated comminuted fractures caused from direct trauma to the orbit that results in a backward displacement of the eyeball and an increase in the intraorbital pressure.

This causes a fracture at the weakest point of the orbital wall which is usually the posteromedial portion of the maxillary bone in the orbital floor. The second most common site for orbital blowout fracture is the medial wall of the orbit.

The patient with blow out fracture usually presents with immediate periorbital edema and blood extravasation in and around the orbit. Proptosis of the eyeball secondary to orbital edema or hemorrhage with paresthesia along the infraorbital nerve are also very common.

Imaging by a plain x-ray in water's view or by CT demonstrates depression of bony fragments and orbital contents herniating through the orbital floor into the maxillary antrum. This is called as the "hanging drop sign" or "tear drop sign".

Definitive treatment is by surgical repair 10-14 days after injury with initial supportive management with analgesics and antibiotics.

Le fort fracture- Is the term used to describe different types of maxillary fractures.

Solution to Question 6:

The likely diagnosis, based on the above image, is pterygium.

Pterygium is a subconjunctival fibrovascular tissue encroaching the cornea. It is a degenerative condition of the sub-conjunctival tissue and is never inflammatory. It occurs due to exposure to UV-B radiation and the most common site is the nasal.

It is made up of three parts, "cap" (an avascular halo-like zone at the advancing edge), a head, and a body.

Clinical features:

- Astigmatism with the rule - most common finding
- Stocker's line - deposition of iron on the advancing edge
- Fuchs islets - small, discrete, whitish flecks consisting of clusters of pterygial epithelial cells, often present at the advancing edge.

The differential diagnosis is pseudo pterygium.

The treatment for pterygium is usually excision. To prevent a recurrence, mitomycin C or 5 fluorouracil is used. Other methods:

- Auto-grafting- Conjunctival limbal autograft (CLAG) from the superior limbal conjunctiva is the best method.
- Bare sclera technique- high recurrence rate of 30 to 80%.
- PERFECT surgery- zero rates of recurrence. (PERFECT stands for Pterygium Extended Removal Followed by Extended Conjunctival Transplant)

Pterygium	Pseudo-ptyerygium
Degenerativeprocess	Inflammatoryprocess
Occurs inelderly	Can occur at any age

Pterygium	Pseudo-ptyerygium
Situated at the palpebral aperture	Can occur at any site
Either progressive, regressive, or stationary	Always stationary
A probe cannot be passed underneath	A probe can be passed under the neck

Solution to Question 7:

Accommodative esotropia is associated with a high degree of hypermetropia (+4 to +7 D).

Accommodative esotropia is the most common type of squint. Typically presents around 2 to 3 years of age.

Types:

- Refractive
- Associated with high hypermetropia
- Fully correctable by spectacles
- Non-refractive type
- Abnormal accommodative convergence/accommodation (AC/A) ratio
- Fully corrected by bifocal glasses
- Miotics useful in cases where glasses are unsuitable
- Echothiophate causes iris cyst
- Prevention of iris cysts- Phenylephrine drops
- Mixed type

The accommodative convergence/accommodation (AC/A) ratio is defined as the amount of convergence measured in prism diopters per unit (diopter) change in accommodation.

The image below shows accommodative esotropia with and without correction.



Solution to Question 8:

Chemotherapy commonly used is a three-drug regimen containing vincristine, carboplatin, and etoposide.

Solution to Question 9:

The clinical vignette points to a diagnosis of glaucoma not responsive to medical treatment and trabeculectomy and the device used is an ex-Press™ glaucoma implant (mini shunt).

It is a form of minimally invasive glaucoma surgery (MIGS) to increase aqueous drainage.

Ex-PRESS glaucoma implant is a small stainless steel device. It is inserted under a scleral flap during a modified trabeculectomy. Complications such as hypotony are lower than with standard trabeculectomy.

Glaucoma drainage devices are devices that allow aqueous outflow by establishing communication between the anterior chamber and sub-Tenon's space/sub-conjunctival space. While these devices were traditionally reserved for high-risk patients, or after trabeculectomy had failed, they are becoming increasingly popular.

The commonly used devices are as follows:

1. Shunts using episcleral explants
 - Ahmed glaucoma valve: Silicone
 - Molteno implant - Polypropylene plate and silicone plate
 - Baerveldt implant - Silicone
2. Mini-shunts
 - Ex-Press™ Mini-Shunt: Stainless steel
 - iStent® - Titanium

Solution to Question 10:

The above instrument is a Maddox rod.

Maddox Rod Test

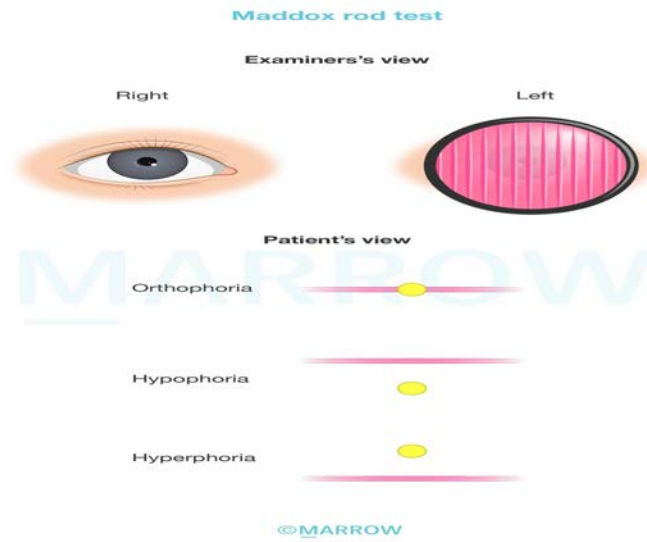
A Maddox rod is a specially constructed device, consisting of a series of parallel cylinders, that converts a point source of light into a line image. It can be used to test for horizontal and vertical deviations, and when used in conjunction with another Maddox rod, for cyclo deviations.

A patient is asked to fix on a point light in the center of Maddox tangent scale placed 6 meters away. A Maddox rod is placed in front of one eye with the axis of the rod parallel to the axis of deviation. The patient will see a point light with one eye and a red line with the other. Due to dissimilar images of the two eyes, fusion is broken and heterophoria becomes manifest. The number on the Maddox tangent scale where the red line falls will be the amount of heterophoria in degrees.

An illustration of a hand mirror with a black handle and a circular frame. The mirror's surface is filled with red diagonal lines. A light blue trapezoidal shape extends from the mirror, leading to a magnified circular inset. This inset shows a top-down view of a head with red diagonal lines, representing the back of the head. The background is white with faint, large, light blue letters spelling 'MARROW' diagonally across it.



3028



Solution to Question 11:

The test shown is the cover - uncover test.

Ophthalmology NEET 2020

Question 1:

Which of the following layers is responsible for maintaining the hydration and transparency of the cornea?

- a) Descemet's membrane
- b) Endothelial cells
- c) Stroma
- d) Corneal epithelium

Question 2:

What is the characteristic feature of a fungal ulcer?

- a) Reverse hypopyon
- b) Dendritic ulcer on a fluorescein dye
- c) Ring abscess
- d) Satellite lesion

Question 3:

What is the term used to describe an extra row of eyelashes behind the grey line?

- a) Trichiasis
- b) Distichiasis
- c) Tylosis
- d) Madarosis

Question 4:

A middle-aged woman presented with bilateral proptosis, restriction of eye movements, and chemosis. Her thyroid profile showed she was euthyroid. What is the most probable cause?

- a) Orbital pseudotumor
- b) Thyroid ophthalmopathy

- c) Orbital lymphoma
- d) Orbital cellulitis

Question 5:

Which of the following drugs increases the uveoscleral outflow?

- a) Timolol
- b) Latanoprost
- c) Pilocarpine
- d) Carbonic anhydrase inhibitor

Question 6:

Photostress test is used to differentiate between:

- a) Macula and optic nerve disease
- b) Cataract and glaucoma
- c) Lens and cornea
- d) Retina and vitreal pathologies

Question 7:

Shifting fluid sign is seen in?

- a) Exudative RD
- b) Traction RD
- c) Rhegmatogenous RD
- d) Retinal dialysis

Question 8:

A 40-year-old diabetic patient presented with the ■squint. On examination, the secondary deviation was found to be more than the primary deviation. Forced duction test is negative. Which of the following types of squint is most likely to be present in this patient?

- a) Comitant squint

- b) Restrictive squint
- c) Paralytic squint
- d) Pseudosquint

Question 9:

A physician observed the following finding in a small child who came for an eye exam. This finding is associated with which of the following conditions?



- a) Myopia
- b) Astigmatism
- c) Hyperopia
- d) Emmetropia

Question 10:

Sympathetic ophthalmia is seen after _____.

- a) Penetrating trauma
- b) Blunt trauma
- c) Chemical injury
- d) Urinary tract infection

Question 11:

A patient presented with unilateral proptosis and bilateral abducent palsy. What is the most probable cause ?

- a) Retinoblastoma
- b) Orbital pseudotumor
- c) Orbital cellulitis
- d) Cavernous sinus thrombosis

Answer Key

Question No.	Correct Option
1	b
2	d
3	b
4	b
5	b
6	a
7	a
8	c
9	c
10	a
11	d

Detailed Explanations**Solution to Question 1:**

The hydration level and transparency of the cornea is maintained mainly by the endothelial cells of the cornea.

The endothelium possesses Na⁺/K⁺ ATPase, which keeps the corneal stroma in a relatively dehydrated state. This helps maintain corneal transparency. Endothelial cells decrease in number with age. If the cornea becomes overhydrated, it begins to scatter light and gradually begins to lose its transparency

Other factors contributing to the transparency of the cornea are as follows:

- Regular arrangement of corneal lamellae (Maurice theory)
- The peculiar refractive index of corneal lamellae (Goldmann theory)
- Avascularity of the cornea
- A balance between the intraocular pressure (IOP) and swelling pressure

The different layers of the cornea are as follows:

- Epithelium
- Bowman's membrane
- Stroma (substantia propria)
- Dua's layer (pre-Descemet's layer)
- Descemet's membrane
- Endothelium

Solution to Question 2:

Satellite lesions are characteristic of fungal ulcer.

The signs of a fungal ulcer include:

- Irregular margins of the ulcer.
- Presence of satellite lesions surrounding the ulcer.
- An immune ring (Wessely) may be seen around the ulcer due to deposition of the immune complexes.
- Slough is usually dry, with feathery borders surrounded by a yellow line of demarcation.
- Hypopyon may be present or absent. If present, it is thick, immobile due to the direct invasion of the fungal hyphae into the anterior chamber.

Solution to Question 3:

The extra row of eyelashes behind the grey line is called distichiasis.

Distichiasis can occur due to congenital or acquired causes. These lashes may irritate the cornea. Management is by lid splitting with cryotherapy to the base of the hair follicles.

Other options:

Option A: Trichiasis is inward misdirection of cilia, with the normal position of the lid margin.

Option C: Tylosis is the thickening of the tarsal plate.

Option D: Madarosis is the loss of eyelashes or eyebrows.

Solution to Question 4:

The given clinical scenario is suggestive of thyroid ophthalmopathy.

Thyroid eye disease (TED) is the most common cause of both bilateral and unilateral proptosis in an adult in the euthyroid state. The main features of TED include lid retraction, proptosis, periorbital edema, chemosis, and restrictive myopathy.

Other options-

Option A- Orbital pseudotumor presents with painful eye movements periorbital edema, erythema, proptosis, ptosis, diplopia.

Option C- Orbital lymphoma usually presents in the 7th decade, most commonly observed as a painless palpable mass in the superolateral quadrant.

Option D- Orbital cellulitis is usually a unilateral condition with pain, proptosis, and chemosis.

NOTE: Thyroid eye disease can also, though less commonly, occur in euthyroid patients.

Solution to Question 5:

Latanoprost is a prostaglandin drug, which acts by increasing the uveoscleral outflow and thereby lowers the intraocular pressure.

Mechanism of action: Latanoprost is a prodrug and gets activated during its passage through the cornea. It acts by stimulating the prostanoic receptors on the ciliary muscles. This leads to the release of matrix metalloproteinases, which increases the uveoscleral outflow.

Solution to Question 6:

Photostress test is used to differentiate between the disease involving the macula and optic nerve.

This test is of particular use in early macular diseases such as central serous retinopathy.

In this test, one eye is covered and the patient is asked to read the smallest line on the near chart. A bright light is then shone into the eye for 15 seconds, and the patient is asked to read the line again, and the time taken to recover is noted. The test is repeated with the other eye.

In normal people and in patients with optic nerve disease, the recovery time is not affected.

Whereas, in patients with macular disease the recovery time is prolonged and is usually more than one-third of the normal recovery time. This is because, in macular disease, the photoreceptors are damaged, and there is a delay in the visual pigment regeneration process.

Solution to Question 7:

The presence of shifting fluid is a hallmark feature of serous or exudative retinal detachment.

In serous retinal detachment, there is a subretinal accumulation of the serous fluid mostly due to the breakdown in the blood-retinal barrier. When the position of the head is altered, the fluid shifts to the dependent portion. Serous retinal detachment has a smooth convex configuration. A retinal tear or preretinal traction is absent.

Other options:

Option B: Traction RD has a concave configuration and breaks are absent. Retinal mobility is severely reduced and shifting fluid is absent.

Option C: Rhegmatogenous RD presents with tobacco dust in anterior vitreous with substantial vitreous blood or inflammatory cells. Retinal breaks are also seen.

Solution to Question 8:

The above clinical scenario is suggestive of paralytic squint. It is characterized by the presence of secondary deviation more than the primary deviation and a negative forced duction test.

The squint may be either comitant or incomitant.

- Comitant squint: The angle of squint measured is same in all gazes.
- Incomitant squint: The angle of squint measured is different in different gazes.

The differences between comitant and incomitant squint are shown in the table below:

The incomitant squint is further divided into two main types: Paralytic and restrictive.

Paralytic squint occurs due to the paralysis of extraocular muscle while restrictive squint is due to the mechanical restriction caused by tight or fibrosed muscle.

Forced duction test(FDT) differentiates between these two types. In this test, the eye is moved mechanically with the help of a forceps.

- The test is negative if there is no resistance to the movement. It is seen in a paralytic squint.
- The test is positive if there is resistance to the movement. It is seen in a restrictive squint.

Features	Comitantsquint	Incomitantsquint
Magnitude of squint	Same in all gazes	Varies with the gaze
Diplopia	Absent	Present
Ocular movements	Full	Restricted
False projection	Absent	Present
Abnormal head posture	Absent	Present
Secondary deviation	Equal to primary deviation	More than primary deviation

Solution to Question 9:

The given image shows convergent squint or esotropia, which is commonly seen in children with hyperopia, also known as hypermetropia.

Esotropia results due to increased accommodative effort in cases where there is under correction of hypermetropia. The condition may be prevented by full hypermetropic correction. If the child is unable to wear the glasses due to flat nasal bridge or if the child is less than 2 years old, the accommodation may be treated with drugs. Miotic drugs with anticholinesterase activity such as di-isopropyl fluorophosphate (DFP) or echothiophate can be used in these children. Further treatment may include orthotropic training or surgical correction of the squint.

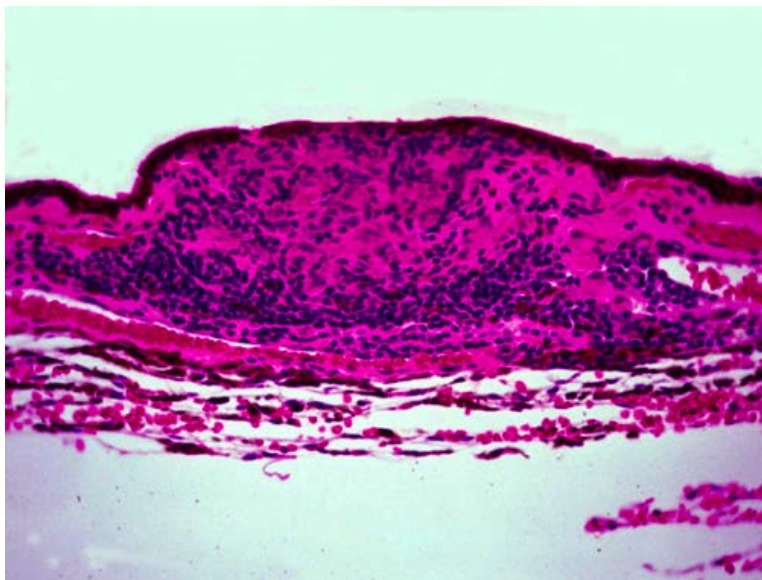
Solution to Question 10:

Sympathetic ophthalmia is seen after a penetrating trauma.

Sympathetic ophthalmia is a bilateral granulomatous panuveitis seen after a penetrating trauma or less frequently, following intraocular surgery, usually multiple vitreoretinal procedures. Patients present between 2 weeks and 3 months after initial injury.

The earliest symptom is blurred near vision in the sympathizing (uninjured) eye. The earliest sign is a retrolental flare in the sympathizing eye. Anterior uveitis develops in both eyes. The fundus shows multifocal choroidal infiltrates in the mid-periphery. Sub-retinal pigment epithelium (RPE) infiltrates corresponding to Dalen–Fuchs nodules (image)(a granuloma situated between Bruch membrane and the RPE) are also present.

Steroids are the basis of treatment. High-dose oral prednisolone is given for several months. Immunosuppressants such as azathioprine, cyclosporine, and methotrexate can be used in resistant cases or as steroid-sparing agents. Enucleation of a severely injured eye with no visual potential within 2 weeks of injury prevents the occurrence of sympathetic ophthalmia.



Solution to Question 11:

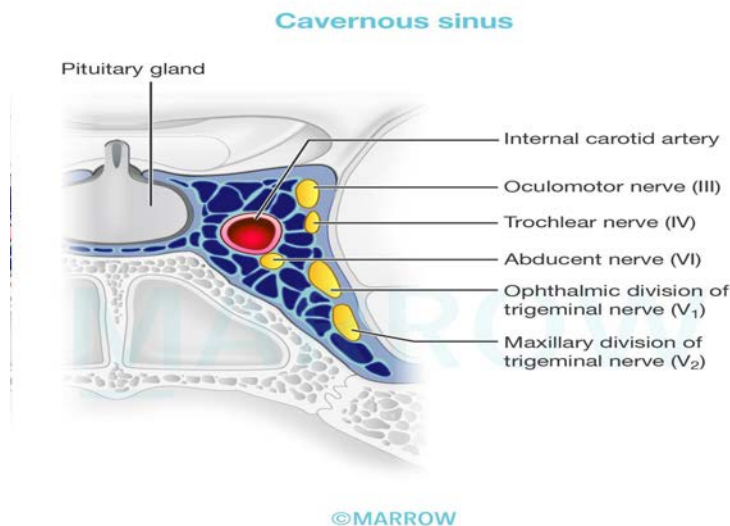
The given clinical scenario is suggestive of cavernous sinus thrombosis.

Features of cavernous sinus thrombosis include early involvement lateral rectus palsy (abducent nerve involvement) since it runs through the cavernous sinus. Oculomotor and trochlear nerve palsy occur later as they are related to the lateral wall of the cavernous sinus. Other signs are proptosis, severe supraorbital pain, edema in the mastoid region, absence of corneal sensation, and dilated pupil.

Options A- Retinoblastoma occurs in young children and presents with unilateral proptosis with leukocoria, buphthalmos, and squint.

Option B- Orbital pseudotumor presents with unilateral proptosis with diplopia, conjunctival chemosis. The inflammation is responsive to systemic corticosteroids.

Option C- Orbital cellulitis presents with unilateral proptosis with pain and decreased ocular motility.



Ophthalmology INI-CET 2021

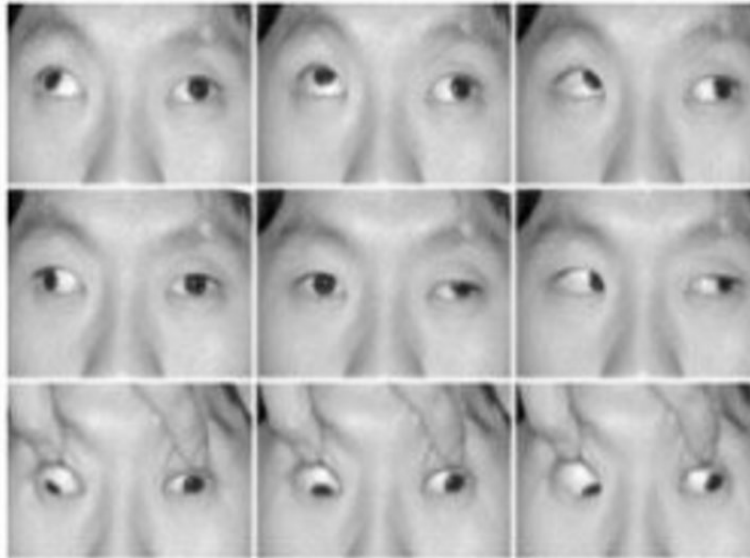
Question 1:

A 30-year-old female presents with redness and pain in the eye. Examination revealed an IOP of 38 mm of Hg, aqueous flare, and corneal precipitates. Which of the following drugs must be avoided in her?

- a) Beta blockers
- b) Mannitol
- c) Carbonic anhydrase inhibitors
- d) Prostaglandin analogs

Question 2:

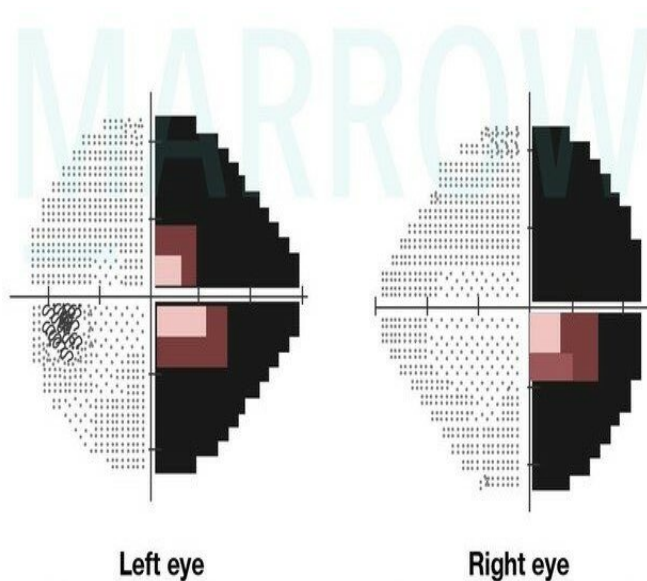
Identify the condition:



- a) Medial rectus palsy
- b) Trochlear nerve palsy
- c) Abducens nerve palsy
- d) Oculomotor nerve palsy

Question 3:

Identify the site of the lesion.



- a) Left occipital lobe
- b) Right occipital lobe
- c) Left geniculate body
- d) Optic chiasma

Question 4:

What is the most common route of the spread of retinoblastoma?

- a) Optic nerve invasion
- b) Hematogenous
- c) Direct invasion
- d) Lymphatic spread

Question 5:

The most common tumor of the lacrimal gland is?

- a) Adenoid cystic carcinoma
- b) Non Hodgkin Lymphoma
- c) Pleomorphic adenoma
- d) Mucoepidermoid carcinoma

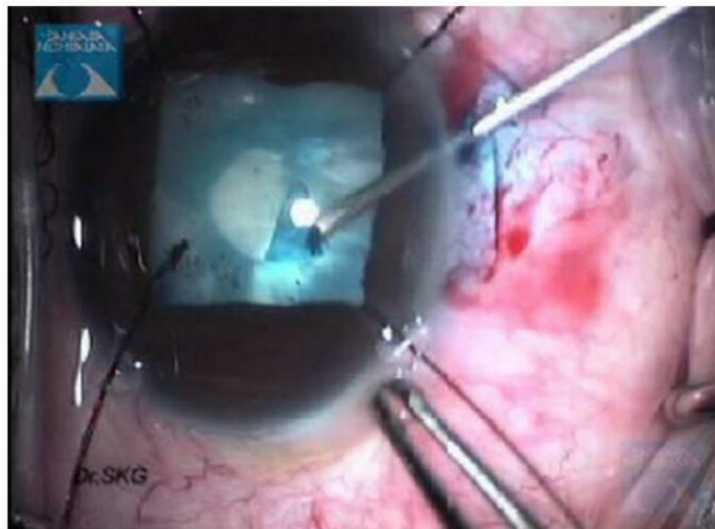
Question 6:

SAFE strategy for trachoma includes all except?

- a) Surgery for trichiasis
- b) Antibiotics
- c) Facial cleanliness
- d) Evaluation of control program

Question 7:

Identify the surgical step shown in the image given below.



- a) Capsulorrhexis
- b) Hydrodissection
- c) Intraocular lens implantation
- d) Lens aspiration

Question 8:

The most common ocular manifestation of COVID-19 is:

- a) Retinal vein thrombosis
- b) Viral keratouveitis
- c) Follicular conjunctivitis

d) Pseudomembranous conjunctivitis

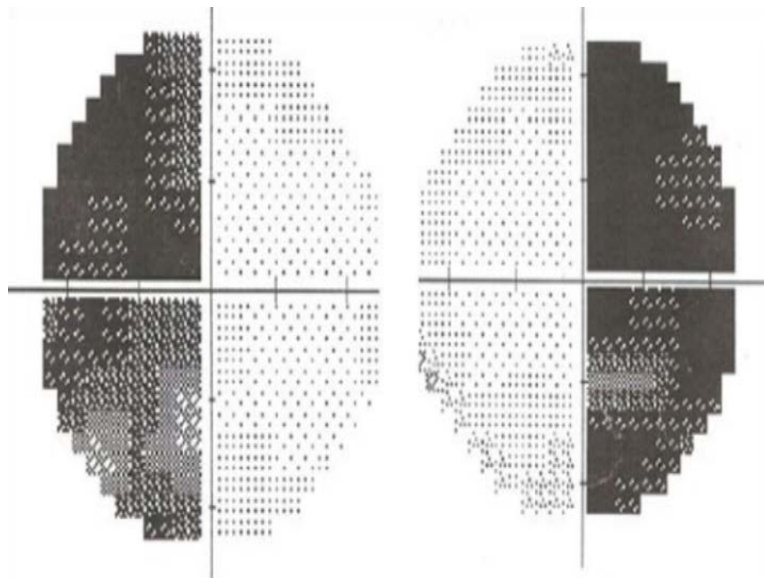
Question 9:

A patient who is a known case of angle-closure glaucoma has been diagnosed with major depressive disorder. Which of the following drugs have to be avoided in this patient?

- a) Amitriptyline
- b) Sertraline
- c) Mirtazapine
- d) Fluvoxamine

Question 10:

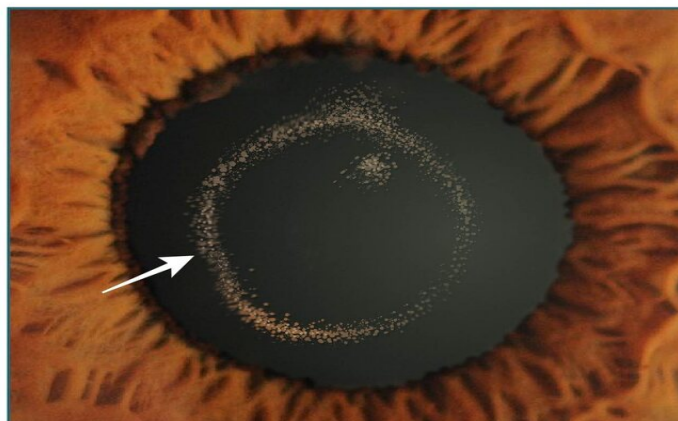
A patient is not able to see the lateral fields. He was having a mild headache for a few weeks and now presented to the clinic. Visual field examination shows the following finding. What is the probable site of the lesion based on the image displayed?



- a) Optic chiasma
- b) Occipital lobe
- c) Optic tract
- d) Optic radiation

Question 11:

Identify the finding marked by the arrow in the below image.

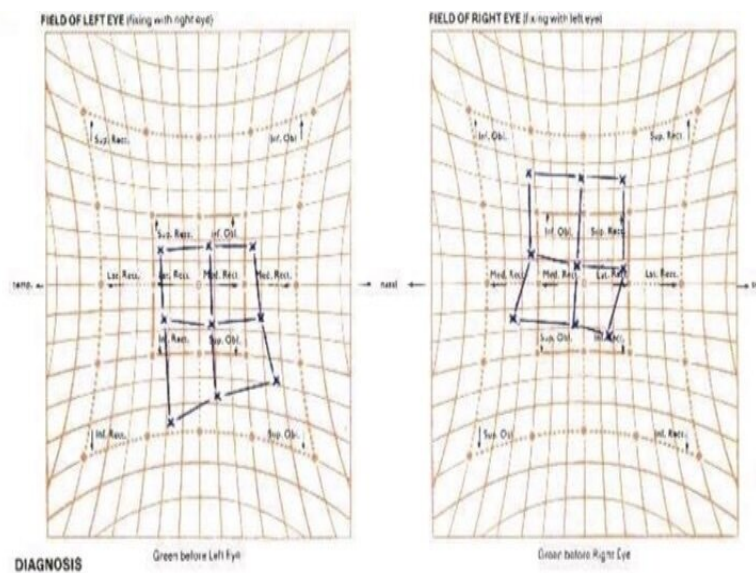


©MARROW

- a) Vossius ring
- b) Weiss ring
- c) Fleischer's ring
- d) Kayser Fleischer ring

Question 12:

Identify the chart given in the picture below.



- a) Hess chart

- b) Snellen chart
- c) Pelli-Robson chart
- d) ETDRS chart

Question 13:

A 14-day-old neonate brought to the outpatient department is found to have bilateral conjunctivitis. Chest X-ray done showed lung infiltrates. Which of the following organisms is most likely responsible for the infection?

- a) Chlamydia trachomatis
- b) Hemophilus influenzae
- c) Streptococcus
- d) Mycoplasma

Answer Key

Question No.	Correct Option
1	d
2	d
3	a
4	c
5	c
6	d
7	a
8	c
9	a
10	a
11	a
12	a
13	a

Detailed Explanations

Solution to Question 1:

The given clinical vignette is suggestive of anterior uveitis with a raised intraocular pressure. Among the given options, latanoprost is the least suited for use in this patient as it can result in worsening of the anterior uveitis.

Latanoprost is a PGF 2 alpha analog that acts by increasing the uveoscleral outflow and thereby lowers the intraocular pressure. Latanoprost is a prodrug and gets activated during its passage through the cornea. It acts by stimulating the prostanoid receptors on the ciliary muscles. This leads to the release of matrix metalloproteinases, which act to increase the uveoscleral outflow. It should be used with caution in those with uveitis, iritis, cystoid macular edema, and herpes simplex keratitis.

Topical prostaglandins like latanoprost, travoprost, bimatoprost, and tafluprost are now used as first-line agents to lower IOP. The most common ocular side effects are conjunctival hyperemia, eyelash thickening, iris hyperpigmentation, and loss of periorbital fat (bimatoprost). The systemic side effects are uncommon and include occasional headache, malaise, myalgia, skin rash, and mild upper respiratory infections.

Solution to Question 2:

The image is suggestive of oculomotor nerve palsy.

Isolated 3rd nerve palsy

Causes

- Microvascular disease such as hypertension and diabetes is the most common cause of third nerve palsy.
- Aneurysm of the posterior communicating artery
- Episodic cases of isolated palsy may be seen due to migraine, compression, ischemia, and alterations in intracranial pressure
- Trauma
- Inflammatory diseases such as syphilis, Lyme disease, sarcoidosis; giant cell arteritis
- Vasculitis

The signs of left oculomotor nerve palsy are summarized in the image attached below.

Description	Image
Profound ptosis due to weakness of levator	
Abduction & depression in 1 st gaze due to unopposed action of lateral rectus & superior oblique	
Intorsion due to intact superior oblique	
Limited depression due to weak inferior rectus	
Limited elevation due to weak superior rectus & inferior oblique	
Dilated pupil due to parasympathetic palsy	
Normal abduction	

©Marrow

Solution to Question 3:

The visual field defect shows right homonymous hemianopia with macular-sparing, which is caused by lesions of the left occipital lobe (contralateral/crossing of fibers).

Such lesions are generally due to the occlusion of the posterior cerebral artery. These are commonly seen in head injury and may be associated with visual hallucinations, denial of blindness (Anton syndrome), and the Riddoch phenomenon (only moving visual targets are perceived). These lesions generally spare the central vision/macular vision. The anatomical reason for this sparing is attributed to the uniqueness in the blood supply. While the part of the occipital lobe representing the peripheral field vision is supplied by the posterior cerebral artery, the area at the tip of the calcarine cortex, representing the central macular vision is supplied by a branch of the middle cerebral artery.

Solution to Question 4:

Direct invasion is the most common mode of spread of retinoblastoma.

Retinoblastoma arises from the neurosensory retina. It is a proliferation of neural cells that have failed to evolve normally. Retinoblastoma is the most common primary intraocular malignancy in children. Tumours are composed of small basophilic cells (retinoblasts), with large hyperchromatic nuclei and scanty cytoplasm.

It is of 4 types:

- Endophytic (into the vitreous) with seeding of tumor cells throughout the eye
- Exophytic (into the subretinal space) leading to retinal detachment
- Mixed type
- Retinocytoma: retina may be diffusely infiltrated

Solution to Question 5:

Pleomorphic adenoma is the most common tumor of the lacrimal gland while adenoid cystic carcinoma is the most common carcinoma of the lacrimal gland.

Solution to Question 6:

E in the SAFE strategy stands for Environmental improvement with better hygiene and sanitation.

Solution to Question 7:

Capsulorrhexis is the step shown in the given image.

Capsulorrhexis is a method of anterior capsulotomy, where an opening is made in the anterior capsule of the lens. It is one of the steps involved in conventional extracapsular cataract extraction (ECCE).

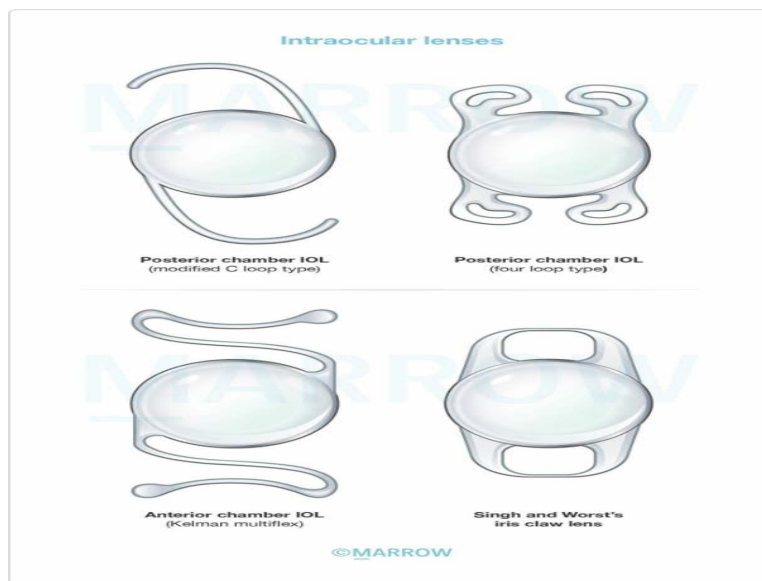
Anterior capsulotomy can be done by one of three methods - can opener's technique, linear capsulotomy or envelope technique, or continuous curvilinear capsulorhexis.

Continuous curvilinear capsulorrhexis (CCC), is considered to be superior to all the preexisting methods because when done correctly, it does not leave any edges or tears. Trypan blue dye can be used to stain the lens capsule for better visibility.

In ECCE, a portion of the anterior capsule of the lens is removed, allowing extraction of the lens nucleus and cortex. This leaves the remainder of the anterior capsule, the posterior capsule, and the zonular support intact. The intact lens nucleus is then removed from the eye through a limbal incision.

An anterior capsulotomy is followed by hydro dissection using a balanced salt solution. This helps to separate the corticonuclear mass from the capsule. Removal of lens nucleus and aspiration of the cortex is done using a two-way irrigation and aspiration cannula. An intraocular lens (IOL) is then implanted and the incision is then closed.

Some of the IOLs used are shown in the image below.



Solution to Question 8:

The most common ocular manifestation of COVID-19 is follicular conjunctivitis.

It has been observed that SARS-CoV-2 needs the ACE-2 receptors for cell invasion. The ACE-2 receptors are found not only in human type 2 alveolar epithelial cells but also in the cornea and conjunctiva. This suggests that ocular surface tissue may be a potential target tissue for SARS-CoV-2.

The ocular manifestations of COVID-19 vary greatly. The most prevalent ocular symptoms were dry eye or foreign body sensation, redness, tearing, itching, eye pain, and discharge.

The most common ocular complication of COVID-19 is mild follicular conjunctivitis. Hemorrhagic conjunctivitis, dry eye disease, episcleritis, chemosis, and photophobia can also occur, but less frequently. Intraocular complications include retinal artery occlusions, hemorrhages, and cotton wool spots caused by complement-mediated thrombotic angiopathy.

Sometimes, conjunctivitis may appear as the only sign and symptom of COVID-19. Patients should therefore undergo a nasopharyngeal RT-PCR test.

All physicians and ophthalmologists should be cautious when addressing a patient with conjunctivitis and should wear eye-protective gear in addition to face masks and other protective devices during clinical examination of these patients.

Solution to Question 9:

Amitriptyline is the drug to be avoided in patients having angle-closure glaucoma with major depressive disorder.

In angle-closure glaucoma, there is apposition of peripheral iris against the trabecular meshwork resulting in obstruction of aqueous outflow by the closure of an already narrow-angle of the

anterior chamber, with an increase in the intraocular pressure being the aftermath.

Amitriptyline is a tricyclic antidepressant that also has a marked anticholinergic activity due to a high affinity for muscarinic acetylcholine receptors and the subsequent blockade of the receptors. These muscarinic cholinergic receptors, specifically M₃ type, upon stimulation, are responsible for contraction of the iris sphincter, ciliary muscle and trabecular meshwork as well as the increased outflow of aqueous humour. Therefore, blocking of these receptors would result in the decreased outflow of aqueous humour, which ultimately leads to an increase in intraocular pressure.

Therefore, amitriptyline when administered in a patient with pre-existing angle-closure type glaucoma can exacerbate the disorder by further impeding the drainage of aqueous humour.

Other options:

Options B and D: Sertraline and Fluvoxamine are selective serotonin reuptake inhibitors with no activity at the muscarinic cholinergic receptors and therefore have no effect on the outflow of aqueous humour and thereby, on the intraocular pressure.

Option C: Mirtazapine is an atypical antidepressant that is a serotonin-norepinephrine reuptake inhibitor with the ability to precipitate angle-closure glaucoma by itself; however, due to lack of any activity at the muscarinic cholinergic receptors, it does not increase the intraocular pressure to the extent that Amitriptyline does.

Solution to Question 10:

The visual field defect in the image is indicative of bitemporal hemianopia, so the probable site of the lesion is the optic chiasma.

Hemianopia or hemianopsia refers to loss of sight in half the visual field. Bitemporal hemianopia refers to loss of sight in the temporal halves of the visual field, seen in lesions affecting the optic chiasma. The most common cause of this is pituitary gland tumours. Other causes include suprasellar aneurysms, chronic arachnoiditis, suprasellar tumours like craniopharyngioma and suprasellar meningioma.

Damage to the middle part of the optic chiasm affects fibres crossing at the nasal retina of each eye, leaving the uncrossed fibres from the temporal retinas intact. The resulting loss of vision is confined to the temporal visual field of each eye and this is known as bitemporal hemianopia. It is also called heteronymous hemianopia to emphasize the fact that the parts of the visual field that are lost in each eye do not overlap.

Individuals with this condition are able to see in both left and right central visual fields, with both eyes open. However, all information from the most peripheral parts of visual fields (which are seen only by the nasal retinas) is lost.

Other options:

Lesions at the occipital lobe, optic tract and optic radiation would all cause homonymous hemianopia. Here, either the right or left half of the binocular field of vision is lost, due to the loss of the temporal half of one field and the nasal half of the other.

Solution to Question 11:

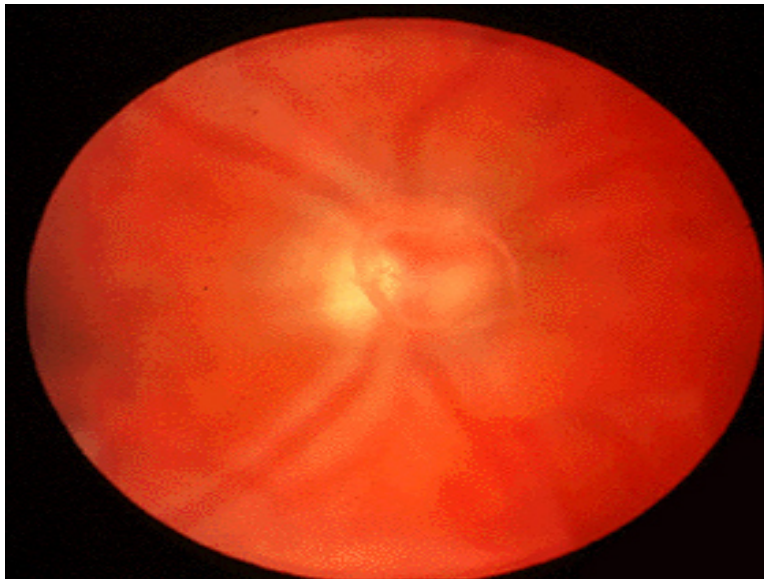
The finding given in the image is that of a Vossius ring.

Vossius ring is a circular ring of faint or stippled opacity, seen on the anterior surface of the lens due to brown amorphous granules of pigment lying on the capsule of the lens. It is usually seen after blunt trauma to the eye and orbit.

It usually has about the same diameter as the contracted pupil. The force of the blow drives the cornea and iris backward and thus producing an impression of the iris on the lens. The resorption of the pigment results in discrete subcapsular opacities.

Other options:

Option B: Weiss ring is a floater which may be seen by the patient as a circle or other large solitary lesion. It is the detached former attachment to the margin of the optic disc. It is seen in posterior vitreous detachment and is not confirmatory. It is seen in the image below.



Option C: A Fleischer's ring is usually observed in keratoconus, representing epithelial iron deposits. It is seen best with a cobalt blue filter.

Option D: Kayser-Fleischer ring is observed in Wilson's disease, due to intraocular deposition of copper. A brownish-yellow zone is seen in the peripheral Descemet membrane of the cornea. Anterior capsular cataract, also known as a sunflower cataract may be seen in some patients.



Solution to Question 12:

The chart given in the picture is Hess's chart.

Hess chart is used to diagnose and monitor patients with incomitant strabismus by assessing the field of a binocular single vision. In its interpretation, a smaller chart indicates the eye with the paretic muscle and a larger chart indicates the eye with an overacting muscle.

Incomitant strabismus is one where squint generally results from the limitation of eye movement associated with paralytic or mechanical etiologies like extraocular muscle palsy, thyroid ophthalmopathy, blow-out fracture, myasthenia gravis, etc. The field of binocular single vision is the area where bifoveal fusion of an object of interest occurs.

Other options:

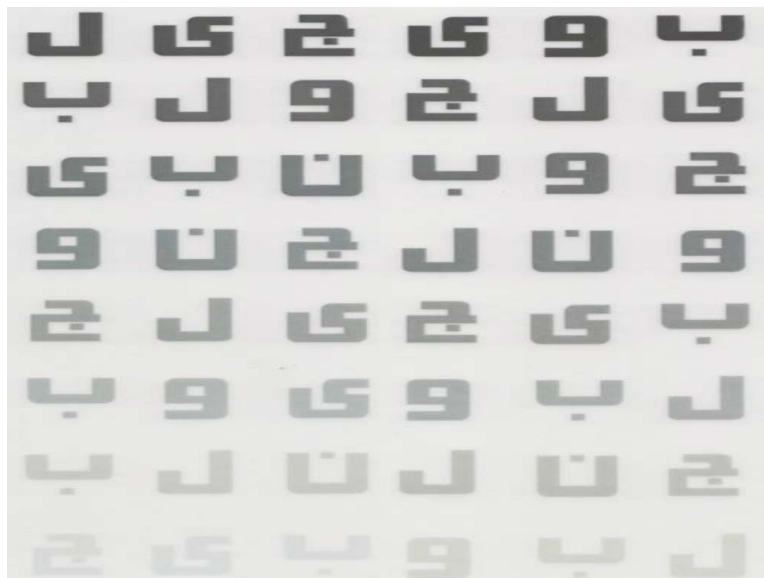
Option B: The Snellen chart is used to test visual acuity. Visual acuity is the eye's ability to discriminate between two points.

A picture of the Snellen chart is given below.

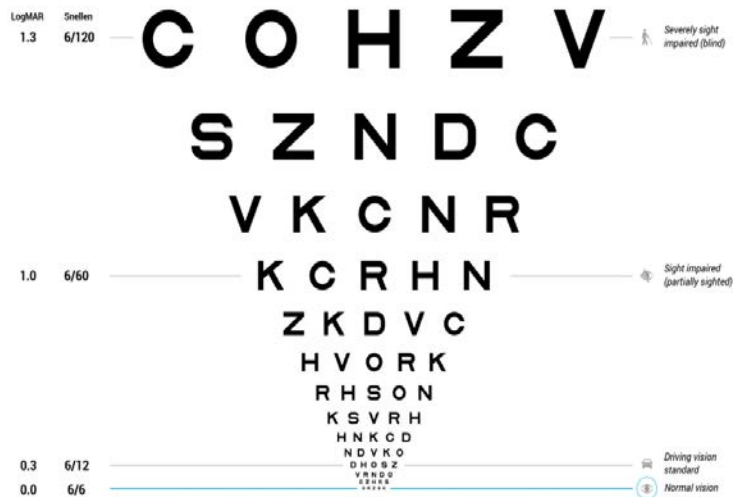
E	1	20/200	0.1
F P	2	20/100	0.2
T O Z	3	20/70	0.3
L P E D	4	20/50	0.4
P E C F D	5	20/40	0.5
E D F C Z P	6	20/30	0.7
F E L O P Z D	7	20/25	0.8
D E F P O T E C	8	20/20	1.0
L E F O D F C T	9	20/15	1.3
F D P L T C E O	10	20/12	1.7
F E Z O L C F T D	11	20/10	2.0

Option C: The Pelli-Robson contrast sensitivity letter chart consists of letters of constant size but decreasing contrast. It is viewed by the patient from a distance of 1 meter until the lowest resolvable group of three letters on the chart is reached. It is used to measure contrast sensitivity, a measure of the ability to distinguish an object against its background.

An image of the Pelli-Robson chart used in the middle east is given below.



Option D: The early treatment diabetic retinopathy study or ETDRS chart presents a series of 5 letters of equal difficulty on each row, with standardized spacing between letters and rows. It consists of a total of 14 lines having 70 letters. It is also used to test visual acuity. An image of the ETDRS chart is given below.



Solution to Question 13:

Chlamydia trachomatis is most likely responsible for the infection.

Chlamydia trachomatis inclusion conjunctivitis usually manifests 1 week after birth. It is usually a venereal infection derived from the cervix or urethra of the mother during delivery.

On examination, the conjunctiva is swollen and edematous. The discharge is initially watery, later becomes purulent or bloody.

Newborns with suspected chlamydia infection should have samples drawn from both the conjunctivae and oropharynx. Since it is an obligate intracellular organism, swabs should also include conjunctival epithelial cells if possible. Subsequent Giemsa staining shows characteristic intracellular inclusion bodies formed by Chlamydia oculogenitalis.

Treatment involves administration of suspension of erythromycin ethylsuccinate 50 mg/kg daily in four divided doses before feeds for 2–3 weeks.

If left untreated, associated systemic diseases such as chlamydial otitis and pneumonia may develop later. In all cases, both parents must receive appropriate treatment for genital infection.

Neonatal conjunctivitis, also called ophthalmia neonatorum, typically presents during the first month of life. The age of the neonate is an important clue towards the etiology of the infection.

Neonates are at higher risk of conjunctivitis due to predisposing factors such as:

- Decreased tear production
- Absence of IgA in tears
- Decreased immune function
- Absence of lymphoid tissue in the conjunctiva
- Decreased lysozyme activity.

Other risk factors for developing neonatal conjunctivitis are premature rupture of membranes, prolonged delivery, prematurity, poor prenatal care, mechanical ventilation, poor hygiene conditions, and history of midwife interference.

Other options :

Option B: Hemophilus influenza is one of the most common causes of acute bacterial conjunctivitis in children < 6 years of age and is responsible for causing conjunctivitis-otitis syndrome where conjunctivitis is followed by acute otitis media.

Option C: Streptococcus usually manifests 48-72 hours after birth. It is a common cause of acute bacterial conjunctivitis in children.

Option D: Mycoplasma, specifically M.genitalium can cause conjunctivitis due to contamination of the eye with genital infections in adults.

Age	Cause of conjunctivitis
First 24 hours	Chemical causes- erythromycin/gentamycin drops,
24 to 48 hours	Neisseria gonorrhea (most common), Staphylococcus aureus, Streptococcus, coliforms
5 to 14 days	Chlamydia trachomatis, Herpes keratoconjunctivitis
5 to 18 days	Pseudomonas aeruginosa

Ophthalmology NEET 2021

Question 1:

A 27-year-old IT professional has presented to you with intense watering, redness, and foreign body sensation in the eye for the past 2 months. He took out his contact lens and applied eye lubricants to help alleviate his ailment, but it didn't provide any relief. What is the most probable diagnosis?



- a) Acute trachoma
- b) Follicular conjunctivitis
- c) Giant papillary conjunctivitis
- d) Spring catarrh

Question 2:

A 60-year-old patient with progressive painless loss of vision presents to the ophthalmology OPD. The fundus image of the patient is given below. What could be the probable finding and the cause of it?



- a) Soft exudate, hypertension
- b) Hard exudate, diabetes mellitus
- c) Flame-shaped hemorrhages, hypertension
- d) Soft exudate, central retinal vein occlusion

Question 3:

A 33-year-old woman presents with complaints of progressive loss of vision in the right halves of both eyes. Where is the lesion located in the optic pathway?

- a) Left optic tract
- b) Right visual cortex
- c) Optic chiasma
- d) Right optic nerve

Question 4:

A 2-month-old infant was brought by his mother with complaints of inability to open the eyes in light and excessive watering. The clinical picture is given below. What is the most probable diagnosis?



- a) Ophthalmia neonatorum
- b) Congenital glaucoma
- c) Mucopolysaccharidosis
- d) Cataract

Question 5:

A 15-year-old girl is not compliant with spectacles for her myopic astigmatism. What would be the appropriate management in her case?

- a) LASIK
- b) Spherical equivalent spectacles
- c) Femto LASIK
- d) Implantable collamer lens

Question 6:

What is the most common complication associated with the following abnormality?



- a) Exposure keratopathy
- b) Restricted eye movements
- c) Diplopia
- d) Amblyopia

Question 7:

A 7-year-old boy presented with a white pupillary reflex for which enucleation was done. The histopathological examination of the specimen showed Flexner Winterstein rosettes. What is the most likely diagnosis?

- a) Retinoblastoma
- b) Medulloblastoma
- c) Rhabdomyosarcoma
- d) Astrocytoma

Question 8:

An industrial worker was admitted to the hospital after an eye injury sustained while cutting stone with a hammer and iron chisel. A foreign body was suspected to be impacted in his eye. Which investigation can be detrimental to this patient?

- a) MRI
- b) CT
- c) X-ray

d) B mode scan

Answer Key

Question No.	Correct Option
1	c
2	b
3	a
4	b
5	b
6	a
7	a
8	a

Detailed Explanations

Solution to Question 1:

The clinical history of watering, redness, and foreign body sensation in a contact lens user, along with the image showing giant papillae in the conjunctiva, suggests a probable diagnosis of giant papillary conjunctivitis.

Giant papillary conjunctivitis is a severe form of mechanically induced papillary conjunctivitis. It is most commonly associated with contact lens wear. It can also be associated with ocular prostheses, exposed sutures, and scleral buckles.

Patients usually present with redness, foreign body sensation, increased mucus production, itching, blurring, and loss of contact lens tolerance. On examination, giant papillae are seen on the upper tarsal conjunctiva with hyperemia. Focal apical ulceration and whitish scarring may develop on larger papillae.

Papillae are reddish, flat-topped, raised areas that give a red and velvety appearance to the tarsal conjunctiva. These are caused by the hyperplasia of the normal vascular system, with glomerulus-like bunches of capillaries growing into the epithelium.

The image below shows papillae in the tarsal conjunctiva.



Treatment includes the removal of the offending cause. Mast cell stabilizers, such as sodium cromoglycate, and drugs like azelastine and olopatadine, which have combined antihistamine and mast cell-stabilizing effects, are effective. Steroids can be used for resistant cases.

Other options:

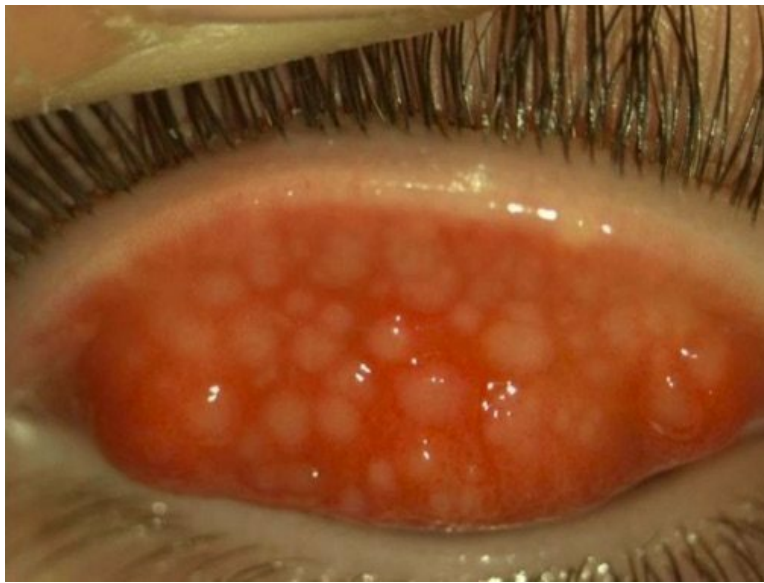
Option A: In acute trachoma, sago-like follicles are characteristically seen. It is caused by *Chlamydia trachomatis*. The SAFE strategy is used in the management of trachoma.

The image below shows sago-like follicles in trachoma.



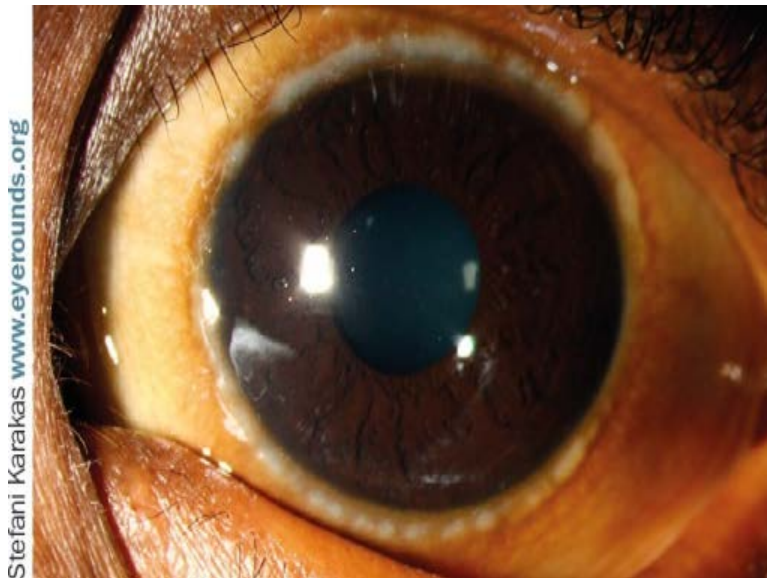
Option B: In follicular conjunctivitis, follicles are seen. It is most commonly caused by herpes and adenoviruses. Follicles are greyish-white raised areas on fornices and palpebral conjunctiva. These are formed due to the localized aggregations of lymphocytes in the sub-epithelial adenoid layer.

The image shows follicles.



Option D: In spring catarrh (vernal catarrh), cobblestone appearance is a papillary reaction in the palpebral conjunctiva, ropy discharge, and Maxwell-Lyon sign are present. Horner-Trantas spots that are discrete, whitish, raised dots along the limbus can also be seen. It is more common during summer.

The image below shows Horner-Trantas spots.



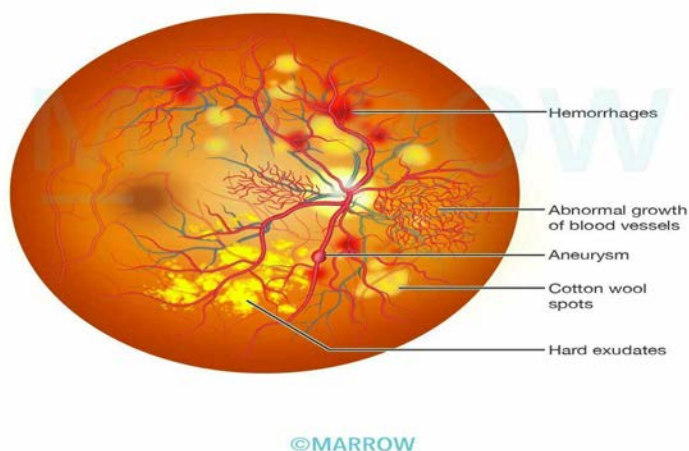
Solution to Question 2:

The given image shows small, discrete yellowish, waxy areas with crenated margins that are suggestive of hard exudates, seen in diabetes mellitus.

In diabetic retinopathy, hard exudates occur due to chronic localized edema and are composed of leaked lipoproteins and lipid-filled macrophages.

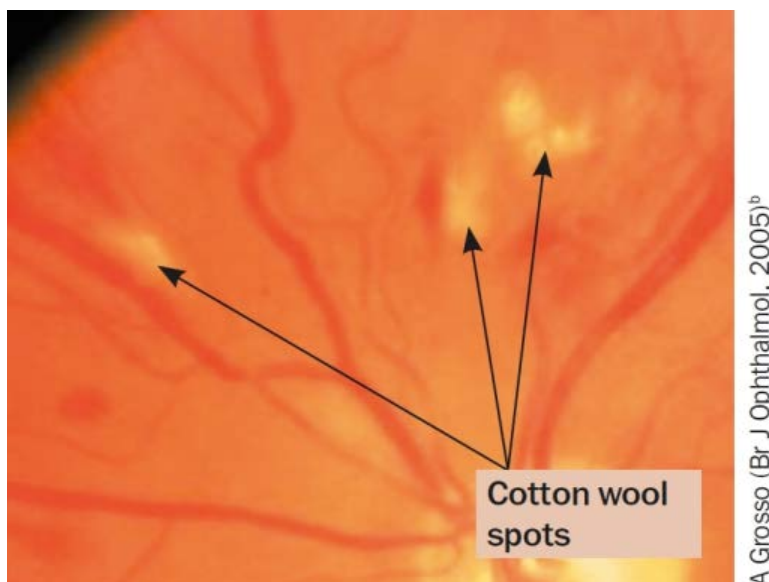
Other common causes of hard exudates include hypertensive retinopathy, Coats' disease, and circinate retinopathy.

Fundal features of Diabetic Retinopathy

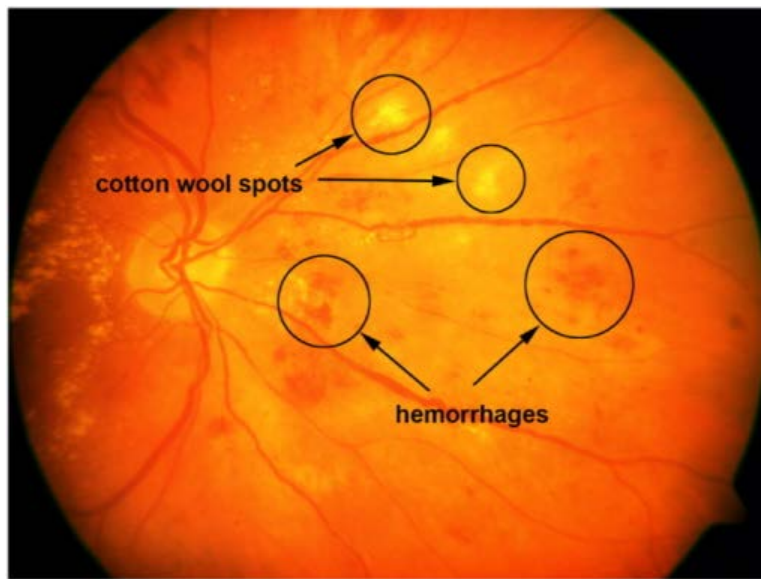


Cotton wool spots (soft exudates) appear as whitish fluffy spots with indistinct margins that represent areas of infarcts in nerve fiber layer. Common causes are hypertensive retinopathy, toxemic retinopathy of pregnancy, diabetic retinopathy, anemias, and collagen disorders like discoid lupus erythematosus, polyarteritis nodosa, and scleroderma.

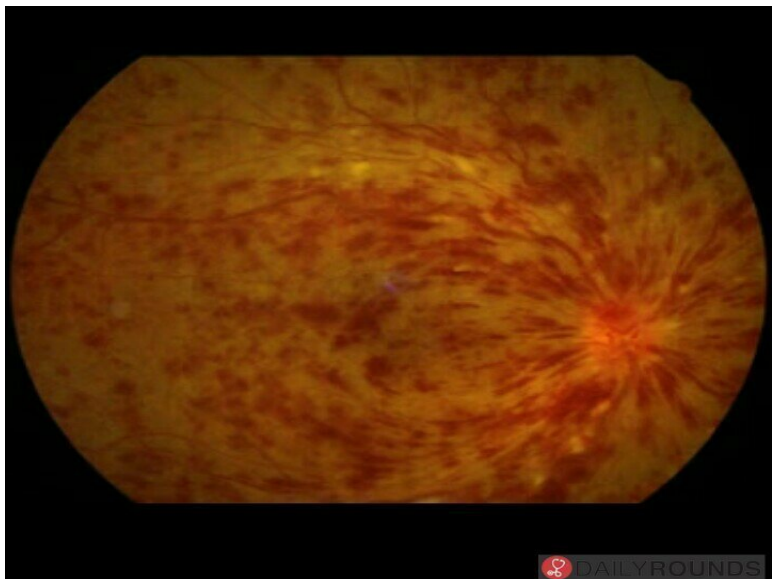
The image shows cotton wool spots



Flame-shaped hemorrhages are superficial retinal hemorrhages that occur due to disruption of the capillaries in the retinal nerve fiber layer. Common causes are diabetic retinopathy, non-ischemic central retinal vein occlusion, traumatic retinopathy, Eale's disease, hypertensive retinopathy, and anemic retinopathy.



Fundus examination of central retinal vein occlusion (CRVO) reveals massive engorgement, congestion, and tortuosity of retinal veins. Ischemic CRVO has massive retinal hemorrhages giving a splashed tomato appearance involving the whole fundus.



Solution to Question 3:

The patient presenting with progressive visual loss in the right halves of both eyes is suggestive of right homonymous hemianopia. The site of the lesion in this case is the left optic tract.

Here, vision is lost in the right half of both eyes, which means:

- in the left eye, the nasal visual field is lost, and
- in the right eye, the temporal visual field is lost.

It means left temporal and right nasal fibers are destroyed which leads to right homonymous hemianopia. Anatomically, this can be explained by the fact that each optic tract contains fibers from the ipsilateral temporal half of the retina and the contralateral nasal half of the retina.

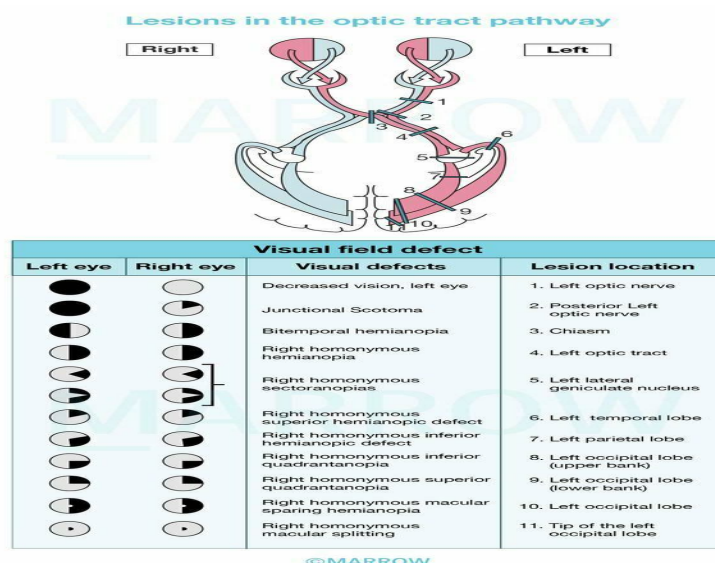
Optic tract lesions can occur due to pituitary adenomas (most common), craniopharyngioma, tumors of optic thalamus, large aneurysm of superior cerebellar or posterior cerebral arteries, demyelinating diseases, and infarction.

Other options:

Option B - Lesion at right visual cortex causes left homonymous hemianopia with macular sparing.

Option C - Lesion at optic chiasma causes bitemporal hemianopia.

Option D - Lesion at the right optic nerve causes blindness in the right eye.



Solution to Question 4:

The clinical history of inability to open eyes in light (photophobia), excessive watering, and the image showing hazy cornea are suggestive of congenital glaucoma.

Primary congenital glaucoma refers to abnormally high intraocular pressure (IOP) which occurs as a result of a developmental anomaly of the angle of the anterior chamber. It is not associated with any other ocular or systemic anomaly. It can be classified as follows:

- True congenital glaucoma: The IOP is elevated during intrauterine life.
- Infantile glaucoma: Manifests prior to age 3.
- Juvenile glaucoma: The IOP rises between 3 and 16 years of age. It is the least common type.

The CYP1B1 gene is implicated in the pathogenesis of congenital glaucoma.

Clinical presentation: Lacrimation, photophobia, and blepharospasm form the classic triad of symptoms of congenital glaucoma. A corneal haze is seen due to diffuse edema secondary to

raised IOP. Buphthalmos occur as a result of stretching due to elevated IOP prior to the age of 3 years. Haab striae that are curvilinear healed breaks in Descemet's membrane are seen. Corneal scarring can also occur.

Evaluation under general anesthesia is generally required. IOP measurement, anterior chamber examination, optic disc examination, corneal diameter measurement, gonioscopy, and refraction should be evaluated.

Management is essentially surgical with goniotomy or trabeculotomy. If angle surgery fails, trabeculectomy, tube shunt implantation, and ciliary body ablative procedures are performed.

Other options:

Option A - In ophthalmia neonatorum, there will be inflammation of the conjunctiva. Clinical presentation includes conjunctival discharge, hyperemia chemosis, swollen eyelids, tender eyeball. Corneal involvement is rare which may occur in the form of superficial punctate keratitis.

Option C - In mucopolysaccharidosis, there will be bilateral cloudy cornea.

Option D - In cataract, there will be opacification of lens presenting as whitish reflex/ leukocoria.

Solution to Question 5:

The appropriate management for myopic astigmatism in a 15-year-old girl (<18 years) would be spherical equivalent spectacles.

Prescription in astigmatism has 3 components:

- Power of the spherical component: Convex (+) or concave (-) in dioptre sphere (DS)
- Power of the cylinder: Convex (+) or concave (-) dioptre cylinder (DC)
- Axis or orientation of the cylinder

The cylinder power and axis are correcting the astigmatic part of the refractive error. If an astigmatic eye were to be corrected with a spherical lens alone, the result would be a blur circle on the retina, called the circle of least confusion.

The spherical lens power places the circle of least confusion on the retina and is called the spherical equivalent. It is defined as the spherical power whose focal point coincides with the circle of least confusion, where one would see minimal blurring in their vision. To calculate the spherical equivalent, add half the cylinder power to the sphere power:

Spherical equivalent= spherical power + 1/2 (cylindrical power)

For example, for spherical power +2 and cylindrical power +3,

$$+2.00 + (+3.00 / 2) = +3.50.$$

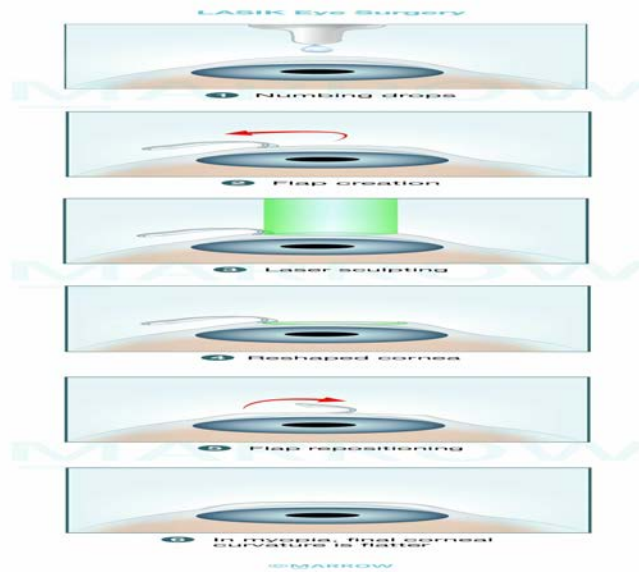
Spherical equivalent spectacles are given if one has to give less cylinder power than was measured.

Other options:

Options A and C - LASIK and Femto LASIK should be done in patients above 18 years of age.

Option D - Implantable Collamer lens is indicated for patients 21-45 years of age.

The image demonstrates the steps of LASIK



Solution to Question 6:

The given image shows a coloboma of the upper eyelid. The most common complication associated with this condition is exposure keratopathy since the defect leaves the cornea exposed. If left untreated, it can lead to corneal ulceration.

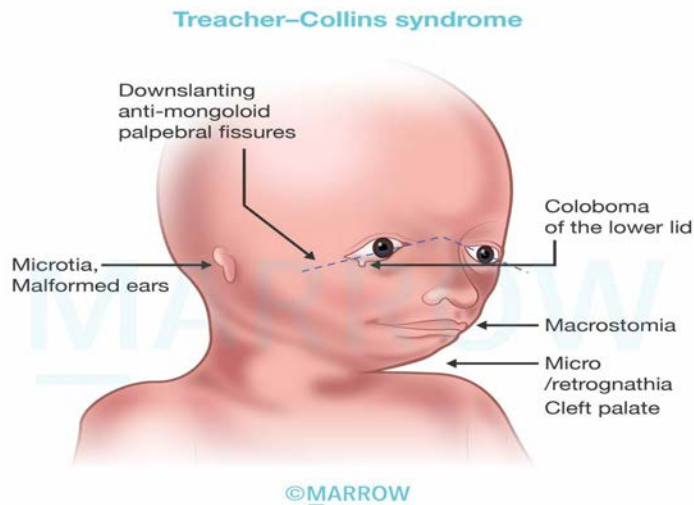
Coloboma of the eyelid is a partial or full-thickness eyelid defect. It usually occurs near the nasal side. It more frequently involves the upper lid than the lower lid. It occurs when eyelid development is incomplete, due to failure of migration of lid ectoderm to fuse with the lid folds.

Small defects can be managed by primary closure, while large defects require skin grafts and rotation flaps.

Upper lid colobomas occur at the junction of the middle and inner thirds. It has a relatively strong association with cryptophthalmos, facial abnormalities, and Goldenhar syndrome. Cryptophthalmos is a congenital anomaly in which the eyelids are absent, replaced by a continuous layer of skin.

Lower lid colobomas occur at the junction of the middle and outer thirds and are frequently associated with systemic conditions.

Treacher Collins syndrome (mandibulofacial dysostosis) is a genetically heterogeneous condition. It is characterized by mandibular and ear anomalies, lower eyelid coloboma, slanted palpebral apertures, cataract, microphthalmos, and lacrimal atresia.



Solution to Question 7:

The most likely diagnosis with a histopathological finding of Flexner Winterstein rosettes in a 7-year-old child with a whitish pupillary reflex is retinoblastoma. Flexner Winterstein rosettes refer to an arrangement of columnar cells around a central lumen, which is highly characteristic of retinoblastoma.

Retinoblastoma is a malignant proliferation of immature retinal neural cells and is the most common intraocular malignancy in children. It is associated with mutations in the RB1 gene at chromosome 13q14. It is a tumour of a group called small round blue cell tumours because the immature retinal neural cells are seen as small round cells with large nuclei.

White pupillary reflex or leukocoria is the most common presenting feature, followed by a squint which is usually convergent. Rarely, the child may present with nystagmus and defective vision.

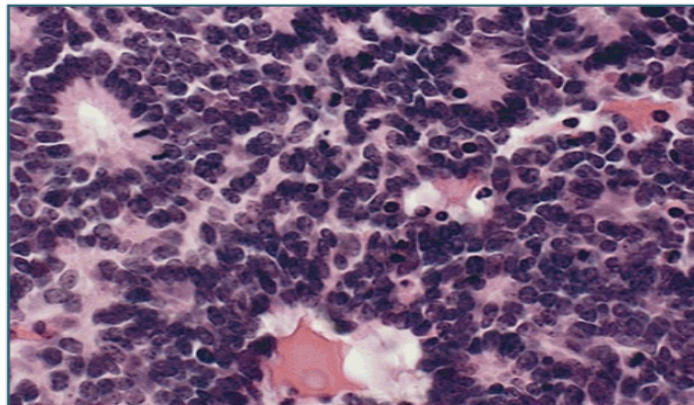
Image showing leukocoria of left eye



Histopathology of the tumor shows small round cells with large nuclei. Microscopic features of a well-differentiated tumor include Flexner-Wintersteiner rosettes (highly specific of retinoblastoma), Homer-Wright rosettes, pseudorosettes, and fleurettes formation. Areas of necrosis and calcification can also be seen.

The histopathological image of retinoblastoma is shown below.

Retinoblastoma



©Marrow

Treatment typically involves chemotherapy plus focal therapy. The commonly used three-drug regimen is the CVE regimen consisting of Vincristine, Carboplatin, Etoposide. Focal therapy includes cryotherapy, laser photocoagulation, thermotherapy, plaque radiotherapy, and external beam radiotherapy.

Enucleation is done when

- the tumor involves more than half of the retina
- the optic nerve is involved

- glaucoma is present and the anterior chamber is involved

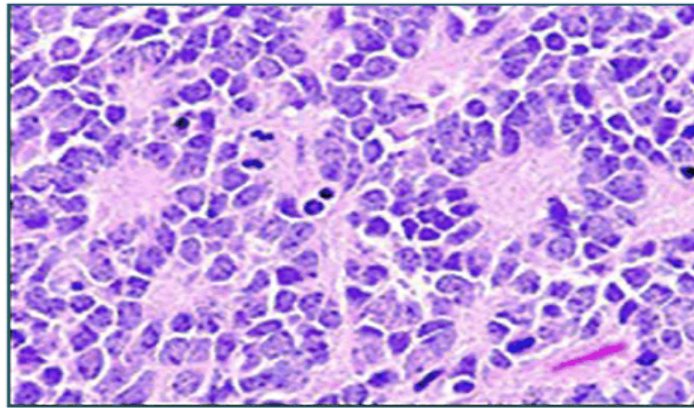
In the stage of extraocular extension, the globe bursts through the sclera, usually near the limbus or near the optic disc. This is followed by rapid fungation and involvement of extraocular tissues results in marked proptosis. Distant metastasis occurs through the lymphatic spread and the direct extension by continuity to the optic nerve and brain.

Other options:

Option B - Medulloblastoma is a malignant embryonal tumor of childhood that occurs exclusively in the cerebellum. Patients usually present with headaches, ataxia, and signs of brainstem involvement.

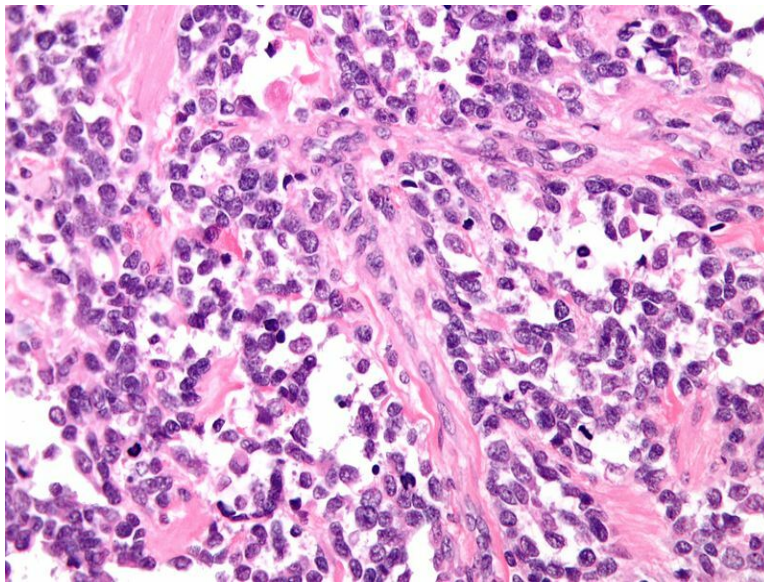
Histologically, the tumor is very densely cellular. Individual tumor cells have scant cytoplasm and elongated or crescent-shaped hyperchromatic nuclei. Tumor cells form Homer Wright rosettes. They express neuronal markers such as synaptophysin.

Medulloblastoma



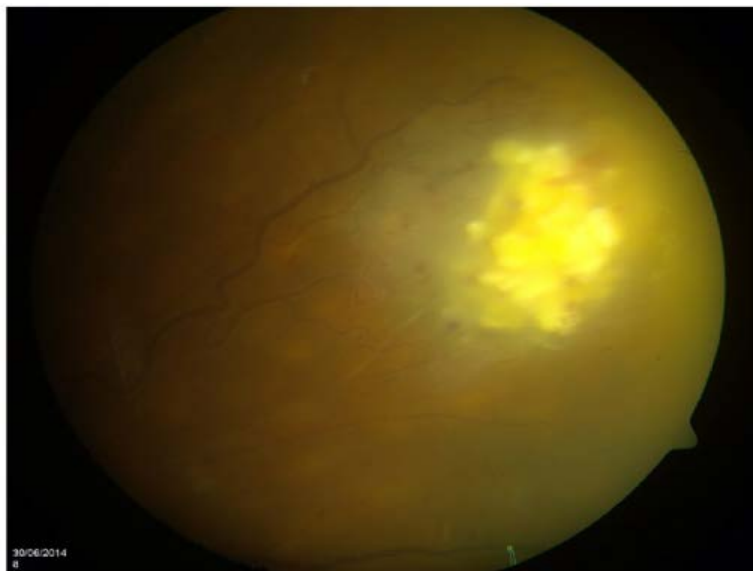
©MARROW

Option C - Rhabdomyosarcoma is the most common soft tissue sarcoma of childhood. The child usually presents with rapidly progressive unilateral proptosis. There are 4 subtypes - embryonal, alveolar, botryoid, and pleomorphic. Cells show light microscopic features of striated muscle differentiation. Immunohistochemistry is positive for desmin.



Option D - Astrocytoma of the retina and optic nerve head is rare. It is usually asymptomatic and is most frequently seen in tuberous sclerosis. Fundus examination reveals a yellowish or white plaque, nodule, or mulberry-like lesion. These lesions are semitransparent peripherally and calcified centrally.

Image showing calcified astrocytoma:



Solution to Question 8:

MRI can be detrimental for a patient with a history of an intraocular foreign body (IOFB) after cutting stone with a hammer and iron chisel. It can cause further damage by producing movements of the foreign body if it is magnetic. The use of MRI is contraindicated for metallic (iron) foreign bodies and is not recommended as a screening procedure.

The foreign bodies most likely to penetrate and be retained in the eye are minute chips of iron or steel. While chipping stone with an iron chisel, the chip of the iron chisel is more likely to enter the eye than the stone. A foreign body entering the eye may cause damage by:

- mechanical effects
- introduction of infection
- reacting with the intraocular tissues

Anteroposterior and lateral view X-rays are recommended for locating an IOFB as most foreign bodies are radio-opaque. Ultrasonographic localization can tell the position of even non-radiopaque foreign bodies. CT scan is the best method of IOFB localization. MRI is contraindicated for metallic IOFB.

Management: A foreign body should always be removed unless:

- it is inert and probably sterile
- little damage has been done to vision
- the process of removal will destroy sight

Siderosis bulbi refers to the ocular degenerative changes produced by an iron foreign body. It usually occurs 2 months to 2 years after the injury. This condition is due to the electrolytic dissociation of the metal by an intrinsic resting current in the eye which disseminates the metal, combining it with cellular proteins and cause atrophy.

Clinical features:

- Iris: first stained greenish and later turns reddish-brown, causing heterochromia iridis.
- Pupil: iron deposition in sphincter leads to mydriasis.
- Lens: iron deposition in a radial fashion on the anterior capsule of the lens, leading to anterior capsular cataract.
- Retina: pigmentary retinopathy; followed by atrophy of the retina and retinal pigmentary epithelium.
- Secondary open-angle Glaucoma: due to damage to the trabecular meshwork.

Ophthalmology INICET 2022

Question 1:

What is the action of the superior oblique muscle?

- a) Intorsion, abduction, depression
- b) Intorsion, adduction, elevation
- c) Extorsion, abduction, depression
- d) Extorsion, adduction, elevation

Question 2:

Which of the following triad is seen after an acute attack of angle-closure glaucoma?

- a) Vogt's triad
- b) Virchow triad
- c) Hutchinson's triad
- d) Gaucher triad

Question 3:

In which of the following conditions do you see punched-out lesions of the retina?

- a) Posterior staphyloma
- b) Fundal coloboma
- c) Congenital rubella
- d) Toxoplasmosis

Question 4:

Which of the following has familial inheritance?

- a) 1, 2, 3, 4
- b) 1, 2, 3
- c) 1, 3

d) 3, 4

Question 5:

A 45-year-old man presented with lagophthalmos, orbicularis oculi palsy, and loss of corneal sensation. Which of the following is the most likely causative organism?

- a) *Acanthamoeba castellanii*
- b) *Mycobacterium leprae*
- c) *Onchocerca volvulus*
- d) *Chlamydia A to C*

Question 6:

A 27-year-old female patient presents with sudden diminishing vision associated with relative afferent pupillary defect in the right eye. On examination, the left eye is normal. Which of the following investigations would you do next?

- a) 1, 2, and 3
- b) 1 and 2
- c) 2 and 3
- d) 3 and 4

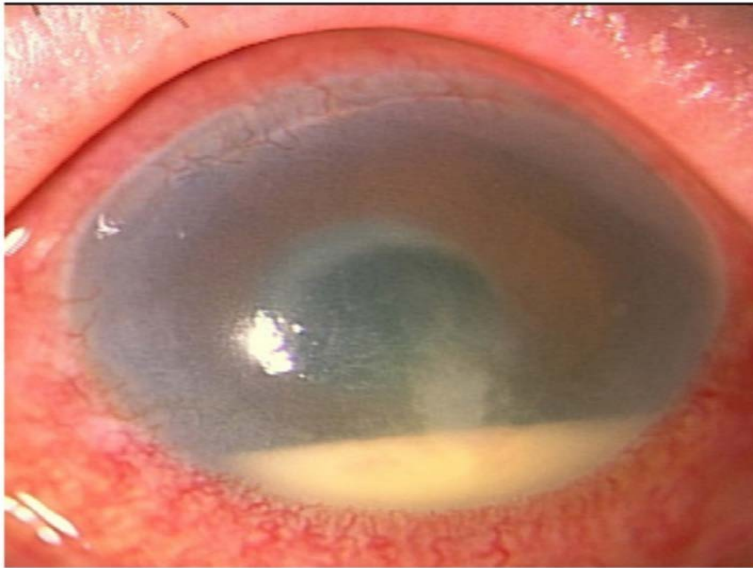
Question 7:

Which of the following is not a feature of the optic disc in glaucoma?

- a) Roenne's nasal step
- b) Bayonetting
- c) Increased cup-disc ratio
- d) Baring of nasal arteries

Question 8:

A farmer presents with pain and discharge in his left eye. He says that he sustained an injury caused by a leaf a few days back. What is the finding seen in the image?



- a) Hyphema
- b) Aqueous flare
- c) Arlt line
- d) Hypopyon

Question 9:

What is the most common cause of moderate-severe visual impairment according to the National Blindness and Visual Impairment Survey 2015-19?

- a) Refractive error
- b) Untreated cataract
- c) Glaucoma
- d) Diabetic retinopathy

Question 10:

Which of the following statements is true about optic neuritis?

- a) Relative afferent pupillary defect is seen.
- b) There is no pain on ocular movement.
- c) Multiple myeloma is a common cause.
- d) Oral corticosteroids are the mainstay of treatment.

Question 11:

The most common ocular manifestation of COVID-19 is _____.

- a) Uveitis
- b) Mucormycosis
- c) Follicular conjunctivitis
- d) Papillary conjunctivitis

Question 12:

Identify the condition given in the image.



- a) Symblepharon
- b) Ankyloblepharon
- c) Congenital blepharophimosis
- d) Congenital ptosis

Question 13:

Match the following:

- a) 1-d, 2-b, 3-a, 4-c
- b) 1-a, 2-c, 3-d, 4-b
- c) 1-b, 2-a, 3-d, 4-c

d) 1-d, 2-c, 3-a, 4-b

Question 14:

In which of the following is atropine contraindicated?

- a)** Acute congestive glaucoma
- b)** Acute anterior uveitis
- c)** Malignant glaucoma
- d)** Corneal ulcer

Question 15:

What is the mechanism of action when the retina is exposed to light?

- a)** Hyperpolarization, increase in neurotransmitter release
- b)** Hyperpolarization, decrease in neurotransmitter release
- c)** Depolarization, decrease in neurotransmitter release
- d)** Depolarization, increase in neurotransmitter release

Question 16:

Which of the following is not a feature of keratoconus?

- a)** Haab's striae
- b)** Munson's sign
- c)** Irregular astigmatism
- d)** Scissoring reflex

Question 17:

A patient has presented for a routine eye evaluation. You have checked his visual acuity using Snellen's chart and found it to be 6/6. What is the minimum angle of resolution?

- a)** 15 minutes of arc
- b)** 5 minutes of arc
- c)** 10 minutes of arc

- d) 20 minute of arc

Question 18:

Which of the following is not a feature of pterygium?

- a) Rate of recurrence is 95%
- b) It is a fibrovascular growth
- c) The parts of a pterygium are head, neck and body
- d) Has progressive and regressive types

Question 19:

Which of the following is not a feature of complicated cataracts?

- a) Krukenberg spindle
- b) Polychromatic lustre
- c) Occurs after uveitis
- d) Breadcrumb appearance

Question 20:

Oculocephalic reflex assesses the integrity of which of the following cranial nerves?

- a) A and C
- b) A and E
- c) A and D
- d) A, C, and D

Question 21:

A pregnant woman presents with fever, retroorbital pain, headache, pulsatile proptosis of the right eye, and tinnitus. BP and fundus were normal. Which of the following structures are involved?

- a) A,B,C
- b) B,C,D

c) A,C,D

d) A,B,C,D

Answer Key

Question No.	Correct Option
1	a
2	a
3	d
4	a
5	b
6	a
7	a
8	d
9	b
10	a
11	c
12	b
13	c
14	a
15	b
16	a
17	b
18	a
19	a
20	d
21	d

Detailed Explanations

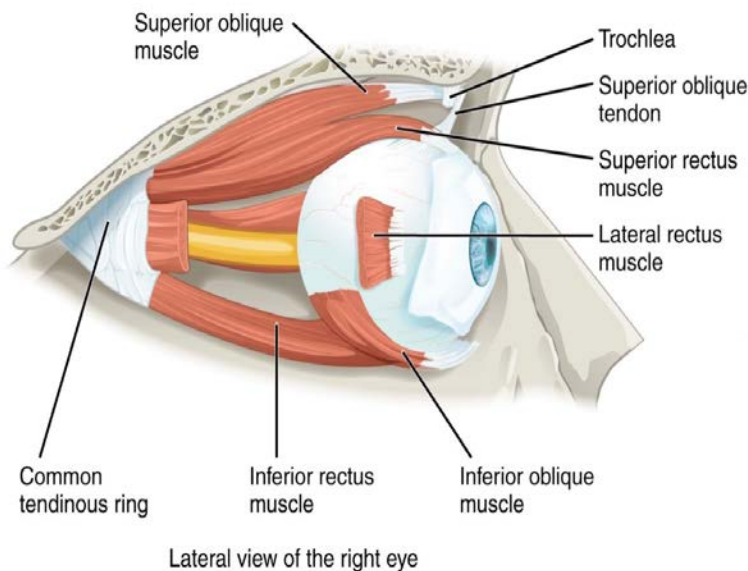
Solution to Question 1:

The actions carried out by the superior oblique muscle are intorsion, depression, and abduction.

There are six extraocular muscles for each eye. Four rectus muscles namely - superior, inferior, medial, and lateral recti. The oblique muscles are two - superior oblique and inferior oblique.

These muscles are responsible for the movements of the eye which can be either unilateral or binocular movements.

The superior oblique muscle originates from the apex of the orbit, turns around the trochlea, and inserts into the sclera behind the equator. The oblique direction of the tendon is responsible for its rotatory effect (intorsion), in addition to depression and abduction.



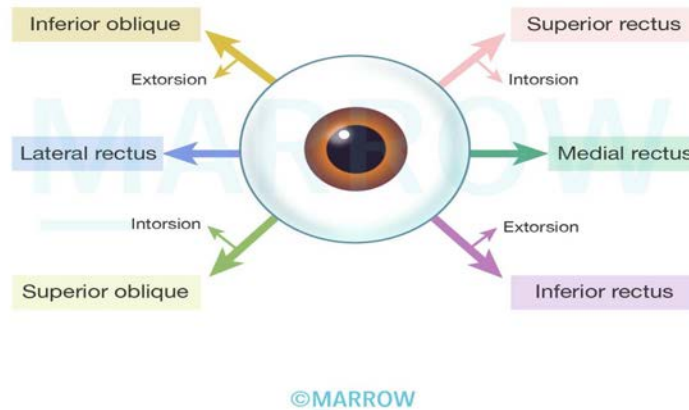
Unilateral movements (ductions) are :

- 1.Adduction: It is the movement towards the nasal side around the vertical axis.
- 2.Abduction: It is the movement towards the temporal side around the vertical axis.
- 3.Supraduction (elevation): It is upward movement around the horizontal axis.
- 4.Infraduction (depression): It is downward movement around the horizontal axis.
- 5.Intorsion (medial rotation): Rotatory movement along the anteroposterior axis where the superior pole of cornea moves medially.
- 6.Extorsion (lateral rotation): Rotatory movement along the anteroposterior axis where the superior pole of cornea moves laterally.

Binocular movements can be of two types - versions and vergences.

The oculomotor nerve supplies all the extraocular muscles except the superior oblique and lateral rectus. The superior oblique muscle is supplied by the trochlear nerve and lateral rectus muscle by the abducent nerve.

Actions of extraocular muscles



Solution to Question 2:

Vogt's triad is seen after an acute attack of primary angle-closure. It includes glaukomflecken, patches of iris atrophy, and a slightly dilated nonreactive pupil. Glaukomflecken is used to describe an anterior subcapsular lenticular opacity. The slightly dilated pupil is attributed to sphincter atrophy.

Glaucoma is used to describe a group of conditions where there is progressive optic neuropathy with irreversible visual field defects associated with raised intraocular pressure. It can be classified into congenital/developmental glaucomas, primary adult glaucomas, and secondary adult glaucomas. Primary adult glaucomas are of three types:

- Primary open-angle glaucoma (POAG)
- Primary angle-closure glaucoma (PACG)
- Primary mixed mechanism glaucoma.

In primary angle-closure disease, there is apposition of the peripheral iris with the trabecular meshwork. This interferes with the drainage of aqueous humor. It is seen more commonly in elderly females. Anatomical risk factors include hypermetropic eyes, anteriorly placed iris-lens diaphragm, and a narrow-angle of the anterior chamber. The international society of geographical and epidemiological ophthalmology (ISGEO) has classified primary angle-closure disease into three types.

- Primary angle-closure suspect (PACS)
- Primary angle-closure (PAC)
- Primary angle-closure glaucoma (PACG).

PACS and PAC are seen before the development of PACG. Vogt triad is seen after an acute attack of PAC.

PAC includes irido-trabecular contact on gonioscopy and elevated intraocular pressure (IOP) and/or peripheral anterior synechiae. The optic disc and visual fields are usually normal. Due to any reason, there can be an acute rise of IOP in patients with PAC. It is a vision-threatening emergency and the sequelae of which can be ciliary body shut down, Vogt's triad, and rarely spontaneous angle reopening.

Other options:

Option B - Virchow triad is used to describe the abnormalities that result in thrombosis. These include endothelial injury, abnormal blood flow, and hypercoagulability of the blood.

Option C - Hutchinson triad is used to describe manifestations of congenital syphilis. It includes interstitial keratitis, Hutchinson teeth, and eighth nerve deafness. Hutchinson teeth are screw-driver or peg shaped small incisors.

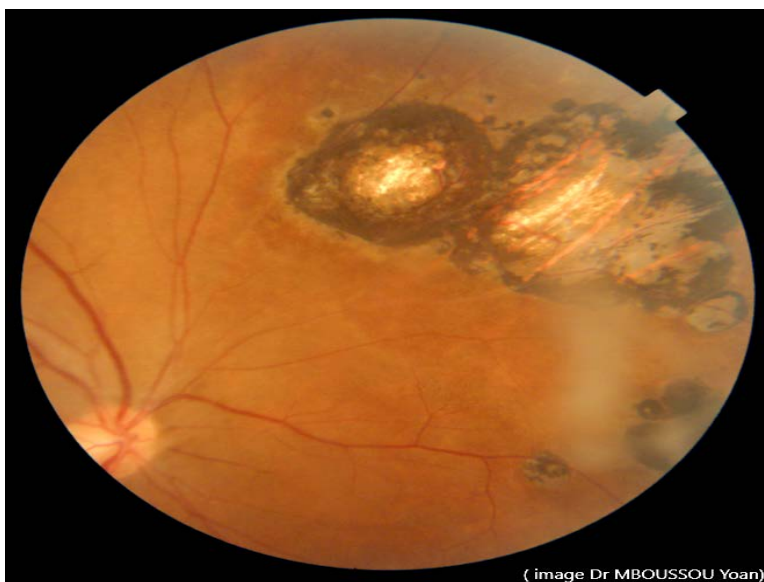
Option D - Gaucher's triad is not a named triad. Hypertonia, strabismus, and stridor usually point to a diagnosis of type 2 Gaucher's disease.

Solution to Question 3:

Toxoplasmosis characteristically causes punched-out lesions in the retina.

Toxoplasmosis is a parasitic infestation due to *Toxoplasma gondii*. It primarily affects the central nervous system and retina. In humans, it occurs as congenital, acquired, or recurrent.

Congenital toxoplasmosis is acquired by the transplacental route. Infants are usually born with inactive disease. Bilateral healed punched-out chorioretinal lesions in the macular area are usually seen. In acquired toxoplasmosis, punched-out lesions similar to congenital toxoplasmosis are seen. In addition, punctate outer retinal toxoplasmosis (PORT) may also be seen.



(image Dr MBOUSSOU Yoan)

Other options:

Option A - Posterior staphyloma is used to describe a bulge in the sclera with the choroid behind the equator. On ophthalmoscopy, there is an excavated area with retinal vessels dipping in it.

Option B - Fundal coloboma occurs due to a defect in the closure of the embryonic cleft. It appears as a comet-shaped defect with the apex towards the disc which may or may not be involved. A few choroidal or retinal vessels may also be seen on the surface.

Option C - Congenital rubella typically produces a salt and pepper appearance of the fundus.

Solution to Question 4:

All the above conditions (1, 2, 3, 4) have a familial inheritance.

Retinoblastoma is a malignancy arising from the retina more commonly seen in children. 10% of the cases are familial. These cases have an autosomal dominant mode of inheritance. The remaining 90% occurs sporadically. The clinical presentation includes leucocoria, strabismus, and painful red eye.

Retinitis pigmentosa is a pigmental retinal dystrophy where the photoreceptors, especially rods are affected. Around 50 - 60% of the cases have a familial inheritance. It has multiple modes of inheritance, autosomal dominant, autosomal recessive and X-linked recessive. Night blindness is a characteristic feature. Tubular vision is seen in advanced cases.

Myopia (shortsightedness) is a refractive error in which the light rays from infinity are focused in front of the retina. It is proven to have familial inheritance. Features include defective vision, impaired night vision, and muscae volitantes.

Primary open-angle glaucoma has a normal anterior chamber angle. Family history is a risk factor implicated in the condition. Symptoms include headache, scotoma, and delayed dark adaptation.

Solution to Question 5:

Mycobacterium leprae is the most likely causative organism based on the clinical features given which points to ocular manifestations of leprosy. These include corneal anesthesia, lagophthalmos, corneal ulceration, episcleritis, and iris atrophy. Patients can also present with cataract, glaucoma, and madarosis.

Lagophthalmos is the inability to close the eyelids. This can be due to the involvement of the zygomatic branch of the facial nerve (as in leprosy). Other causes of lagophthalmos include extreme proptosis and symblepharon. The decreased corneal sensation seen is due to the involvement of the ophthalmic branch of the trigeminal nerve which innervates the cornea.

Other options:

Option A - *Acanthamoeba castellanii* can lead to acanthamoeba keratitis which presents with pain, lacrimation, photophobia, and foreign body sensation. It results from direct corneal contact with infected material. Contact lens users are at an increased risk. Radial keratoneuritis is pathognomonic.

Option C - *Onchocerca volvulus* causes onchocerciasis commonly seen in African countries. Symptoms include sclerosing keratitis, uveitis, chorioretinitis, and optic neuritis. Ultimately it can lead to optic atrophy.

Option D - Chlamydia trachomatis serovars A, B, Ba, and C are responsible hyperendemic (blinding) trachoma. The source of infection is infected conjunctival discharge. If there is no secondary bacterial infection, symptoms include lacrimation, scanty mucoid discharge and foreign body sensation. Conjunctival congestion and follicles, papillary hyperplasia, Herbert follicles (seen in the limbal area), and progressive pannus are the characteristic features. Serovars D to K causes neonatal and adult inclusion conjunctivitis. Neonates present with conjunctival discharge, chemosis and swollen lids. Adult inclusion conjunctivitis more commonly seen in sexually active adults present with foreign body sensation, mild photophobia, and mucopurulent discharge.

Solution to Question 6:

The clinical vignette is suggestive of optic neuritis. Sudden, progressive, and profound visual loss associated with relative afferent pupillary defect (RAPD) is characteristic of optic neuritis. MRI brain & orbit, screening for circulating myelin oligodendrocyte glycoprotein (MOG) & neuromyelitis optica (NMO) antibodies, and blood sugar levels are the next best investigations for optic neuritis.

A brain MRI is recommended in every patient after the first attack of optic neuritis. The 15-year cumulative probability of developing multiple sclerosis after an attack of optic neuritis is about 50%. It may remain an isolated event in some patients.

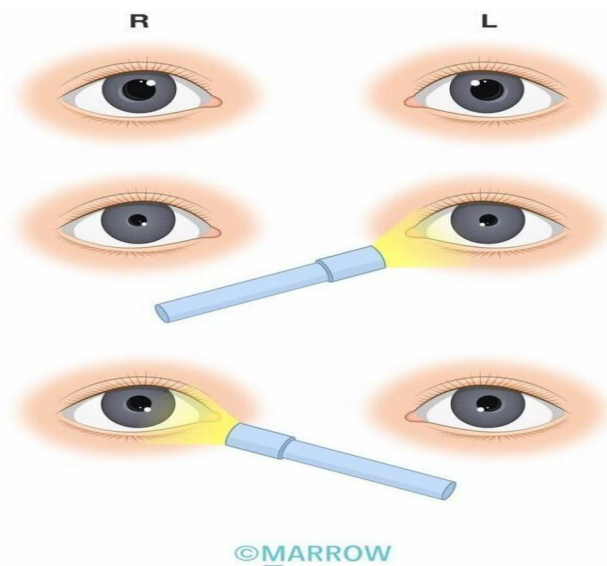
Neuromyelitis optica (NMO) is a severe type of optic neuritis which may occur bilaterally and is associated with myelitis. Detection of antibodies like MOG & NMO (also known as aquaporin-4 antibody) is diagnostic of the condition.

Optic neuritis is primarily treated with steroids. The blood sugar levels must be checked as steroids can worsen the glycemic control in individuals with DM which may precipitate hyperglycemia.

Optic neuritis comprises inflammatory and demyelinating disorders of the optic nerve. It can be idiopathic, hereditary (Leber's disease), or due to demyelinating disorders such as multiple sclerosis or neuromyelitis optica (Devic's disease). Anatomically, it can be papillitis, neuroretinitis, or retrobulbar neuritis. It is more common in women.

Other symptoms include lowered dark adaptation and visual obscuration in bright light. Transient obscuration of vision on exposure to light (Uhthoff's symptom) and impaired depth perception (Pulfrich's phenomenon) may also be seen.

Marcus Gunn pupil which points to RAPD is a characteristic sign. It is elicited by the swinging flashlight test. One eye is illuminated with the flashlight and the response is noted. The same is done for the other eye. This is repeated multiple times (swinging). The normal response is constriction in both eyes. If there is RAPD in one eye, it will dilate when light is shone on it.



Solution to Question 7:

Roenne's nasal step is not a feature of the optic disc in glaucoma.

Roenne's nasal step is a visual field defect. It is the result of two arcuate scotomas running in different arcs and meeting at the horizontal meridian. This forms a right-angled defect in the nasal side of the visual field.

Early glaucomatous changes of the optic disc include an increase in cup-disc ratio (C/D ratio) to more than 0.4, splinter hemorrhages near the optic disc margin, and atrophy of the retinal nerve fiber layer.

Advanced glaucomatous changes of the optic disc include a marked increase in C/D ratio (0.7 or more), thinning of the neuroretinal rim, nasal shifting of the retinal vessels or bayonetting, and pulsations of retinal arterioles at the disc margin.

If the damage progresses, all neural tissue is destroyed and the optic nerve appears white and deeply excavated.

Solution to Question 8:

The finding seen in the image is hypopyon.

Hypopyon is a purulent whitish exudate consisting of inflammatory cells in the inferior part of the anterior chamber. It can be associated with bacterial or fungal corneal ulcers and HLA-B27-associated anterior uveitis.

In bacterial corneal ulcers, the hypopyon is sterile. In these lesions, the outpouring of polymorphonuclear cells is due to the toxins and not due to actual invasion by bacteria. The hypopyon is absorbed when the ulcerative process is controlled.

In fungal corneal ulcers, the hypopyon is not sterile. The fungi can penetrate into the anterior chamber without perforation.

The hypopyon associated with anterior uveitis is immobile due to high fibrin content. In Behcet disease, there is minimal fibrin content, so the hypopyon shifts according to the patient's head position.

Treatment of hypopyon involves correcting the underlying conditions.

Other options:

Option A: Hyphema is a hemorrhage in the anterior chamber. It commonly occurs after a blunt ocular injury. The bleeding is from the iris root or ciliary body face. The blood settles inferiorly forming a fluid level.

Option B: Aqueous flare refers to the haziness of the normally clear fluid in the anterior chamber. It reflects the presence of protein due to the breakdown of the blood-aqueous barrier. The presence of flare indicates active inflammation.

Option C: Arlt line is the scarring on the conjunctiva in the area of sulcus subtarsalis. It can be a single line, irregularly shaped, or star-shaped. Common causes are trachoma, healed traumatic wounds, or surgical scars.

Solution to Question 9:

According to the National Blindness and Visual Impairment Survey 2015-19, the most common cause of moderate-severe visual impairment (MSVI) is untreated cataract.

MSVI is defined as presenting visual acuity of $\leq 6/18$ to $3/60$ in the better eye with available correction.

Most commonly, cases of visual impairment and blindness are more prevalent in patients aged 50 years and above.

Other options:

Option A: Refractive errors are the leading cause of MSVI in patients aged 49 years and below.

Options C and D: Glaucoma and diabetic retinopathy account for a very low number of visual impairment cases in patients aged 50 years and above.

Solution to Question 10:

In optic neuritis, relative afferent pupillary defect (RAPD) or Marcus Gunn pupil is seen. It is elicited by the swinging flashlight test. Normally, both pupils remain constricted. In the presence of RAPD, both pupils dilate on moving the flashlight from the normal to the abnormal eye. This paradoxical response of the pupil to light is called Marcus Gunn pupil.

Other options:

Option B: Pain is aggravated by ocular movement, especially in the upward or downward direction due to attachment of the superior rectus fibers to the dura mater.

Option C: Multiple sclerosis, and not multiple myeloma, is a known demyelinating cause of optic neuritis.

Option D: Oral corticosteroid therapy alone is contraindicated because it does not improve the outcome and is associated with an increased risk of new attacks of optic neuritis. Intravenous methylprednisolone can be used if the patient shows radiological evidence of multiple sclerosis.

Solution to Question 11:

The most common ocular manifestation of COVID-19 is follicular conjunctivitis.

Hemorrhagic conjunctivitis, dry eye disease, episcleritis, chemosis, and photophobia may also occur, but less frequently.

The most prevalent ocular symptoms are dry eye or foreign body sensation, redness, tearing, itching, eye pain, and discharge.

Intraocular complications include retinal artery occlusions and hemorrhages, in addition to cotton-wool spots caused by complement-mediated thrombotic angiopathy.

The ACE-2 receptors present in the cornea and conjunctiva make the ocular surface a potential target tissue for SARS-CoV-2.

Sometimes, conjunctivitis may appear as the only sign and symptom of COVID-19. Patients should, therefore, undergo a nasopharyngeal RT-PCR test.

Solution to Question 12:

The condition given in the image is ankyloblepharon. It is the adhesion between the margins of the upper and lower eyelids.

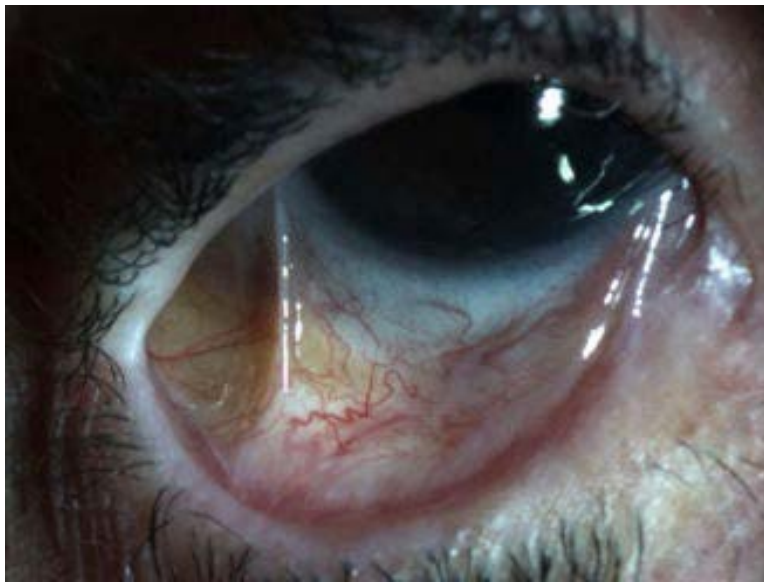
It may occur as a congenital anomaly or as acquired adhesions after healing of chemical burns, ulcers, or traumatic wounds of the lid margins.

Treatment is the excision of the adhesions between lid margins. Epithelial grafts should be used if adhesions extend to angles.

Other options:

Option A: Symblepharon is the adhesion between the palpebral and bulbar conjunctiva in which the eyelids become adherent to the eyeball.

The image given below shows symblepharon.



Option C: Congenital blepharophimosis is a condition in which the extent of the palpebral fissure is decreased. It can be congenital or acquired and is due to a vertical skin fold at the lateral canthus. No treatment is required. In some cases, canthoplasty is done.

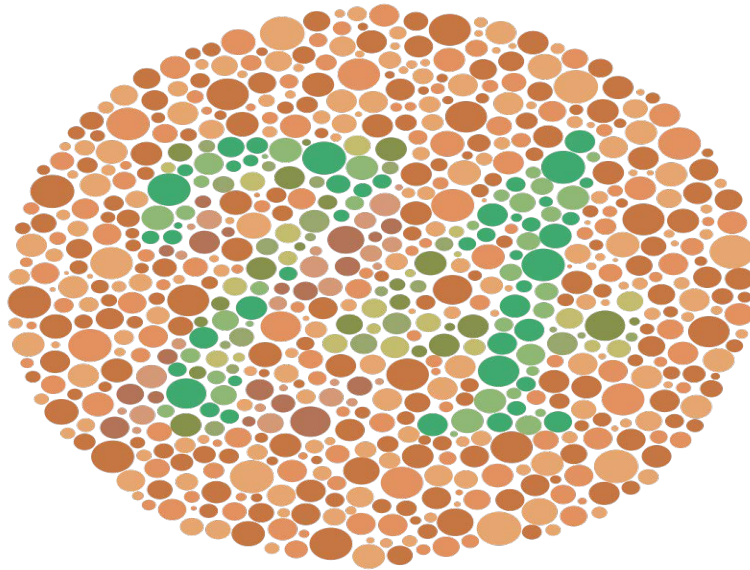
Option D: Congenital ptosis is associated with developmental defect and weakness of the levator palpebrae superioris (LPS) muscle. Drooping of one or both of the upper eyelids is seen since birth. Lid lag on downward gaze is present. Treatment involves surgical correction.

Solution to Question 13:

The correct option is Option C: 1-b, 2-a, 3-d, 4-c.

Color vision is tested with Ishihara plates, which is a pseudoisochromatic chart. There are patterns of colored and grey dots that reveal one pattern to persons with normal color vision and another to those with the color deficiency.

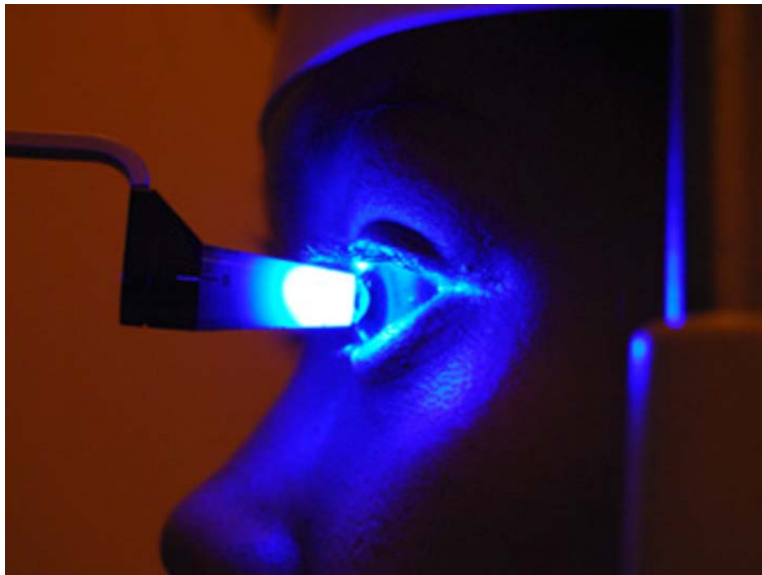
The below image shows Ishihara chart.



An applanation tonometer is used to measure the intraocular pressure. It is based on the Imbert-Fick law, which states that the pressure inside a sphere (P) is equal to the force (W) required to flatten its surface divided by the area of flattening (A).

$$P = W/A$$

The following is an image of the applanation tonometer:



Perimetry is the procedure of estimating the extent of the visual fields. There are 2 types of perimetry: kinetic and static perimetry.

In kinetic perimetry, a target of given luminance is moved from a non-seeing area in the periphery towards the center, where it is perceived. In the static method, the location of a stimulus remains fixed, with the intensity increased until it is seen by the subject (threshold is reached) or decreased until it is no longer detected.

The tangent screen perimetry evaluates the central 30 degrees field at 1 m, thereby testing the central vision.

Solution to Question 14:

Atropine is contraindicated in acute congestive glaucoma.

Acute congestive or angle closure glaucoma is caused by the opposition of the peripheral iris against the trabecular meshwork. This obstructs the aqueous outflow by closing an already narrow-angle of the anterior chamber leading to an acute increase in intraocular pressure(IOP). It can be precipitated by dim light, mydriatics like atropine.

It presents as sudden onset of severe unilateral eye pain or a headache associated with blurred vision, rainbow-colored halos around bright lights, nausea, and vomiting. The physical exam will reveal a fixed midpoint pupil and a hazy or cloudy cornea with marked conjunctival injection.

Management of acute condition includes rapid lowering of IOP by mannitol or glycerol, acetazolamide, and topical antiglaucoma drugs. Definitive management is done by laser peripheral iridotomy or filtration surgery.

Atropine is indicated as cycloplegic to reduce pain and prevent synechiae in corneal ulcers and acute anterior uveitis. It is the initial management in malignant glaucoma.

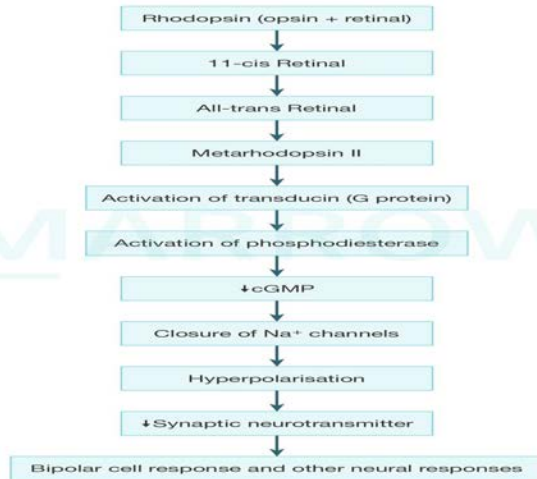
Solution to Question 15:

The exposure of the retina to light causes hyperpolarization and a decreased release of neurotransmitters in rods and cones.

When photoreceptors are exposed to light, a series of biochemical reactions result in the generation of electrical charge. This phenomenon of conversion of light energy into electrical impulse is known as phototransduction.

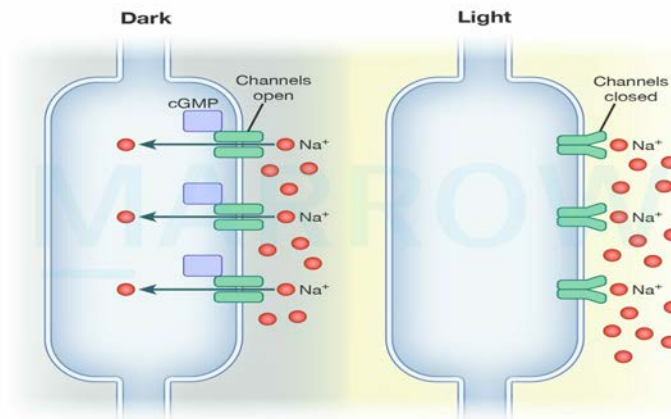
In dark, cyclic GMP (cGMP) gated channels are open leading to a steady release of neurotransmitters (glutamate). When light falls on the retina, 11-cis-retinal is converted to all-trans-retinal. This activates transducin which is a G protein-coupled receptor with phosphodiesterase activity breaking down cGMP. With a decrease in cGMP levels sodium channels close causing hyperpolarization. Hyperpolarization reduces glutamate release, which leads to the formation of the action potential.

Phototransduction in rods and cones



©MARROW

Rod receptors - Sodium flow



©MARROW

Solution to Question 16:

The incorrect about keratoconus is Haab's striae. Haab's striae are seen in cases of buphthalmos where the intraocular pressure is raised and the eyeball is enlarged as a whole. The enlarged cornea is associated with tears in Descemet's membrane called Haab's striae.

Keratoconus or the conical cornea is a bilateral non-inflammatory progressive condition of the axial part of the cornea. Central or paracentral thinning of the cornea and apical protrusion of the anterior cornea are pathognomonic. This is attributed to mucopolysaccharide and collagen tissue synthesis defects.



Clinical features of the condition include defective vision due to progressive myopia and irregular astigmatism. Light sensitivity and ocular irritation may also be present. Complications include acute hydrops following the rupture of Descemet's membrane.

Slit-lamp examination reveals Fleischer's ring at the base of the cone formed. Deep, vertical stromal striae (Vogt lines) that disappear with external pressure are characteristic. Retinoscopy shows a scissoring reflex. The bulging of the lower lid in the downgaze, called Munson's sign, is shown in the image below.



The severity of the condition is graded based on keratometry. Corneal topography is a diagnostic modality with the highest sensitivity for early cases.

Intacs or intracorneal ring segments are useful in moderate cases due to their flattening effect. Rigid gas-permeable contact lenses, corneal collagen cross-linking with riboflavin, and keratoplasty are the other treatment modalities.

Solution to Question 17:

The angle subtended for a patient with 6/6 vision is 5 minutes of arc.

Visual acuity is a measure of the spatial resolution of the eye or an estimation of its ability to discriminate between two points.

Snellen's chart consists of a series of letters arranged in lines, each diminishing in size.

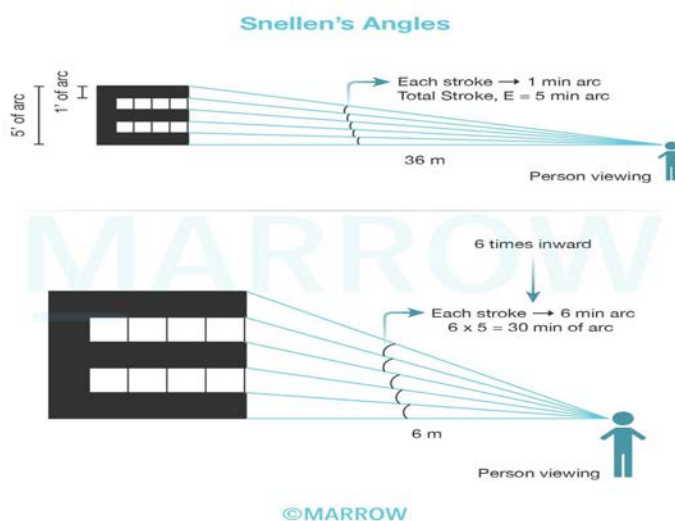
Each whole letter will subtend an

the angle of 5 minutes at the nodal point of the eye at the given distance.

In the Snellen chart, the largest letter will subtend an angle of 5 minutes at the nodal point if it is 60 m from the eye. Those in the subsequent lines will subtend an angle of 5 minutes if they are 36, 24, 18, 12, 9, and 6 m from the eye.

The patient is placed at 6 m and asked to read the visual acuity chart from the largest letters downwards to the smallest that can be visualized with each eye separately. A person with normal 6/6 vision sitting at a distance of 6 m from the types should be able to read every letter from the top to the end of the 6 m line. So the angle subtended by the 6 m line at the eye a distance of 6 meters will be 5 minutes of arc.

If a patient can only read up to the 12m line, his distant vision is defective. His visual acuity will be recorded as 6/12. The numerator denotes the distance at which the patient is able to read the particular line and the denominator denotes the distance at which a person with normal visual acuity will be able to read the given line.



Solution to Question 18:

The recurrence rate of 95% is not a feature of pterygium. The recurrence rate of pterygium is around 80% after simple surgical excision.

Pterygium is a wing-shaped, triangular fibrovascular ingrowth (Option B) of degenerative conjunctival tissue that can encroach onto the cornea. It is a response to prolonged exposure to

ultraviolet rays, dry heat, etc. It has 3 parts mainly, a head (apical part on the cornea), a body (the region between the limbus and canthus), and a cap(semilunar white infiltrate in front of the head)(Option C).. Some authors mention the neck as the constricted part near the limbal area.

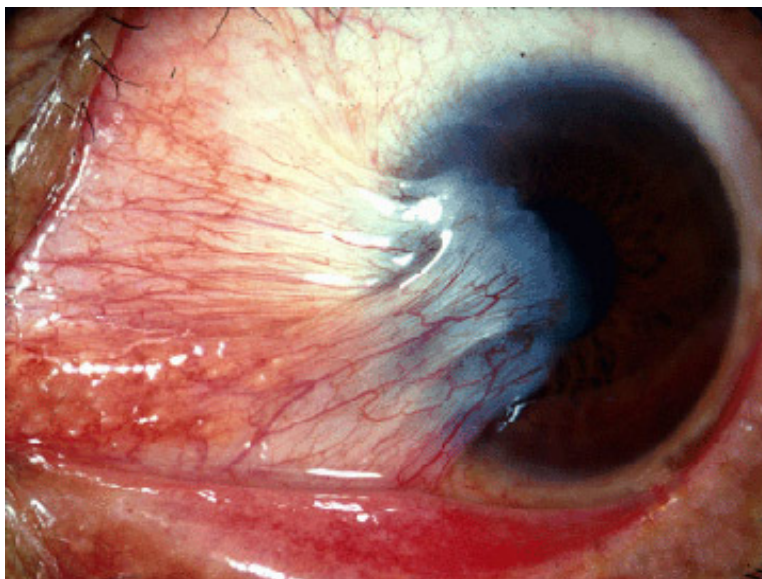
Patients may complain of itching and grittiness due to Dellen (localized drying effect). If the pterygium encroaches onto the pupillary area, it may interfere with vision and also induce astigmatism. Stocker's line may be associated with a pterygium. It is a linear deposition of iron seen in the corneal epithelium, anterior to the head of a pterygium. Pterygium is of 2 types (Option D):

- Progressive which is thick, fleshy, and vascular.
- Regressive which is thin, atrophic, and attenuated.

This condition can be treated symptomatically with tear substitutes for dry eye symptoms and patients are advised to avoid ultraviolet rays. Surgical excision with free conjunctival limbal autograft is the preferred treatment to avoid recurrence.

The recurrence rate of pterygium is around 80% after simple surgical excision. Excision together with adjunctive therapies such as mitomycin C or 5-fluorouracil can reduce the risk of recurrence to approximately 10%

The image below shows a pterygium.



Solution to Question 19:

Krukenberg spindle is not a feature of complicated cataracts. The slit lamp examination in pigmentary glaucoma shows a pattern of pigment deposition on the back of the cornea-endothelium called Krukenberg spindles.

Complicated or secondary cataracts develop secondary to inflammatory/degenerative conditions of the lens or other parts of the eye like iridocyclitis and ciliary body tumors which lead to disturbances in the nutrition of the lens. These cataracts present with a breadcrumb appearance and polychromatic luster.

Even in the early stages, vision is impaired. Inflammation of the posterior segment leads to the characteristic posterior subcapsular cataract while the anterior segment inflammation causes opacification throughout the cortex.

Solution to Question 20:

The oculocephalic reflex or doll's eye reflex is used to assess the integrity of the cranial nerves III, VI, and VIII.

It is a brainstem reflex examined by observing the motion of eyes during the sudden rotation of the head of the patient by the examiner. It is done in both directions laterally, followed by flexion and extension. In a normal positive reflex, the eyes move conjugately in the opposite direction of the movement of the head to maintain their position in space.

The reflex has afferents in the vestibular nerve (VIII) that relay to the brainstem nuclei and bring about conjugate eye movements through the oculomotor nerve (III) and abducens nerve (VI).

A negative reflex occurs in any of the above cranial nerve palsies or brainstem injuries, where the eyes show no movement or movement in the same direction as head movement. This is used to assess brainstem reflexes in a comatose patient as a criterion of brain death.

Solution to Question 21:

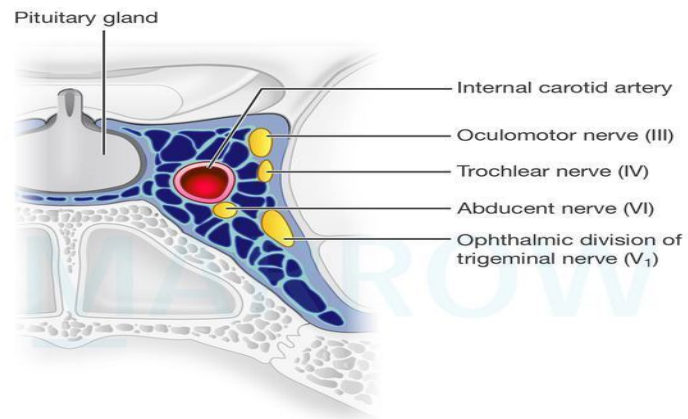
The structures affected in this patient are A, B, C and D.

Cavernous sinus thrombosis (CST) is a septic thrombus formed in the cavernous sinus. It spreads from infection of the sinuses, teeth, ears, nose, and skin of the face.

It presents as fever, severe headache, supraorbital pain, seizures, and edema in the mastoid region. Eye signs include proptosis, chemosis, and diplopia due to ophthalmoplegia from the involvement of oculomotor (CN-III), trochlear (CN-IV), and abducens nerve (CN-VI). Absence of corneal sensation due to the involvement of the ophthalmic division of trigeminal nerve (CN-Va). Ipsilateral ptosis, dilated pupil, and absence of direct and consensual pupillary right reflex due to oculomotor nerve palsy. Pulsatile proptosis occurs with the involvement of the internal carotid artery.

Investigation using brain magnetic resonance imaging or magnetic resonance venography should be done urgently. Following early diagnosis, management of cavernous sinus thrombosis typically includes intravenous broad-spectrum antibiotics, analgesics, anti-inflammatory drugs, and anticoagulants.

Cavernous sinus



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Ophthalmology NEET 2022

Question 1:

What is the diagnosis?



- a) Intraocular foreign body
- b) Pseudoexfoliation syndrome
- c) Ocular trauma
- d) Vossius ring

Question 2:

A patient presents with a history of penetrating injury to the eye. A diagnosis of sympathetic ophthalmitis was confirmed. Which of the following will be seen?

- a) Acute anterior uveitis
- b) Pars planitis
- c) Panuveitis
- d) Chronic anterior uveitis

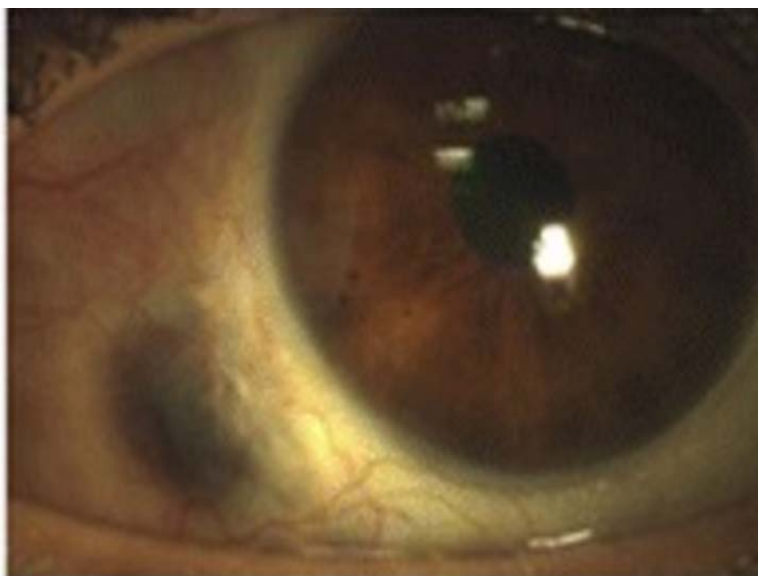
Question 3:

A diabetic patient presents to you with visual acuity of 6/9 in one eye. Further investigations revealed pre-retinal hemorrhages with neovascularization at the optic disc. What is the next step in management?

- a) Focal laser photocoagulation
- b) Pan-retinal photocoagulation
- c) Grid laser photocoagulation
- d) Scleral buckling

Question 4:

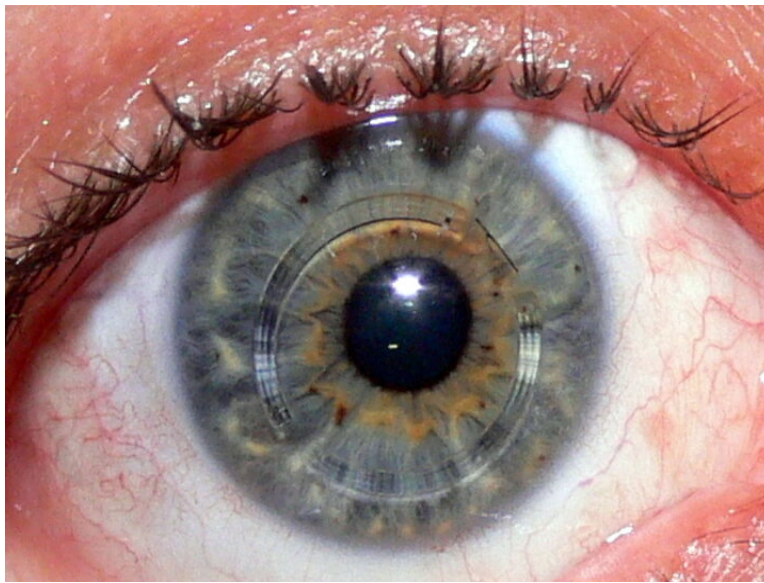
A 35-year-old woman is diagnosed with rheumatoid arthritis. What associated complication is shown in the image?



- a) Scleromalacia perforans
- b) Ciliary staphyloma
- c) Coloboma
- d) Malignant melanoma

Question 5:

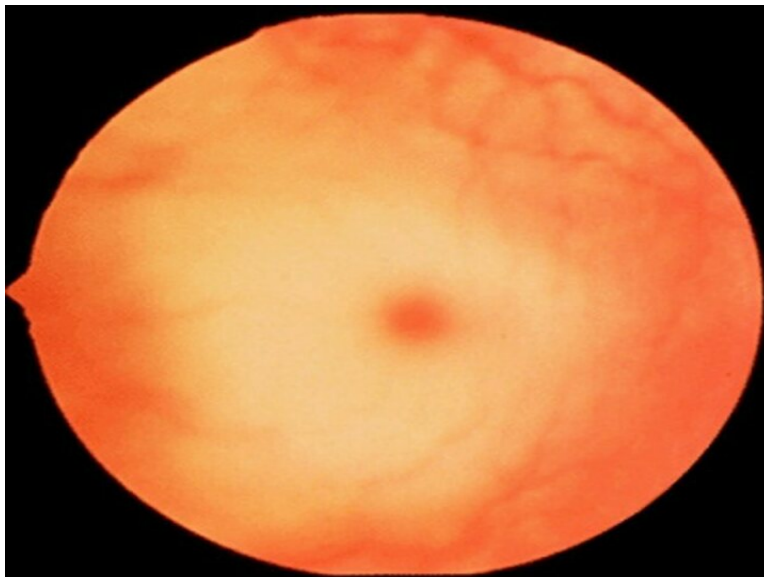
What is the indication of this procedure?



- a) Keratoconus
- b) Vogt's limbal girdle
- c) Keratoglobus
- d) Corneal dystrophy

Question 6:

A 3-year-old boy presents with mental retardation and an inability to walk. The funduscopy image is given below. What is the most likely diagnosis?



- a) Tay-Sachs disease

- b) Hunter disease
- c) Hurler syndrome
- d) Gaucher disease

Answer Key

Question No.	Correct Option
1	b
2	c
3	b
4	a
5	a
6	a

Detailed Explanations

Solution to Question 1:

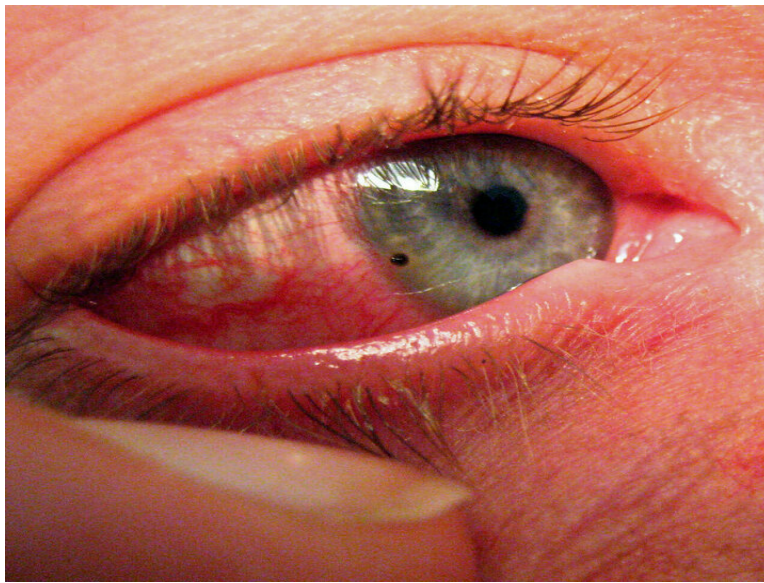
The diagnosis based on the image given is pseudoexfoliation syndrome.

Pseudoexfoliation syndrome is an age-related degenerative disorder, more commonly seen in women. It is marked by the deposition of exfoliative material on the iris, ciliary region, and lens capsule. They appear as flakes on the anterior lens capsule and edge of the iris.

The material can get collected in the anterior chamber angle and impede the aqueous humor drainage. Patients affected by this condition are at an increased risk of developing open-angle glaucoma. Rarely, angle-closure glaucoma can also be seen. Other ocular symptoms include iris sphincter atrophy, anisocoria, and heterochromia. Systemic manifestations due to abnormal fibrillary deposition may also be seen.

Other options:

Option A - Common intraocular foreign bodies include glass particles or metal chips. It can lodge in various sites like the anterior chamber, lens, posterior chamber, and vitreous. This can lead to cataract formation, retinal hemorrhages, and tears. Organic foreign bodies can cause infection. Iron foreign bodies can cause siderosis while copper ones can result in chalcosis. An entry wound can be identified with topical fluorescein. In cases of metallic intraocular foreign bodies, MRI is contraindicated. CT scan can be done instead.

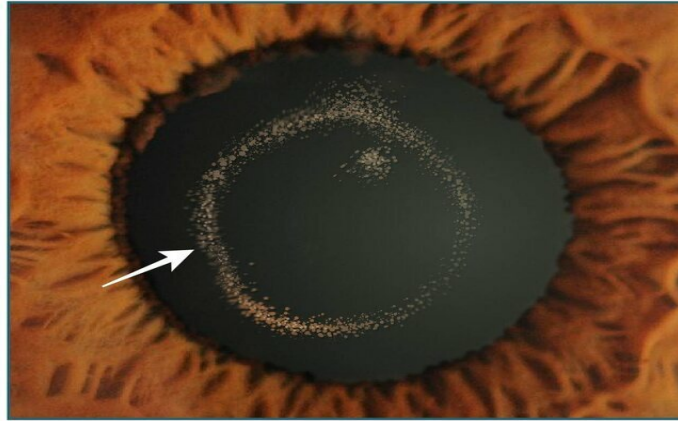


Option C - Ocular trauma can be a closed-globe or open-globe injury. Closed-globe injury can vary from simple corneal abrasions to widespread intraocular damage. Open-globe injuries include globe rupture or laceration.

A corneal abrasion highlighted with fluorescein staining is shown below.



Option D - Vossius ring is used to describe a ring-shaped pigment dispersal observed on the anterior surface of the lens due to the deposition of brown amorphous pigment granules. It is close to the miosed pupil in diameter. The force of the blow drives the cornea and iris backward, creating an impression of the iris on the lens.



©Marrow

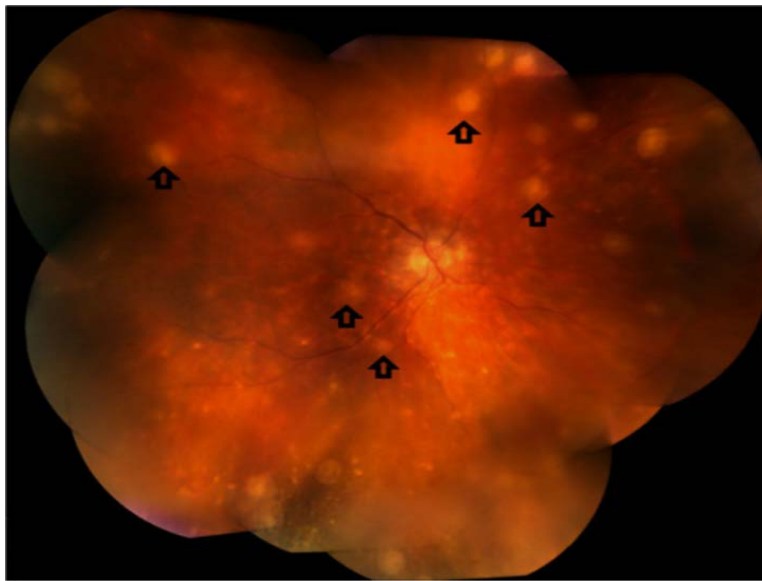
Solution to Question 2:

Panuveitis is the usual clinical manifestation seen in sympathetic ophthalmitis. Panuveitis is inflammation of the whole uvea.

Sympathetic ophthalmitis is a granulomatous pan-uveitis that usually occurs after penetrative trauma. It can also develop after intraocular surgery. Both the injured eye (exciting eye) and fellow eye (sympathizing eye) develop panuveitis. Trauma to the ciliary region (dangerous zone) poses an increased risk.

It is a type 4 hypersensitivity reaction. According to the allergic theory, the uveal pigment in the injured eye acts as an allergen that causes uveitis in the other eye. Dalen-Fuchs nodules (proliferation of pigment epithelium), exudative retinal detachment, and optic disc swelling can be seen on fundoscopy. A sunset glow appearance can persist due to residual chorioretinal scarring.

Dalen-Fuchs nodules are depicted in the image below.



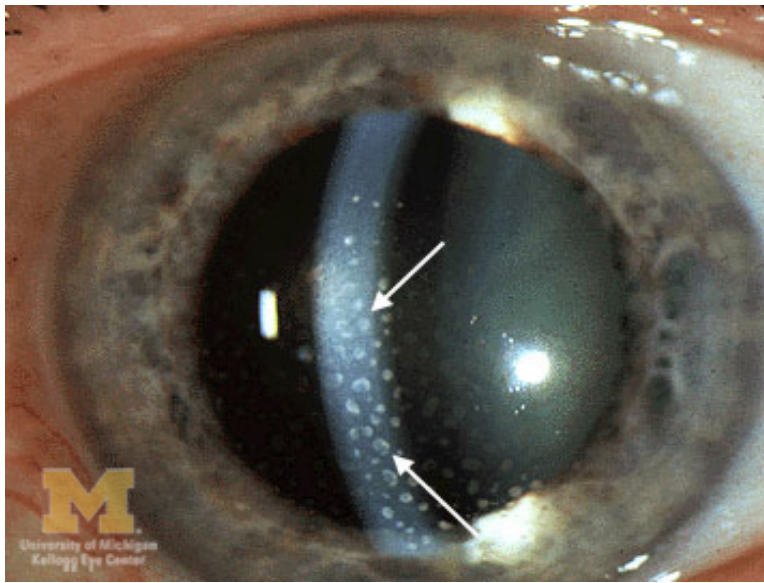
The exciting eye shows features of low-grade uveitis such as ciliary congestion, lacrimation, and tenderness. The sympathizing eye is usually involved 4-8 weeks later. It has a prodromal stage with photophobia, ciliary congestion, and keratic precipitates. In the fully developed stage, symptoms of acute iridocyclitis supervene.

Prophylactic measures to be taken before the condition sets in include topical or systemic steroids and antibiotics. Early excision of the injured eye is carried out if it cannot be salvaged. Once sympathetic ophthalmitis has already set in, enucleation of the injured eye must be done. Atropine, immunosuppressants, and corticosteroids should also be administered.

Other options:

Options A and D: Anterior uveitis is inflammation of the iris up to pars plicata. Keratic precipitates are characteristic of the condition. Acute anterior uveitis, which can last for 6 weeks to 3 months, presents with pain, photophobia, and lacrimation. Chronic anterior uveitis lasts for longer than 6 months. Anterior uveitis is associated with juvenile idiopathic arthritis, sarcoidosis, and ankylosing spondylitis.

The image below shows keratic precipitates.



Option B: Pars planitis or intermediate uveitis is an inflammatory condition that affects the pars plana of the ciliary body, the peripheral retina, the choroid, and the vitreous base. Snowbanking is the hallmark feature. It is a grey-white fibrovascular plaque formed due to the coalescence of snowball opacities over the pars plana. Pars planitis can be seen in multiple sclerosis, sarcoidosis, and syphilis.

Solution to Question 3:

The clinical vignette is suggestive of proliferative diabetic retinopathy with high-risk characteristics and the next step in management is pan-retinal photocoagulation.

Diabetic retinopathy is used to describe retinal changes seen in diabetic patients. It is classified into the following types:

- Non-proliferative diabetic retinopathy (NPDR)
- Proliferative diabetic retinopathy (PDR)
- Diabetic maculopathy
- Advanced diabetic eye disease

Fundoscopy in NPDR reveals hard exudates, cotton-wool spots, and microaneurysms. Hard exudates are small, discrete yellowish, waxy areas with crenated margins.



Proliferative diabetic retinopathy is marked by NVD or NVE which is the hallmark of the condition. Condensation of connective tissue surrounding the newly formed vessels results in the formation of a fibrovascular epiretinal membrane. Vitreous detachment and hemorrhage may also occur. Preretinal hemorrhages are considered a high-risk characteristic. In addition, changes seen in NPDR may also be present.

The condition usually develops 25 years after the onset of the disease. Hence, it is more commonly seen in patients with type I diabetes.

The image below shows NVD.



Clare Gilbert

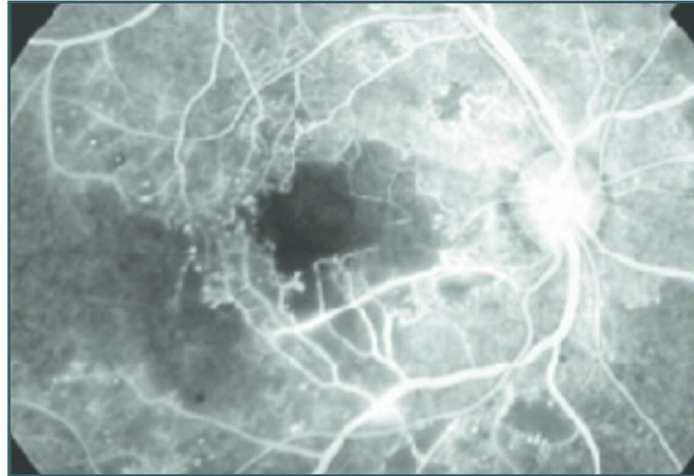
PDR with high-risk characteristics is managed with pan-retinal photocoagulation. Laser treatment helps by inhibiting the release of vascular endothelial growth factors (VEGF). Anti-VEGFs are recommended before pan-retinal photocoagulation in patients with PDR.

Diabetic maculopathy can be classified as:

- Focal exudative maculopathy: Microaneurysms, hard exudates, and hemorrhages can be seen.

- Diffuse exudative maculopathy: Diffuse retinal edema with few hard exudates.
- Ischemic maculopathy: Microaneurysms, hemorrhages, few hard exudates. Fluorescein angiography reveals areas of non-perfusion.
- Mixed maculopathy: Features of both exudative and ischemic maculopathy.

An FFA image showing features of macular ischemia is given below.



Other options:

Option A: Focal laser photocoagulation is the treatment of choice for focal diabetic macular edema (DME) that does not involve the fovea.

Option C: Grid laser photocoagulation can be used in diffuse DME not responding to anti-VEGFs and intravitreal steroids.

Option D: Scleral buckling can be used in cases of advanced PDR with tractional retinal detachment.

Solution to Question 4:

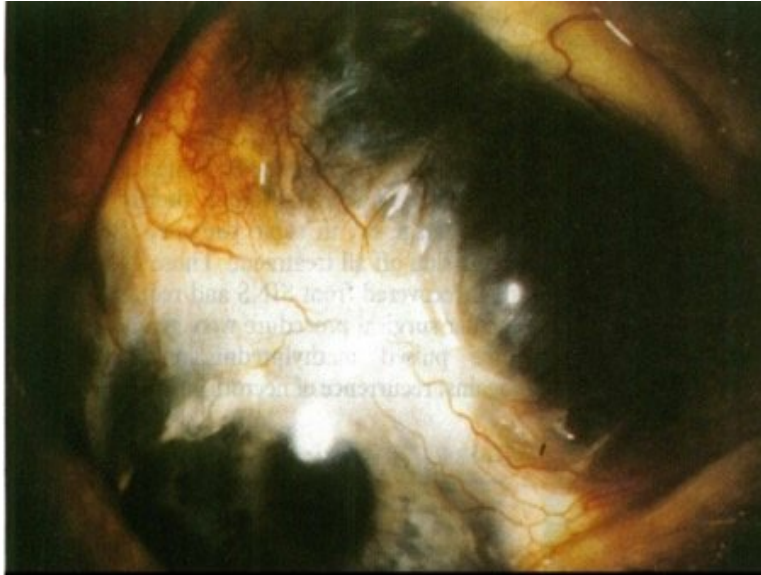
Scleromalacia perforans is the associated complication of rheumatoid arthritis that is shown in the image.

Scleromalacia perforans is a specific type of necrotizing scleritis without inflammation, characterized by progressive scleral thinning. The uveal tissue appears to shine through it. Spontaneous perforation is a rare complication of the condition. It is typically seen in elderly women with long-standing rheumatoid arthritis. A yellowish patch of melting sclera can be seen. This is due to the obliteration of blood supply.

Clinical features include pain, photophobia, and lacrimation. The affected eye appears red. Medical management includes steroids and immunosuppressants. In severe cases, surgical treatment by scleral patch graft may be done.

Other options:

Option B - Ciliary staphyloma is used to describe the bulge of an area of weak sclera lined by the ciliary body. Common causes include scleritis, developmental glaucoma, and end-stage primary or secondary glaucoma. Thinning of the sclera, secondary to perforating injury, can also result in the condition.



Option C - Coloboma is a congenital anomaly, that develops due to a defect in the closure of the embryonic cleft. It is characterized by poorly developed uveal tissue. In some cases, it might be deficient. The lens, choroid, and retina may also be involved.

The image below shows coloboma of the right eyelid.



Option D - Malignant melanoma of the conjunctiva appears as a mass near the limbus. It can be pigmented or non-pigmented. It is usually seen in elderly patients. The choroid, iris, ciliary body, and eyelids can also develop the tumor.

The image below shows malignant melanoma of the choroid.



Solution to Question 5:

The given image shows the placement of Intacs, which is done for keratoconus.

Keratoconus or conical cornea is a non-inflammatory condition of the axial cornea. Central or paracentral thinning of the cornea and apical protrusion of the anterior cornea are pathognomonic. This is attributed to mucopolysaccharide and collagen tissue synthesis defects.



Clinical features of the condition include defective vision, which develops due to progressive myopia and irregular astigmatism. Light sensitivity and ocular irritation may also be present. Complications include acute hydrops due to rupture of Descemet's membrane.

Slit-lamp examination reveals Fleischer's ring at the base of the cone formed. Deep, vertical stromal striae (Vogt lines) that disappear on external pressure are characteristic. Retinoscopy

shows a yawning reflex.

The image below shows Munson's sign i.e. bulging of the lower lid on looking down.



The severity of the condition is graded based on keratometry. Corneal topography is a diagnostic modality with the highest sensitivity for early cases.

Intacs or intracorneal ring segments are useful in moderate cases, due to their flattening effect. Rigid gas permeable contact lenses, corneal collagen cross-linking with riboflavin, and keratoplasty are the other treatment modalities.

Other options:

Option B - Vogt's limbal girdle is a degenerative corneal disease that is more commonly seen in women aged ≥ 40 years. A white crescentic limbal band can be seen at 9 and/or 3 'o'clock position, more frequently on the nasal side. In type I, a swiss-cheese hole pattern with a clear zone separating it from the scleral margin can be observed. Type II is more common and lacks a clear zone and holes.

Option C - Keratoglobus is a familial disorder, marked by thinning and hemispherical protrusion of the entire cornea. It can be congenital or acquired. Globular protrusion is seen here, in contrast to the conical ectasia seen in keratoconus. Intrastromal ring segments and cross-linking can be used for management.



Option D - Corneal dystrophy is used to describe a group of non-inflammatory conditions, with bilateral opacification of the cornea with no vascularisation. These are divided into epithelial, Bowman layer, stromal, Descemet membrane, and endothelial dystrophies. Most conditions have genetic abnormalities.

Solution to Question 6:

The given clinical scenario, along with the fundoscopy image showing a macular cherry-red spot, indicates the most likely diagnosis to be Tay-Sachs disease.

Tay-Sachs disease is a lysosomal storage disease caused by the deficiency of β -hexosaminidase. It can present during infancy with macrocephaly, increased startle reaction, gradual loss of motor skills, and a macular cherry-red spot. Affected children usually die by the age of 10–15 years.

It is of two types—juvenile-onset and adult-onset. The symptoms of the juvenile-onset form include ataxia and dementia. Dysarthria, lower motor neuron signs, and psychiatric disorders are common in the adult-onset type.

Other options:

Options B - Hunter disease (mucopolysaccharidosis II) is an X-linked disorder that arises due to the deficiency of iduronate 2-sulfatase. Short stature, joint stiffness, and intellectual disability are the features.

Option C - Hurler syndrome (mucopolysaccharidosis I) is an autosomal recessive condition that affects multiple organs. Presentation is usually at 6–24 months with corneal clouding, coarse facial features, joint stiffness, and skeletal dysplasia. Glaucoma, retinal degeneration, and nasal discharge are the other features. The condition is generally fatal.

Option D - Gaucher disease is a glycolipid storage disorder that can present with hepatosplenomegaly, mental retardation, and erosion of long bones. It is caused by the deficiency of β -glucosidase. The Gaucher cell has a wrinkled-paper appearance due to intracytoplasmic inclusions, which is a characteristic feature.

Ophthalmology NEET 2023

Question 1:

A female presents with loss of vision in the right halves of both eyes. Where is the lesion located in the optic pathway?

- a) Left optic tract
- b) Optic radiation
- c) Optic chiasma
- d) Right optic tract

Question 2:

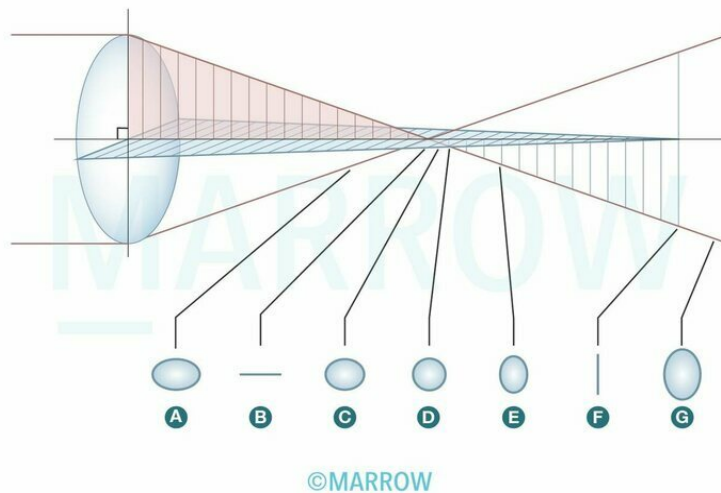
A 65-year-old diabetic woman presents with painless loss of vision. The ocular findings are shown in the image below. What is the likely diagnosis and management for this patient?



- a) Zonular cataract and lensectomy
- b) Rosette cataract and sugar control
- c) Sunflower cataract and lesion extraction
- d) Immature senile cataract and phacoemulsification with IOL

Question 3:

Which of the following refractive errors is associated with the image given below?



- a) Presbyopia
- b) Astigmatism
- c) Hypermetropia
- d) Myopia

Question 4:

A 5-year-old boy presents with a small nodule on the sclera as shown in the image below. What is the likely diagnosis?



- a) Pterygium
- b) Pinguecula
- c) Dermolipoma
- d) Dermoid

Question 5:

The eye examination of a 20-year-old male shows a golden ring on the iris. What is the next investigation to diagnose the condition?

- a) Alpha-1-antitrypsin
- b) Alpha-fetoprotein
- c) Serum iron
- d) Serum ceruloplasmin

Question 6:

What is the dye and filter used to diagnose corneal ulcers?



- a) Lissamine dye - green filter
- b) Lissamine dye - cobalt blue filter
- c) Fluorescein dye - visualized under green filter
- d) Fluorescein dye - visualized under cobalt blue filter

Question 7:

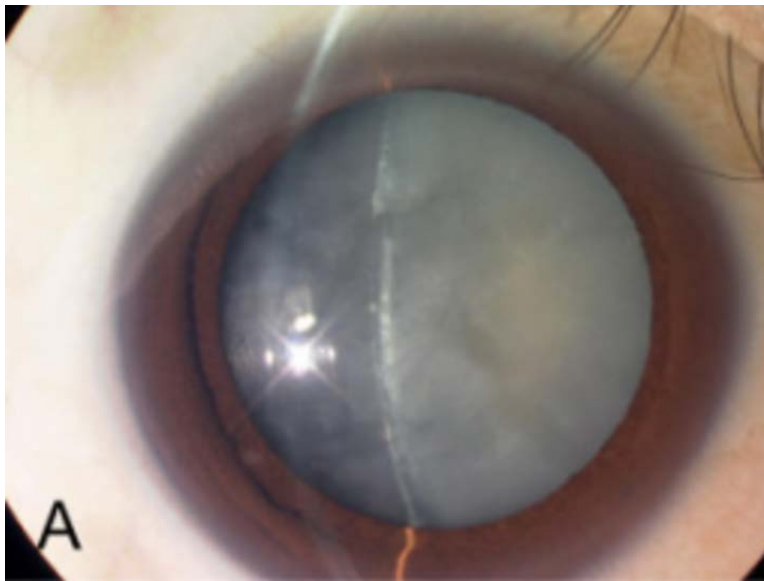
An infant is brought with complaints of excessive watering of the eyes and photophobia. The image is given below. What is the likely diagnosis?



- a) Congenital glaucoma
- b) Congenital cataract
- c) Mucopolysaccharidosis
- d) Ophthalmia neonatorum

Question 8:

A 60-year-old patient complains of decreased distant vision but now he does not use spectacles for near vision. The image of his ocular examination is given below. What is this type of refractive error called?



- a) Index myopia
- b) Axial myopia
- c) Curvatural myopia
- d) Positional myopia

Answer Key

Question No.	Correct Option
1	a
2	d
3	b
4	d
5	d
6	d
7	a
8	a

Detailed Explanations

Solution to Question 1:

Visual loss in the right halves of both eyes is suggestive of right homonymous hemianopia. The site of the lesion in this case is the left optic tract.

Here, vision is lost in the right half of both eyes, which means:

- in the left eye, the nasal visual field is lost, and
- in the right eye, the temporal visual field is lost.

It means left temporal and right nasal fibers are destroyed, which leads to right homonymous hemianopia. Anatomically, this can be explained by the fact that each optic tract contains fibers from the ipsilateral temporal half of the retina and the contralateral nasal half of the retina.

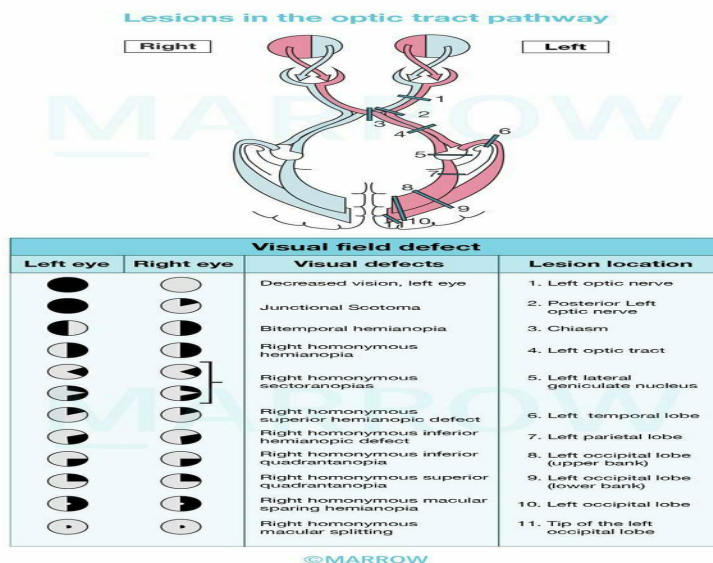
Optic tract lesions can occur due to pituitary adenomas (most common), craniopharyngioma, tumors of the optic thalamus, large aneurysm of superior cerebellar or posterior cerebral arteries, demyelinating diseases, and infarction.

Other options:

Option B: Optic radiation lesions result in contralateral homonymous hemianopia, which means that the visual loss would occur in the left halves of both eyes rather than the right halves, as described in the question.

Option C: Lesions at the optic chiasma typically cause bitemporal hemianopia, which involves visual loss in the outer (temporal) halves of the visual field in both eyes. This is not consistent with the patient's described symptoms.

Option D: A lesion in the right optic tract would result in left homonymous hemianopia, causing visual loss in the left halves of both eyes. This does not match the patient's presentation of vision loss in the right halves of both eyes.



Solution to Question 2:

The given clinical scenario and the above image are suggestive of an immature senile cataract and the appropriate treatment is phacoemulsification with posterior intraocular lens implantation.

An immature senile cataract is characterized by partial opacification of the lens. The cuneiform and cupuliform patterns of opacification are seen and the lens appears greyish-white. Iris shadow is visible. Slit-lamp examination shows areas of the normal and cataractous cortex.

The treatment of choice for immature senile cataracts is treated by extracapsular cataract extraction by phacoemulsification and foldable posterior chamber intraocular lens implantation.

Other options:

Option A: Zonular cataract is a developmental type of cataract in which opacification occupies a discrete zone in the lens. It is the most common type of congenital cataract presenting with visual impairment. Lensectomy is the treatment of choice for congenital cataracts.

Option B: Rosette cataract presents after close globe injuries (blunt trauma) resulting in damage to the lens fibers and capsule resulting in hydration of lens fibers by aqueous humor. This results in opacification of the posterior subcapsular cortex along the posterior sutures. It presents as a flower-shaped opacification. Iridodialysis occurs due to dehiscence of the iris from the ciliary body at its root. The pupil becomes D-shaped. Surgery is recommended in case of visual impairment.

Option C: Sunflower cataract is seen in Wilson disease due to the deposition of cuprous oxide in the anterior lens capsule and subcapsular cortex in a stellate pattern. D-penicillamine is an effective treatment modality.

Solution to Question 3:

The given image is a Sturm's conoid. It represents the configuration of a bundle of light rays formed by a regularly astigmatic (toric) surface and helps in understanding the optics of regular astigmatism.

Astigmatism is a type of refractive error that occurs when light rays enter the eye at different planes and converge at different points forming focal lines.

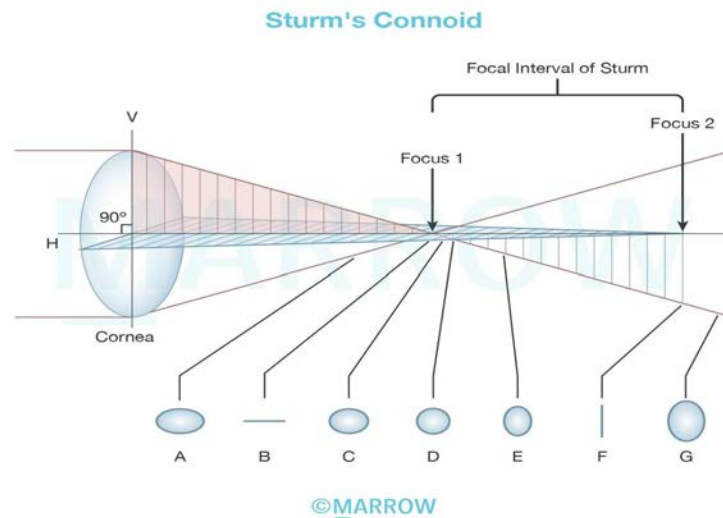
When light passes through a regularly astigmatic surface (surface with toric curvature), vertical and horizontal rays converge at different points and form two focal lines. The shape of a bundle of light rays at different points is as follows:

For the given Sturm's conoid,

- At point A, a horizontal oval is formed because vertical rays converge more rapidly than horizontal rays.
- At point B (first focus), vertical rays come to a focus, while horizontal rays are still converging; so, it will be a horizontal straight line.
- At points C, D, and E, the vertical rays are diverging, and horizontal rays are still converging.
- At point D, the vertical rays have diverged as much as the horizontal rays have converged; so, we get a perfect circle called the circle of least diffusion.
- At point E, a vertical oval is formed.
- At point F (second focus), a vertical straight line is formed
- At point G and beyond, both rays are diverging, and we always get a vertical oval/ prolate ellipse.

- The focal interval of Sturm refers to the distance between two foci (B and F).

The image is clearest or least confusing as point D, the circle of least confusion/ circle of diffusion. The circle of diffusion lies at the mid-point of this interval.



- Option A: Presbyopia is a refractive error caused by the loss of elasticity of the crystalline lens, leading to difficulty in focusing on near objects. Presbyopia is not associated with Sturm's conoid.
- Option C: Hypermetropia, or farsightedness, is a refractive error in which the eye does not focus incoming light properly, causing near objects to appear blurry. It is not associated with Sturm's conoid.
- Option D: Myopia, or nearsightedness, is a refractive error in which distant objects appear blurry because the eye focuses light in front of the retina rather than on it. Myopia is not associated with Sturm's conoid.

Solution to Question 4:

A dermoid is a likely diagnosis based on the ocular finding and the given image.

An orbital dermoid is a choristoma (a mass of histologically normal tissue in an abnormal location) formed due to subcutaneous displacement of the ectoderm.

It is lined by keratinized stratified squamous epithelium and contains dermal appendages like sweat glands, sebaceous glands, and hair follicles. It can be located anteriorly (superficial) or posteriorly (deep) to the orbital septum.

Clinical features: Superficial dermoid presents in infancy as a painless nodule in the superotemporal part of the orbit whereas deep dermoid presents in adulthood as a gradually protruding eye or acutely as an inflamed orbit. CT imaging would show a well-circumscribed cystic lesion.

Treatment is by excision in toto.

Other options:

Option A: Pterygium is a triangular fibrovascular ingrowth of degenerative bulbar conjunctival tissue on the limbus as a result of chronic exposure to UV rays.

Option B: Pinguecula is an asymptomatic yellow-white mound of degenerated conjunctival stroma seen adjacent to the limbus. It is more common on the nasal limbus.

Option C: Dermolipoma presents in adulthood as a subconjunctival yellowish mass near the outer canthus of the eye. It contains dermal appendages and fatty tissue.

Solution to Question 5:

The given image shows a Kayser-Fleischer ring, which is seen in Wilson's disease. Low serum ceruloplasmin level is diagnostic of Wilson's disease.

Wilson's disease (hepatolenticular degeneration) is an inherited metabolic disorder resulting due to abnormal copper deposition in tissues (mainly the liver and brain). It is caused by a deficiency of ceruloplasmin (copper-carrying blood protein).

Kayer-Fleischer ring is a brownish-yellow zone of copper deposition that is present in the peripheral Descemet membrane.



The ring is best seen on gonioscopy. Penicillamine therapy may cause the ring to disappear.

Another ocular finding in Wilson's disease is anterior capsular sunflower cataract.

Other options:

Option A: Alpha-1 antitrypsin deficiency leads to lung and liver disease.

Option B: Alpha-fetoprotein is a diagnostic marker in hepatocellular carcinoma.

Option C: Serum iron level is assessed in hemochromatosis which occurs due to increased iron deposition in various tissues. The ocular manifestation includes dry eye and rusty-brown perilimbal and conjunctival discoloration.

Solution to Question 6:

The image given is of a dendritic ulcer seen in recurrent herpetic keratitis. Corneal ulcers can be detected with the help of fluorescein sodium dye and a cobalt-blue filter. The epithelial defect of the cornea appears opaque green with a cobalt blue filter.

Fluorescein staining of the cornea is carried out with a disposable autoclaved filter paper strip impregnated with the dye. The denuded epithelial area seen in corneal abrasions or ulcers appears brilliant green when stained with fluorescein. Under a cobalt blue light, the stained area appears opaque green.

Other options:

Lissamine green dye is used to stain devitalized cells of the cornea and conjunctiva. It is used to diagnose ocular surface disorders, like dry eye.

Solution to Question 7:

The clinical scenario of excessive watering and photophobia in an infant and the given image with corneal enlargement and haze suggest a likely diagnosis of congenital glaucoma.

Primary congenital glaucoma (PCG) is caused by impaired aqueous outflow due to a maldeveloped anterior chamber angle.

It can be classified as

- True congenital glaucoma - IOP is elevated during intrauterine life.
- Infantile glaucoma - manifests before 3 years of age.
- Juvenile glaucoma (least common) - IOP rises between 3 and 16 years of age.

Clinically, the child presents with lacrimation, photophobia, and blepharospasm.

PCG may also present with:

- Corneal haze (due to diffuse edema secondary to raised IOP)
- Buphthalmos (a large eye that results from stretching due to raised IOP before 3 years of age). The thin sclera appears blue due to underlying uveal tissue.
- Haab striae (curvilinear healed breaks in Descemet's membrane)
- Corneal scarring
- Optic disc cupping

Evaluation is done under general anesthesia. Initially, IOP measurement is performed (10–12 mmHg is normal). A corneal diameter measuring greater than 12 mm is highly suggestive of an infant under the age of 1 year.

Management is essentially surgical and goniotomy and/or trabeculectomy may be done.

Other options:

Option B: Congenital cataract occurs due to disturbance in lens growth leading to the development of opaque lens fibers. It can be idiopathic, hereditary, or secondary to metabolic disorders and infections. It presents with leukocoria (white reflex in the pupillary area), wandering eye movement, squint, and nystagmus.

Option C: Mucopolysaccharidoses are inherited metabolic disorders presenting with corneal opacification, pigmentary retinopathy, glaucoma, and optic atrophy.

Option D: Ophthalmia neonatorum presents in neonates with conjunctival discharge and chemosis, and tenderness of the eyeball secondary to infections.

Solution to Question 8:

The most likely diagnosis based on the given clinical scenario and the image is index myopia seen in nuclear senile cataracts.

Index myopia occurs due to an increase in the refractive index of the crystalline lens due to nuclear sclerosis. Nuclear sclerosis is an ageing process in which the lens becomes hard and inelastic resulting in an increase in the refractive index, causing progressive myopia (decreased distant vision). A patient with presbyopia may be able to read again, without the aid of spectacles. This improvement in vision is termed second sight.

Other options:

Option B: Axial myopia is the most common type of myopia resulting due to an increase in the anteroposterior length of the eyeball.

Option C: Curvatural myopia results due to increased curvature of the cornea, lens, or both.

Option D: Positional myopia results due to anterior placement of the crystalline lens in the eye.

Ophthalmology INI-CET 2023

Question 1:

The vestibulo-ocular reflex (VOR) is seen to stabilize the image on the retina while the head moves. What is the gain of VOR (eye velocity/head velocity)?

- a) 4
- b) 2
- c) 1
- d) 3

Question 2:

In ocular surface examination and dry eye staining, which of the following order of dyes is used?

- a) Fluorescein -> Rose Bengal -> Lissamine green
- b) Rose Bengal -> Lissamine green -> Fluorescein
- c) Lissamine green -> Rose Bengal -> Fluorescein
- d) Fluorescein-> Lissamine green -> Rose Bengal

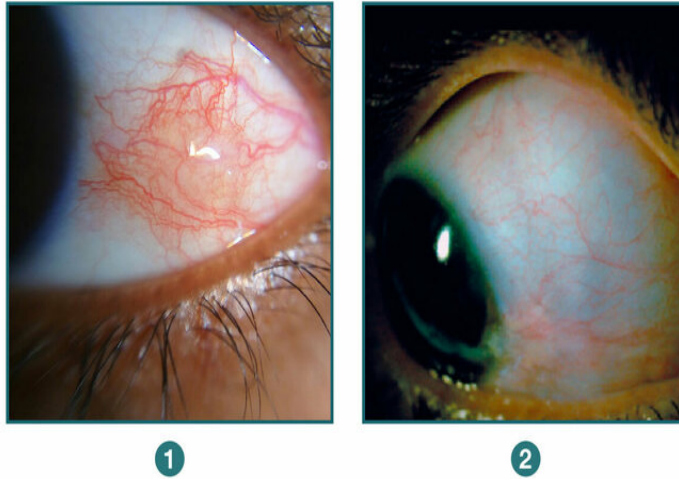
Question 3:

Which of the following are direct gonioscopes?

- a) 1, 3
- b) 2, 4
- c) 4
- d) 1, 2, 3

Question 4:

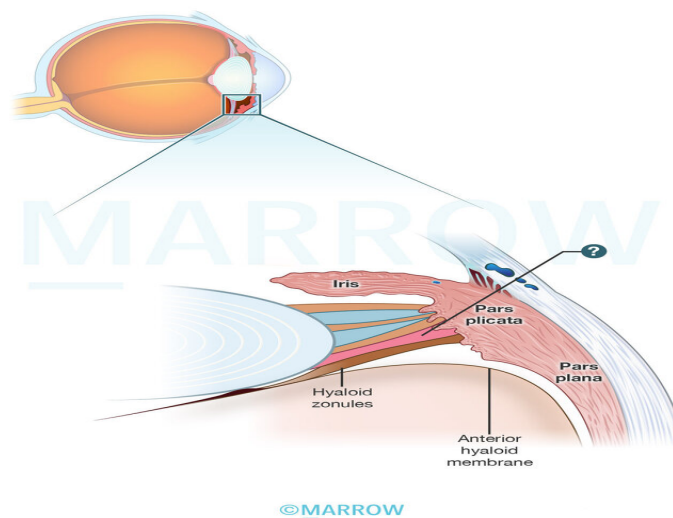
On administration of phenylephrine, the following change was noted as shown in the images below. What is the probable diagnosis?



- a) Nodular scleritis
- b) Nodular episcleritis
- c) Tenon's cyst
- d) Nodular phlycten

Question 5:

Identify the marked structure in the eye.



- a) Eggers line
- b) Lens capsule

- c) Canal of Hannover
- d) Canal of Petit

Question 6:

Which of the following is true about perimetry?

- a) Kinetic perimetry is used in quantification and tracking of visual changes in established glaucoma
- b) In Static perimetry, stimulus intensity is held constant and moved centrally with detection
- c) Visual field defect to be classified as glaucoma, it should have the corresponding optic nerve head abnormalities
- d) Early signs include generalised constriction of isopters and baring of blind spot

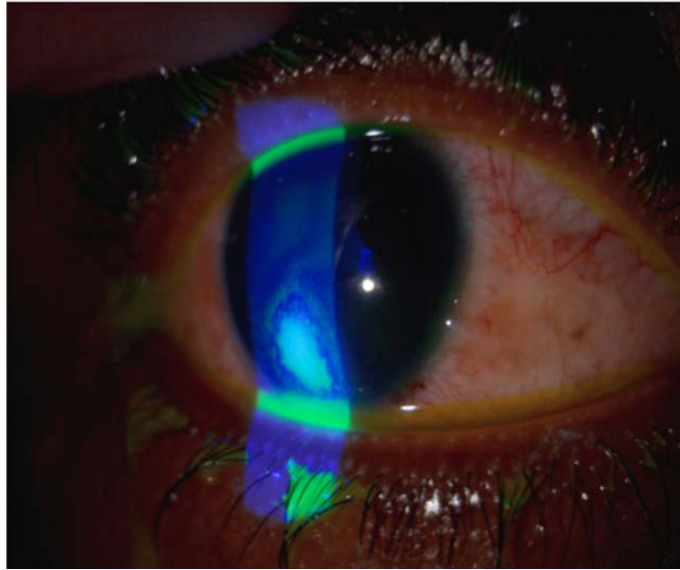
Question 7:

A man with known history of diabetes has come to your clinic. His vision is 6/6. However, on retinal examination, he is classified to have severe NPDR. Which of the following is not a feature of this condition?

- a) Retinal microvascular changes
- b) Retinal hemorrhages
- c) Disc neovascularisation
- d) Venous beading

Question 8:

Identify the filter, the dye and the condition shown below



- a) Cobalt green - Fluorescein dye - Corneal degeneration
- b) Cobalt blue - Fluorescein dye - Corneal ulcer
- c) Cobalt blue - Lissamine green dye - Corneal opacity
- d) Cobalt green - Rose Bengal dye - Interstitial keratitis

Question 9:

A professional boxer wants vision correction surgery. Which of the following is not recommended?

- a) RK
- b) PRK
- c) LASIK
- d) SMILE

Question 10:

A child went to an ophthalmologist and is found to have a power of -1 D spherical and -2 D cylindrical at 180 degrees. What kind of astigmatism does the child have?

- a) Simple astigmatism
- b) With the rule astigmatism
- c) Against the rule astigmatism
- d) Mixed astigmatism

Question 11:

The following instrument is used to measure which of the following?



- a) Proptosis
- b) Squint
- c) Diplopia
- d) Stereopsis

Answer Key

Question No.	Correct Option
1	c
2	a
3	b
4	b
5	d
6	d
7	c
8	b
9	a

10	b
11	a

Detailed Explanations

Solution to Question 1:

The gain of the vestibulo-ocular reflex (VOR) is 1, which means there is a compensatory eye velocity equal to the head velocity and in the opposite direction.

VOR prevents the slipping of the retinal images when the head moves; it moves the eyes at the same speed but in the opposite direction as the head. If an object is being pursued with eye and head movement, vision cancels the vestibulo-ocular reflex to prevent the eyes from moving in the opposite direction as the head.

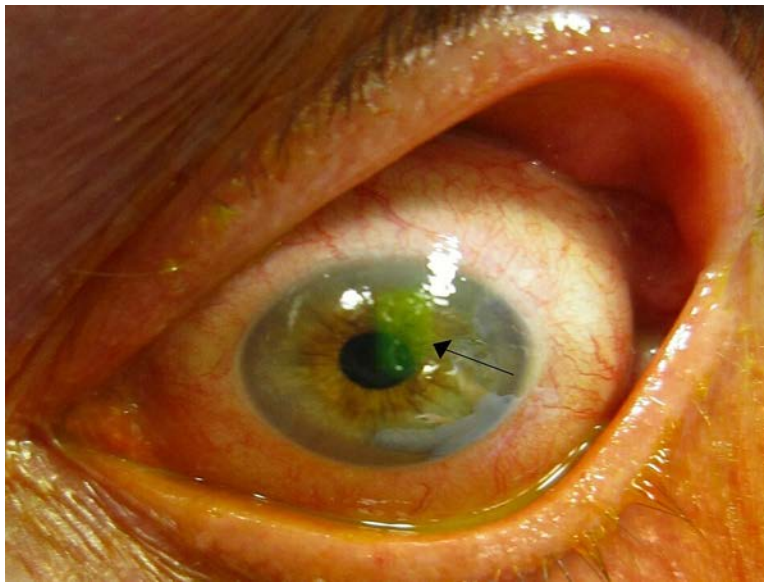
VOR is a physiological vestibular nystagmus that forms the basis of the caloric test. If cold water is poured into the right ear the patient develops left jerk nystagmus (rapid phase towards the left), while the reverse happens with warm water, i.e. patient develops right jerk nystagmus. It can be remembered by the mnemonic 'COWS' (Cold- Opposite, Warm-Same).

Solution to Question 2:

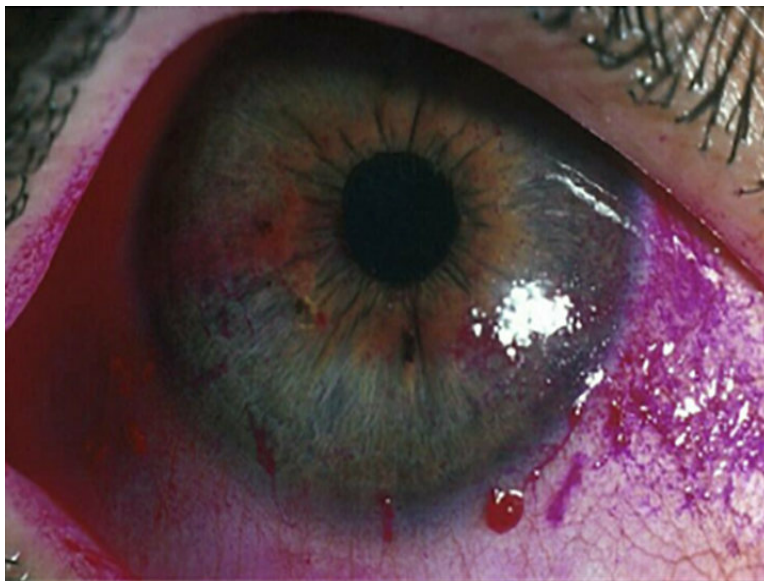
Amongst the given options, the better order of dyes to pick for use in ocular surface examination and dry eye staining is Fluorescein > Rose Bengal > Lissamine green.

Fluorescein staining of the cornea is carried out with a disposable autoclaved filter paper strip impregnated with the dye. The denuded epithelial area seen in corneal abrasions or ulcers appears brilliant green when stained with fluorescein. Under a cobalt blue light, the stained area appears opaque green.

A corneal abrasion highlighted with fluorescein staining is shown below.



Rose Bengal stains dead and devitalized cells including mucous strands on ocular surface. The dye causes a stinging and burning sensation soon after instillation.



Lissamine green stains dead and degenerate cells, but does not stain healthy epithelial cells. It is used to diagnose ocular surface disease. It stains the conjunctiva. It does not cause stinging like Rose Bengal. The Green staining is best observed two minutes after instillation.

The image below shows the lissamine staining of the eye.



Note: The answer to this question is controversial and there is no clearcut evidence based information on the sequence. The above mentioned answer is the best amongst the given options based on consensus.

Solution to Question 3:

Richardson and Koeppe are direct gonioscopes.

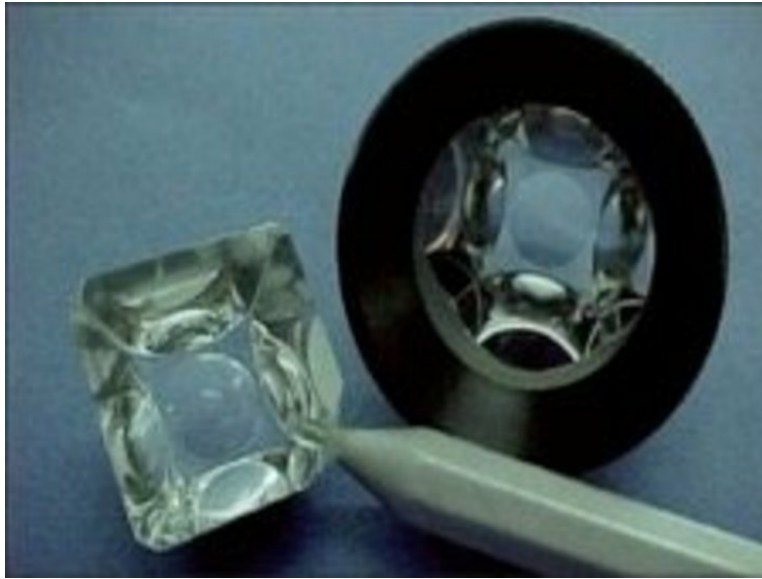
A gonioscope is used for the measurement of the angle of the anterior chamber. The types of gonioscopes include:

- Indirect gonioscopy: Provides a mirror image of the opposite angle. Indirect gonioscopes include Goldmann, Zeiss, and Posner.
- Direct gonioscopy: Provides a direct view of the angle. Direct gonioscopes include the Koeppe, Richardson, Medical Workshop, Barkan, and Swan–Jacob.

Applications:

- To view the angle of the anterior chamber. The structures visualized in order from front to back are:
 - Schwalbe's line
 - Trabecular meshwork
 - Scleral spur
 - Ciliary body band
 - Root of iris
- The localization of foreign bodies, abnormal blood vessels or tumors in the angle.
- Demonstration of the extent of peripheral anterior synechiae.
- During goniotomy, a surgical procedure for the treatment of congenital glaucoma.

The below instrument is an indirect gonioscope of four-mirror type, which is used to visualize the angle of the anterior chamber.



Solution to Question 4:

The image shows a nodule surrounded by redness. On administration of phenylephrine, blanching of superficial episcleral vessels is seen. Hence, the probable diagnosis is nodular episcleritis.

2.5% phenylephrine can be used to differentiate episcleritis and scleritis. Since the vessels of the episclera are superficial, they blanch. Whereas, in scleritis, the vessels do not blanch.

Episcleritis is benign, recurrent, inflammation of the episclera, involving the overlying Tenon's capsule but not the underlying sclera. It is more common in women than in men. It is an allergic reaction to an endogenous toxin. It is commonly associated with rheumatoid arthritis. Episcleritis is characterized by redness, mild ocular discomfort described as a gritty, burning, or foreign body sensation

Clinically, two types of presentations may occur:

- Simple episcleritis is characterized by sectorial or diffuse inflammation of the episclera.
- Nodular episcleritis is characterized by a pink or purple flat nodule surrounded by injection, usually situated 2-3 mm away from the limbus. The nodule is firm, and tender and can be moved separately from the sclera, and the overlying conjunctiva also moves freely.

Treatment includes topical NSAIDs, corticosteroid eye drops, cold compress, and artificial tears. In recurrent cases, systemic NSAIDs are used.

Scleritis refers to an inflammation of the sclera proper. It usually occurs in elderly patients (40-70 years) involving females more than males. It is frequently associated with an underlying systemic inflammatory condition. The patient presents with moderate to severe pain, redness, lacrimation, photophobia, and diminution of vision.

Scleritis can be classified as immune-mediated non-infectious scleritis and infectious scleritis. Immune-mediated non-infectious scleritis is further classified as:

I. Anterior scleritis (98%)

a. Non-necrotizing scleritis (85%)

- Diffuse
- Nodular

b. Necrotizing scleritis(13%)

1. With inflammation:

- Vasa-occlusive
- Granulomatous
- Surgically induced

2. Without inflammation(scleromalacia perforans).

II. Posterior scleritis (2%)

Treatment options include:

- Non-necrotizing scleritis- Topical steroids and systemic indomethacin.
- Necrotizing scleritis- Topical and oral steroids, immunosuppressive agents and surgical treatment like scleral patch graft.
- Infectious scleritis- Antimicrobial therapy, and surgical debridement.

Other options-

Option C: Tenon's cyst or conjunctival inclusion cyst is a benign condition filled with serous fluid and slimy mucous secretions of congenital or acquired origin. Acquired cysts can occur following strabismus surgery, pars plana vitrectomy, scleral buckling, and Ahmed glaucoma valve insertion. The patient presents with swelling near the limbus with no pain or redness. The consistency is cystic to firm, non-tender with no mobility suggesting that the cyst is adherent to the underlying scleral tissue.

Option D: Phlycten is a white-pinkish nodule surrounded by hyperemia on the bulbar conjunctiva near the limbus. Phlyctenular conjunctivitis is a type 4 hypersensitivity reaction due to endogenous allergens like Staphylococcus aureus (most common) and Mycobacterium tuberculosis. Treatment includes topical steroids; atropine if the cornea is involved.

Solution to Question 5:

The marked structure represents the Canal of Petit.

In front of the anterior hyaloid membrane and behind the ciliary body and zonule is the capsulohyaloid sinus (or space), or Petit's canal. This is one of the sites of retrolental hemorrhages. This is also the route along which blood sometimes enters the lens, through some rupture in the posterior lens capsule.

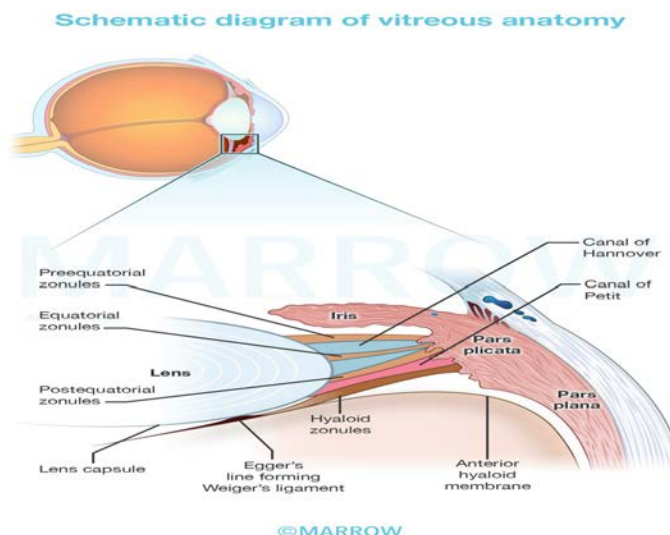
Other options-

Option A: Posterior capsule of lens has circular adhesions with anterior hyaloid surface called as the hyaloideocapsular ligaments of Weiger or retrolental ligament. The circular attachment is

known as Egger's line. The potential space between the anterior hyaloid face and the posterior lense capsule with the circular attachment is referred to as the Berger's space.

Option B: Lens capsule is a thin, transparent, hyaline membrane surrounding the lens which is thicker over the anterior than the posterior surface

Option C: The canal of Hannover lies between orbiculo-anterocapsular and orbiculo-posterocapsular parts of zonular fibers.



Solution to Question 6:

Early signs of glaucoma in perimetry include generalized constriction of isopters and baring of blind spot.

Progression of visual field changes seen in glaucoma:

Perimetry is the procedure of estimating the extent of the visual fields. There are 2 types of perimetry - kinetic and static. In kinetic perimetry, a target of given luminance is moved from a non-seeing area in the periphery towards the center where it is perceived. In the static method, the location of a stimulus remains fixed, with the intensity increased until it is seen by the subject (threshold is reached) or decreased until it is no longer detected.

Charting of the visual fields is useful in the diagnosis of many disease conditions such as glaucoma, neurological disorders, and retinal diseases (e.g. retinitis pigmentosa) as well as following laser treatment of diabetic retinopathy.

Other options-

Option A: Static perimetry (not kinetic perimetry) is used in the quantification and tracking of visual changes in established glaucoma.

Option B: In Kinetic perimetry (not static perimetry), stimulus intensity is held constant and moved centrally with detection.

Option C: Optic disc changes are much greater than visual field changes. Therefore, visual field defects need not correspond to optic nerve head abnormalities to be classified as glaucoma.

Early non-specific changes	Isopter contraction Baring of blind spot: Exclusion of blind spot from the central field
Early significant changes	Peripheral nasal step(may be early or late) Paracentral scotoma Bjerrum's area. Siedel's scotoma: Paracentral scotoma joins the blind spot
Late significant changes	Arcuate or Bjerrum scotoma Ring or Double arcuate scotoma: When 2 arcuate scotomas join together Roenne's central nasal step
Advanced glaucomatous field defects	Tubular vision No light perception: Complete visual field defect.

Solution to Question 7:

Disc neovascularization is not a feature of nonproliferative diabetic retinopathy (NPDR), and is seen in proliferative diabetic retinopathy.

The important findings in severe NPDR are one or more of severe retinal hemorrhages in 4 quadrants, significant venous beading in 2 or more quadrants, and moderate intraretinal microvascular anomalies (microaneurysms) in 1 or more quadrants. This is known as the 4-2-1 rule.

Diabetic retinopathy is used to describe retinal changes seen in diabetic patients. It is classified into the following types:

- Non-proliferative diabetic retinopathy (NPDR).
- Proliferative diabetic retinopathy (PDR).
- Diabetic maculopathy.
- Advanced diabetic eye disease.

Management of diabetic retinopathy includes intravitreal anti-VEGF drugs, intravitreal steroids, laser therapy, and or surgical management.

Surgical management is done in the following cases:

- Tractional diabetic macular edema with NPDR: pars plana vitrectomy (PPV) with the removal of posterior hyaloid.
- Advanced PDR with dense vitreous hemorrhage/ fibrovascular epiretinal membrane: PPV along with the removal of opaque vitreous gel/ membrane and endophotocoagulation.

- Advanced PDR with tractional retinal detachment: PPV with endophotocoagulation and reattachment of the detached retina by scleral buckling.

Early Treatment of Diabetic Retinopathy Study classification:

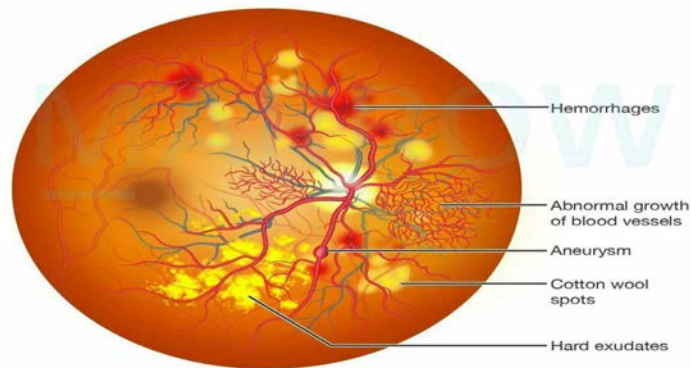
Nonproliferative diabetic retinopathy (NPDR) -

- Very mild NPDR :
 - Microaneurysms only.
 - Management - Review in 12 months.
- Mild NPDR :
 - Any or all of : microaneurysms, retinal hemorrhages, exudates, cotton wool spots, up to the level of moderate NPDR. No intraretinal microvascular anomalies (IRMA) or significant beading.
 - Management - Medical control of diabetes with follow-up in 9-12 months.
- Moderate NPDR :
 - Severe retinal hemorrhages (about 20 medium-large per quadrant) in 1–3 quadrants or mild IRMA.
 - Significant venous beading can be present in no more than 1 quadrant.
 - Cotton wool spots commonly present.
 - Management - Review in approximately 6 months.
- Severe NPDR : The 4–2–1 rule; one or more of:
 - Severe hemorrhages in all 4 quadrants
 - Significant venous beading in 2 or more quadrants
 - Moderate IRMA in 1 or more quadrants
- Management - Review in 4 months.
- Very severe NPDR : The 4–2–1 rule, two or more of the criteria for severe NPDR
- Review in 2–3 months.

Proliferative diabetic retinopathy (PDR)-

- Mild to moderate PDR :
 - New vessels on the disc (NVD) or new vessels elsewhere (NVE), but extent insufficient to meet the high-risk criteria.
 - Management- Prompt Pan-retinal photocoagulation is done if type 2 DM in a patient >40 years of age, otherwise defer PRP.
- High-risk PDR :
 - New vessels on the disc (NVD) (about 1/3 disc area).
 - Any NVD with vitreous hemorrhage.
 - NVE greater than 1/2 disc area with vitreous hemorrhage.
 - Management - Prompt pan-retinal photocoagulation.

Fundal features of Diabetic Retinopathy



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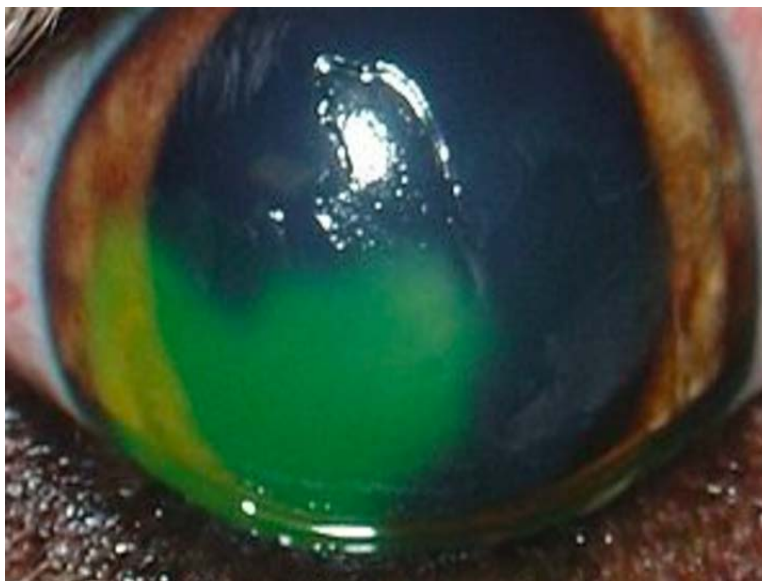
Solution to Question 8:

The image depicts cobalt blue filter, fluorescein dye and corneal ulcer.

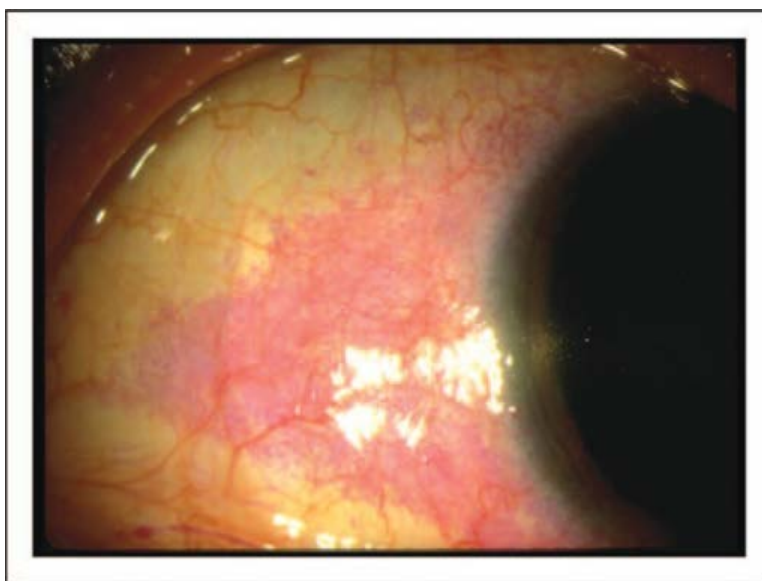
Corneal ulcers can be detected with the help of fluorescein sodium dye and a cobalt blue filter. Fluorescein staining of the cornea is carried out with a disposable autoclaved filter paper strip impregnated with the dye. The denuded epithelial area seen in corneal abrasions or ulcers appears brilliant green when stained with fluorescein. The epithelial defect of the cornea appears opaque green with a cobalt blue filter.

Corneal epithelial lesions are visualised using vital dye staining techniques. Dyes used are:

1. Fluorescein - to delineate areas with epithelial damage. Fluorescein Staining is shown below.



2. Rose Bengal - Stain diseased and devitalised cells. Rose bengal staining is shown below.



3. Alcian Blue - Stains excess mucus selectively.

Other options :

Option C : Lissamine green dye is used to stain devitalized cells of the cornea and conjunctiva. It is used to diagnose ocular surface disorders, like dry eye.

Option D : Interstitial keratitis is a form of non ulcerative keratitis. It is a non suppurative inflammation of the corneal stroma without primary involvement of the epithelium or endothelium. It is characterized by cellular infiltration of the corneal stroma. Some causes of interstitial keratitis include herpes simplex virus, syphilis, cogan syndrome and rheumatoid arthritis.

Slit-lamp examination of the cornea and conjunctiva after rose bengal staining reveals punctate corneal ulcerations and attached filaments of corneal epithelium in keratoconjunctivitis sicca. The image attached below sums up the various patterns of staining and their diagnostic utility.

Patterns of corneal staining and diagnostic clenchers to the aetiology

	Pattern	Example
	Diffuse	• Viral conjunctivitis • Trauma • Toxicity
	Inferior	• Blepharoconjunctivitis • Lagophthalmos • Trichiasis
	Interpalpebral	• Dry eye disease • Exposure • Neurotrophic keratopathy
	Superior	• Superior limbic keratoconjunctivitis • Foreign body under eyelid • Trichiasis
	Superior conjunctivitis	• Superior limbic keratoconjunctivitis
	3 and 9 o'clock	• Contact lens
	Lower conjunctivitis	• Mechanical • Meibomian gland dysfunction

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Solution to Question 9:

RK (Radial Keratotomy) would not be recommended to this boxer who wants vision correction surgery.

Radial keratotomy (RK) involves making deep (90% of corneal thickness) radial incisions in the peripheral part of cornea leaving the central 4 mm optical zone. These incisions on healing flatten the central cornea reducing its refractive power. This procedure was done to correct low to moderate myopia (2 to 6 D). However, because of its disadvantages and advent of safe techniques (LASIK and PRK), RK is now obsolete.

Disadvantages of RK:

- Cornea is weakened, increasing chances of globe rupture following trauma (RK more than PRK). Hence this procedure should not be done for patients who are at high risk of blunt trauma, such as sports persons and military personnel.
- Irregular astigmatism from uneven healing.
- Glare at night.

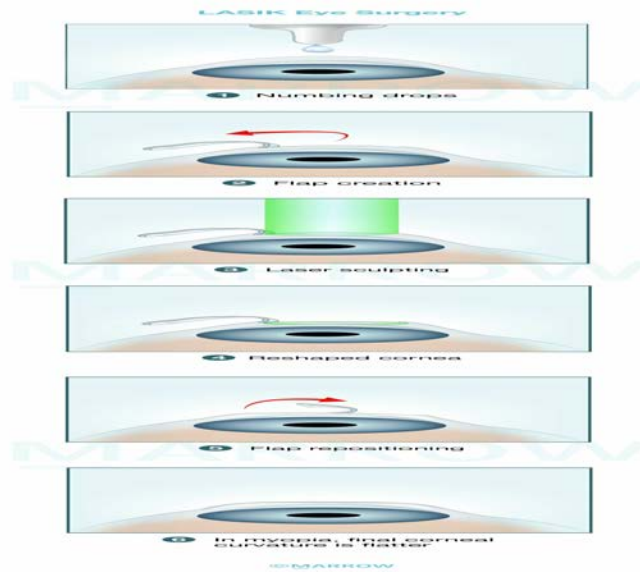
Laser-assisted in situ keratomileusis (LASIK) (Option C) is the most commonly done surgery for myopia. An excimer laser beam reshapes the cornea by ablating the superficial stroma to a predetermined extent after lifting a flap of the cornea with a sharp microkeratome. It is safer than RK.

Patient selection criteria:

- In individuals aged 21 years and above.
- In individuals who have a stable refractive error for at least 1 year.
- In individuals who have had unsatisfactory results with non-surgical treatment.
- In individuals in whom corneal-thinning disorders such as keratoconus have been ruled out.

Complications:

- Tear instability (dry eye)
- Wrinkling, distortion, or dislocation of the flap
- Subepithelial haze with resultant glare, especially at night.
- Persistent epithelial defects.
- Epithelial ingrowth under the flap.
- Diffuse lamellar keratitis
- Corneal ectasia
- Keratoconus



Other options :

Option B : PRK (Photorefractive keratectomy) corrects myopia by using an excimer laser to photoablate a central optical zone of anterior corneal stroma. This causes flattening of the central cornea. Like RK, the PRK also gives very good correction for up to -6 D of myopia but is safer.

Option D : SMILE (Small incision lenticule extraction) is a technique in which a lenticule of corneal stroma is extracted using a femtosecond laser. This can correct myopia, with and without astigmatism up to 10 D.

Solution to Question 10:

The child with a power of -1 D spherical and -2 D cylindrical at 180 degrees has with the rule astigmatism, as the vertical meridian is greater and more curved than the horizontal meridian.

Step 1 : - 1 D spherical would mean a power of -1 on vertical and horizontal axes.

$\begin{array}{c} | \text{ (-1)} \\ | \\ \text{-----} | \text{----- (-1)} \\ | \\ | \end{array}$

Step 2 : - 2 D cylindrical at 180 degrees, would mean a power of -2 D at 90 degrees (perpendicular to 180).

$\begin{array}{c} | \text{ (-1 - 2 = -3)} \\ | \\ \text{-----} | \text{----- (-1)} \\ | \end{array}$

|

Hence, as the vertical meridian (-3D) is greater than horizontal meridian (-1D), this would be with the rule astigmatism.

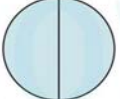
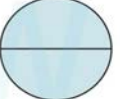
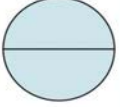
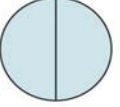
Astigmatism refers to the condition where there is different refractory power in both the principal meridians of the eye. Astigmatism is classified into two broad varieties - regular and irregular astigmatism.

- Regular astigmatism: there are two principal meridians.
- With the rule - principal meridians are vertical and horizontal; both meridians at 90°; and the vertical meridian is more curved than horizontal.
- Against the rule (Option C) - principal meridians are vertical and horizontal; both meridians at 90°; and the horizontal meridian is more curved than vertical.
- Oblique - the principal meridians are not vertical and horizontal, but they are at 90° to each other; e.g., at 45° and 135°.
- Bi-oblique - the two principal meridians are not vertical and horizontal, and they are not at 90° each other; e.g., at 40° and 100°.
- Irregular astigmatism: there are multiple meridians.

Clinical presentation of astigmatism include asthenopia (tiredness of eyes relieved by closing the eyes), blurred vision and defective vision, elongation of objects, keeping reading material closer, and half closure of the lid.

Tilted or oval optic disc is seen on ophthalmoscopy. Retinoscopy shows different power on the two meridians. Other investigations include keratometry, astigmatic fan test, and Jackson's cross-cylinder test.

Clinical types of astigmatism

	Cornea	Typical age	Axis of correcting plus cylinder (± 20°)	Axis of correcting minus cylinder (± 20°)
With the rule astigmatism		Young		
Against the rule astigmatism		old		

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Prescription in astigmatism: There are 3 components.

- 1) Power of the spherical component: Convex (+) or concave (-) dioptr sphere (DS)
- 2) Power of the cylinder: Convex (+) or concave (-) dioptr cylinder (DC)

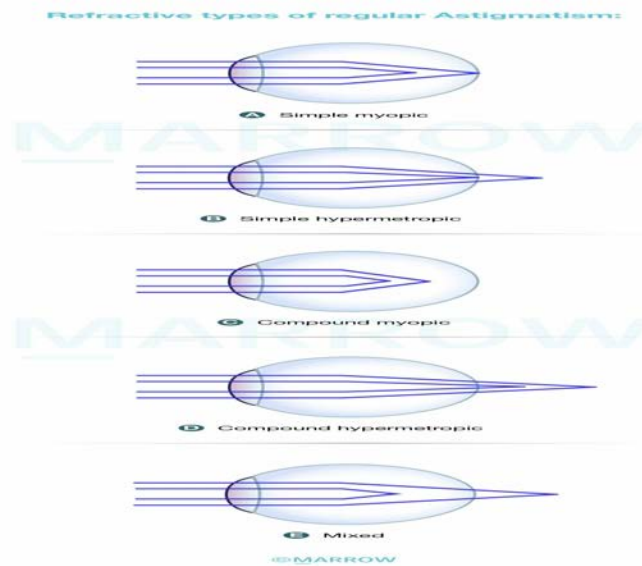
3) Axis (orientation) of the cylinder

Other options :

Option A : In simple astigmatism, one of the meridians focuses on the retina. The other meridian focuses either in front of the retina (simple myopic) or behind the retina (simple hyperopic).

Example, -1.00 DC $\times$ 180 degree.

Option D : Mixed astigmatism: Light rays in one meridian are focused in front of the retina and those in the other meridian are focused behind the retina. Example, -1DS with +2DC at 180 degree.



Solution to Question 11:

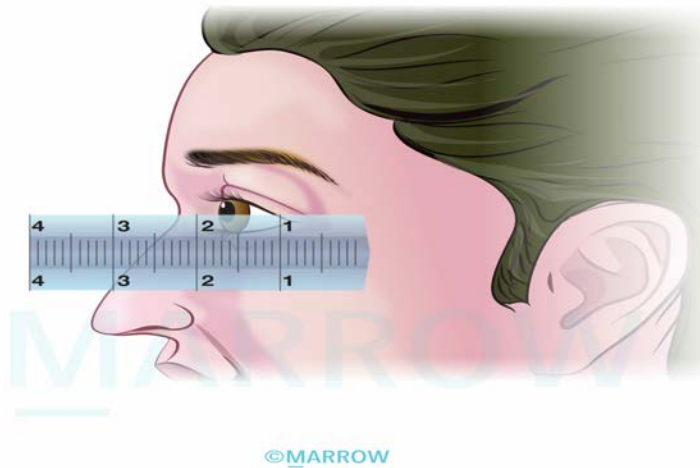
The image shows Hertel's exophthalmometer used to measure proptosis.

An abnormal protrusion of the globe because of a space-occupying lesion in the orbit is called proptosis. Exophthalmometry measures protrusion of the apex of the cornea from the outer orbital margin (with the eyes looking straight ahead). An absolute protrusion of ≥ 20 mm or a difference of ≥ 2 mm between the two eyes is labelled as proptosis.

Instruments used to measure proptosis:

- Hertel's exophthalmometer: the most common instrument used. Its advantage is that it measures both eyes simultaneously.
- Luedde exophthalmometer : The simplest instrument to measure proptosis

Luedde exophthalmometer



- Naugle Exophthalmometer : This is a superior and inferior orbital rim-based instrument.

Pseudoproptosis occurs when the eye has a proptotic anatomic appearance that is not caused by a mass effect from a lesion displacing the globe. Pseudoproptosis is seen in high myopia (due to increase in size of the eyeball), ipsilateral lid retraction (resultant sclera creates the impression of the globe protruding forward) and contralateral enophthalmos (the normal orbit, which is situated more anteriorly, may simulate proptosis when there is none).

Ophthalmology INI-CET 2024

Question 1:

A 3 months old baby was brought with photophobia and lacrimation. Which of the following cannot be the cause?

- a) Congenital entropion
- b) Congenital glaucoma
- c) Congenital distichiasis
- d) Congenital nasolacrimal duct obstruction

Question 2:

A construction site worker comes in with red eyes and lacrimation. He got chuna particles in his eyes an hour ago. What is the initial step in the treatment?

- a) Irrigation with normal saline
- b) Topical antibiotics and cycloplegics
- c) Patch the eye after topical lubricants
- d) Temporary tarsoraphy

Question 3:

What is anisometropia ?

- a) Difference in visual acuity
- b) Difference in refractive power
- c) Difference in axial length
- d) Difference in size of eye ball

Question 4:

The following are features of third cranial nerve palsy except

- a) Elevation
- b) Adduction

- c) Limited abduction
- d) Ptosis

Question 5:

What is the most common cause of proptosis in middle aged individuals?

- a) Cavernous hemangioma
- b) Orbital varix
- c) Thyroid eye disease
- d) Retinoblastoma

Question 6:

A 40 year old with high myopia wearing thick glasses from childhood came with complaints of sudden onset of flashes of light in left eye. What could be the likely cause ?

- a) Retinal detachment
- b) CRAO
- c) CRVO
- d) Subarachnoid haemorrhage

Question 7:

Which of the following disease results in weakness of the levator palpebrae superioris, corneal anaesthesia, lagophthalmos, and ultimately blindness?

- a) Lyme's disease
- b) Leprosy
- c) Ophthalmia neonatorum
- d) Trachoma

Question 8:

All of the following are actions of superior oblique except?

- a) Depression

- b) Adduction
- c) Abduction
- d) Intorsion

Question 9:

A patient presents with diminished night vision and the following finding was seen on fundoscopy. Which of the following statements is not true regarding the condition?



- a) Dystrophy of rods more common than that of cones
- b) ERG shows decreased amplitude of A and B waves
- c) Sometimes patients present in later stages with loss of peripheral vision.
- d) Night blindness is a late feature

Question 10:

A patient of rheumatic heart disease presents with superior visual field defect in left eye since 1 week, what is the probable cause?

- a) Central retinal artery occlusion
- b) Central retinal vein occlusion
- c) Branch retinal vein occlusion
- d) Branch retinal artery occlusion

Question 11:

An 80 yrs old patient presents with a painful, red right eye. On examination, the IOP of his right eye is 50mmHg and left eye is 16mmHg. Slit lamp examination reveals deep anterior chamber, presence of flare cells in the anterior chamber and cataractous lens in the right eye. What is the most likely diagnosis?

- a) Phacolytic glaucoma
- b) Malignant glaucoma
- c) Fuch heterochromic iridocyclitis
- d) Phacomorphic glaucoma

Question 12:

Which of the following is a complication of diabetic retinopathy?

- a) Posterior staphyloma
- b) Retinoschisis
- c) Tractional retinal detachment
- d) Progressive myopia

Question 13:

Which of the following is true about macular opacity?

- a) Iris and pupil are visible
- b) Only the pupillary margin is visible
- c) Iris details are visible
- d) Iris and pupillary margins are not visible

Question 14:

Which of the following is true about viscoelastic substances?

- a) Causes mechanical injury to corneal endothelium
- b) Prevents the Anterior Chamber from collapsing during crucial steps like capsulorhexis and IOL placement
- c) It is inert and can be left behind after surgery

- d) Silicone oil, perfluoropropane, and SF6 are the most commonly used substances

Question 15:

In pterygium, there is

- a) Eye pain
- b) Macular edema
- c) Purulent discharge
- d) Astigmatism

Question 16:

A 6-month-old baby, born preterm at 33 weeks of gestation, is brought by parents with complaints of white pupillary reflex since 2 weeks. CT revealed calcification of the globe in both eyes. What is the probable diagnosis?

- a) ROP
- b) Retinoblastoma
- c) Congenital Glaucoma
- d) Congenital Cataract

Question 17:

Which of the following is a cause of bilateral sudden painless loss of vision?

- a) Acute congestive glaucoma
- b) Diabetic retinopathy
- c) Keratoconus
- d) Iridocyclitis

Question 18:

In FFA, the microaneurysms appear as:

- a) Hypofluorescent dots
- b) Hyperfluorescent dots

- c) Hypofluorescent lines
- d) Hyperfluorescent branching-like structures in the vessel

Question 19:

Normal diameter of artery to vein at the optic disc margin

- a) 1:3
- b) 2:3
- c) 2:5
- d) 1:5

Question 20:

In the case of Best disease, which of the following is the diagnostic investigation?

- a) EOG
- b) ERG
- c) Multifocal ERG
- d) Visual evoked potential

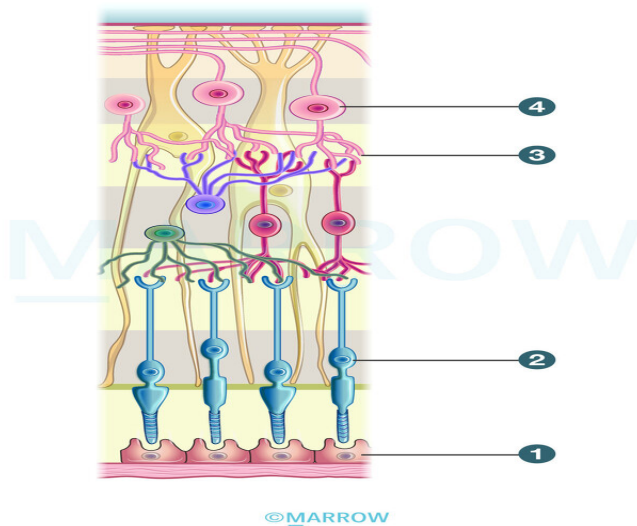
Question 21:

A factory worker presented with a chemical injury in both his eyes a few hours back. What is the first immediate management?

- a) Copious irrigation of eyes
- b) Take history
- c) Check vision in both eyes
- d) Check IOP and plan for surgery

Question 22:

Which of the following layers of the retina are correctly labeled?



- a)** 1- Retinal pigment epithelium; 2- Outer nuclear layer; 3- Inner plexiform layer; 4-Ganglion layer
- b)** 1- Retinal pigment epithelium; 2-Inner nuclear layer; 3- Outer plexiform layer; 4- Ganglion layer
- c)** 1- Ganglion layer; 2- Inner nuclear layer; 3- Outer plexiform layer; 4- Retinal pigment epithelium
- d)** 1- Ganglion layer; 2- Outer plexiform layer; 3- Inner nuclear layer; 4- Retinal pigment epithelium

Answer Key

Question No.	Correct Option
1	d
2	a
3	b
4	c
5	c
6	a
7	b
8	b
9	d
10	d
11	a

12	c
13	b
14	b
15	d
16	b
17	b
18	b
19	b
20	a
21	a
22	a

Detailed Explanations

Solution to Question 1:

Congenital nasolacrimal duct obstruction (NLDO) is least likely to cause watering and photophobia.

Nasolacrimal duct obstruction is the most common disease of the lacrimal system. It usually presents in the first few weeks or months of life as excessive tearing and ocular discharge. It mostly occurs due to incomplete canalization of the duct.

Congenital NLDO mainly results in persistent tearing and sometimes a discharge from the eyes but not photophobia, as the light sensitivity is not directly related to this condition.

Typically, diagnosis is established through a combination of patient history and physical examination. In cases where symptoms are intermittent, the presence of an obstruction can be confirmed using the fluorescein dye disappearance test.

The initial treatment involves performing nasolacrimal massage, typically 2-3 times daily, along with cleansing the eyelids using warm water and applying topical antibiotics. If it does not resolve after massaging by the age of 6 to 10 months, lacrimal duct probing is done.

If probing fails, other treatment options include intubation with silastic tubing with or without balloon dilatation of the nasolacrimal duct and dacryocystorhinostomy.

Other options:

Option A: Congenital entropion: This is a condition where the eyelid is turned inward, causing the eyelashes to rub against the cornea. This can lead to irritation, photophobia, and excessive tearing (lacrimation).

Option B: Primary congenital glaucoma (PCG) is caused by impaired aqueous outflow due to a maldeveloped anterior chamber angle. Clinically, the child presents with lacrimation, photophobia, and blepharospasm.

Option C: Congenital distichiasis: This is a condition where an extra row of eyelashes grows from the meibomian glands on the eyelid margin. These extra lashes can irritate the cornea, leading to photophobia and tearing.

Solution to Question 2:

The initial step in the management of a patient presenting with chemical burns is immediate and thorough irrigation with the available clean water or normal saline delivered through an IV tubing. This is done to prevent further damage.

Along with irrigation, contaminant elimination through mechanical means shall be done. In the event of residue, especially with lime (chuna), it is essential to meticulously eliminate them using a swab stick.

After the initial steps, maintenance of favorable conditions for rapid and uncomplicated healing by following measures:

- Topical antibiotic drops, for example, moxifloxacin 4-6 times a day to prevent infection.
- Steroid eye drops to decrease inflammation, neutrophil infiltration, and treat anterior uveitis. Caution is advised after 2 weeks, as steroids could disrupt collagen synthesis.
- Cycloplegics, such as atropine, might enhance comfort.

Grading of severity (Roper–Hall system) of chemical injuries is based on corneal clarity and severity of limbal ischemia:

- Grade 1 is characterized by a clear cornea (epithelial damage only) and no limbal ischemia (excellent prognosis).
- Grade 2 shows a hazy cornea but with visible iris detail and less than one-third of the limbus being ischemic (good prognosis).
- Grade 3 manifests as a total loss of the corneal epithelium, stromal haze obscuring iris detail, and between one-third and half of the limbus showing ischemia (guarded prognosis).
- Grade 4 manifests with an opaque cornea and more than 50% of the limbus showing ischemia (poor prognosis).

Solution to Question 3:

Anisometropia is the condition where there is difference in refractive powers of two eyes.

Anisometropia occurs when the total refraction of both eyes are unequal. A difference of 1D in the two eyes causes a 2% difference in the size of retinal images formed. A difference up to 2.5D is well-tolerated, but beyond 4D cannot be tolerated. Anisometropia can occur as a combination of emmetropia, myopia, hypermetropia, and astigmatism.

It is diagnosed after retinoscopic examination and autorefractometry.

Treatment includes the usage of spectacles (it can be tolerated up to a maximum difference of 4D), contact lenses, intraocular lenses (for unioocular aphakia), refractive corneal surgery, or refractive

lens exchange methods (used in high degree anisometropia).

Solution to Question 4:

Limited abduction is not a feature of third cranial nerve palsy.

Abduction is carried out by the lateral rectus, which is supplied by the abducens (sixth) nerve.

Third cranial nerve (oculomotor nerve) palsy presents with:

- Profound ptosis due to the weakness of the levator muscle
- Limited adduction due to medial rectus weakness
- Limited elevation due to the weakness of the superior rectus and inferior oblique
- Abduction and depression in the primary position ("down and out")

Signs of Oculomotor Nerve Palsy	
Description	Image
Profound ptosis due to weakness of levator	
Abduction & depression in 1° gaze due to unopposed action of lateral rectus & superior oblique	
Intorsion due to intact superior oblique	
Limited depression due to weak inferior rectus	
Limited elevation due to weak superior rectus & inferior oblique	
Dilated pupil due to parasympathetic palsy	
Normal abduction	

Solution to Question 5:

The most common cause of proptosis in middle age is thyroid eye disease.

Graves ophthalmopathy is one of the most serious ocular manifestations of Graves disease.

Between the ages of 25 and 50, it's the leading cause of unilateral and bilateral proptosis in adults.

Proptosis, eyelid retraction, restricted myopathy, and perhaps compressive optic neuropathy are clinical manifestations that could potentially be used for diagnosis.

The mechanism proposed for this is the acute precipitation of a hypothyroid state which leads to an increase in TSH secretion. This leads to stimulation of the retro-orbital fibroblasts and adipocytes which have the receptor for TSH.

An abnormal protrusion of the globe is called proptosis. Exophthalmometry measures protrusion of the apex of the cornea from the outer orbital margin (with the eyes looking straight ahead). An

absolute protrusion of ≥ 20 mm or a difference of ≥ 2 mm between the two eyes is labelled as proptosis.

Option A: Cavernous hemangioma is the most common primary benign orbital tumor in adults. They are composed of large, dilated vascular channels; are more infiltrative; frequently involve deep structures; and do not spontaneously regress. On histologic examination, the mass is sharply delineated but unencapsulated with large, cavernous blood-filled vascular spaces, separated by connective tissue stroma.

Option B: The type of proptosis seen in orbital varix is non-pulsatile, intermittent, and reducible proptosis. On activities such as forward bending or lowering of the head, turning the head forcefully, hyperextension of the neck, and coughing, the varices fill up, causing proptosis, and then, they regress on shifting to a non-dependent position.

Option D: Retinoblastoma is the most common intraocular malignancy in children. It is associated with mutations in the RB1 gene at chromosome 13q14. White pupillary reflex or leukocoria is the most common presenting feature, followed by a squint which is usually convergent. Histopathology of the tumor shows small round cells with large nuclei. Microscopic features include Flexner-Wintersteiner rosettes (highly specific of retinoblastoma), Homer-Wright rosettes, pseudorosettes, and fleurettes formation. Areas of necrosis and calcification can also be seen.

Solution to Question 6:

The given scenario of patient with severe myopia presenting sudden onset of flashes of light is likely retinal detachment.

Retinal detachment is the separation of the neurosensory retina and the retinal pigment epithelium (RPE).

Risk factors include myopia, previous intraocular surgery such as aphakia or pseudophakia, a family history of retinal detachment, and trauma.

Clinical features include sudden painless loss of vision, flashes of light or photopsia, floaters in the eye, and curtains in the field of vision.

Examination findings will reveal:

- Normal or slightly low intraocular pressure
- Marcus Gunn pupil (relative afferent pupillary defect)
- Altered red reflex on the distant direct ophthalmoscope
- Indirect ophthalmoscopy may show grey reflex of the freshly detached retina or breaks in the retina
- Vitreous pigments on the anterior vitreous (Shaffer sign)
- Shifting fluid sign- Change in the position of the detached area with a change in gravity is the hallmark of exudative retinal detachment.

Management of rhegmatogenous retinal detachment includes sealing of retinal breaks, reducing the vitreous traction on the retina, and flattening of the retina by draining the subretinal fluid.

Scleral buckling, pneumatic retinopexy, and pars plana vitrectomy can help in maintaining the chorioretinal apposition.

Treating the cause helps in exudative retinal detachment. Enucleation can be done if intranuclear tumors are present.

Solution to Question 7:

Leprosy could result in paresis of the levator palpebrae superioris, corneal anaesthesia, lagophthalmos, and ultimately blindness.

Lagophthalmos is the inability to close the eyelids. This can be due to the involvement of the zygomatic branch of the facial nerve (as in leprosy). Other causes of lagophthalmos include extreme proptosis and symblepharon. The decreased corneal sensation seen is due to the involvement of the ophthalmic branch of the trigeminal nerve which innervates the cornea.

Leprosy or Hansen disease is a chronic granulomatous disease caused by infection with *Mycobacterium leprae*. Ocular features in leprosy occur due to bacterial invasion and include corneal anesthesia, lagophthalmos, corneal ulceration, episcleritis, and iris atrophy. Other features seen are:

- Keratitis with thickened, beaded corneal nerves, punctate subepithelial lesions, pannus, and vascularization
- Chronic low-grade anterior uveitis
- Miosis
- Iris atrophy
- Iris pearls
- Retinal pearls

Option B: The most common ocular manifestation in Lyme's disease is conjunctivitis. Other manifestations like keratitis, iridocyclitis, retinal vasculitis, choroiditis, optic neuropathy, and uveitis occurs very rarely.

Option C: Ophthalmia neonatarum also called neonatal conjunctivitis is inflammation of the conjunctiva occurring in an infant <30 days after birth. The clinical features include conjunctival redness, eyelid edema, and ocular discharge (purulent or mucopurulent).

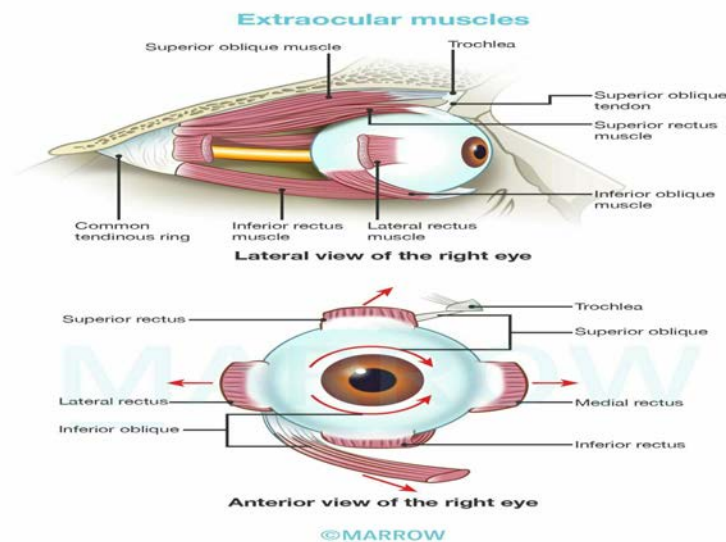
Option D: In trachoma, the symptoms include lacrimation, scanty mucoid discharge and foreign body sensation. Conjunctival congestion and follicles, papillary hyperplasia, Herbert follicles (seen in the limbal area), and progressive pannus are the characteristic features.

Solution to Question 8:

Adduction is not an action of the superior oblique muscle.

The superior oblique muscle's primary, secondary and tertiary actions are intorsion, depression and abduction respectively.

It originates from the apex of the orbit, turns around the trochlea, and inserts into the sclera behind the equator. The oblique direction of the tendon is responsible for its rotatory effect (intorsion), in addition to depression and abduction. It is supplied by the trochlear nerve.



Solution to Question 9:

Diminished night vision along with fundoscopy features point to the diagnosis of retinitis pigmentosa. Option D is not true regarding retinitis pigmentosa. Night blindness is an early feature.

Retinitis pigmentosa is retinal dystrophy that results in degeneration of photoreceptors (rods more than cones). It can be inherited (autosomal dominant, autosomal recessive, or X-linked recessive) or acquired due to sporadic mutation in the rhodopsin gene. Fundoscopy shows bilateral, mid-peripheral intraretinal perivascular 'bone-spicule' changes and atrophy of the retinal pigment epithelium.

Patients present with worsening night blindness initially. However, patients do not often notice night blindness and may present after much disease progression. They also develop tunnel vision, which manifests as difficulty performing activities of daily living such as reading, walking, and driving.

Ocular examination reveals normal visual activity and central vision (though central vision will be involved later in disease progression). Fundoscopy shows bilateral, mid-peripheral intraretinal perivascular 'bone-spicule' changes and atrophy of the retinal pigment epithelium. There is severe peripheral pigmentation, arteriolar narrowing, and pallor of the disc.

Retinitis pigmentosa can lead to posterior subcapsular cataracts, open-angle glaucoma, keratoconus, and posterior vitreous detachment as possible complications.

There is no specific treatment for retinitis pigmentosa.

Option B: Electroretinography (ERG) is the record of changes in the resting potential of the eye induced by a flash of light.

Waves of normal ERG

- a-wave: negative wave possibly arising from the rods and cones
- b-wave: large positive wave which is generated by Muller cells, but represents the activity of the bipolar cells
- c-wave: positive wave representing the metabolic activity of pigment epithelium

In retinitis pigmentosa, it is abnormal and shows reduced scotopic, subnormal responses in early disease and becomes completely extinguished in advanced disease.

Solution to Question 10:

A patient with rheumatic heart disease (RHD) presenting with superior visual defects for the past one week is likely due to branch retinal artery occlusion (BRAO).

Retinal artery occlusion usually occurs due to an embolus. Emboli can originate between the vasculature from the heart to the ophthalmic artery. Risk factors include Buerger's disease, hypertension, elevated lipid levels, cigarette smoking, RHD, and diabetes mellitus.

Retinal emboli can be of 3 types: cholesterol (Hollenhorst plaque), platelet-fibrin, and calcific. The emboli associated with cardiac thrombosis as seen in RHD are platelet-fibrin.

Features of BRAO:

- Sudden painless loss of vision in one eye corresponding to the territory of the obstructed artery. Amaurosis fugax can be seen in patients with preexisting carotid disease.
- Retinal whitening is the most notable finding seen due to ischaemia of the inner layers of the retina. Cotton wool spots can also appear.
- Fluorescein angiography (FA) shows diminution in the dye distal to the site of obstruction. There may be some leaks visible at the site of the embolus.
- OCT (optical coherence tomography) initially shows hyperreflective and thickened inner layers of the retina in the obstructed territory, which start thinning out with time.
- Visual field testing is used to assess the extent of loss of vision.

No effective treatment has been found for BRAO. Ocular massage, lasers, or paracentesis can be used to disrupt the embolus. Vasodilation with inhaled amyl nitrite can be done.

In hypercoagulable states, prophylactic anticoagulation can be started to prevent future episodes.

Other options:

Option A: Central retinal artery occlusion occurs due to obstruction at the level of lamina cribrosa. Patient complains of sudden painless loss of vision occurring over seconds.

Option B: Patients with CRVO usually complain of the acute onset of painless blurred vision in one eye. CRVO is seldom asymptomatic.

Option C: Patients with BRVO typically present with peripheral visual field defects in the defective eye. However, in settings of RHD, arterial occlusion is more likely.

Solution to Question 11:

The given clinical scenario of an 80 year old presenting with painful red eyes, increased IOP and presence of a deep anterior chamber points towards the diagnosis of secondary open-angle glaucoma - phacolytic glaucoma.

Phacolytic glaucoma is secondary open-angle glaucoma occurring in association with hyper mature morgagnian cataracts.

Trabecular obstruction is caused by high molecular-weight lens proteins that leak through the intact capsule in cases of the hyper mature cataract into the aqueous humour; macrophages containing lens proteins may also contribute to this.

Presentation:

- Eye pain
- Diminished vision
- Photophobia
- Lacrimation
- Headache
- Increased intraocular pressure
- Deep anterior chamber with fine white protein particles which may settle as pseudohypopyon
- Open-angle seen on gonioscopy
- Lens: Milky white cortex with sunken brownish nucleus (Morgagnian cataract)

Treatment: Medical therapy to lower the intraocular pressure, Surgery to extract the lens

Other options:

Option B: Malignant glaucoma is seen after intraocular surgery such as trabeculectomy. The forward flow of aqueous is blocked as the tips of ciliary processes rotate forward and press against the lens/IOL/anterior hyaloid of the vitreous. Hence, aqueous humor collects in the vitreous cavity. There is persistent shallow or flat anterior chamber

Option C: In Fuchs heterochromic iridocyclitis, the IOP is not usually so high.

Option D: Phacomorphic glaucoma is an acute secondary angle-closure glaucoma caused by an intumescent cataractous lens due to rapid maturation of cataract or following traumatic capsular rupture. The swollen lens obliterates the angle by forcing the root of the iris against the cornea. The patient presents with features of acute glaucoma with a shallow anterior chamber and a swollen cataractous lens.

Solution to Question 12:

Tractional retinal detachment is a complication of diabetic retinopathy.

Diabetic retinopathy is a complication of diabetes that affects the retina due to prolonged high blood sugar levels. Over time, it can lead to several complications, including:

- **Diabetic Macular Edema (DME):** Diabetic retinopathy weakens the walls of the blood vessels in your retina, causing the leak of fluid and blood into the macula.
- **Vitreous Hemorrhage:** New blood vessels can bleed into the vitreous. This bleeding prevents light from reaching the retina. If the bleeding is minor, may present with floaters. However, in more severe cases, the blood can fill the vitreous cavity, completely obstructing your vision.
- **Neovascular glaucoma:** Abnormal blood vessels can grow out of the iris and can block fluid from draining properly, leading to a type of glaucoma.
- **Tractional retinal detachment** occurs due to the formation of fibrovascular tissue from proliferative diabetic retinopathy. The fibrous tissue exerts traction on the retina, pulling it away from the underlying layers and causing a retinal detachment. The most common cause of tractional retinal detachment is proliferative diabetic retinopathy.

Other options

Option A: Posterior staphyloma is associated with progressive myopia.

Option B: Retinoschisis refers to the splitting of the retinal layers and is seen in degenerative retinal diseases.

Option D: Progressive myopia is unrelated to diabetic retinopathy and occurs due to elongation of the eyeball, genetic factors, environmental factors and too much near work such as reading and using digital devices for long hours.

Solution to Question 13:

Among the given options, 'only the pupillary margin is visible' is the true statement regarding macular opacity. Macular opacity is a type of corneal opacity.

Corneal macular opacity refers to a moderate to dense opacity in the cornea. In such cases, light transmission through the cornea is partially obstructed. This limits the visibility of deeper structures like the iris and its fine details(options A & C). However, because the pupillary margin contrasts with its surrounding structures and lies directly behind the corneal opacity and the obstruction selectively allows the pupillary margin to be visible(options B & D).

Solution to Question 14:

Viscoelastic substances prevent the anterior chamber from collapsing during crucial steps like capsulorhexis and IOL (Intra-ocular lens) placement.

Viscoelastic substances are commonly used in ocular surgery, particularly in cataract surgery, to maintain the anterior chamber (AC), protect the corneal endothelium, and facilitate surgical procedures. They prevent the collapse of the AC during crucial steps such as capsulorhexis and IOL (intraocular lens) placement by providing a cushioning effect.

Other options

Option A: Viscoelastic substances are typically used to protect the corneal endothelium, not to cause any mechanical injury.

Option C: While some viscoelastics are inert, not all can be safely left behind. Some types of viscoelastic substances (especially those that are not fully biodegradable) should be removed from the eye after surgery to prevent complications.

Option D: Sodium hyaluronate and hydroxypropyl methylcellulose are the most commonly used viscoelastic substances, compared to silicone oil, perfluoropropane, and SF6. Silicone oil, perfluoropropane, and SF6 are the substances used for vitreoretinal surgery (such as tamponade agents for retinal detachment), not for routine cataract surgery. Viscoelastic used in cataract surgery are typically substances like sodium hyaluronate and hydroxypropyl methylcellulose.

Solution to Question 15:

In pterygium, there is astigmatism. It causes astigmatism by flattening the cornea along the axis of the pterygium.

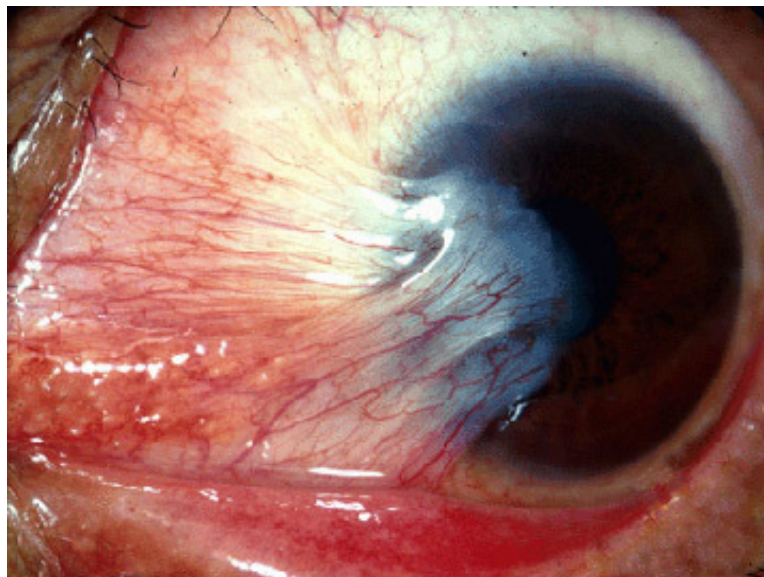
Pterygium is a subconjunctival fibrovascular tissue encroaching on the cornea. It is a degenerative condition of the sub-conjunctival tissue and is never inflammatory. It occurs due to exposure to UV-B radiation and the most common site is the nasal.

It is made up of three parts - a "cap" (an avascular halo-like zone at the advancing edge), a head, and a body.

Clinical features:

- Astigmatism with the rule - most common finding
- Stocker's line - deposition of iron on the advancing edge
- Fuchs islets - small, discrete, whitish flecks consisting of clusters of pterygial epithelial cells, often present at the advancing edge

The image below shows a pterygium.

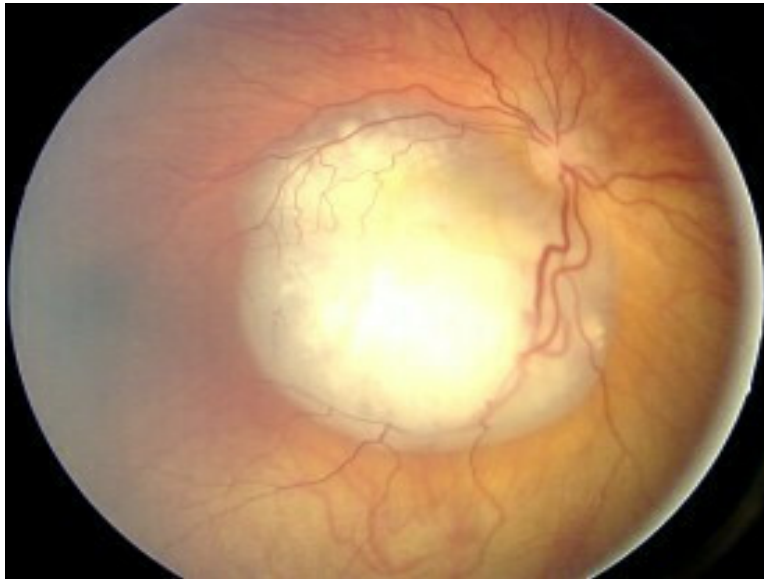


Solution to Question 16:

The given clinical presentations of white pupillary reflex (leukocoria) and CT revealing calcification of the globe in both eyes is suggestive of retinoblastoma.

Retinoblastoma is a proliferation of neural cells that have failed to evolve normally. It arises from the neurosensory retina. It is a malignant tumour of the retina that typically presents in children, often before the age of 5. Leukocoria and squint are the most common presentation of retinoblastoma. It may also present as painful red eyes, cataracts, buphthalmos, proptosis(not microphthalmos) and pseudohypopyon.

In this case, the presence of calcification within the globe on a CT scan supports the diagnosis of retinoblastoma, as it is known to frequently show calcifications on imaging.



Other options

Option A: ROP (Retinopathy of Prematurity) is a concern in preterm infants (especially those born before 32 weeks), and it can also present with white pupillary reflex, it does not show calcification of the globe on a CT scan. ROP can lead to retinal detachment or fibrosis, but calcification is not a typical feature.

Option C: Congenital glaucoma may present with symptoms such as enlarged corneas, increased intraocular pressure, and a shallow anterior chamber. White pupillary reflex or calcification on CT are not typical presentations of congenital glaucoma.

Option D: Congenital cataracts can present with leukocoria (white pupillary reflex) but would not show calcification on a CT scan. They are more likely to be detected by a clinical examination and would typically not result in calcifications.

Solution to Question 17:

Diabetic retinopathy is a cause of bilateral sudden painless loss of vision due to vascular complications.

Bilateral sudden painless loss of vision is a hallmark of advanced diabetic retinopathy, particularly when associated with complications such as:

- Vitreous haemorrhage: Leakage of blood into the vitreous cavity due to rupture of fragile neovascular vessels.
- Tractional retinal detachment: Occurs when fibrovascular tissue pulls the retina away from its underlying structures.
- Macular oedema: Severe swelling in the macula can lead to central vision loss.

These conditions can cause sudden and significant visual impairment, often affecting both eyes in patients with long-standing, poorly controlled diabetes.

Other options

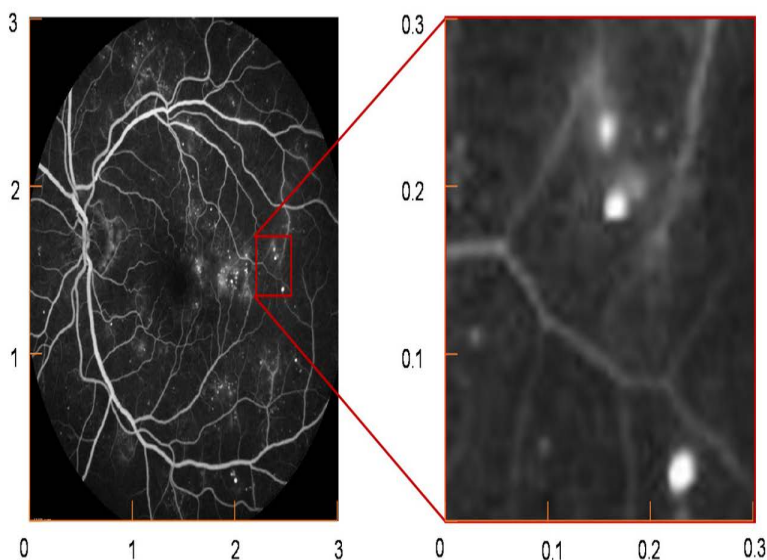
Option A: Acute congestive glaucoma: Typically causes sudden vision loss, but it is painful and unilateral in most cases.

Option C: Keratoconus is a progressive condition that leads to gradual vision loss due to corneal thinning and irregular astigmatism, not sudden or bilateral loss.

Option D: Iridocyclitis is associated with pain, redness, and photophobia, and it is usually unilateral.

Solution to Question 18:

In Fundus Fluorescein Angiography (FFA), microaneurysms (a hallmark of diabetic retinopathy) appear as hyperfluorescent dots due to the accumulation of fluorescein dye in the localized dilations of retinal capillaries.



Other options:

Option A: Hypofluorescent dots indicate blocked fluorescence, often due to hemorrhage or ischemia.

Option C: Hypofluorescent line might occur in cases of vascular occlusion.

Option D: Hyperfluorescent branching-like structure in the vessel suggests leakage from neovascularization or vascular abnormalities.

Solution to Question 19:

The normal diameter ratio of the central retinal artery to the central retinal vein at the optic disc margin is 2:3.

The ratio 2:3 means that the artery's diameter is approximately two-thirds of the vein's diameter. This is considered the physiological normal. The ratio is assessed during the fundoscopic examination, and significant deviations can indicate pathological conditions:

- Decreased ratio (arteriolar narrowing) may suggest hypertension or arteriosclerosis.
- Increased ratio (relative venous narrowing) is less common but may occur in rare conditions.

Solution to Question 20:

EOG (electrooculogram) is the diagnostic investigation in the case of Best disease. EOG measures the function of the retinal pigment epithelium (RPE), which is typically abnormal in Best disease, even in asymptomatic carriers. The Arden ratio is significantly reduced (<1.8) in affected individuals.

Best's disease/Vitelliform macular dystrophy is an inherited macular dystrophy in which lipofuscin accumulates in the central macula causing progressive central vision loss. It is autosomal-dominant and linked to the mutation in the bestrophin (VMD2) gene.

It is characterized by bilateral yellow, yolk-like (vitelliform) macular lesions.

The stages of evolution of the macular lesions are described as progressing from:

- Pre-vitelliform stage
- Vitelliform stage- scrambled egg stage with or without hypopyon
- Vitelliruptive- the lesion breaks and visual acuity drops
- The atrophic stage

Patients with Best's disease usually have good visual acuity when the "yolk" remains intact.

Diagnosis: This is the only disease with relatively normal ERG (electroretinogram) results associated with an abnormal EOG. Optical coherence tomography (OCT) reveals that the vitelliform material appears as a dome-shaped, hyperreflective, and homogenous lesion, located below the hyperreflective photoreceptor layer.

The image below shows Best's disease - the typical vitelliform lesion



Other options

Option B: ERG (electroretinogram) assesses retinal activity and is mainly used in diagnosing pigmentary retinal dystrophy, complete occlusion of the retinal artery, early glaucomatous damage, macular degeneration in cataract, complete old retinal detachment or advanced siderosis.

Option C: Multifocal ERG assesses retinal activity and mainly provides a detailed evaluation of the central retina as in Stargardt disease or chloroquine toxicity.

Option D: Visual evoked potential (VEP) is used to evaluate the visual pathway's function

Solution to Question 21:

Copious irrigation of the eyes is the first immediate management in this case.

The initial step in the management of a patient presenting with chemical burns is immediate and thorough irrigation with the available clean water or normal saline delivered through an IV tubing. This is done to prevent further damage.

Along with irrigation, contaminant elimination through mechanical means shall be done. In the event of residue, especially with lime (chuna), it is essential to meticulously eliminate them using a swab stick.

After the initial steps, maintenance of favourable conditions for rapid and uncomplicated healing by following measures:

- Topical antibiotic drops, for example, moxifloxacin 4-6 times a day to prevent infection.
- Steroid eye drops to decrease inflammation, and neutrophil infiltration, and treat anterior uveitis. Caution is advised after 2 weeks, as steroids could disrupt collagen synthesis.
- Cycloplegics, such as atropine, might enhance comfort.

Grading of severity (Roper–Hall system) of chemical injuries is based on corneal clarity and severity of limbal ischemia:

- Grade 1 is characterized by a clear cornea (epithelial damage only) and no limbal ischemia (excellent prognosis).
- Grade 2 shows a hazy cornea but with visible iris detail and less than one-third of the limbus being ischemic (good prognosis).
- Grade 3 manifests as a total loss of the corneal epithelium, stromal haze obscuring iris detail, and between one-third and half of the limbus showing ischemia (guarded prognosis).
- Grade 4 manifests with an opaque cornea and more than 50% of the limbus showing ischemia (poor prognosis).

Solution to Question 22:

The layers of the retina marked above are:

- 1- Retinal pigment epithelium
- 2- Outer nuclear layer
- 3- Inner plexiform layer
- 4- Ganglion layer

Ophthalmology NEET 2024

Question 1:

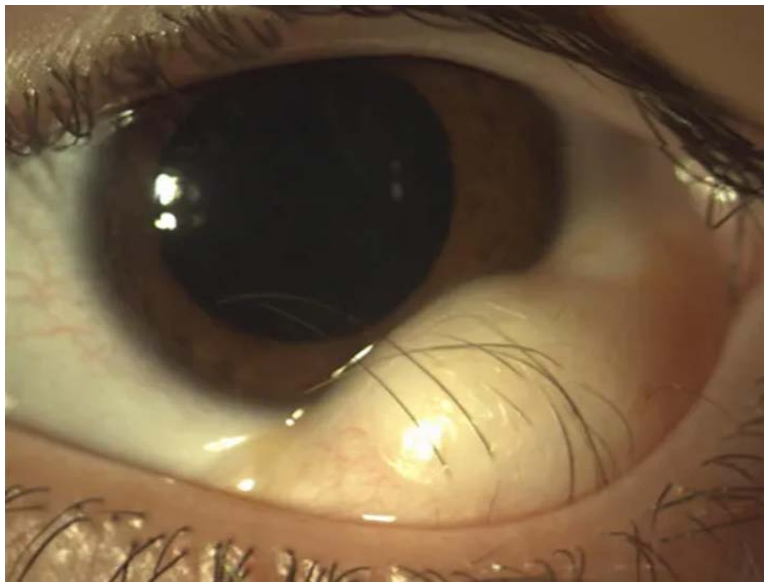
Which of the following muscle is responsible for ptosis in horner's syndrome?



- a) Levator palpebrae superioris muscle
- b) Orbicularis oculi
- c) Horner's muscle
- d) Muller's muscle

Question 2:

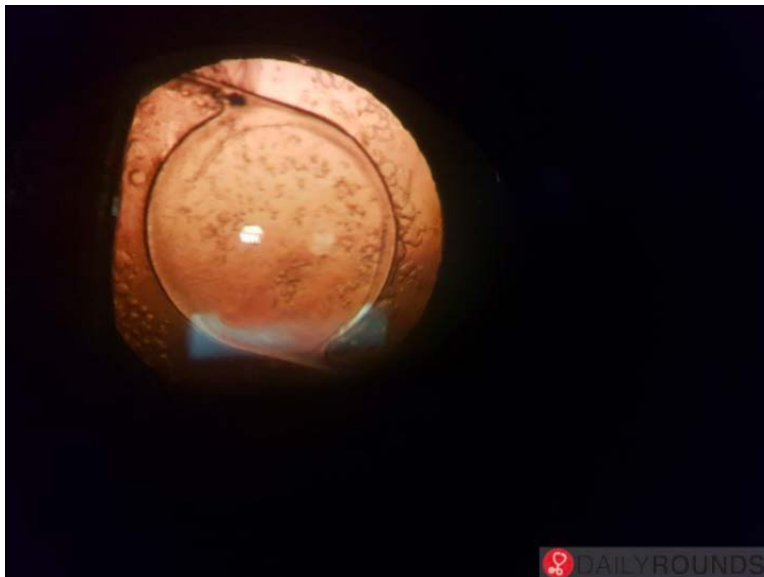
An 11-year-old presented with the following growth in the eye as shown below. What is the diagnosis?



- a) Papilloma
- b) Myxoma
- c) Pyogenic granuloma
- d) Limbal dermoid

Question 3:

Patient undergoes lens implantation following which there's worsening of vision after 1 year. Slit lamp examination finding is shown in the image below. Identify the complication.

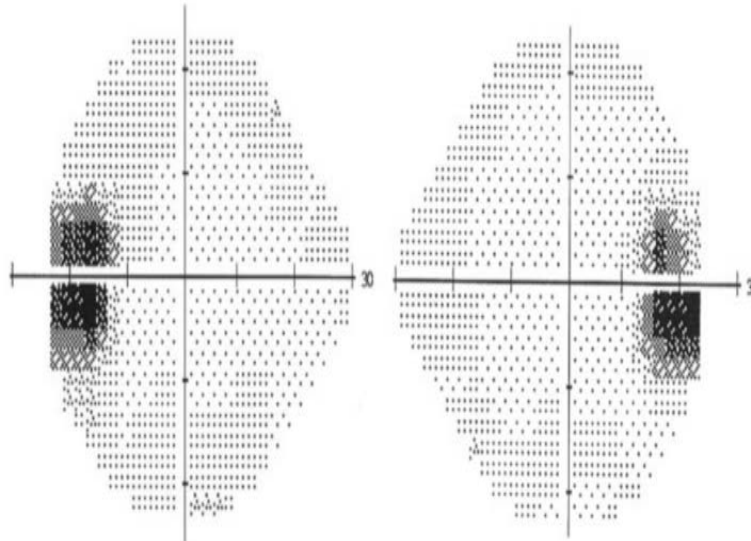


- a) Posterior subcapsular cataract

- b) Bullous keratopathy
- c) Subluxation of lens
- d) Corneal opacity

Question 4:

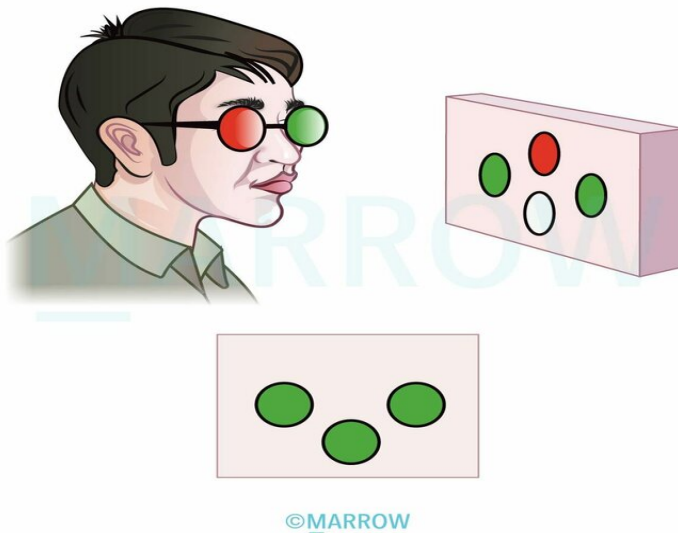
Identify the perimetry finding.



- a) Arcuate scotoma
- b) Blind spot enlargement
- c) Altitudinal defect
- d) Roenne nasal step

Question 5:

A patient undergoes worth's four-dot test . The results are as shown in the image below. Identify the condition.



- a) Crossed diplopia
- b) Left suppression
- c) Right suppression
- d) Uncrossed diplopia

Question 6:

A 20-year-old presented with gradual loss of vision at night and a funduscopy image is shown. What is the diagnosis?



- a) Retinitis pigmentosa

- b) Retinitis punctata albicans
- c) Retinal hemorrhage
- d) Fundus flavimaculatus

Question 7:

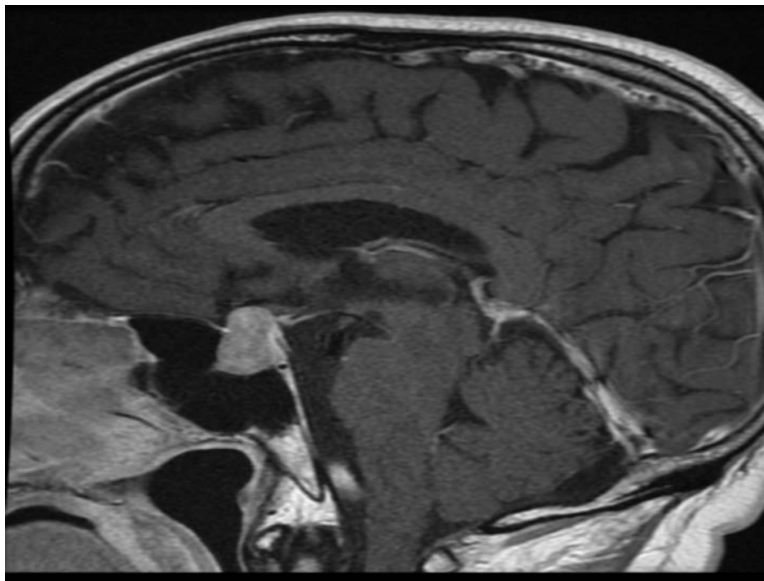
A 2-month-old girl child was brought to the hospital by her parents with the following presentation. What is the appropriate management?



- a) Immediate surgery
- b) Wait until 4 months of age
- c) Wait until 2 years of age
- d) Trial of medical therapy

Question 8:

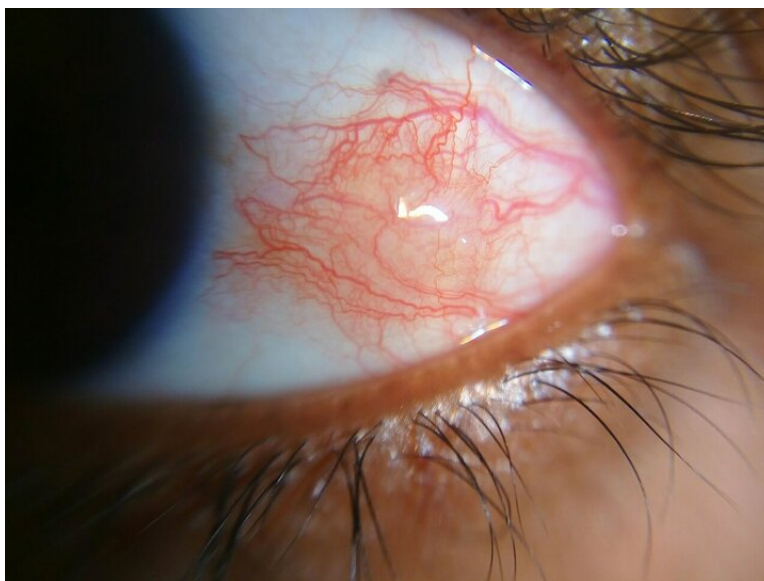
The following MRI image shows a malignancy. What is the visual field defect observed in this case?



- a) Complete blindness
- b) Homonymous hemianopia
- c) Binasal hemianopia
- d) Bitemporal hemianopia

Question 9:

A 26-year-old teacher presents with a red nodule over her sclera, which resolves with medications but is recurrent. Identify the condition and what else will you look for in this patient?

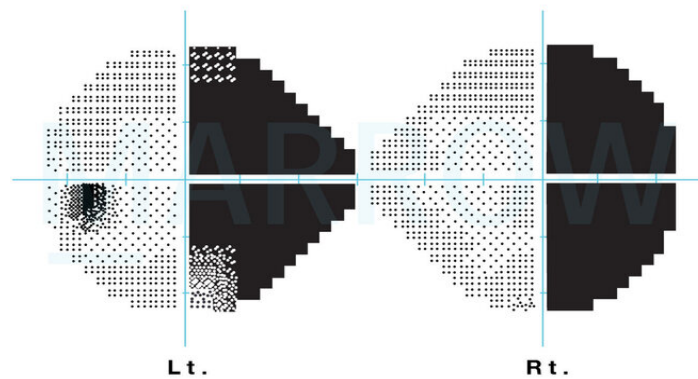


- a) Pingecula; Dry eye

- b) Episcleritis; Rheumatoid arthritis
- c) Pterygium; Astigmatism
- d) Scleritis; Rheumatoid arthritis

Question 10:

Identify the visual field defect in the given image:

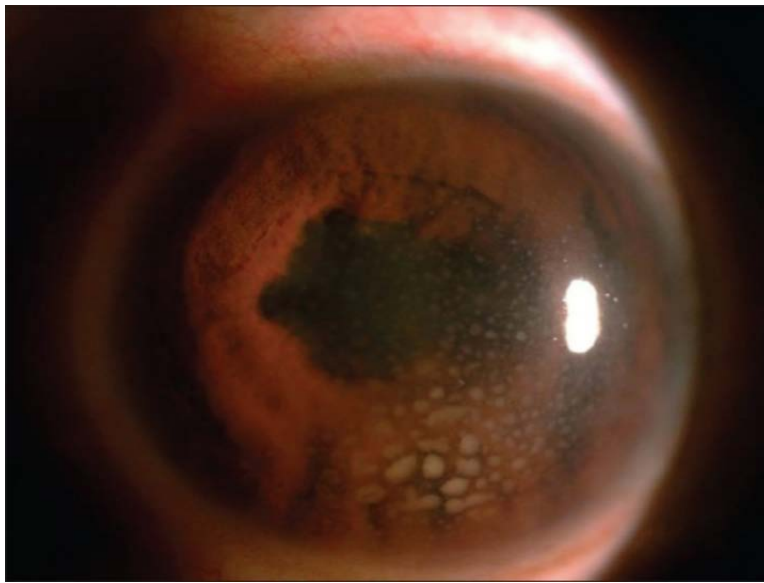


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- a) Altitudinal hemianopia
- b) Complete binasal hemianopia
- c) Bitemporal hemianopia
- d) Homonymous hemianopia

Question 11:

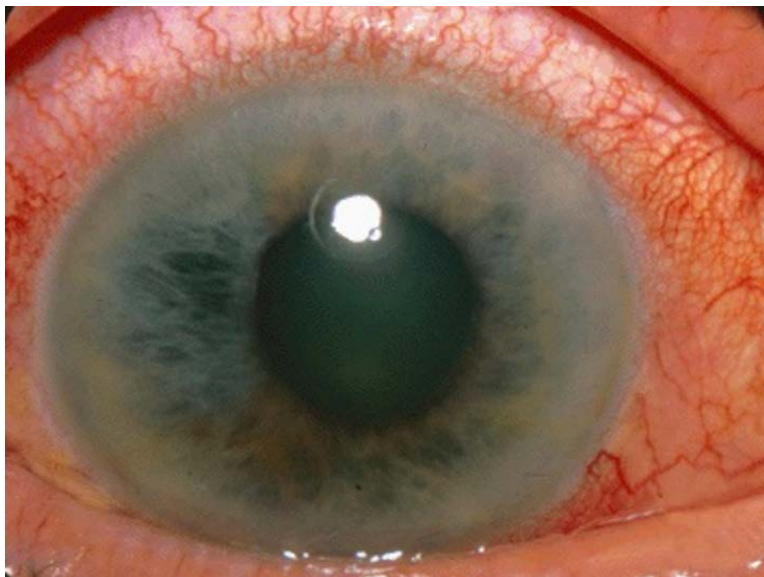
A patient complained of acute painful diminution of vision in the left eye for the past 2 days. On ophthalmic examination, circumcorneal congestion and irregularly shaped pupil were noted as shown below, with the left eye visual acuity at 6/60 and the right eye at 6/12. What is the probable diagnosis?



- a) Angle closure glaucoma
- b) Acute anterior uveitis
- c) Acute conjunctivitis
- d) Episcleritis

Question 12:

A patient who was watching a movie suddenly developed acute pain in the eye and diminution in vision. He was brought to the emergency department and on examination, a shallow anterior chamber was noted. What would be the appropriate management for this condition?



- a) IV Mannitol + atropine e/d
- b) IV Mannitol + pilocarpine e/d
- c) IV Mannitol+ methyl cellulose e/d
- d) IV Mannitol+ moxifloxacin e/d

Question 13:

Identify the defect shown in the image:



- a) Herbert pits
- b) Horner tranta spots
- c) Gelatinous papillae
- d) Pannus

Question 14:

A patient wearing contact lenses develops pain in the orbital region. He has had a very casual attitude towards cleaning his contact lenses. He presented with corneal ulcer and keratitis. Scrapings showed an organism with a spine-like pseudopodia. Which organism is the likely cause of this patient's symptoms?

- a) Balamuthia
- b) Acanthamoeba
- c) Naegleria fowleri

d) Staphylococcus aureus

Answer Key

Question No.	Correct Option
1	d
2	d
3	a
4	b
5	c
6	a
7	a
8	d
9	b
10	d
11	b
12	b
13	a
14	b

Detailed Explanations

Solution to Question 1:

Ptosis in Horner's syndrome is due to the paralysis of Muller's muscle. Muller's muscle(superior tarsal muscle) maintains the elevation of the upper eyelid. It is innervated by the sympathetic nervous system.

Fasanella-Servat operation is indicated in cases having mild ptosis(< 2 mm) and good levator function. In this surgery, the upper tarsal border along with its attached Muller's muscle and conjunctiva are resected.

Solution to Question 2:

The image shows the presence of a limbal dermoid.

Limbal Dermoid is a choristoma (a mass of histologically normal tissue in an abnormal location). It consists of collagenous tissue containing dermal elements and covered by stratified squamous epithelium. Limbal dermoid present in early childhood as a subconjunctival mass in the infratemporal limbus. Treatment is indicated for chronic irritation, amblyopia, vision impairment and corneal dellen formation. Small dermoids can be excised, larger lesions may need lamellar keratosclerectomy.

Solution to Question 3:

The given clinical vignette supported with the image of vacuolated clusters seen behind IOL is suggestive of Elschnig's pearls which is a type of posterior subcapsular cataract (after cataract).

Posterior subcapsular cataracts occur mainly due to ageing, other causes include the use of topical steroids, trauma, excessive consumption of alcohol, and exposure to ionizing radiation. Visual impairment is more during the day or exposure to bright light as the pupil constricts which blocks light rays.

After cataract also known as secondary cataract or posterior capsular opacification is a late postoperative complication that follows extracapsular extraction of the lens.

Different types of after cataract include:

- Elschnig's pearls: where the subcapsular epithelial cells proliferate and develop into vacuolated soap bubble like clusters along the posterior capsule.
- Soemmering's ring: A peripheral dense ring is formed behind the iris and is enclosed between layers of the capsule.
- Dense membranous after cataract: formed in the pupillary area impairing the vision.
- Thin membranous after cataract : is thickened posterior capsule.

Types of after-cataract



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Treatment of after cataract is by

- Nd-YAG laser capsulotomy of the posterior capsule which works on the principle of photodisruption
- surgical posterior capsulotomy.

Solution to Question 4:

The perimetry image in the question above shows blind spot enlargement.

Blind spot enlargement is a rare finding in the visual field examination. The most common cause of blind spot enlargement is Idiopathic intracranial hypertension. Blind spot enlargement occurs due to displacement of the peripapillary retina and refractive changes due to significant optic nerve head swelling. Other common causes of blind spot enlargement are Syphilis, West Nile virus, or Rubella vaccination

Progression of visual field changes seen in glaucoma :

Early non-specific changes	Isopter contraction Baring of a blind spot: Exclusion of blind spot from the central field
Early significant changes	Peripheral nasal step(may be early or late) Paracentral scotoma in Bjerrum's area. Siedel's scotoma: Paracentral scotoma joins the blind spot
Late significant changes	Arcuate or Bjerrum scotoma Ring or Double arcuate scotoma: When 2 arcuate scotomas join together Roenne's central nasal step
Advanced glaucomatous field defects	Tubular vision No light perception: Complete visual field defect.

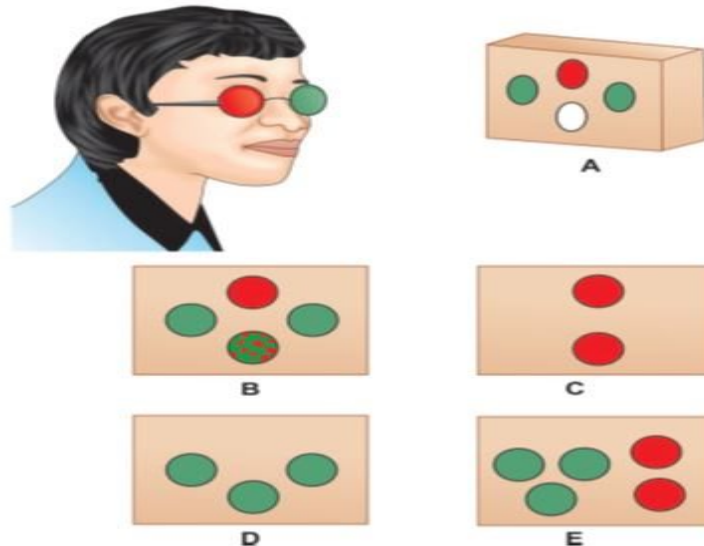
Solution to Question 5:

The given image with 3 green dots likely suggests that the patient is suffering from right suppression.

Worth's four-dot test

- Sensory function test
- Patient wears goggles with
- Red lens: Right eye
- Green lens: Left eye
- A box is viewed having 4 lights: one red, two green, and one white

- Inference:
- A: Normal—All 4 lights seen
- B: Abnormal retinal correspondence—All 4 lights seen in the presence of manifest squint
- C: Left Suppression—2 red lights seen
- D: Right suppression—3 green lights seen
- E: Diplopia—5 lights seen (2 red, 3 green)



Solution to Question 6:

The fundus shows a pale waxy disc, attenuated arteries, and bony spicules along with gradual loss of vision which implies that it is case of retinitis pigmentosa (RP).

Retinitis pigmentosa (RP) is a dystrophy of rods and cones, with rods being primarily affected. It is a genetic disease. The non-hereditary/sporadic type is most common. Among the hereditary type, it can occur in an autosomal dominant (AD), autosomal recessive (AR), or X-linked recessive (XLR) pattern. AD disease generally has the best prognosis.

The presenting feature is night blindness.

Investigations:

- Perimetry- the initial stage will have ring scotoma, which further develops into tunnel vision.
- Electroretinography- reveals a decreased amplitude of both "a" and "b" waves.

Atypical types of RP are as follows:

- Sectoral RP
- Pericentric/Inverse RP
- RP sine pigmento
- RP punctata albescens

There is no effective treatment for RP.

Solution to Question 7:

For the 2-month-old girl presenting with leukocoria in the left eye, likely indicative of a congenital infantile cataract, immediate surgery is the most appropriate management.

Congenital infantile cataract refers to lens opacities present at birth, which can obstruct vision and potentially affect the child's overall development. If left untreated, congenital cataracts, especially unilateral cataracts, can lead to amblyopia, a condition where one eye fails to develop normal vision due to insufficient visual input. Immediate surgical intervention is crucial not only to remove the cataract and restore visual clarity but also to minimize the risk of amblyopia, ensuring the best possible outcomes for the child's visual and developmental health. Prompt diagnosis and treatment can significantly enhance the chances of preserving and improving the child's vision.

Leukocoria, or the white pupillary reflex or cat's eye reflex, is a critical sign that often indicates serious underlying eye disorders requiring prompt evaluation and management. Common causes of leukocoria include:

- Retinoblastoma
- Congenital cataract
- Coloboma of the choroid
- Persistent hyperplastic primary vitreous
- Toxocara infestation
- Coat's exudative retinopathy.

Note: Retinoblastoma, a malignant tumour of the retina, often affects children under 6 years of age and can present with symptoms such as leukocoria, strabismus (crossed eyes), or vision problems. Treatment options vary widely depending on factors like the tumour's size, location, and whether it is unilateral or bilateral.

Solution to Question 8:

The given MRI shows a tumour protruding out of sella turcica. This is likely to be a pituitary tumour (pituitary macroadenoma) compressing the optic chiasma, resulting in bitemporal hemianopia.

Pituitary macroadenoma can compress the optic chiasm, causing bitemporal hemianopia, which leads to the loss of vision in the temporal halves of the visual fields in both eyes. This occurs when damage to the middle part of the optic chiasm affects the crossing fibres from the nasal retina of each eye while the uncrossed fibres from the temporal retinas remain intact. As a result, individuals retain central vision but lose peripheral vision on the sides. The most common cause of bitemporal hemianopia is pituitary adenoma. Other common causes include suprasellar aneurysms, chronic arachnoiditis, and suprasellar tumours like craniopharyngioma and

suprasellar meningioma.

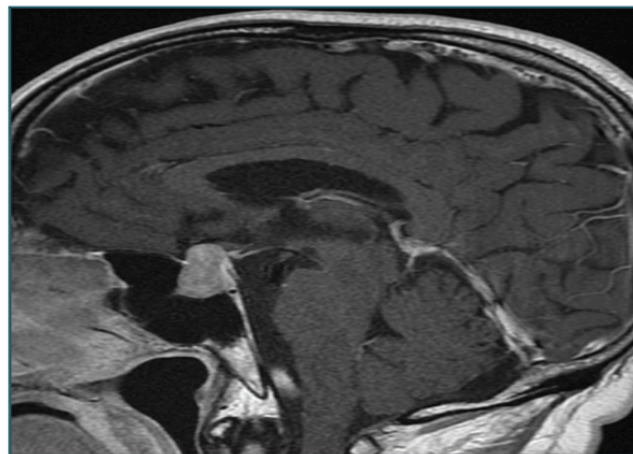
Individuals with this condition can see in both left and right central visual fields, with both eyes open. However, all information from the most peripheral parts of visual fields (which are seen only by the nasal retinas) is lost.

A pituitary adenoma is the most common tumour of the sellar region. Depending on their size they are broadly classified into:

- Pituitary microadenoma: $\leq$ 1 cm in size
- Pituitary macroadenoma: $>$ 1 cm in size and usually extending superiorly.

Clinical features include bi-temporal hemianopia, headache, and vomiting. Skull X-ray finding is ballooning of the sellae. MRI is the investigation of choice. Indentation at the diaphragma sellae gives rise to a snowman/figure of eight appearance, which is seen in the pituitary adenoma as shown below.

Snowman appearance – Pituitary adenoma



Other options:

Option A: Complete blindness refers to a total lack of vision, which can result from various causes, including severe damage to the eyes themselves (like retinal detachment, severe glaucoma, or optic nerve damage) or the visual pathways leading to the brain. Individuals who are completely blind have no visual perception at all.

Option B: Homonymous hemianopia (HH) involves loss of vision on the same side of the visual field in both eyes. HH is characteristically seen in lesions involving the visual pathway posterior to the optic chiasma.

Option C: Binasal hemianopia is seen in lesions affecting bilateral lateral chiasma where the temporal fibres of both the retina continue forward without crossing. This affects vision in the nasal quadrants of both eyes.

Solution to Question 9:

A red nodule on the sclera, as shown in the image, with a history of recurrence, suggests a diagnosis of nodular episcleritis. It is often associated with systemic conditions like rheumatoid arthritis.

Episcleritis is benign, recurrent, inflammation of the episclera, involving the overlying Tenon's capsule but not the underlying sclera. It is more common in women than in men. It is an allergic reaction to an endogenous toxin. Clinically, two types of presentations may occur:

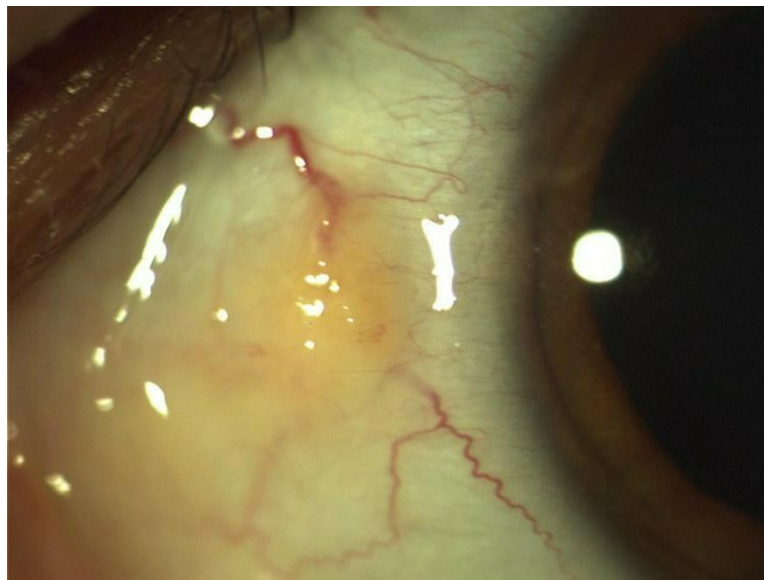
- Simple episcleritis is characterized by sectorial or diffuse inflammation of the episclera.
- Nodular episcleritis is characterized by a pink or purple flat nodule surrounded by injection, usually situated 2-3 mm away from the limbus. The nodule is firm, and tender and can be moved separately from the sclera, and the overlying conjunctiva also moves freely.

Management of episcleritis typically includes topical NSAIDs, corticosteroid eye drops, cold compresses, and artificial tears. For recurrent cases, systemic NSAIDs may be required.

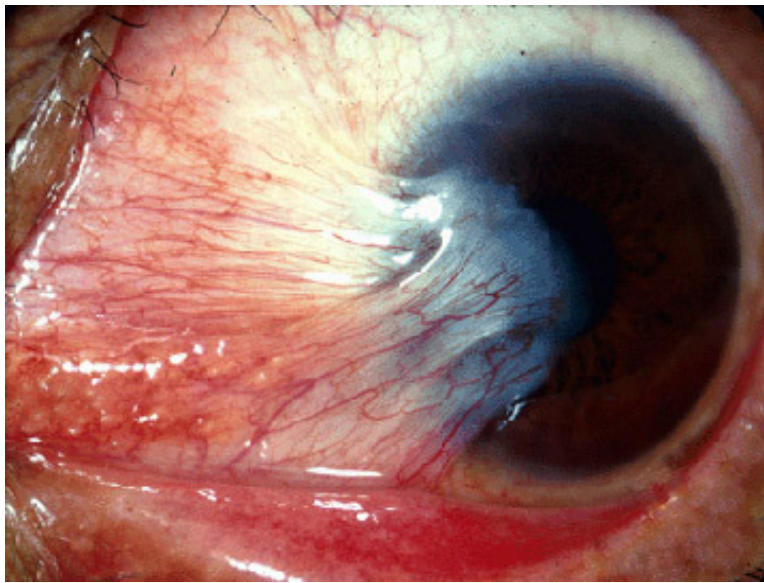
2.5% phenylephrine can be used to differentiate episcleritis and scleritis. Since the vessels of the episclera are superficial, they blanch, whereas, in scleritis, the vessels do not blanch.

Other options:

Option A: Pinguecula is a degenerative condition of the conjunctiva, characterized by the formation of a yellowish-white patch on the bulbar conjunctiva near the limbus. It usually presents as a triangular patch on the nasal side, with the triangle's apex commonly pointed away from the cornea.



Option C: Pterygium is a wing-shaped, triangular fibrovascular ingrowth of degenerative conjunctival tissue that can encroach onto the cornea. It is a response to prolonged exposure to ultraviolet rays, dry heat, etc. This condition can be treated symptomatically with tear substitutes for dry eye symptoms and patients are advised to avoid ultraviolet rays.



Option D: Scleritis refers to an inflammation of the sclera proper. It usually occurs in elderly patients (40-70 years) involving females more than males. It is frequently associated with an underlying systemic inflammatory condition. The patient presents with moderate to severe pain, redness, lacrimation, photophobia, and diminution of vision.

Solution to Question 10:

The visual field defect in the given image is the right homonymous hemianopia.

Right homonymous hemianopia, marked by loss of vision in the right half of the visual field of both eyes, indicates a lesion in the left optic tract. This occurs because the optic tract transmits visual information from the left nasal retina and right temporal retina, and disruption leads to right-sided visual field loss.

This condition highlights a post-chiasmal visual pathway issue, primarily affecting the optic tract, which processes contralateral visual fields. Common causes of lesions include pituitary adenomas, craniopharyngiomas, optic thalamus tumours, and aneurysms of the superior cerebellar or posterior cerebral arteries, as well as demyelinating diseases and infarctions.

Complete homonymous hemianopia affects the complete visual hemifields of both eyes due to large lesions posterior to the optic chiasma. In contrast, smaller lesions can cause incomplete homonymous hemianopia, categorized as congruous or incongruous based on the similarity of defects in each half field. Lesions in the immediate post-chiasmal region often produce incongruent defects, while more posterior lesions in the occipital cortex result in congruous homonymous hemianopia.

Altitudinal hemianopia is defined as hemianopia affecting the upper or lower half of the visual field. It is most commonly seen in anterior ischemic optic neuropathy (AION). Also seen in suprasellar tumors and occipital lobe lesions.

Solution to Question 11:

Given the patient's symptoms—acute pain, decreased vision in the left eye, circumcorneal congestion, irregularly shaped pupil and keratic precipitates, as shown in the image—the most likely diagnosis is acute anterior uveitis. This condition, also known as iritis, involves inflammation of the iris and anterior chamber.

Key features of acute anterior uveitis include unilateral eye pain, circumcorneal redness, photophobia, and reduced visual acuity. The pupil may appear irregular or constricted, and keratic precipitates (KP) on the corneal endothelium are pathognomonic for this condition. Inflammation leads to pseudo-rubeosis iridis and aqueous flare, a hazy appearance in the anterior chamber due to inflammatory exudate.

Anterior uveitis involves inflammation of the uvea, which can cause sluggish pupillary reactions and a "muddy" appearance of the iris. Aqueous flare is often the earliest sign seen during slit-lamp examination, alongside Koeppe and Busacca nodules and potential posterior synechiae formation.

Management focuses on reducing inflammation and managing symptoms, typically using topical corticosteroids and cycloplegics like homatropine or atropine to relieve pain and prevent synechiae. Monitoring intraocular pressure is also essential, as cycloplegics are contraindicated in angle-closure glaucoma due to the risk of exacerbating the condition.

Other options:

Option A: Angle-closure glaucoma involves a sudden increase in intraocular pressure due to blockage of the anterior chamber angle. It is characterized by severe eye pain, blurred vision, and a mid-dilated pupil, with diffuse redness rather than the circumcorneal congestion seen in uveitis.

Option C: Acute conjunctivitis typically presents with diffuse redness, discharge, and mild irritation, without significant pupil changes or severe vision loss.

Option D: Episcleritis features sectoral or diffuse redness with mild tenderness but does not cause severe pain, irregular pupil, or substantial visual changes seen in anterior uveitis.

Solution to Question 12:

The given clinical scenario, supported by the image of the mid-dilated pupil, with a shallow anterior chamber (AC), is suggestive of acute angle-closure glaucoma and the appropriate management for this patient includes intravenous mannitol and topical medications like pilocarpine, which is given to reduce the IOP, and constrict the pupil and help relieve the acute angle closure attack.

The primary goal in managing Primary Angle-Closure Glaucoma (PACG) is to prevent optic nerve damage by reducing IOP. In acute attacks, treatment involves aggressively lowering IOP using intravenous mannitol and oral acetazolamide to quickly control symptoms. Topical medications such as pilocarpine, beta-blockers, cholinergic agents, and carbonic anhydrase inhibitors are also used to address the acute episode.

Once the immediate crisis is managed, YAG laser iridotomy (YAG PI) is performed to create an alternate drainage pathway for the aqueous humor, both in the affected eye and the fellow eye. For ongoing management, patients with PACG may require glaucoma medications to maintain target IOP. In cases where maximal medical therapy fails, surgical options such as trabeculectomy

or glaucoma drainage devices may be necessary, particularly if there is progressive visual field loss or non-compliance with treatment. Cyclodestructive procedures may be considered for patients with a painful, blind eye or those with recurrent surgical failures.

Acute primary angle-closure glaucoma refers to the occlusion of the trabecular meshwork by the peripheral iris (iridotrabecular contact [ITC]), obstructing aqueous outflow.

Angle closure can be primary when it occurs in an anatomically predisposed eye or secondary to another ocular or systemic factor.

Classification:

Symptoms: Pain, redness, photophobia, impaired vision, watering

Signs:

- Corneal edema- coloured halos
- Conjunctiva- congestion, chemosis
- IOP- elevated 40–70 mm Hg
- AC very shallow
- Pupil- vertically oval, mid-dilated
- Angle- completely closed, Grade o

Primary angle-closure suspect (PACS)	Gonioscopy shows posterior meshwork ITC in three or more quadrants but no peripheral anterior synechiae (PAS). Intraocular pressure (IOP), optic disc, and visual field are normal.No PAS are seenThe risk of PACG at 5 years may be around 30%.
Primary angle closure (PAC)	Gonioscopy shows three or more quadrants of ITC with raised IOP and/or PAS or excessive pigment smudging on the TM.Normal optic disc and field.
Primary angle-closure glaucoma (PACG)	ITC in three or more quadrants, with glaucomatous optic neuropathy.Optic nerve damage from an episode of severe IOP elevation, such as acute angle closure, may not appear as typical glaucomatous cupping.

Solution to Question 13:

The image shows tiny lesions at the limbus, indicating Hebert's pits. This finding is seen in trachoma, which is caused by *Chlamydia trachomatis*, more specifically the trachoma inclusion conjunctivitis (TRIC) serovars A, B, Ba, C.

Trachoma is a form of chronic infective conjunctivitis and is one of the leading causes of blindness. It has an 'active' inflammatory stage and a 'cicatricial' chronic stage.

The active inflammatory stage is more prevalent in children and presents with nonspecific symptoms of ocular irritation, foreign body sensation, and discharge (often mild symptoms). Examination findings at this stage show:

- Conjunctival follicles contain Leber cells (multinucleated lymphocytes)
- corneal Herbert's follicles (similar to conjunctival follicles)

Cicatricial chronic stage is seen in middle age. And when the chronic infection with sequelae sets in, the patient complains of pain, lacrimation, photophobia, and later blurring, and finally loss of vision. Conjunctiva shows concretions (whitish deposits of inspissated mucus in depressions called glands of Henle). Corneal signs include:

- Herbert pits (pitted scars in the limbal which are remnants of healed Herbert follicles)
- corneal opacity is seen in later stages.

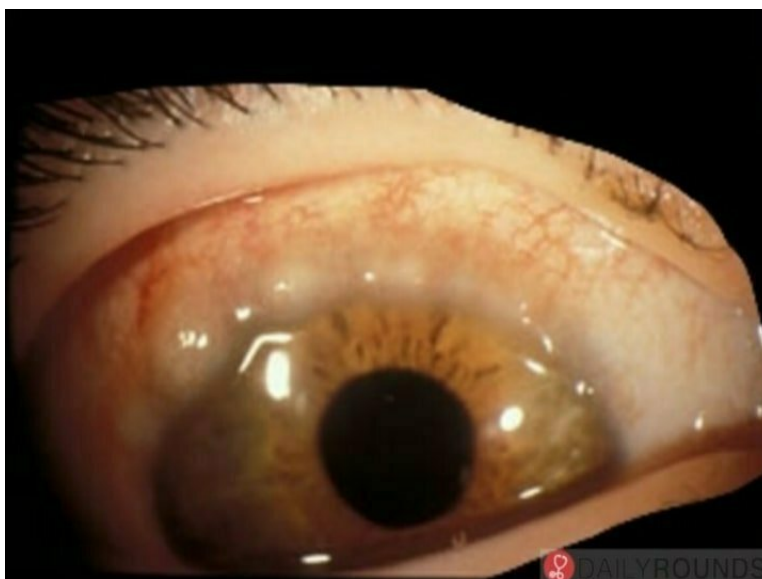
There can be associated trichiasis (when at least one eyelash rubs the eyeball) and entropion.

At present WHO grading with 'FISTO' is in use.

Trachoma is treated according to the WHO classification using the SAFE strategy (Surgery, Antibiotics, Facial hygiene, and Environmental improvement). Azithromycin is the antibiotic of choice, and surgery involves the correction of trichiasis and entropion. In case of significant corneal scarring, penetrating keratoplasty is done.

Other options:

Option B: Horner–Trantas dots (image below) are distinct white, chalky, gelatinous nodules made up of eosinophils and epithelial debris that appear at the limbus, and they are characteristic of limbal vernal keratoconjunctivitis (VKC).

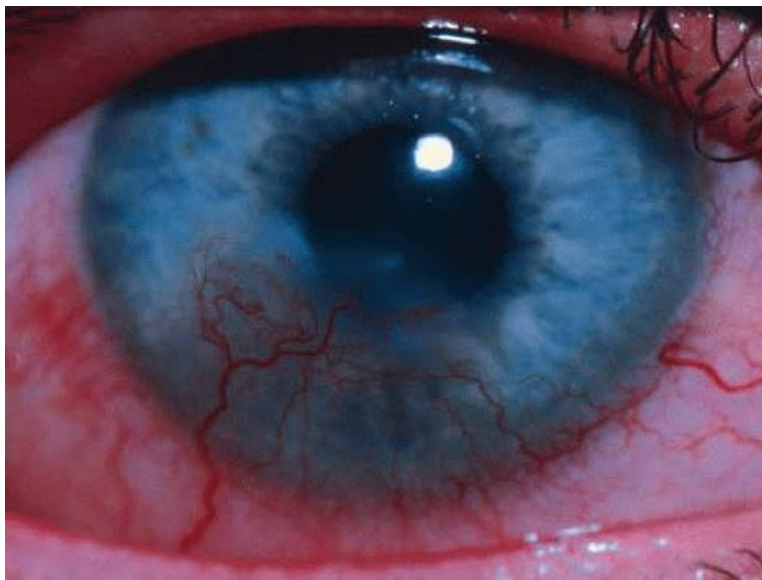


Option C: In giant papillary conjunctivitis (GPC), gelatinous papillae initially appear as small, elevated nodules on the conjunctiva. These papillae eventually grow larger, forming macropapillae and giant papillae, and may be accompanied by Trantas dots at the limbus.

The image below shows giant papillae in the conjunctiva:



Option D: Progressive pannus in trachoma involves the infiltration of the cornea with new blood vessels developing in the upper part, lying between the epithelium and Bowman's membrane. As the condition advances, these superficial vessels contribute to the destruction of Bowman's membrane, with corneal infiltration occurring before significant vascularization.



Solution to Question 14:

The given clinical history of orbital pain in a contact lens wearer and the demonstration of corneal ulcer and keratitis with scrapings showing an organism with spine-like pseudopodia suggest Acanthamoeba keratitis and the organism implicated in causing this patient's condition is Acanthamoeba.

Acanthamoeba exists as an active trophozoite form and a resistant cystic form, both of which are infective. The large, 20–50 μm trophozoite form is distinguished by spine-like pseudopodia called acanthopodia. Human beings acquire infection by inhalation of cysts or trophozoites, ingestion of cysts or through traumatized skin or eyes. Following inhalation, the trophozoites reach the lungs and invade the central nervous system through the bloodstream, producing granulomatous amebic encephalitis (GAE). It presents chiefly as two chronic conditions: (1) keratitis and (2) encephalitis.

Acanthamoeba keratitis is an infection of the eyes that typically develops from the entry of the amebic cyst through abrasions on the cornea. Most of the cases are associated with the use of contact lenses and individuals who swim or bathe in contaminated water.

Patients usually present with photophobia, excessive tearing, redness, and foreign body sensation, which can be bilateral in contact lens users. Pain is disproportionate to the degree of keratitis due to perineural invasion. It presents as a ring lesion or ring stromal infiltrate as shown below.

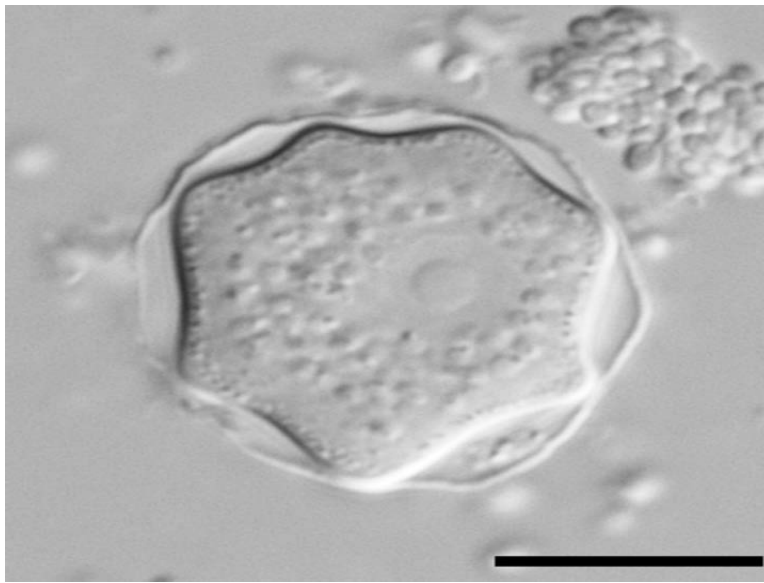


 DAILYROUNDS

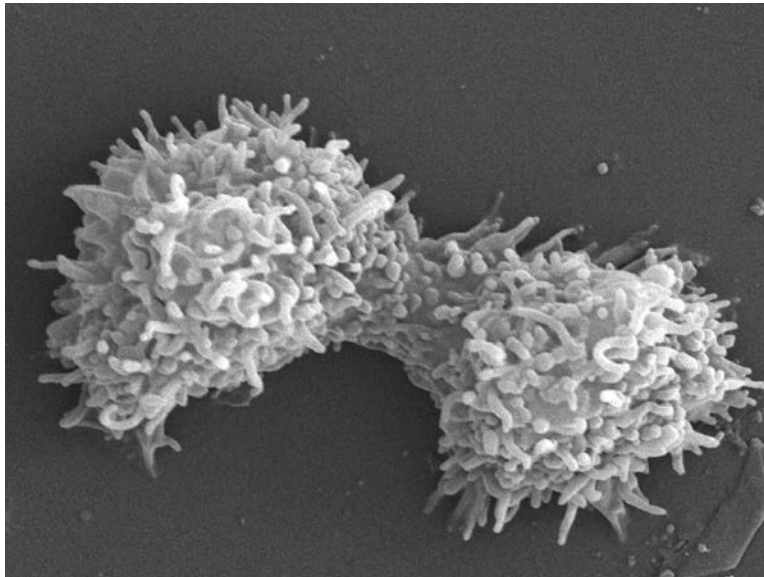
Diagnosis of amoebic keratitis is made by the demonstration of the cyst in corneal scrapings by wet mount, histology, and culture. The cysts are smaller (12-25 micrometres) and polygonal or star-shaped.

Topical administration of biguanides or chlorhexidine with or without diamidine agents can be used for the management of acanthamoeba keratitis. In severe cases penetrating keratoplasty can be done.

Polygonal cysts of acanthamoeba in interface contrast microscopy:



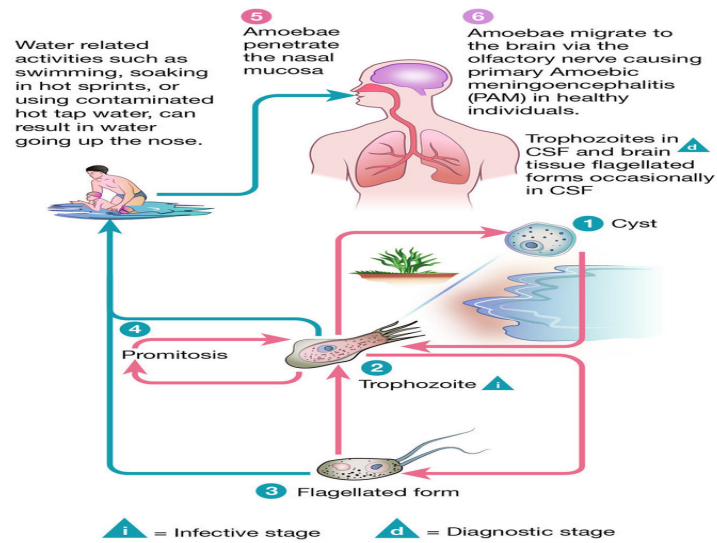
Spine like pseudopodia of acanthamoeba:



Other options:

Option A: Balamuthia is a leptomixid free-living amoeba, reported to cause GAE. It exists in the ameboid trophozoite stage and is not associated with acanthopodia as seen in Acanthamoeba species.

Option C: The electron microscopy image of the infective form of Naegleria fowleri shows rounded pseudopodia called lobopodia. The infection usually follows swimming in freshwater rivers or ponds. The amoebae invade the nasal mucosa and pass through the olfactory nerve branches in the cribriform plate into the meninges. It is mainly associated with primary amoebic meningoencephalitis. The disease advances rapidly, causing fever, headache, vomiting, stiff neck, ataxia, seizure and coma.



Option D: *Staphylococcus aureus* is a common bacterium(not associated with spine-like pseudopodia), that can cause phlyctenular keratoconjunctivitis, characterized by eye redness, pain, and the development of a white-pinkish nodule on the conjunctiva. This condition results from a delayed hypersensitivity reaction to the bacterial proteins.